Dell PowerFlex 4.5.x

CLI Reference Guide

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DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

This document provides a reference for using the PowerFlex command line interface (CLI) to work with your system.

The PowerFlex CLI is intended for users who want to interactively manage a PowerFlex system, or to use commands in scripts for automating routine tasks.

For additional PowerFlex software documentation, go to PowerFlex software technical documentation.

2

Revision history

Table 1. Revisions

Date	Document revision	Description of changes
August 2026	1.7	Added procedure to rollback V-Tree migration
March 2026	1.6	Technical updates
October 2024	1.5	Technical updates
August 2024	1.4	Technical updates
March 2024	1.3	Technical updates
January 2024	1.2	Editorial and technical updates
November 2023	1.1	Added new commands for: • Alerts • Alert templates • Directory services • Dispatch destination • Dispatch policies • Dispatch sources • Events • File DNS servers • File ftp servers • File interface routes • File interfaces • File Kerberos servers • File LDAP servers • File NDMP servers • File NIS servers • SMB servers • SMB shares • SMTP services • Login banners • File user quotas • File user checkers • Files • Groups • Users • Roles • Protection policies • Snapshot rules • NFS exports • PowerFlex file services
September 2023	1.0	Initial release

I

CLI

This section explains the basic concepts of using the PowerFlex CLI, including how to access the SCLI and log in.

3

Accessing CLI

The `powerflex_cli` file is available in the Installer node in the folder `/opt/dell/pfmp`. You must download the bundle from the Dell support site and open it to locate the file. The PFMP bundle includes Meta Data Manager and other components. PowerFlex CLI is a tool that connects to the MDM, which in turn, sends commands or requests information about the system. You can use the PowerFlex CLI to run commands that are related to the management and operations cluster—the management layer, that is, events and alerts, and services in the management and operations cluster.

Logging in to CLI

To log in to CLI, it is mandatory to have a server address and enter the username and password. The command options are:

- `server_address`—(Mandatory) For other commands, the `server_address` is taken from the session file.
- `server_port`—(Optional) Default value: M&O Default port: 443
- `session_name`—(Optional) Default value is "default". It is a user-provided name for the session.
- `cert_file`—(Optional) It is possible to log in using an M&O certificate that exists from the prior session to the system.

The CLI commands are derived from the RESTful API. For example:

- Accessing help

```
powerflex_cli -help
```

- Running a query

```
powerflex_cli alerts query --session_token de98deb6b9927fc5 --id "3af3ab8277783029"
```

Login command examples

```
./powerflex_cli login --username scaleio --password scaleio --cert_file /tmp/
my_ca_cert.crt --server_address dellpowerflex.com
```

To log in to the CLI:

1. Copy `/tmp/my_ca_cert.crt` to `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`
2. Try to log in to dellpowerflex.com using the certificate. If the login succeeds:
   - An encrypted session file `~/.powerflex_cli/cli/default.token` is generated.

     A session token is returned.

Login command with session name

An example of a login command with a session name is given below.

```
./powerflex_cli login --username scaleio --password scaleio --cert_file /tmp/
my_ca_cert.crt --server_address dellpowerflex.com -session_name some_script_name
```

1. Copy `/tmp/my_ca_cert.crt` to `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`
2. Try to log in to dellpowerflex.com using the certificate. If the login succeeds:
   - An encrypted session file `~/.powerflex_cli/cli/some_script_name.token` is generated.

     A session token is returned.

Login command without certificate

An example of a login command with a session name is given below.

```
./powerflex_cli login --username scaleio --password scaleio --server_address
dellpowerflex.com
```

1. Try to log in to dellpowerflex.com using the certificate `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`.
2. If the certificate does not exist, the login without the certificate.

Certificate

The CLI must provide a certificate once per server. The certificates can be provided more than once too. When the certificate is provided, the CLI saves it in `~/.powerflex_cli/certificates/<server>/ca.crt`. The certificate is used again if a user logs in to the same server without a certificate.

You can set the certificate using a flag in the login command.

To get the certificate from the M&O server, run the command:

```
kubectl get secret platform-ca -o json | jq -r '.data["ca-bundle"]' | base64 --decode
```

Sessions

If the login succeeds, the CLI generates an encrypted session file and saves it in `~/.powerflex/cli/`. The name of the session token file is in the format, `<session_name>.token` if the session name is given or `default.token` if the session name is not given. The session name allows you to run more than once script that calls login command simultaneously and to be logged in to more than one server at the same time. The session file contains the name of the server, server port, and keycloak access token in an encrypted format.

Session token

The CLI returns a user session token, which is the key to open the session file. The session token is mandatory in any call to the CLI, except login. It expires if the session is idle for 30 minutes (default value). While in use, the session token is valid for 10 hours (default value). The CLI refreshes the session automatically. If a session name other than `default`, is passed in the log in command, the same session name must be passed in every command with the session token.

4

Alerts commands

This section contains commands for managing alerts.

Acknowledge alerts

Acknowledges an alert.

Parameters

-h, --help

Help for acknowledge

--id [string]

ID of an alert that needs to be acknowledged

Clear alerts

Clears an alert.

Parameters

-h, --help

Help for clear

--id [string]

ID of an alert that needs to be cleared

Disacknowledge alerts

Disacknowledges an alert

Parameters

-h, --help

Help for disacknowledge

--id [string]

ID of an alert that needs to be disacknowledged

Query alerts

Returns a list of alerts

Parameters

-h, --help

Help for querying alerts.

--filter [string]

Conditions for filtering alerts.

--id [string]

ID of alert to return.

--limit [int32]

Limit number of returned objects.

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order [string]

Sorts the result set by the properties specified. Ascending order is default if not specified

5

Alert template commands

This section contains commands for managing alert templates.

Modify alert template

Updates an alert template.

Parameters

-h, --help

Help for modifying alert templates.

--id *[string]*

ID of alert template to return.

--limit *[int32]*

limit number of returned objects.

--offset *[int]*

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order *[string]*

Sorts the result set by the properties specified. Ascending order is default if not specified.

Query alert template

Returns a list of alert templates.

Parameters

-h, --help

Help for query

--filter *[string]*

Conditions for filtering alerts

--id [string]

ID of alert template to return

--limit *[int32]*

Limit number of returned objects

--offset *[int]*

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order *[string]*

Sorts the result set by the properties specified. Ascending order is default if not specified

6

Directory service commands

This section contains commands for managing directory services.

Create directory services

Creates a directory service.

Parameters

-h, --help
Help for creating directory services

--configuration [string]

--directory_service_type [string]

--group_configuration [string]

--name [string]
Name of the directory service

--state [string]

--user_search_setting [string]

Modify directory services

Modifies a directory service by ID.

Parameters

-h, --help
Help for modifying directory services

--configuration [string]

--group_configuration [string]

--id [string]
ID of an existing directory service

--name [string]
Name of the directory service

--state [string]

--user_search_setting [string]

Query directory services

Query directory services.

Parameters

-h, --help

Help for querying directory services.

--id [string]

ID of an existing directory service.

Sync directory services

Synchronizes directory services

Parameters

-h, --help

Help for synchronizing directory services

--id [string]

Delete directory services

Deletes a directory service by ID.

Parameters

-h, --help

Help for deleting directory services

--id [string]

Id of an existing directory service

7

Dispatch destination commands

This section contains commands for managing dispatch destinations.

Query dispatch destinations

Returns a list of dispatch destinations.

Parameters

-h, --help

Help for querying dispatch destinations

--id [string]

ID of source

Modify dispatch destinations

Parameters

-h, --help

Help for modifying dispatch destinations

--id [string]

--data [string]

--description [string]

--name [string]

--type [string]

Options:

- SNMPV2C
- SNMPV3
- SYSLOG
- WEBHOOKS
- SUPPORTASSIST
- EMAIL

Delete dispatch destinations

Deletes dispatch destinations.

Parameters

-h, --help

Help for deleting dispatch destinations

--id [string]

8

Dispatch policies commands

This section contains commands for managing dispatch policies.

Query dispatch policies

Parameters

-h, --help

Help for querying dispatch policies.

--id [string]

Unique identifier of dispatch policy.

Modify dispatch policies

Parameters

-h, --help

Help for modifying dispatch policies.

--id [string]

Unique identifier of dispatch policy.

--description [string]

--domain [string]

Values are:

- MANAGEMENT
- BLOCK
- NETWORK
- PLATFORM
- COMPUTE
- FILE
- SECURITY

--name [string]

--severity [string]

--source [string]

Values are:

- SNMPV2C
- SNMPV3
- SYSLOG

Delete dispatch policies

Parameters

-h, --help

Help for deleting dispatch policies.

--id [string]

Unique identifier of dispatch policy.

9

Dispatch sources commands

This section contains commands for managing dispatch sources.

Query dispatch sources

Parameters

-h, --help

Help for querying dispatch sources.

--id [string]

Unique identifier of dispatch sources.

Modify dispatch sources

Parameters

-h, --help

Help for modifying dispatch sources.

--id [string]

Unique identifier of dispatch sources.

--data [string]

--description [string]

--name [string]

--type [string]

Values are:

- SNMPV2C
- SNMPV3
- SYSLOG

Delete dispatch sources

Parameters

-h, --help

Help for deleting dispatch sources.

--id [string]

Unique identifier of dispatch sources.

10

Event commands

This section contains commands for managing events.

Query events

Returns a list of events.

Parameters

-h, --help

Help for querying events

--id [string]

--filter [string]

Conditions for filtering alerts

--limit [int32]

Limit number of returned objects

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order [string]

Sorts the result set by the properties specified. Ascending order is default if not specified

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File commands

This section contains commands to manage various file servers.

Kerberos servers

Parameters

-h, --help

Help for file Kerberos servers.

LDAP Servers

Parameters

-h, --help

Help for file LDAP servers.

Virus checkers

Parameters

-h, --help

Help for file virus checkers.

PowerFlex file service servers

Parameters

-h, --help

Help for PowerFlex file services servers.

12

File DNS server commands

This section contains commands for managing file DNS servers.

Create file DNS servers

Create a DNS server configuration for a PowerFlex file services server. Only one object can be created per PowerFlex file services server.

Parameters

-h, --help

Help for creating DNS file servers.

--domain [string]

Name of the DNS domain, where the PowerFlex file services server does host names lookup when an FQDN is not specified in the request.

--ip_addresses [string]

The list of DNS server IP addresses. The addresses may be IPv4 or IPv6.

--nas_server_id [string]

Unique identifier of the associated PowerFlex file services server instance that uses this DNS object. Only one DNS object per PowerFlex file services server is supported.

--transport [string]

Transport used when connecting to the DNS server. Options are:

- UDP—DNS uses the UDP protocol (default)
- TCP—DNS uses the TCP protocol

Modify file DNS servers

Modify the DNS settings of a PowerFlex file services server.

Parameters

-h, --help

Help for modifying DNS file servers.

--id [string]

Unique identifier of the DNS object.

--domain [string]

Name of the DNS domain, where the PowerFlex file services server does host names lookup when an FQDN is not specified in the request.

--ip_addresses [string]

A new list of DNS server IP addresses to replace the existing list. The addresses may be IPv4 or IPv6.

--add_ip_addresses [string]

IP addresses to add to the current list. The addresses may be IPv4 or IPv6. Error occurs if an IP address already exists. Cannot be combined with ip_addresses.

--remove_ip_addresses [string]

IP addresses to remove from the current list. The addresses may be IPv4 or IPv6. Error occurs if IP address is not present. Cannot be combined with ip_addresses.

--transport [string]

Transport used when connecting to the DNS server. Options:

- UDP—DNS uses the UDP protocol (default)
- TCP—DNS uses the TCP protocol (UDP,TCP)

Query file DNS servers

Query a specific DNS settings object of a PowerFlex file services server.

Parameters

-h, --help

Help for querying DNS file servers.

--id [string]

Unique identifier of the DNS object.

Delete file DNS server

Delete the DNS settings of a PowerFlex file services server.

Parameters

-h, --help

Help for deleting DNS file servers.

--id [string]

Unique identifier of the DNS object.

13

File FTP server commands

This section contains commands for managing FTP servers.

Create file FTP servers

Create an FTP or SFTP server.

Parameters

-h, --help

Help for creating FTP or SFTP server

--audit_dir [string]

Directory of FTP/SFTP audit files. It is applicable when the value of `is_audit_enabled` is true. Logs are saved in '/' directory (default) or in a mounted file system (absolute path of the file system directory which should already exist).

--audit_max_size [int]

Maximum size of all (current plus archived) FTP or SFTP audit files, in bytes. There is a maximum of five audit files, one current audit file (ftp.log) and four archived audit files. The maximum value for this setting is 5GB (each file of 1GB) if the audit directory belongs to a user file system of the PowerFlex file services server. If the audit directory is '/', the maximum value is 5MB (each file of 1MB). The minimum value is 40kB (each file of 8KB) on any file system. It is applicable when the value of `is_audit_enabled` is true.

--default_homedir [string]

Default directory of FTP and SFTP clients who have a home directory that is not defined or accessible. It is applicable when the value of `is_homedir_limit_enabled` is false.

--groups [string]

Allowed or denied user groups, depending on the value of the is_allowed_groups attribute.

- If allowed groups exist, only users who are members of these groups and no others can connect to the PowerFlex file services server through FTP or SFTP.
- If denied groups exist, all users who are members of those groups always have access denied to the PowerFlex file services server through FTP or SFTP.
- If the list is empty, there is no restriction to the PowerFlex file services server access through FTP or SFTP based on the user group.

--hosts [string]

Allowed or denied hosts, depending on the value of the is_allowed_hosts attribute. A host is defined using its IP address. Subnets using CIDR notation are also supported.

- If allowed hosts exist, only those hosts and no others can connect to the PowerFlex file services server through FTP or SFTP.
- If denied hosts exist, they always have access denied to the PowerFlex file services server through FTP or SFTP.
- If the list is empty, there is no restriction to PowerFlex file services server access through FTP or SFTP based on the host IP address.
- The addresses may be IPv4 or IPv6.

--is_allowed_groups

Indicates whether the groups attribute contains allowed or denied user groups. Values are:

- true—groups contains allowed user groups.
- false—groups contains denied user groups.

--is_allowed_hosts

Indicates whether the hosts attribute contains allowed or denied hosts. Values are:

- true—hosts contains allowed hosts.
- false—hosts contains denied hosts.

--is_allowed_users

Indicates whether the users attribute contains allowed or denied users. Values are:

- true—users contains allowed users.
- false—users contains denied users.

--is_anonymous_authentication_enabled

Indicates whether FTP clients can be authenticated anonymously. Values are:

- true—Anonymous user name is accepted.
- false—Anonymous user name is not accepted.

--is_audit_enabled

Indicates whether the activity of FTP and SFTP clients is tracked in audit files. Values are:

- true—FTP/SFTP activity is tracked.
- false—FTP/SFTP activity is not tracked.

is_ftp_enabled

Indicates whether the FTP server is enabled on the PowerFlex file services server specified in the `nasServer` attribute. Values are:

- true—FTP server is enabled on the specified PowerFlex file services server.
- false—FTP server is disabled on the specified PowerFlex file services server.

--is_homedir_limit_enabled

Indicates whether an FTP or SFTP user access is limited to the home directory of the user. Values are:

- true—An FTP or SFTP user can access only the home directory of the user.
- false—FTP and SFTP users can access any PowerFlex file services server directory, according to PowerFlex file services server permissions.

--is_sftp_enabled

Indicates whether the SFTP server is enabled on the PowerFlex file services server specified in the `nasServer` attribute. Values are:

- true—SFTP server is enabled on the specified PowerFlex file services server.
- false—SFTP server is disabled on the specified PowerFlex file services server.

--is_smb_authentication_enabled

Indicates whether FTP and SFTP clients can be authenticated using an SMB user name. These user names are defined in a Windows domain controller, and their formats are user@domain or domain\user. Values are:

- true—SMB user names are accepted for authentication.
- false—SMB user names are not accepted for authentication.

--is_unix_authentication_enabled

Indicates whether FTP and SFTP clients can be authenticated using a Unix user name. Unix user names are defined in LDAP, NIS servers or in local passwd file. Values are:

- true—Unix user names are accepted for authentication.
- false—Unix user names are not accepted for authentication.

--is_smb_authentication_enabled

Indicates whether FTP and SFTP clients can be authenticated using an SMB user name. These user names are defined in a Windows domain controller, and their formats are user@domain or domain\user. Values are:

- true—SMB user names are accepted for authentication.
- false—SMB user names are not accepted for authentication.

--message_of_the_day [string]

Message of the day displayed on the console of FTP clients after their authentication. The length of this message is limited to 511 bytes of UTF-8 characters, and the length of each line is limited to 80 bytes.

--nas_server_id [string]

Unique identifier of the PowerFlex file services server that is configured with the FTP server.

--users [string]

Allowed or denied users, depending on the value of the is_allowed_user attribute.

--nas_server_id [string]

Unique identifier of the PowerFlex file services server that is configured with the FTP server.

- If allowed users exist, only those users and no others can connect to the PowerFlex file services server through FTP or SFTP.
- If denied users exist, they have always access denied to the PowerFlex file services server through FTP or SFTP.
- If the list is empty, there is no restriction to the PowerFlex file services server access through FTP or SFTP based on the user name.

--welcome_message [string]

Welcome message displayed on the console of FTP and SFTP clients before their authentication. The length of this message is limited to 511 bytes of UTF-8 characters, and the length of each line is limited to 80 bytes.

Modify file FTP server

Modify FTP or SFTP server settings.

Parameters

-h, --help

Help for modifying FTP or SFTP server

--add_groups [string]

Groups to add to the current groups. Error occurs if the group already exists. Cannot be combined with groups.

--add_hosts [string]

Host IP addresses to add to the current hosts. The addresses may be IPv4 or IPv6. Error occurs if the IP address already exists. Cannot be combined with hosts.

--add_users [string]

Users to add to the current users. Error occurs if the user already exists. Cannot be combined with users.

--audit_dir [string]

Directory of FTP/SFTP audit files. Logs are saved in '/' directory (default) or in a mounted file system (Absolute path of the file system directory which should already exist). It is applicable when the value of `is_audit_enabled` is true.

--audit_max_size [string]

Applies when the value of is_audit_enabled is true. Maximum size of all (current plus archived) FTP/SFTP audit files, in bytes. There is a maximum of five audit files, one current audit file (ftp.log) and four archived audit files. The maximum value for this setting is 5GB (each file of 1GB) if the audit directory belongs to a user file system of the PowerFlex file services server. If the audit directory is '/', the maximum value is 5MB (each file of 1MB). The minimum value is 40kB (each file of 8KB) on any file system.

--default_homedir [string]

Default directory of FTP and SFTP clients that have a home directory which is not defined or accessible. Applies when the value of is_homedir_limit_enabled is false.

--groups [string]

Allowed or denied user groups, depending on the value of the `is_allowed_groups` attribute.

- If allowed groups exist, only users who are members of these groups and no others can connect to the PowerFlex file services server through FTP or SFTP.
- If denied groups exist, all users who are members of those groups always have access denied to the PowerFlex file services server through FTP or SFTP.
- If the list is empty, there is no restriction to the PowerFlex file services server access through FTP or SFTP based on the user group.

--hosts [string]

Allowed or denied hosts, depending on the value of the is_allowed_hosts attribute. A host is defined using its IP address. Subnets using CIDR notation are also supported.

- If allowed hosts exist, only those hosts and no others can connect to the PowerFlex file services server through FTP or SFTP.
- If denied hosts exist, they always have access denied to the PowerFlex file services server through FTP or SFTP.
- If the list is empty, there is no restriction to PowerFlex file services server access through FTP or SFTP based on the host IP address.
- The addresses may be IPv4 or IPv6.

--id [string]

Unique identifier of the FTP/SFTP server object.

--is_allowed_groups [string]

Indicates whether the activity of FTP and SFTP clients is tracked in audit files. Values are:

- true—groups contains allowed user groups.
- false—groups contains denied user groups.

--is_allowed_hosts [string]

Indicates whether the hosts attribute contains allowed or denied hosts. Values are:

- true—hosts contains allowed hosts.
- false—hosts contains denied hosts.

--is_allowed_users [string]

Indicates whether the users attribute contains allowed or denied users. Values are:

- true—users contains allowed users.
- false—users contains denied users.

--is_anonymous_authentication_enabled [string]

Indicates whether FTP clients can be authenticated anonymously. Values are:

- true—Anonymous user name is accepted.
- false—Anonymous user name is not accepted.

--is_audit_enabled [string]

Indicates whether the activity of FTP and SFTP clients is tracked in audit files. Values are:

- true—FTP/SFTP activity is tracked.
- false—FTP/SFTP activity is not tracked.

--is_ftp_enabled

Indicates whether the FTP server is enabled on the PowerFlex file services server specified in the `nasServer` attribute. Values are:

- true—FTP server is enabled on the specified PowerFlex file services server.
- false—FTP server is disabled on the specified PowerFlex file services server.

--is_homedir_limit_enabled

Indicates whether an FTP or SFTP user access is limited to the home directory of the user. Values are:

- true—An FTP or SFTP user can access only the home directory of the user.
- false—FTP and SFTP users can access any PowerFlex file services server directory, according to PowerFlex file services server permissions.

--is_sftp_enabled

Indicates whether the SFTP server is enabled on the PowerFlex file services server specified in the `nasServer` attribute. Values are:

- true—SFTP server is enabled on the specified PowerFlex file services server.
- false—SFTP server is disabled on the specified PowerFlex file services server.

--is_smb_authentication_enabled

Indicates whether FTP and SFTP clients can be authenticated using an SMB user name. These user names are defined in a Windows domain controller, and their formats are user@domain or domain\user. Values are:

- true—SMB user names are accepted for authentication.
- false—SMB user names are not accepted for authentication.

--is_unix_authentication_enabled

Indicates whether FTP and SFTP clients can be authenticated using a UNIX user name. UNIX user names are defined in LDAP, NIS servers or in local passwd file. Values are:

- true—UNIX user names are accepted for authentication.
- false—UNIX user names are not accepted for authentication.

--message_of_the_day [string]

Message of the day displayed on the console of FTP clients after their authentication. The length of this message is limited to 511 bytes of UTF-8 characters, and the length of each line is limited to 80 bytes.

--remove_groups [string]

Groups to remove from the current groups. Error occurs if the group is not present. Cannot be combined with groups.

--remove_hosts [string]

Host IP addresses to remove from the current hosts. The addresses may be IPv4 or IPv6. Error occurs if the IP address is not present. Cannot be combined with hosts.

--remove_users [string]

Users to remove from the current users. Error occurs if the user is not present. Cannot be combined with users.

--users [string]

Allowed or denied users, depending on the value of the is_allowed_users attribute.

- If allowed users exist, only those users and no others can connect to the PowerFlex file services server through FTP or SFTP.
- If denied users exist, they always have access denied to the PowerFlex file services server through FTP or SFTP.
- If the list is empty, there is no restriction to the PowerFlex file services server access through FTP or SFTP based on the user name.

--welcome_message [string]

Welcome message displayed on the console of FTP and SFTP clients before their authentication. The length of this message is limited to 511 bytes of UTF-8 characters, and the length of each line is limited to 80 bytes.

Query file FTP servers

Query FTP or SFTP instances.

Parameters

-h, --help

Help for querying an FTP or SFTP server

--id [string]

Unique identifier of the FTP/SFTP Server object.

Delete file FTP servers

Delete an FTP or SFTP Server.

Parameters

-h, --help

Help for deleting an FTP or SFTP server

--id [string]

Unique identifier of the FTP/SFTP Server object.

14

File interface commands

This section contains commands for managing file interfaces.

Create file interfaces

Create a Kerberos configuration. The operation fails if a Kerberos configuration exists.

Parameters

-h, --help

Help for creating file interface.

--kdc_addresses [string]

Fully qualified domain names of the Kerberos key distribution center (KDC) servers. IPv4 and IPv6 addresses are not supported.

--nas_server_id [string]

Unique identifier of the associated PowerFlex file services server instance that uses this Kerberos object. Only one Kerberos object per PowerFlex file services server is supported.

--port_number [int32]

KDC servers TCP port.

--realm [string]

Realm name of the Kerberos service.

Modify file interfaces

Modify the Kerberos service settings of a PowerFlex file services server.

Parameters

-h, --help

Help for modifying file interface.

--add_kdc_addresses [string]

Fully qualified domain names of the Kerberos key distribution center (KDC) servers to add to the current list. Error occurs if name already exists. Cannot be combined with kdc_addresses. IPv4 and IPv6 addresses are not supported.

--id [string]

Unique identifier of the Kerberos service object.

--kdc_addresses [string]

Fully qualified domain names of the Kerberos key distribution center (KDC) servers. IPv4 and IPv6 addresses are not supported.

--port_number [int32]

KDC servers TCP port.

--realm [string]

Realm name of the Kerberos service.

--remove_kdc_addresses [string]

Fully qualified domain names of the Kerberos key distribution center (KDC) servers to remove from the current list. Error occurs if name is not in the existing list. Cannot be combined with `kdc_addresses`. IPv4 and IPv6 addresses are not supported.

Query file interfaces

Query of the Kerberos service settings of PowerFlex file services servers.

Parameters

`-h, --help`

Help for querying file interface.

`--id [string]`

Unique identifier of the Kerberos service object.

Delete file interfaces

Delete Kerberos configuration of a PowerFlex file services server.

Parameters

`-h, --help`

Help for deleting file interface.

`--id [string]`

Unique identifier of the Kerberos service object.

15

File interface route commands

This section contains commands for managing file interface routes.

Create file interface routes

Create and configure a new file interface route.

Parameters

-h, --help

Help for creating file interface route. There are three route types— subnet, default, and host.

`-h, --help help for file-interface-routes`

- The default route establishes a static route to a default gateway. To create a default route, provide only the default gateway IP address. `-h, --help help for file-interface-routes`
- The host route establishes a static route to a specific host. To create a host route, provide the IP address of the specific host in the destination field, and the gateway. `-h, --help help for file-interface-routes`
- The subnet route establishes a static route to a particular subnet. To create a subnet route, provide the IP address of the target subnet in the destination, the prefix length for that subnet, and the gateway. `-h, --help help for file-interface-routes`

--file_interface_id [string]

Unique identifier of the associated file interface.

--destination [int32]

IPv4 or IPv6 address of the target network node based on the specific route type. Values are:

- For a default route, there is no value because the system will use the specified gateway IP address.
- For a host route, the value is the host IP address.
- For a subnet route, the value is a subnet IP address.

--gateway [string]

IP address of the gateway associated with the route.

--prefix_length [int32]

IPv4 or IPv6 prefix length for the route.

Modify file interface routes

Modify file interface route settings.

Parameters

-h, --help

Help for modifying file interface route.

--id [string]

Unique identifier of the file interface route object.

--destination [string]

IPv4 or IPv6 address of the target network node based on the specific route type. Values are:

- For a default route, there is no value because the system will use the specified gateway IP address.
- For a host route, the value is the host IP address.
- For a subnet route, the value is a subnet IP address.

--id [string]

Unique identifier of the file interface route object.

--prefix_length [int32]

IPv4 or IPv6 prefix length for the route.

Query file interface routes

Query file interface routes.

Parameters

-h, --help

Help for querying file interface route.

--id [string]

Unique identifier of the file interface route object.

Delete file interface routes

Delete file interface route.

Parameters

-h, --help

Help for deleting file interface route.

--id [string]

Unique identifier of the file interface route object.

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File Kerberos server commands

This section contains commands for managing Kerberos servers.

Create file Kerberos servers

Create a Kerberos configuration. The operation fails if a Kerberos configuration exists.

Parameters

-h, --help

Help for creating Kerberos servers.

--kdc_addresses [string]

Fully qualified domain names of the Kerberos key distribution center (KDC) servers. IPv4 and IPv6 addresses are not supported.

--nas_server_id [string]

Unique identifier of the associated PowerFlex file services server instance that uses this Kerberos object. Only one Kerberos object per PowerFlex file services server is supported.

--port_number [int32]

KDC servers TCP port.

--realm [string]

Realm name of the Kerberos service.

Modify file Kerberos servers

Modify the Kerberos service settings of a PowerFlex file services server.

Parameters

-h, --help

Help for querying Kerberos servers.

--id [string]

Unique identifier of the Kerberos service object.

--add_kdc_addresses [string]

Fully Qualified domain names of the Kerberos Key Distribution Center (KDC) servers to add to the current list. Error occurs if name already exists. Cannot be combined with kdc_addresses. IPv4 and IPv6 addresses are not supported.

--kdc_addresses [string]

Fully Qualified domain names of the Kerberos Key Distribution Center (KDC) servers. IPv4 and IPv6 addresses are not supported.

--port_number [int32]

KDC servers TCP port.

--realm [string]

Realm name of the Kerberos service.

--remove_kdc_addresses [string]

Fully Qualified domain names of the Kerberos Key Distribution Center (KDC) servers to remove from the current list. Error occurs if name is not in the existing list. Cannot be combined with kdc_addresses. IPv4 and IPv6 addresses are not supported.

Query file Kerberos servers

Query of the Kerberos service settings of PowerFlex file services servers.

Parameters

-h, --help

Help for querying Kerberos servers.

--id [string]

Unique identifier of the Kerberos service object.

Delete file Kerberos servers

Delete Kerberos configuration of a PowerFlex file services server.

Parameters

-h, --help

Help for deleting Kerberos servers.

--id [string]

Unique identifier of the Kerberos service object.

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File LDAP server commands

This section contains commands for managing file LDAP servers.

Create file LDAP servers

Create an LDAP service on a PowerFlex file services server. Only one LDAP service object can be created per PowerFlex file services server.

Parameters

-h, --help

Help for creating file LDAP servers.

--addresses [string]

The list of LDAP server IP addresses. The addresses may be IPv4 or IPv6.

--authentication_type [string]

Authentication type for the LDAP server. Options are:

- ANONYMOUS—ANONYMOUS authentication means no authentication occurs and the PowerFlex file services server uses an anonymous login to access the LDAP-based directory server.
- SIMPLE—SIMPLE authentication means the PowerFlex file services server must provide a bind distinguished name and password to access the LDAP-based directory server.
- KERBEROS—KERBEROS authentication means the PowerFlex file services server uses a KDC to confirm the identity when accessing the Active Directory. (ANONYMOUS,SIMPLE,KERBEROS)

--base_DN [string]

Name of the LDAP base DN. Base distinguished name (BDN) of the root of the LDAP directory tree. The appliance uses the DN to bind to the LDAP service and locate in the LDAP directory tree to begin a search for information. The base DN can be expressed as a fully-qualified domain name or in X.509 format by using the attribute dc=. For example, if the fully-qualified domain name is mycompany.com, the base DN is expressed as dc=mycompany,dc=com.

--bind_DN [string]

Bind distinguished name (DN) to be used when binding.

--bind_password [string]

The associated password to be used when binding to the server.

--is_smb_account_used [string]

Indicates whether SMB authentication is used to authenticate to the LDAP server. Values are:

- True—Indicates that the SMB settings are used for Kerberos authentication.
- False—Indicates that Kerberos uses its own settings.

--is_verify_server_certificate [string]

Indicates whether certification authority certificate is used to verify the LDAP server certificate for secure SSL connections. Values are:

- True—verifies the certificate of the LDAP server.
- False—does not verify the certificate of the LDAP server.

--is_verify_server_certificate [string]

Indicates whether certification authority certificate is used to verify the LDAP server certificate for secure SSL connections. Values are:

- True—verifies the certificate of the LDAP server.
- False—does not verify the certificate of the LDAP server.

--nas_server_id [string]

Unique identifier of the associated PowerFlex file services server instance that will use this LDAP object. Only one LDAP object per PowerFlex file services server is supported.

--password [string]

The associated password for Kerberos authentication.

--port_number [int32]

The TCP/IP port used by the PowerFlex file services server to connect to the LDAP servers. The default port number for LDAP is 389 and LDAPS is 636.

--principal [string]

Specifies the principal name for Kerberos authentication.

--profile_DN [string]

For an iPlanet LDAP server, specifies the DN of the entry with the configuration profile.

--protocol [string]

Indicates whether the LDAP protocol uses SSL for secure network communication. SSL encrypts data over the network and provides message and server authentication.

- LDAP - LDAP protocol without SSL.
- LDAPS - (Default) LDAP protocol with SSL. When you enable LDAPS, make sure to specify the appropriate LDAPS port (usually port 636) and to upload an LDAPS trust certificate to the LDAP server. (LDAP,LDAPS)

--realm [string]

Specifies the realm name for Kerberos authentication.

Modify file LDAP servers

Modify the LDAP settings object of a PowerFlex file services server.

Parameters

-h, --help

Help for modifying file LDAP server.

--id [string]

Unique identifier of the LDAP settings object ID.

--add_addresses [string]

IP addresses to add to the current server IP addresses list. The addresses may be IPv4 or IPv6. Error occurs if an IP address already exists in the addresses list. Cannot be combined with addresses.

--addresses [string]

The list of LDAP server IP addresses. The addresses may be IPv4 or IPv6.

--authentication_type [string]

Authentication type for the LDAP server. Options are:

- ANONYMOUS —ANONYMOUS authentication means no authentication occurs and the PowerFlex file services server uses an anonymous login to access the LDAP-based directory server.
- SIMPLE—SIMPLE authentication means the PowerFlex file services server must provide a bind distinguished name and password to access the LDAP-based directory server.
- KERBEROS—KERBEROS authentication means the PowerFlex file services server uses a KDC to confirm the identity when accessing the Active Directory. (ANONYMOUS,SIMPLE,KERBEROS)

--base_DN [string]

Bind distinguished name (DN) to be used when binding.

--bind_password [string]

The associated password to be used when binding to the server.

--is_smb_account_used [string]

Indicates whether SMB authentication is used to authenticate to the LDAP server. Values are:

- True—Indicates that the SMB settings are used for Kerberos authentication.

- False—Indicates that Kerberos uses its own settings.

--is_verify_server_certificate [string]

Indicates whether certification authority certificate is used to verify the LDAP server certificate for secure SSL connections. Values are:

- True—verifies the certificate of the LDAP server.
- False—Does not verify the certificate of the LDAP server.

--password [string]

The associated password for Kerberos authentication.

--port_number [int32]

The TCP/IP port used by the PowerFlex file services server to connect to the LDAP servers.

--principal [string]

Specifies the principal name for Kerberos authentication.

--profile_DN [string]

For an iPlanet LDAP server, specifies the DN of the entry with the configuration profile.

--protocol [string]

Indicates whether the LDAP protocol uses SSL for secure network communication. SSL encrypts data over the network and provides message and server authentication. Options are:

- LDAP—LDAP protocol without SSL.
- LDAPS—(Default) LDAP protocol with SSL. When you enable LDAPS, make sure to specify the appropriate LDAPS port (usually port 636) and to upload an LDAPS trust certificate to the LDAP server. (LDAP, LDAPS)

--realm [string]

Specifies the realm name for Kerberos authentication.

--remove_addresses [string]

IP addresses to remove from the current server IP addresses list. The addresses may be IPv4 or IPv6. Error occurs if an IP address does not exist in the addresses_list. Cannot be combined with addresses.

Query file LDAP servers

List LDAP service instances.

Parameters

-h, --help

Help for querying file LDAP server.

--id [string]

Unique identifier of the LDAP settings object.

Delete file LDAP servers

Delete the LDAP settings of a PowerFlex file services server.

Parameters

-h, --help

Help for deleting file LDAP server.

--id [string]

LDAP settings object ID.

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File NDMP server commands

This section contains commands to manage file NDMP servers.

Create file NDMP servers

Add an NDMP service configuration to a PowerFlex file services server. Only one NDMP service object can be configured per PowerFlex file services server.

Parameters

-h, --help
Help for creating file NDMP server.

--nas_server_id [string]
Unique identifier of the PowerFlex file services server to be configured with these NDMP settings.

--password [string]
Password for the NDMP service user.

--user_name [string]
User name for accessing the NDMP service.

Modify file NDMP servers

Modify an NDMP service configuration instance.

Parameters

-h, --help
Help for modifying file NDMP server.

--id [string]
Unique identifier of the NDMP service object.

--password [string]
Password for the NDMP service user.

--user_name [string]
User name for accessing the NDMP service.

Query file NDMP servers

List configured NDMP service instances.

Parameters

-h, --help
Help for querying file NDMP server.

--id [string]

Unique identifier of the NDMP service object.

Delete NDMP servers

Delete an NDMP service configuration instance of a PowerFlex file services server.

Parameters

-h, --help

Help for deleting file NDMP server.

--id [string]

Unique identifier of the NDMP service object.

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File NIS server commands

This section contains commands to manage an NIS service on a PowerFlex file service server.

Create file NIS servers

Create a NIS service on a PowerFlex file services server. Only one NIS setting object can be created per PowerFlex file services server.

Parameters

-h, --help

Help for creating NIS service on a PowerFlex file services server.

--nas_server_id [string]

Unique identifier of the associated PowerFlex file services server instance that uses this NIS service object. Only one NIS service per PowerFlex file services server is supported.

--ip_addresses [string]

The list of NIS server IP addresses.

--domain [string]

Name of the NIS domain.

Modify file NIS servers

Modify the NIS settings of a PowerFlex file services server.

Parameters

-h, --help

Help for modifying NIS service on a PowerFlex file services server.

--id [string]

Unique identifier of the NIS object.

--add_ip_addresses [string]

IP addresses to add to the current list. The addresses may be IPv4 or IPv6. Error occurs if the IP address already exists. Cannot be combined with `ip_addresses`.

--domain [string]

Name of the NIS domain.

--ip_addresses [string]

A new list of NIS server IP addresses to replace the existing list. The addresses may be IPv4 or IPv6.

--remove_ip_addresses [string]

IP addresses to remove from the current list. The addresses may be IPv4 or IPv6. Error occurs if the IP address is not present. Cannot be combined with `ip_addresses`.

Query file NIS servers

Query a specific NIS settings object of a PowerFlex file services server.

Parameters

-h, --help

Help for querying NIS service on a PowerFlex file services server.

--id [string]

Unique identifier of the NIS object.

Delete file NIS servers

Delete NIS settings of a PowerFlex file services server.

Parameters

-h, --help

Help for deleting NIS settings of a PowerFlex file services server.

--id [string]

Unique identifier of the associated PowerFlex file services server instance that uses this NIS Service object. Only one NIS Service per PowerFlex file services server is supported.

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File system commands

This section contains commands to manage file systems.

Create file systems

Create a file system.

Parameters

-h, --help

Help for creating file systems.

--access_policy [string]

File system security access policies. Each file system uses its access policy to determine how to reconcile the differences between NFS and SMB access control. Selecting an access policy determines which mechanism is used to enforce file security on the particular file system. File system access policy is set only for PROTOCOL snapshots. Options are:

- NATIVE—Native Security.
- UNIX—UNIX Security.
- WINDOWS—Windows Security. (NATIVE,UNIX,WINDOWS)

--description [string]

File system description. (255 UTF-8 characters).

--folder_rename_policy [string]

File system folder rename policies for the file system with multiprotocol access enabled. These policies control whether the directory can be renamed from NFS or SMB clients when at least one file is opened in the directory, or in one of its child directories.File system folder rename policy will be set only for PROTOCOL snapshots. Options are:

- ALL_ALLOWED—All protocols are allowed to rename directories without any restrictions.
- SMB_FORBIDDEN—A directory rename from the SMB protocol will be denied if at least one file is opened in the directory or in one of its child directories.
- ALL_FORBIDDEN—Any directory rename request will be denied regardless of the protocol used, if at least one file is opened in the directory or in one of its child directories. (ALL_ALLOWED,SMB_FORBIDDEN,ALL_FORBIDDEN)

--is_async_MTime_enabled [string]

Indicates whether asynchronous MTIME is enabled on the file system or protocol snaps that are mounted writeable. Values are:

- True—Asynchronous MTIME is enabled on the file system.
- False—Asynchronous MTIME is disabled on the file system.

--is_smb_no_notify_enabled [string]

Indicates whether notifications of changes to directory file structure are enabled. Values are:

- True—Change directory notifications are enabled.
- False—Change directory notifications are disabled.

--is_smb_notify_on_access_enabled [string]

Indicates whether file access notifications are enabled on the file system. Values are:

- True—File access notifications are enabled on the file system.
- False—File access notifications are disabled on the file system.

--is_smb_notify_on_write_enabled [string]

Indicates whether file writes notifications are enabled on the file system. Values are:

- True—File writes notifications are enabled on the file system.
- False—File writes notifications are disabled on the file system.

--is_smb_op_locks_enabled [string]

Indicates whether opportunistic file locking is enabled on the file system. Values are:

- True—Opportunistic file locking is enabled on the file system.
- False—Opportunistic file locking is disabled on the file system.

--is_smb_sync_writes_enabled [string]

Indicates whether the synchronous writes option is enabled on the file system. Values are:

- True—Synchronous writes option is enabled on the file system.
- False—Synchronous writes option is disabled on the file system.

--locking_policy [string]

File system locking policies. These policy choices control whether the NFSv4 range locks are honored. Because NFSv3 is advisory by design, this policy specifies that the NFSv4 locking feature behaves like NFSv3 (advisory mode), for backward compatiblity with applications expecting an advisory locking scheme. File system locking policy is set only for PROTOCOL snapshots. Values are:

- ADVISORY—No lock checking for NFS and honor SMB lock range only for SMB.
- MANDATORY—Honor SMB and NFS lock range (ADVISORY,MANDATORY).

--name [string]

Name of the file system. (255 UTF-8 characters).

--nas_server_id [string]

ID of the PowerFlex file services server on which the file system is mounted.

--size_total [int]

Size that the file system presents to the host or end user (bytes). Value is always rounded up to next MB.

--smb_notify_on_change_dir_depth [int32]

Lowest directory level to which the enabled notifications apply, if any.

--storage_pool_id [string]

ID of the storage pool on which the file system has to be provisioned.

Modify file systems

Modify a file system.

Parameters

-h, --help

Help for modifying file systems.

--id [string]

File system ID.

--access_policy [string]

File system security access policies. Each file system uses its access policy to determine how to reconcile the differences between NFS and SMB access control. Selecting an access policy determines which mechanism is used to enforce file security on the particular file system. File system access policy is set only for PROTOCOL snapshots. Values are:

- NATIVE—Native Security.
- UNIX—UNIX Security.
- WINDOWS—Windows Security (NATIVE,UNIX,WINDOWS).

--default_hard_limit [int]

Default hard limit of user quotas and tree quotas (bytes). The hard limit value is always rounded up to match the file system's physical block size. (0 means 'No limitation'. This value can be used to compute the amount of space consumed without limiting the space).

--default_soft_limit [string]

Default soft limit of user quotas and tree quotas (bytes). Value is always rounded up to match the file system's physical block size. (0 means 'No limitation'.)

--description [string]

File system description (255 UTF-8 characters).

--expiration_timestamp [string]

Time when the snapshot will expire. Use 1970-01-01T00:00:00.000Z to set expiration timestamp to null.

--folder_rename_policy [string]

File system folder rename policies for the file system with multiprotocol access enabled. These policies control whether the directory can be renamed from NFS or SMB clients when at least one file is opened in the directory, or in one of its child directories. File system folder rename policy is set only for PROTOCOL snapshots. Values are:

- ALL_ALLOWED—All protocols are allowed to rename directories without any restrictions.
- SMB_FORBIDDEN—A directory rename from the SMB protocol is denied if at least one file is opened in the directory or in one of its child directories.
- ALL_FORBIDDEN—Any directory rename request is denied regardless of the protocol used, if at least one file is opened in the directory or in one of its child directories (ALL_ALLOWED,SMB_FORBIDDEN,ALL_FORBIDDEN).

--grace_period [int32]

Grace period of soft limits (seconds):

- 1—default: Infinite grace (Windows policy).
- 0—Use system default of 1 week.
- Positive—Grace period after which the soft limit is treated as a hard limit (seconds).

--is_async_MTime_enabled [string]

Indicates whether asynchronous MTIME is enabled on the file system or protocol snaps that are mounted writeable. Values are:

- True—Asynchronous MTIME is enabled on the file system.
- False—Asynchronous MTIME is disabled on the file system.

--is_quota_enabled [string]

Indicates whether quota is enabled. Quotas are not supported for read-only file systems. Default value for the grace period is set to infinite=-1 to match Windows' quota policy Values are:

- True—Start tracking usages for all users on a file system or a quota tree, and user quota limits are enforced.
- False—Stop tracking usages for all users on a file system or a quota tree, and user quota limits are not enforced.

--is_smb_no_notify_enabled [string]

Indicates whether notifications of changes to a directory file structure are enabled. Values are:

- True—Change directory notifications are enabled.
- False—Change directory notifications are disabled.

--is_smb_notify_on_access_enabled [string]

Indicates whether file access notifications are enabled on the file system. Values are:

- True—File access notifications are enabled on the file system.
- False—File access notifications on file access are disabled on the file system.

--is_smb_notify_on_write_enabled [string]

Indicates whether notifications on file writes are enabled on the file system. Values are:

- True—File writes notifications are enabled on the file system.
- False—File writes notifications are disabled on the file system.

--is_smb_op_locks_enabled [string]

Indicates whether opportunistic file locking is enabled on the file system. Values are:

- True—Opportunistic file locking is enabled on the file system.
- False—Opportunistic file locking is disabled on the file system.

--is_smb_sync_writes_enabled [string]

Indicates whether the synchronous writes option is enabled on the file system. Values are:

- True—Synchronous writes option is enabled on the file system.
- False—Synchronous writes option is disabled on the file system.

--locking_policy [string]

File system locking policies. These policy choices control whether the NFSv4 range locks are honored. Because NFSv3 is advisory by design, this policy specifies that the NFSv4 locking feature behaves like NFSv3 (advisory mode), for backward compatiblity with applications expecting an advisory locking scheme. File system locking policy is set only for PROTOCOL snapshots. Values are:

- ADVISORY—No lock checking for NFS and honor SMB lock range only for SMB.
- MANDATORY—Honor SMB and NFS lock range (ADVISORY,MANDATORY).

--smb_notify_on_change_dir_depth [int32]

Lowest directory level to which the enabled notifications apply, if any.

Query file systems

List file systems.

Parameters

-h, --help

Help for querying file systems.

--id [string]

File system ID.

Clone file systems

Create a clone of a file system.

Parameters

-h, --help

Help for cloning file file systems.

--id [string]

Unique identifier of the virus checker instance.

--access_policy [string]

File system security access policies. Each file system uses its access policy to determine how to reconcile the differences between NFS and SMB access control. Selecting an access policy determines which mechanism is used to enforce file security on the particular file system. File system access policy is set only for PROTOCOL snapshots. Options are:

- NATIVE—Native Security.
- UNIX—UNIX Security.
- WINDOWS—Windows Security (NATIVE,UNIX,WINDOWS).

--description [string]

Description of the clone.

--folder_rename_policy [string]

File system folder rename policies for the file system with multiprotocol access enabled. These policies control whether the directory can be renamed from NFS or SMB clients when at least one file is opened in the directory, or in one of its child directories. File system folder rename policy is set only for PROTOCOL snapshots. Options are:

- ALL_ALLOWED—All protocols are allowed to rename directories without any restrictions.
- SMB_FORBIDDEN—A directory rename from the SMB protocol will be denied if at least one file is opened in the directory or in one of its child directories.
- ALL_FORBIDDEN—Any directory rename request will be denied regardless of the protocol used, if at least one file is opened in the directory or in one of its child directories (ALL_ALLOWED,SMB_FORBIDDEN,ALL_FORBIDDEN).

--is_async_MTime_enabled [string]

Indicates whether asynchronous MTIME is enabled on the file system. Values are:

- True—Asynchronous MTIME is enabled on the file system.
- False—Asynchronous MTIME is disabled on the file system.

--is_smb_no_notify_enabled [string]

Indicates whether notifications of changes to directory file structure are enabled. Values are:

- True—Change directory notifications are enabled.
- False—Change directory notifications are disabled.

--is_smb_notify_on_access_enabled [string]

Indicates whether file access notifications are enabled on the file system. Values are:

- True—File system notifications are enabled on the file system.
- False—File system notifications are disabled on the file system.

--is_smb_notify_on_write_enabled [string]

Indicates whether file writes notifications are enabled on the file system. Values are:

- True—File writes notifications are enabled on the file system.
- False—File writes notifications are disabled on the file system.

--is_smb_op_locks_enabled [string]

Indicates if opportunistic file locking is enabled on the file system. Values are:

- True—Opportunistic file locking is enabled on the file system.
- False—Opportunistic file locking is disabled on the file system.

--is_smb_sync_writes_enabled [string]

Indicates whether the synchronous writes option is enabled on the file system. Values are:

- True—Synchronous writes option is enabled on the file system.
- False—Synchronous writes option is disabled on the file system.

--locking_policy [string]

File system locking policies. These policy choices control whether the NFSv4 range locks are honored. Because NFSv3 is advisory by design, this policy specifies that the NFSv4 locking feature behaves like NFSv3 (advisory mode), for backward compatiblity with applications expecting an advisory locking scheme. File system locking policy is set only for PROTOCOL snapshots. Options are:

- ADVISORY—No lock checking for NFS and honor SMB lock range only for SMB.
- MANDATORY—Honor SMB and NFS lock range (ADVISORY,MANDATORY).

--name [string]

Name of the clone.

--smb_notify_on_change_dir_depth [int32]

Lowest directory level to which the enabled notifications apply, if any.

Delete file systems

Delete a file system.

Parameters

-h, --help

Help for deleting file systems.

--id [string]

File system ID.

Refresh file systems

Refresh a snapshot of a file system. The content of the snapshot is replaced with the current content of the parent file system.

Parameters

-h, --help

Help for refreshing file systems.

--id [string]

File system snapshot ID.

Refresh quota file systems

Refresh the content of tree and user quotas objects.

Parameters

-h, --help

Help for refresh-quota of file systems.

--id [string]

File system snapshot ID.

Restore file systems restore

Restore from a snapshot of a file system.

Parameters

-h, --help

Help for restoring of file systems.

--id [string]

File system ID.

--copy_name [string]

Name of the backup snap to be created before the restore operation occurs. If no name is specified no backup copy is made.

--snapshot_id [string]

Snapshot ID to restore from set-protection-policy associate PowerFlex file services protection policy with file-system. Also used to dis-associate protection policy from file-system by either passing no body or passing null `protection_policy_id`.

--protection_policy_id [string]

ID of PowerFlex file services protection policy.

Create snapshot of file systems

Create a snapshot of a file system.

Parameters

-h, --help

Help for creating snapshot of file systems.

--id [string]

File system ID.

--access_policy [string]

File system security access policies. Each file system uses its access policy to determine how to reconcile the differences between NFS and SMB access control. Selecting an access policy determines which mechanism is used to enforce file security on the particular file system. File system access policy is set only for PROTOCOL snapshots. Values are:

- NATIVE—Native Security
- UNIX—UNIX Security
- WINDOWS—Windows Security (NATIVE,UNIX,WINDOWS)

--access_type [string]

Indicates whether the snapshot directory or protocol access is granted to the file system snapshot.

- SNAPSHOT—Snapshot access is via the .snapshot folder in the file system.
- PROTOCOL_READ_WRITE—Protocol read/write access is via normal file shares. Protocol access is not provided by default - the NFS and/or SMB share must be created explicitly for the snapshot.
- PROTOCOL_READ_ONLY—Protocol read only access is via normal file shares. Protocol access is not provided by default - the NFS and/or SMB share must be created explicitly for the snapshot. (SNAPSHOT,PROTOCOL_READ_WRITE,PROTOCOL_READ_ONLY)

--description [string]

Snapshot description.

--expiration_timestamp [string]

Time, when the snapshot expires.

--folder_rename_policy [string]

File system folder rename policies for the file system with multiprotocol access enabled. These policies control whether the directory can be renamed from NFS or SMB clients when at least one file is opened in the directory, or in one of its child directories. File system folder rename policy is set only for PROTOCOL snapshots. Values are:

- ALL_ALLOWED—All protocols are allowed to rename directories without any restrictions.
- SMB_FORBIDDEN—A directory rename from the SMB protocol is denied if at least one file is opened in the directory or in one of its child directories.
- ALL_FORBIDDEN—Any directory rename request is denied regardless of the protocol used, if at least one file is opened in the directory or in one of its child directories (ALL_ALLOWED,SMB_FORBIDDEN,ALL_FORBIDDEN).

--is_async_MTime_enabled [string]

Indicates whether asynchronous MTIME is enabled on the protocol snaps that are mounted writeable. Values are:

- True—Asynchronous MTIME is enabled on the file system.
- False—Asynchronous MTIME is disabled on the file system.

--is_smb_no_notify_enabled [string]

Indicates whether notifications of changes to a directory file structure are enabled. Values are:

- True—Change directory notifications are enabled.
- False—Change directory notifications are disabled.

--is_smb_notify_on_access_enabled [string]

Indicates whether file access notifications are enabled on the file system. Values are:

- True—File access notifications are enabled on the file system.

- False—File access notifications are disabled on the file system.

--is_smb_notify_on_write_enabled [string]

Indicates whether file writes notifications are enabled on the file system. Values are:

- True—File writes notifications are enabled on the file system.
- False—File writes notifications are disabled on the file system.

--is_smb_op_locks_enabled [string]

Indicates whether opportunistic file locking is enabled on the file system. Values are:

- True—Opportunistic file locking is enabled on the file system.
- False—Opportunistic file locking is disabled on the file system.

--is_smb_sync_writes_enabled [string]

Indicates whether the synchronous writes option is enabled on the file system. Values are:

- True—Synchronous writes option is enabled on the file system.
- False—Synchronous writes option is disabled on the file system.

--locking_policy [string]

File system locking policies. These policy choices control whether the NFSv4 range locks are honored. Because NFSv3 is advisory by design, this policy specifies that the NFSv4 locking feature behaves like NFSv3 (advisory mode), for backward compatiblity with applications expecting an advisory locking scheme. File system locking policy is set only for PROTOCOL snapshots. Values are:

- ADVISORY—No lock checking for NFS and honor SMB lock range only for SMB.
- MANDATORY—Honor SMB and NFS lock range (ADVISORY,MANDATORY).

--name [string]

Name of the snapshot. The default name of the snapshot is the date and time when the snapshot is taken.

--smb_notify_on_change_dir_depth [int32]

Lowest directory level to which the enabled notifications apply, if any.

21

File tree quotas commands

This section contains commands to manage file tree quotas.

Create file tree quotas

Create a tree quota instance.

Parameters

-h, --help

Help for creating file tree quota instances.

--description [string]

Description of the tree quota.

--file_system_id [string]

Unique identifier of the associated file system.

--grace_period [int32]

Grace period of tree quota, in seconds. No hard limit when set to -1. This will override the default grace period set at filesystem level.

--hard_limit [int]

Hard limit of the tree quota, in bytes. No hard limit when set to 0. This value can be used to compute amount of space that is consumed without limiting the space. Value is always rounded up to match the physical block size of the filesystem.

--is_user_quotas_enforced [string]

Whether the quota must be enabled for all users, and whether user quota limits, if any, are enforced. Values are:

- True—start tracking usage for all users on the quota tree, and enforce user quota limits.
- False—stop tracking usage for all users on the quota tree, and do not enforce user quota limits.

--path [string]

Path relative to the root of the associated filesystem.

--soft_limit [int]

Soft limit of the tree quota, in bytes. No hard limit when set to 0. Value is always rounded up to match the physical block size of the filesystem.

Query file tree quota

Query a user quota instance.

Parameters

-h, --help

Help for querying file tree quota instances.

--id [string]

Unique identifier of the tree quota.

--description [string]

Description of the tree quota.

Modify file tree quota

Modify a tree quota instance.

Parameters

-h, --help

Help for modifying file tree quota instances.

--id [string]

Unique identifier of the tree quota.

--description [string]

Description of the tree quota.

--grace_period [int32]

Grace period of tree quota, in seconds. No hard limit when set to -1. This will override the default grace period set at filesystem level.

--hard_limit [int]

Hard limit of the tree quota, in bytes. No hard limit when set to 0. This value can be used to compute amount of space that is consumed without limiting the space. Value is always rounded up to match the physical block size of the filesystem.

--is_user_quotas_enforced [string]

Whether the quota must be enabled for all users, and whether user quota limits, if any, are enforced. Values are:

- True—start tracking usage for all users on the quota tree, and enforce user quota limits.
- False—stop tracking usage for all users on the quota tree, and do not enforce user quota limits.

--soft_limit [int]

Soft limit of the tree quota, in bytes. No hard limit when set to 0. Value is always rounded up to match the physical block size of the filesystem.

Delete file tree quota

Delete a tree quota instance.

Parameters

-h, --help

Help for deleting file tree quota instances.

--id [string]

Unique identifier of the tree quota.

22

File user quotas commands

This section contains commands for managing file user quotas.

Create file user quotas

Create a user quota instance.

Parameters

-h, --help

Help for creating file user quota.

--file_system_id [string]

Unique identifier of the filesystem in which the new user quota will be created.

--hard_limit [int]

Hard limit of the user quota, in bytes. No hard limit when set to 0. This value can be used to compute amount of space that is consumed without limiting the space. Value is rounded up to match the physical block size of the filesystem.

--soft_limit [int]

Soft limit of the user quota, in bytes. No hard limit when set to 0. Value is rounded up to match the physical block size of the filesystem.

--tree_quota_id [string]

Unique identifier of the tree quota in which the new user quota is created.

--uid [int]

UNIX user identifier (UID) of the user. Preferred identifier.

--unix_name [string]

UNIX username. Identifers are exclusive. Only one of the four identifiers among 'user uid' / 'unix username' / 'windows username' / 'windows SID' can be used at a time.

--windows_name [string]

Windows username. The format is domain\user for the domain user. Identifers are exclusive. Only one of the four identifiers among 'user uid' / 'unix username' / 'windows username' / 'windows SID' can be used at a time.

--windows_sid [string]

Windows Security Identifier of the user. Identifers are exclusive. Only one of the four identifiers among 'user uid' / 'unix username' / 'windows username' / 'windows SID' can be used at a time.

Modify file user quotas

Modify a user quota instance.

Parameters

-h, --help

Help for modifying file user quota.

--id [string]

Unique identifier of the file user quota.

--hard_limit [int]

Hard limit of the user quota, in bytes. No hard limit when set to 0. This value can be used to compute amount of space that is consumed without limiting the space. Value is rounded up to match the physical block size of the filesystem.

--soft_limit [int]

Soft limit of the user quota, in bytes. No hard limit when set to 0. Value is rounded up to match the physical block size of the filesystem.

Query file user quotas

Query a user quota instance.

Parameters

-h, --help

Help for querying file user quota.

--id [string]

Unique identifier of the file user quota.

23

File virus checker commands

This section contains commands to manage file virus checkers.

Create file virus checkers

Add a new virus checker setting to a PowerFlex file services server. Only one instance can be created per PowerFlex file services server.

Parameters

-h, --help

Help for creating file virus checkers.

--nas_server_id [string]

Unique identifier of an associated PowerFlex file services server instance that uses this virus checker configuration. Only one virus checker configuration per PowerFlex file services server is supported.

Workflow to enable file virus checker settings

About this task

The workflow to enable the virus checker settings on the PowerFlex file services server is described below.

Steps

1. Create a virus checker instance on PowerFlex file services server.
2. Download template virus checker configuration file.
3. Edit the configuration file with virus checker configuration details.
4. Upload the configuration file.
5. Enable the virus checker on the PowerFlex file services server.

Modify file virus checker

Modify the virus checker settings of a PowerFlex file services server.

Parameters

-h, --help

Help for modifying file virus checkers.

--id [string]

Unique identifier of the virus checker instance.

--is_enabled [string]

Indicates whether the anti-virus service is enabled on this PowerFlex file services server. Value are:

- True—Anti-virus service is enabled. Each file created or modified by an SMB client is scanned by the third-party anti-virus servers. If a virus is detected, the access to the file system is denied. If third-party anti-virus servers are not available, according the policy, the access to the file systems is denied to prevent potential viruses propagation.

- False—Anti-virus service is disabled. File systems of the PowerFlex file services servers are available for access without virus checking.

Query file virus checker

Query a specific virus checker setting of a PowerFlex file services server.

Parameters

-h, --help
Help for querying file virus checkers.

--id [string]
Unique identifier of the virus checker instance.

Delete file virus checker settings

Delete the virus checker settings of a PowerFlex file services server.

Parameters

-h, --help
Help for deleting file virus checkers.

--id [string]
Unique identifier of the virus checker instance.

24

Group commands

This section contains commands to manage groups.

Create groups

Creates groups.

Parameters

-h, --help

Help for creating groups.

--name [string]

--role [string]

Options are:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support
- NA

Modify groups

Modify a group by ID.

Parameters

-h, --help

Help for modifying groups.

--id [string]

ID of an existing group.

--role [string]

Options are:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- ReplicationManager
- SnapshotManager

- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support
- NA

Query groups

Get a group by ID.

Parameters

-h, --help

Help for querying groups.

--id [string]

ID of an existing group.

--limit [int32]

Limit number of returned objects.

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

Delete groups

Delete a group by ID.

Parameters

-h, --help

Help for deleting groups.

--id [string]

ID of an existing group.

25

Login banner commands

This section contains commands to manage login banners.

Create login banners

Creates login banners.

Parameters

-h, --help

Help for creating groups.

--login_banner [string]

Modify login banner

Modify a login banner by ID.

Parameters

-h, --help

Help for modifying groups.

--id [string]

ID of an existing login banner.

--login_banner [string]

Query login banner

Queries login banner.

Parameters

-h, --help

Help for querying groups.

--id [string]

ID of an existing login banner.

Delete login banner

Delete a login banner by ID.

Parameters

-h, --help

Help for deleting groups.

--id [string]

ID of an existing login banner.

26

NFS export commands

This section contains commands to manage NFS exports.

Create NFS exports

Create an NFS export for a snapshot.

Parameters

-h, --help

Help for creating NFS export for a snapshot.

--default_access [string]

Default access level for all hosts that can access the export. Values are:

- NO_ACCESS—Deny access to the export for the hosts.
- READ_ONLY—Allow read only access to the export for the hosts.
- READ_WRITE—Allow read write access to the export for the hosts.
- ROOT—Allow read write access to the export for the hosts. Allow access to the export for root user.
- READ_ONLY_ROOT—Allow read only root access to the export for the hosts (NO_ACCESS,READ_ONLY,READ_WRITE,ROOT,READ_ONLY_ROOT).

--description [string]

User defined NFS export description.

--file_system_id [string]

Unique identifier of the file system on which the NFS export must be created.

--is_no_SUID [string]

If set, do not allow access to set SUID. Otherwise, allow access.

--min_security [string]

NFS enforced security type for users accessing an NFS export. Values are:

- SYS—Allow the user to authenticate with any NFS security types: UNIX, Kerberos, Kerberos with integrity, or Kerberos with encryption.
- KERBEROS—Allow only Kerberos security for user authentication.
- KERBEROS_WITH_INTEGRITY—Allow only Kerberos with integrity and Kerberos with encryption security for user authentication.
- KERBEROS_WITH_ENCRYPTION—Allow only Kerberos with encryption security for user authentication (SYS,KERBEROS,KERBEROS_WITH_INTEGRITY,KERBEROS_WITH_ENCRYPTION).

--name [string]

NFS Export name.

--no_access_hosts [string]

Hosts with no access to the NFS export or its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--path [string]

Local path to export relative to the file system root directory. With NFS, each export of a file_system or file_snap must have a unique local path. Before you can create additional exports within an NFS shared folder, you must create directories within it from a Linux/UNIX host that is connected to the file system. After a directory has been created from a mounted host, you can create a corresponding export and set access permissions accordingly.

--no_access_hosts [string]

Hosts with no access to the NFS export or its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--read_only_hosts [string]

Hosts with read-only access to the NFS export and its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--read_only_root_hosts [string]

Hosts with read-only and ready-only for root user access to the NFS export and its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--read_write_hosts [string]

Hosts with read and write access to the NFS export and its snapshots.Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--read_write_root_hosts [string]

Hosts with read and write and read and write for root user access to the NFS export and its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

Modify NFS exports

Modify NFS export properties.

Parameters

-h, --help

Help for modifying NFS export.

--id [string]

NFS export object ID.

--add_no_access_hosts [string]

Hosts to add to the no_access_host list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if the host already exists in the list. Cannot be combined with `no_access_hosts`.

--add_read_only_hosts [string]

Hosts to add to the current read_only_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if the host already exists. Cannot combine with `read_only_hosts`.

--add_read_only_root_hosts [string]

Hosts to add to the current read_only_root_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if the host already exists. Cannot combine with `read_only_root_hosts`.

--add_read_write_hosts [string]

Hosts to add to the current read_write_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if Host is already exists. Cannot combine with `read_write_hosts`.

--add_read_write_root_hosts [string]

Hosts to add to the current read_write_root_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if the host already exists. Cannot combine with `read_write_root_hosts`.

--anonymous_GID [int32]

Specifies the group ID of the anonymous account.

--anonymous_UID [int32]

Specifies the user ID of the anonymous account.

--default_access [string]

Default access level for all hosts that can access the export. Values are:

- NO_ACCESS—Deny access to the export for the hosts.
- READ_ONLY—Allow read only access to the export for the hosts.
- READ_WRITE—Allow read write access to the export for the hosts.
- ROOT—Allow read write access to the export for the hosts. Allow access to the export for root user.
- READ_ONLY_ROOT—Allow read only root access to the export for the hosts (NO_ACCESS,READ_ONLY,READ_WRITE,ROOT,READ_ONLY_ROOT).

--description [string]

NFS export description.

--is_no_SUID [string]

If set, do not allow access to set SUID. Otherwise, allow access.

--min_security [string]

NFS enforced security type for users accessing an NFS export. Values are:

- SYS—Allow the user to authenticate with any NFS security types: UNIX, Kerberos, Kerberos with integrity, or Kerberos with encryption.
- KERBEROS—Allow only Kerberos security for user authentication.
- KERBEROS_WITH_INTEGRITY—Allow only Kerberos with integrity and Kerberos with encryption security for user authentication.
- KERBEROS_WITH_ENCRYPTION—Allow only Kerberos with encryption security for user authentication (SYS,KERBEROS,KERBEROS_WITH_INTEGRITY,KERBEROS_WITH_ENCRYPTION).

--no_access_hosts [string]

Hosts with no access to the NFS export or its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--read_only_hosts [string]

Hosts with read-only access to the NFS export and its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--read_only_root_hosts [string]

Hosts with read-only and ready-only for root user access to the NFS export and its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), netgroups prefixed with @.

--read_write_hosts [string]

Hosts with read and write access to the NFS export and its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask) or, netgroups prefixed with @.

--read_write_root_hosts [string]

Hosts with read and write and read and write for root user access to the NFS export and its snapshots. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @.

--remove_no_access_hosts [string]

Hosts to remove from the current no_access_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if the host is not present. Cannot combine with `no_access_hosts`.

--remove_read_only_hosts [string]

Hosts to remove from the current read_only_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if the host is not present. Cannot combine with `read_only_hosts`.

--remove_read_only_root_hosts [string]

Hosts to remove from the current read_only_root_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if The host is not present. Cannot combine with `read_only_root_hosts`.

`--remove_read_write_hosts [string]`

Hosts to remove from the current read_write_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if Host is not present. Cannot combine with read_write_hosts.

`--remove_read_write_root_hosts [string]`

Hosts to remove from the current read_write_root_hosts list. Hosts can be entered by hostname, IP addresses (IPv4, IPv6, IPv4/PrefixLength, IPv6/PrefixLenght, or IPv4/subnetmask), or netgroups prefixed with @. Error if the host is not present. Cannot combine with `read_write_root_hosts`.

Query NFS exports

Get NFS export properties.

Parameters

`-h, --help`

Help for querying NFS export.

`--id [string]`

NFS export object ID.

Delete NFS exports

Delete NFS export.

Parameters

`-h, --help`

Help for deleting NFS export.

`--id [string]`

NFS export object ID.

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NFS server commands

This section contains commands for managing NFS servers.

Create NFS servers

Create an NFS server.

Parameters

-h, --help

Help for creating NFS servers.

--host_name [string]

The name that is used by NFS clients to connect to this NFS server. This name is required when using secure NFS, except when is_use_smb_config_enabled is true. In this case host_name is forced to the SMB server computer name, and must not be specified.

--credentials_cache_TTL [int32]

Sets the time-to-live (in minutes) expiration time in minutes for a Windows entry in the credentials cache. When failed mapping entries expire, the system retries mapping the UID to the SID.

--is_extended_credentials_enabled [string]

Indicates whether the NFS server supports more than 16 Unix groups in a UNIX credential. Valid values are:

- True—NFS server supports more than 16 Unix groups in a UNIX credential. The NFS server sends additional request to UNIX Directory service to identify UNIX groups.
- False—NFS server supports more than 16 Unix groups in a UNIX credential. The NFS server sends additional request to UNIX Directory service to identify UNIX groups.

NOTE: The NFS server builds its own Unix credential when it supports more than 16 groups. This process can slow performance.

--is_nfsv3_enabled [string]

Indicates whether NFSv3 is enabled on the PowerFlex file services server. When enabled, NFS shares can be accessed with NFSv3. When disabled, NFS shares can not be accessed with NFSv3 protocol. Valid values are:

- True—NFSv3 is enabled on the specified PowerFlex file services server.
- False—NFSv3 is disabled on the specified PowerFlex file services server.

--is_nfsv4_enabled [string]

Indicates whether NFSv4 is enabled on the PowerFlex file services server. When enabled, NFS shares can be accessed with NFSv4. When disabled, NFS shares can not be accessed with NFSv4 protocol. Valid values are:

- True—NFSv4 is enabled on the specified PowerFlex file services server.
- False—NFSv4 is disabled on the specified PowerFlex file services server.

--is_secure_enabled [string]

Indicates whether secure NFS is enabled on the NFS server. Valid values are:

True—Secure NFS is Enabled.

False—Secure NFS is disabled.

--is_use_smb_config_enabled [string]

Indicates whether SMB authentication is used to authenticate to the KDC. Values are:

- True—Indicates that the configured SMB server settings are used for Kerberos authentication.

- False—Indicates that Kerberos uses its own settings.

--nas_server_id [string]

Unique identifier of the PowerFlex file services server.

Modify NFS servers

Modify NFS server settings.

Parameters

-h, --help

Help for modifying NFS servers.

--id [string]

Unique identifier of the NFS server.

--host_name [string]

The name that will be used by NFS clients to connect to this NFS server. This name is required when using secure NFS, except when is_use_smb_config_enabled is true. In this case host_name is forced to the SMB server computer name, and must not be specified.

--is_extended_credentials_enabled [string]

Indicates whether the NFS server supports more than 16 Unix groups in a Unix credential. Valid values are:

- True—NFS server supports more than 16 Unix groups in a Unix credential. The NFS server sends additional request to Unix Directory service to identify Unix groups.
- False—NFS server supports more than 16 Unix groups in a Unix credential. The NFS server sends additional request to Unix Directory service to identify Unix groups.

NOTE: The NFS server builds its own Unix credential when it supports more than 16 groups. This process can slow performance.

--is_nfsv3_enabled [string]

Indicates whether NFSv3 is enabled on the PowerFlex file services server. When enabled, NFS shares can be accessed with NFSv3. When disabled, NFS shares cannot be accessed with NFSv3 protocol. Valid values are:

- True—NFSv3 is enabled on the specified PowerFlex file services server.
- False—NFSv3 is disabled on the specified PowerFlex file services server.

--is_nfsv4_enabled [string]

Indicates whether NFSv4 is enabled on the PowerFlex file services server. When enabled, NFS shares can be accessed with NFSv4. When disabled, NFS shares cannot be accessed with NFSv4 protocol. Valid values are:

- True—NFSv4 is enabled on the specified PowerFlex file services server.
- False—NFSv4 is disabled on the specified PowerFlex file services server.

--is_secure_enabled [string]

Indicates whether secure NFS is enabled on the NFS server. Valid values are:

- True—Secure NFS is enabled.
- False—Secure NFS is disabled.

--is_use_smb_config_enabled [string]

Indicates whether SMB authentication is used to authenticate to the KDC. Values are:

- True—Indicates that the the configured SMB Server settings are used for Kerberos authentication.
- False—Indicates that Kerberos uses its own settings.

Query NFS servers

Query all NFS servers.

Parameters

-h, --help

Help for querying NFS servers.

--id [string]

Unique identifier of the NFS server.

Join NFS servers

Join the secure NFS server to the Active Directory domain of the PowerFlex file services server, which is necessary for secure NFS.

Parameters

-h, --help

Help for joining NFS servers.

--id [string]

Unique identifier of the NFS server.

--domain_user_name [string]

Name of a domain-user with privileges to join the Active Directory domain.

--domain_password [string]

Password of the domain-user specified to join the Active Directory domain.

Skip unjoin and delete NFS servers

Unjoin domain from NFS server and delete the NFS server.

Parameters

-h, --help

Help for skipping, unjoining, and deleting NFS servers.

--id [string]

Unique identifier of the NFS server.

Unjoin NFS servers

Unjoin the secure NFS server from the Active Directory domain of the PowerFlex file services server. If you unjoin with secure NFS exports active, exports are unavailable to the clients.

Parameters

-h, --help

Help for unjoining NFS servers.

--id [string]

Unique identifier of the NFS server.

--domain_user_name [string]

Name of a domain-user with privileges to unjoin from the Active Directory domain.

--domain_password [string]

Password of the domain-user specified to unjoin from the Active Directory domain.

Delete NFS servers

Delete an NFS server.

Parameters

-h, --help

Help for deleting NFS export.

--id [string]

Unique identifier of the NFS server.

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PowerFlex file service commands

This section contains commands for managing PowerFlex file services.

Query PowerFlex file services gateway

Displays PowerFlex file services-gateway software version, uptime, user_readable_uptime, user_readable_starttime.

Parameters

-h, --help

Help for querying PowerFlex file services jobs.

--id [string]

Unique identifier of the PowerFlex file services job.

Query PowerFlex file services jobs

Query PowerFlex file services jobs.

Parameters

-h, --help

Help for querying PowerFlex file services jobs.

--id [string]

Unique identifier of the PowerFlex file services server.

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PowerFlex file service protection policy commands

This section contains commands for managing PowerFlex file service protection policies.

Create PowerFlex file services protection policies

Create a protection policy.

Parameters

-h, --help
Help for creating PowerFlex file services protection policies.

--description [string]
Description about protection policy.

--id [string]
Unique identifier of the protection policy.

--name [string]
Protection Policy name. Use only alphanumeric characters.

--snapshot_rule_ids [string]
Array of rules associated with this policy. Use only alphanumeric characters.

Modify PowerFlex file services protection policies

Modify a protection policy.

Parameters

-h, --help
Help for modifying PowerFlex file services protection policies.

--description [string]
Description about protection policy.

--id [string]
Unique identifier of the policy to be modified.

--name [string]
Protection policy name. Use only alphanumeric characters.

--add_snapshot_rule_ids [string]
Snapshot rule identifiers to be added to this policy.

--remove_snapshot_rule_ids [string]
Snapshot rule identifiers to be removed from this policy.

Query PowerFlex file services protection policies

Query PowerFlex file services protection policies.

Parameters

-h, --help

Help for querying PowerFlex file services protection policies.

--id [string]

Unique identifier of the protection policy.

Delete PowerFlex file services protection policies

Delete a PowerFlex file services protection policy. You can delete protection policies that are used in any storage resources.

Parameters

-h, --help

Help for deleting PowerFlex file services protection policies.

--id [string]

Unique identifier of the PowerFlex file services protection policy to be deleted.

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PowerFlex file service servers commands

This section contains commands for managing PowerFlex file service servers.

Create PowerFlex file services servers

Create a PowerFlex file services server.

Parameters

-h, --help

Help for creating a PowerFlex file services server.

--current_unix_directory_service [string]

Define the UNIX directory service used for looking up identity information for Unix such as UIDs, GIDs, net groups, and so on. The options are:

- None
- NIS
- LDAP
- Local_Files
- Local_Then_NIS
- Local_Then_LDAP

--default_unix_user [string]

Default UNIX user name used for granting access in case of Windows to Unix user mapping failure. When empty, access in such case is denied.

--default_windows_user [string]

Default Windows user name used for granting access in case of Unix to Windows user mapping failure. When empty, access in such case is denied.

--description [string]

Description of the PowerFlex file services server.

--is_auto_user_mapping_enabled [string]

A Windows user must have a corresponding matching Unix user (uid) in order to connect. This attribute enables you to automatically generates this UNIX user (uid), if that Windows user does not have any in the configured UNIX directory service (UDS). In a pure SMB or non multi-protocol environment, this should be set to true.

--is_username_translation_enabled [string]

Enable the possibility to match a Windows account with an UNIX account with different names.

--name [string]

Name of the PowerFlex file services server. Use alphanumeric characters only.

--protection_domain_id [string]

Unique identifier of the PowerFlex file services enabled protection domain.

Modify PowerFlex file services servers

Modify the settings of a PowerFlex file services server.

Parameters

-h, --help

Help for modifying a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

--backup_IPv4_interface_id [string]

Unique identifier of the preferred IPv4 backup interface.

--backup_IPv6_interface_id [string]

Unique identifier of the preferred IPv6 backup interface.

--current_unix_directory_service [string]

Define the Unix directory service used for looking up identity information for UNIX such as UIDs, GIDs, net groups, and so on.

- None
- NIS
- LDAP
- Local_Files
- Local_Then_NIS
- Local_Then_LDAP

--default_unix_user [string]

Default Unix user name used for granting access in case of Windows to UNIX user mapping failure. When empty, access in such case is denied.

--default_windows_user [string]

Default Windows user name used for granting access in case of UNIX to Windows user mapping failure. When empty, access in such case is denied.

--description [string]

Description of the PowerFlex file services server.

--is_auto_user_mapping_enabled [string]

A Windows user must have a corresponding matching UNIX user (uid) in order to connect. This attribute enables you to automatically generate this UNIX user (uid), if that Windows user does not have any in the configured UNIX directory service (UDS). In a pure SMB or non multi-protocol environment, this should be set to true.

--is_username_translation_enabled [string]

Enable the possibility to match a windows account to a UNIX account with different names.

--production_IPv4_interface_id [string]

Unique identifier of the preferred IPv4 production interface.

--production_IPv6_interface_id [string]

Unique identifier of the preferred IPv6 production interface.

Query PowerFlex file services servers

Query a specific PowerFlex file services server.

Parameters

-h, --help

Help for querying a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

Ping PowerFlex file services server

Ping destination from PowerFlex file services server.

Parameters

-h, --help

Help for pinging a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

--destination_address [string]

Destination to ping, can be either an IP address (IPv4 or IPv6) or a hostname being resolved by the PowerFlex file services server name resolvers.

--is_ipv6 [string]

When true, and the destination is a hostname, it will be resolved to IPv6. Otherwise the hostname is resolved preferably to an IPv4 address.

Move PowerFlex file services servers

Move PowerFlex file services server to required node in PowerFlex file services cluster.

Parameters

-h, --help

Help for moving a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

--primary_node [string]

New primary node to where the PowerFlex file services servers to be moved on.

--backup_node [string]

New backup node to be assigned to PowerFlex file services server.

Swap nodes of PowerFlex file services servers

Swap the primary and backup nodes of a PowerFlex file services server.

Parameters

-h, --help

Help for swapping nodes of a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

Update user mappings of PowerFlex file services servers

Fix the user mappings for all file systems associated with the PowerFlex file services server. This process updates file ownership on the file systems of the PowerFlex file services server to reflect changes to SIDs of users. A new UID or GID is obtained from a UNIX Directory Service for the username of the object owner. A user-mapping report is also generated. This operation can take a significant amount of time, depending on the size of the file systems.

Parameters

-h, --help

Help for updating user-mappings of a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

Unjoin SMB and delete PowerFlex file services servers

Unjoin SMB from PowerFlex file services server and delete the server.

Parameters

-h, --help

Help for unjoining SMB and deleting a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

--domain_user_name [string]

Administrator login used to unjoin the associated SMB servers from the Active Directory (AD) domain before deleting the PowerFlex file services server.

--domain_password [string]

Administrator password used to unjoin the associated SMB servers from the Active Directory (AD) domain before deleting the PowerFlex file services server.

Delete PowerFlex file services servers

Delete a PowerFlex file services server.

Parameters

-h, --help

Help for deleting a PowerFlex file services server.

--id [string]

Unique identifier of the PowerFlex file services server.

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PowerFlex file services snapshot rules commands

This section contains commands for managing PowerFlex file services snapshot rules.

Create snapshot rules

Create a PowerFlex file services snapshot rule.

Parameters

-h, --help

Help for creating a PowerFlex file services snapshot rule.

--days_of_week [string]

Days of the week on which the rule should be applied. Applies only for rules where the `time_of_day` parameter is set.

--interval [string]

Interval between snapshots. If interval specified, `time_of_day` and `days_of_week` are not valid. Values are:

- FIVE_MINUTES
- FIFTEEN_MINUTES
- THIRTY_MINUTES
- ONE_HOUR
- TWO_HOURS
- THREE_HOURS
- FOUR_HOURS
- SIX_HOURS
- EIGHT_HOURS
- TWELVE_HOURS
- ONE_DAY

--name [string]

Snapshot rule name. Use only alphanumeric characters.

--retention [int32]

Number of hours since creation of snapshot, must be retained in the system. 0 means no expiration.

--time_of_day [string]

Time of the day to take a snapshot, with format "hh:mm" in 24 hour time format. Either the interval parameter or the time_of_day parameter may be set, but not both. if `time_of_day` specified, `days_of_week` also need to be specified.

--type [string]

Indicates whether the snapshot directory or protocol access is granted to the file system snapshot. Values are:

- SNAPSHOT—Snapshot access is via the .snapshot folder in the file system.
- PROTOCOL_READ_WRITE—(Not supported) Protocol read/write access is via normal file shares. Protocol access is not provided by default - the NFS and/or SMB share must be created explicitly for the snapshot.

- PROTOCOL_READ_ONLY—Protocol read only access is via normal file shares. Protocol access is not provided by default - the NFS and/or SMB share must be created explicitly for the snapshot (SNAPSHOT,PROTOCOL_READ_ONLY).

Modify PowerFlex file services snapshot rules

Modify a snapshot rule. If the rule is associated with a policy that is currently applied to a storage resource, the modified rule is immediately applied to that associated storage resource.

Parameters

-h, --help

Help for modifying PowerFlex file services snapshot rules.

--id [string]

Unique identifier of the snapshot rule.

--interval [string]

Interval between snapshots. If interval specified, `time_of_day` and `days_of_week` are not valid. Options are:

- FIVE_MINUTES
- FIFTEEN_MINUTES
- THIRTY_MINUTES
- ONE_HOUR
- TWO_HOURS
- THREE_HOURS
- FOUR_HOURS
- SIX_HOURS
- EIGHT_HOURS
- TWELVE_HOURS
- ONE_DAY

--name [string]

Snapshot rule name. Use only alphanumeric characters.

--retention [int32]

Number of hours since creation of snapshot, must be retained in the system. 0 means no expiration.

--time_of_day [string]

Time of the day to take a snapshot, with format "hh:mm" in 24 hour time format. Either the interval parameter or the `time_of_day` parameter may be set, but not both. if `time_of_day` specified, `days_of_week` also need to be specified.

--type [string]

Indicates whether the snapshot directory or protocol access is granted to the file system snapshot. Options are:

- SNAPSHOT—Snapshot access is via the .snapshot folder in the file system.
- PROTOCOL_READ_WRITE—Protocol read/write access is via normal file shares. This option is not supported. Protocol access is not provided by default - the NFS and/or SMB share must be created explicitly for the snapshot.
- PROTOCOL_READ_ONLY—Protocol read only access is through normal file shares. Protocol access is not provided by default - the NFS and/or SMB share must be created explicitly for the snapshot (SNAPSHOT,PROTOCOL_READ_ONLY).

Query PowerFlex file services snapshot rules

Query snapshot rules.

Parameters

-h, --help

Help for querying PowerFlex file services snapshot rules.

--id [string]

Unique identifier of the snapshot rule.

Delete PowerFlex file services snapshot rules

Delete a snapshot rule. Snapshot rules that are used by any storage resources cannot be deleted.

Parameters

-h, --help

Help for deleting PowerFlex file services snapshot rules.

--id [string]

Unique identifier of the snapshot rule to be deleted.

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Roles commands

This section contains commands for managing roles.

Query roles

Queries all roles.

Parameters

-h, --help

Help for querying roles.

33

SMB server commands

This section contains commands to manage SMB servers.

Create SMB servers

Create an SMB server.

Parameters

-h, --help

Help for creating SMB servers.

--computer_name [string]

DNS name of the associated computer account when the SMB server is joined to an Active Directory domain. This name is limited to 63 bytes and must not contain the following characters -

- Comma (.)
- Tilde (~)
- Colon (:)
- Exclamation point (!)
- At sign (@)
- Number sign (#)
- Dollar sign ($)
- Percent (%)
- Caret (^)
- Ampersand (&)
- Apostrophe (')
- Period (.)—Note that if you enter string with period only the first word is kept.
- Parentheses (())
- Braces ({})
- Underscore (_)
- White space (blank)

as defined by the Microsoft naming convention (see https://support.microsoft.com/en-us/help/909264/)

--description [string]

Description of the SMB server in UTF-8 characters.

--domain [string]

Domain name where SMB server is registered in Active Directory, if applicable.

--is_standalone [string]

Indicates whether the SMB server is standalone. Values are:

- True—SMB server is standalone.
- False—SMB server is joined to the Active Directory.

--local_admin_password [string]

Regardless of the type of the SMB server, standalone or in the domain, a local administrator user must be created. local_admin_password is the password of this user.

--nas_server_id [string]

Unique identifier of the PowerFlex file services server.

--netbios_name [string]

NetBIOS name is the network name of the standalone SMB server. SMB servers joined to Active Directory also have NetBIOS name, defaulted to the 15 first characters of the computer_name attribute. Administrators can specify a custom NetBIOS Name for a SMB server using this attribute. NetBIOS name is limited to 15 characters and cannot contain the following characters -

- Backslash (\)
- Slash mark (/)
- Colon (:)
- Asterisk (*)
- Question mark (?)
- Quotation mark ("")
- Less than sign (<)
- Greater than sign (>)
- Vertical bar (|)

as defined by the Microsoft naming convention (see https://support.microsoft.com/en-us/help/909264/)

--workgroup [string]

Unique identifier of the PowerFlex file services server. Applies to standalone SMB servers only. Windows network workgroup for the SMB server. Workgroup names are limited to 15 alphanumeric ASCII characters.

Modify servers

Modify the settings of an SMB server.

Parameters

-h, --help

Help for modifying settings of SMB servers.

--id [string]

Unique identifier of the SMB server.

--is_standalone [string]

Indicates whether the SMB server is standalone. Values are:

- True—SMB server is standalone.
- False—SMB server is joined to the Active Directory.

--domain [string]

Domain name where SMB server is registered in Active Directory, if applicable.

--computer_name [string]

DNS name of the associated computer account when the SMB server is joined to an Active Directory domain. This name is limited to 63 bytes and must not contain the following characters -

- Comma (.)
- Tilde (~)
- Colon (:)
- Exclamation point (!)
- At sign (@)
- Number sign (#)
- Dollar sign ($)
- Percent (%)
- Caret (^)
- Ampersand (&)
- Apostrophe (')
- Period (.)—Note that if you enter string with period only the first word is kept
- Parentheses (())
- Braces ({})
- Underscore (_)

- White space (blank)

as defined by the Microsoft naming convention (see Naming conventions in Active Directory for computers, domains, sites, and OUs)

--local_admin_password [string]

Password for the local administrator account of the SMB server.

--netbios_name [string]

NetBIOS name is the network name of the standalone SMB server. SMB servers joined to Active Directory also have NetBIOS name, defaulted to the 15 first characters of the computer_name attribute. Administrators can specify a custom NetBIOS name for an SMB server using this attribute. NetBIOS name is limited to 15 characters and cannot contain the following characters -

- Backslash (\)
- Slash mark (/)
- Colon (:)
- Asterisk (*)
- Question mark (?)
- Quotation mark ("")
- Less than sign (<)
- Greater than sign (>)
- Vertical bar (|)

as defined by the Microsoft naming convention (see Naming conventions in Active Directory for computers, domains, sites, and OUs)

--workgroup [string]

Applies to standalone SMB servers only. Windows network workgroup for the SMB server. Workgroup names are limited to 15 alphanumeric ASCII characters.

Query SMB servers

Query settings of a specific SMB server.

Parameters

-h, --help

Help for querying SMB servers.

--id [string]

Unique identifier of the SMB server.

Join SMB server

Join the SMB server to an Active Directory domain.

Parameters

-h, --help

Help for joining SMB servers.

--id [string]

Unique identifier of the SMB server.

--domain_user_name [string]

Name of a domain-user with sufficient privileges to join the Active Directory domain.

--domain_password [string]

Password of the domain-user specified to join the Active Directory domain.

--organizational_unit [string]

Organizational unit of the SMB server in Active Directory, if applicable.

--reuse_computer_account

If set to yes: try to reuse the existing SMB server account in the Active Directory when joining.

Skip unjoin and delete SMB servers

Skip unjoin from AD domain and Delete a SMB server.

Parameters

-h, --help

Help for skipping unjoin from AD domain and deleting SMB servers.

--id [string]

Unique identifier of the SMB server.

Unjoin SMB servers

Unjoin the SMB server from an Active Directory domain.

Parameters

-h, --help

Help for unjoining SMB servers from Active Directory domain.

--id [string]

Unique identifier of the SMB server.

--domain_user_name [string]

Name of a domain-user with sufficient privileges to unjoin from the Active Directory domain.

--domain_password [string]

Password of the domain-user specified to unjoin from the Active Directory domain.

--is_skip_AD_unjoin [string]

If set to yes: Will not remove the account from the Active Directory. This is to be used in case that no DC is available.

Delete SMB server

Delete an SMB server. The SMB server must not be joined to a domain to be deleted.

Parameters

-h, --help

Help for deleting SMB servers.

--id [string]

Unique identifier of the SMB server.

34

SMB shares commands

This section contains commands for managing SMB shares.

Create SMB shares

Create an SMB share.

Parameters

-h, --help

Help for creating SMB shares.

--description [string]

SMB share description.

--file_system_id [string]

Unique identifier of the file system on which the SMB share is created.

--is_ABE_enabled [string]

Indicates whether access-based enumeration (ABE) is enabled. ABE filters the list of available files and folders on a server to include only those to which the requesting user has access. Values are:

- True—ABE is enabled.
- False—ABE is disabled.

--is_branch_cache_enabled [string]

Indicates whether BranchCache optimization is enabled. BranchCache optimization technology copies content from your main office or hosted cloud content servers and caches the content at branch office locations, allowing client computers at branch offices to access the content locally rather than over the WAN. Values are:

- True—BranchCache is enabled.
- False—BranchCache is disabled.

--is_continuous_availability_enabled [string]

Indicates whether continuous availability for server message block (SMB) 3.0 is enabled for the SMB share. Values are:

- True—Continuous availability for SMB 3.0 is enabled for the SMB share.
- False—Continuous availability for SMB 3.0 is disabled for the SMB share.

--is_encryption_enabled [string]

Indicates whether encryption for server message block (SMB) 3.0 is enabled at the shared folder level. Values are:

- True—Encryption for SMB 3.0 is enabled.
- False—Encryption for SMB 3.0 is disabled.

--name [string]

SMB share name.

--offline_availability [string]

Defines valid states of offline availability. Values are:

- MANUAL—Only specified files is available offline.
- DOCUMENTS—All files that users open is available offline.
- PROGRAMS—Program will preferably run from the offline cache even when connected to the network. All files that users open are available offline.
- NONE—Prevents clients from storing documents and programs in offline cache (default) (MANUAL,DOCUMENTS,PROGRAMS,NONE).

--path [string]

Local path to the file system or any existing sub-folder of the file system that is shared over the network. This path is relative to the PowerFlex file services server and must start with the filesystem's mountpoint path, which is the filesystem name. For example to share the top-level of a filesystem named svr1fs1, which is on the /svr1fs1 mountpoint of the PowerFlex file services server, use /svr1fs1 in the path parameter. SMB shares allow you to create multiple network shares for the same local path.

--umask [string]

The default UNIX umask for new files created on the share. If not specified the umask defaults to 022.

Modify SMB shares

Modify SMB share properties.

Parameters

-h, --help

Help for modifying SMB shares.

--id [string]

SMB share object ID.

--description [string]

NFS share description.

--is_ABE_enabled [string]

Indicates whether access-based enumeration (ABE) is enabled. ABE filters the list of available files and folders on a server to include only those, that the requesting user has access to. Values are:

- True—ABE is enabled.
- False—ABE is disabled.

--is_branch_cache_enabled [string]

Indicates whether BranchCache optimization is enabled. BranchCache optimization technology copies content from your main office or hosted cloud content servers and caches the content at branch office locations, allowing client computers at branch offices to access the content locally rather than over the WAN. Values are:

- True—BranchCache is enabled.
- False—BranchCache is disabled.

--is_continuous_availability_enabled [string]

Indicates whether continuous availability for server message block (SMB) 3.0 is enabled for the SMB share. Values are:

- True—Continuous availability for SMB 3.0 is enabled for the SMB share.
- False—Continuous availability for SMB 3.0 is disabled for the SMB share.

--is_encryption_enabled [string]

Indicates whether encryption for server message block (SMB) 3.0 is enabled at the shared folder level. Values are:

- True—Encryption for SMB 3.0 is enabled.
- False—Encryption for SMB 3.0 is disabled.

--offline_availability [string]

Defines valid states of offline availability, * MANUAL - Only specified files are available offline.

- DOCUMENTS—All files that users open are available offline.
- PROGRAMS—Program preferably runs from the offline cache even when connected to the network. All files that users open are available offline.
- NONE—Prevents clients from storing documents and programs in offline cache (default) (MANUAL,DOCUMENTS,PROGRAMS,NONE).

--umask [string]

The default UNIX umask for new files created on the share.

Query SMB share

Get an SMB share.

Parameters

-h, --help

Help for querying SMB shares.

--id [string]

SMB share object ID.

Delete SMB shares

Delete an SMB share.

Parameters

-h, --help

Help for deleting SMB shares.

--id [string]

SMB share object ID.

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SMTP service commands

This section contains commands to manage SMTP services.

SMTP services to send test email

Parameters

-h, --help

Help for sending test emails.

--password [string]

--recipient_1 [string]

--recipient_2 [string]

--recipient_3 [string]

--recipient_4 [string]

--recipient_5 [string]

--sender_address [string]

--server_type [string]

--username [string]

SMTP services to test connection

Parameters

-h, --help

Help for testing connections.

--password [string]

--recipient_1 [string]

--recipient_2 [string]

--recipient_3 [string]

--recipient_4 [string]

--recipient_5 [string]

--sender_address [string]

--server_type [string]

--username [string]

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SupportAssist configuration commands

This section contains commands to manage SupportAssist configuration.

SupportAssist configuration connection state

Parameters

-h, --help

Help for connection state.

SupportAssist configuration connectivity

Parameters

-h, --help

Help for connectivity.

Modify SupportAssist configuration

Parameters

-h, --help

Help for modifying SupportAssist configuration.

--contacts [string]

--enabled [string]

--eventMuted [string]

--systemMode [string]

Query SupportAssist configuration

Parameters

-h, --help

Help for querying SupportAssist configuration.

Send SupportAssist configuration test alert

Parameters

-h, --help

Help for sending SupportAssist configuration test alert.

--message [string]

--symptomid [string]

--type [string]

SupportAssist configuration status

Parameters

-h, --help

Help for SupportAssist configuration status.

SupportAssist configuration system mode

Parameters

-h, --help

Help for SupportAssist configuration system mode.

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Syslogs commands

This section contains commands to manage syslogs.

Modify syslogs

Modifies the syslogs.

Parameters

-h, --help
Help for modifying the syslogs.

--id [string]
Unique Identifier of the syslogs.

--description [string]

--destination_address [string]

--name [string]

Query syslogs

Queries syslogs.

Parameters

-h, --help
Help for modifying the syslog.

--id [string]
Unique Identifier of the syslog.

Delete syslogs

Deletes syslogs.

Parameters

-h, --help
Help for deleting the syslog.

--id [string]
Unique Identifier of the syslog.

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User management commands

This section contains commands to manage users.

Create users

Parameters

-h, --help

Help for creating users.

--email_address [string]

--first_name [string]

--last_name [string]

--is_enabled [string]

--is_temporary [string]

--username [string]

--password [string]

--role [string]

Values are:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician,Monitor
- Support
- NA

--type [string]

Modify users

Modify a user by ID.

Parameters

-h, --help

Help for modifying users.

--id [string]

Unique Identifier of an existing user.

--is_enabled [string]

--first_name [string]

--last_name [string]

--role [string]

Values are:

- SuperUser
- SystemAdmin
- StorageAdmi
- LifecycleAdmin
- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support
- NA

--username [string]

Query users

Parameters

-h, --help

Help for querying users.

--id [string]

Unique Identifier of an existing user.

--filter [string]

Conditions for filtering users (username, first_name, last_name, type, is_enabled, email_address).

--limit [int32]

Limit number of returned objects.

--offset [string]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

Delete users

Delete a user by ID.

Parameters

-h, --help

Help for deleting users.

--id [string]

Unique identifier of an existing user.

II

SCLI

This section explains the basic concepts of using the PowerFlex SCLI, including how to access the SCLI and log in and lists the SCLI commands.

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SCLI basics

This section explains the basic concepts of using the PowerFlex SCLI, including how to access the SCLI and log in.

Accessing SCLI

PowerFlex CLI (SCLI) is installed as part of the MDM component. It can be found in the following path:

Table 2. Operating systems and CLI paths

Operating system	Path
Linux	`scli`
VMware	`scli`
Citrix Hypervisor (XenServer)	`siocli`

Before using most SCLI commands, you must log in using the following command:

```
scli --login --management_system_ip <MNO IP> --username <username> --password <password>
```

`--management_system_ip <MNO_IP>`—IP address of the management system.

`--username <NAME>`—Username to log in.

`--password <PASSWORD>`—User password. If the password is missing, the user is prompted to enter the password.

Command format

All SCLI commands use the following format:

Table 3. SCLI command format by operating system

Operating system	Command format
Linux	`scli [--mdm_ip <IP>] --<cmd_name>`
VMware	`scli [--mdm_ip <IP>] --<cmd_name>`
Windows	`scli [--mdm_ip <IP>] --<cmd_name>`
Xen	`siocli [--mdm_ip <IP>] --<cmd_name>`

Where:

- `--mdm_ip <IP>`

  is one or more IP addresses of the servers running the primary MDM and secondary MDM. In a non-clustered environment, use the MDM IP address.

  If the CLI does not reside on the MDM, the `--mdm_ip` parameter must be added to every CLI command.

- `--<cmd_name>`

Is the command to be run.

Example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --query_all
```

Additional rules:

- All parameters are preceded by --
- The order of the parameters in the command is insignificant.
- SCLI commands are lowercase and case-sensitive.

--mdm_ip

The `mdm_ip` parameter indicates the MDM designated to receive and run the command. If the command is run from the primary MDM, this switch may be omitted.

Example:

```
scli --query_all
```

To avoid using the `--mdm_ip` parameter in every command, or avoid having to install the CLI on other servers, use SSH or RDM to log in to the shell running on the management server.

You cannot run SCLI commands on the secondary MDM. However, you can send a command from the secondary MDM by adding the `--mdm_ip` parameter with the primary MDM IP address to the command.

Syntax

SCLI syntax follows these rules:

Format

The following command syntax is mandatory:

```
scli --cmd_name
```

Parentheses ( ) indicate a group of required parameters, from which you must choose one.

Angle brackets <> indicate an argument.

Braces { } indicate a series of options for parameter values, from which you must choose one.

A pipe | is used to separate items in a series of options (parameters or parameter values). Select one of them.

Square brackets [ ] indicate an optional parameter.

Usage

Each command entry uses the above syntax and looks similar to the following example:

```
scli --cmd_example --r1 (--r2 | --r3 <V1>) [o1 <V2> | o2 {--so1 <V3> | --so2 }] [Options]
```

Parameter description

`--r1`

The `r1` parameter description

`--r2`

The `r2` parameter description

--r3<*V1*>

The `r3` parameter description with possible <*V1*> input values

--o1<*V2*>

The `o1` description with possible <*V2*> input values

--o2 {--so1 <V3> | --so2 }

The `o2` description

[Options]

Additional optional parameters:

so1 <V3>

The `so1` description with possible *V3* input values

so2

The `so2` description

Interpretation

- The text `scli --cmd_example r1` is mandatory. In commands that have no parameters, just `scli --cmd_example` is necessary.
- `(--r2 | --r3 <V1>)` indicates that you must choose one of the options separated by |. Selecting an option is required, as indicated by ( ).
- `[o1 <V2>| o2 {--so1 <V3> | --so2}]` indicates that you may choose one of the options separated by |. Selecting one of these options is optional, as indicated by [ ].
- `[Options]` indicates that you may choose from the following additional parameters. It is optional, as indicated by [ ].

Object names

Define object names according to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID using a query or the property sheet of the object, available in the PowerFlex Manager user interface.

It is recommended that you give each object a meaningful name that is associated with its operational role.

NOTE: The system has a unique name as it is a single object.

Using SCLI in non-secure mode

If PowerFlex is running in non-secure mode, you must disable secure communications on every MDM server to enable execution of commands.

- To disable secure communications in Windows, on each MDM open the SCLI `conf.txt` file, and add the following line:

```
cli_use_secure_communication=0
```

- To disable secure communications in Linux, run the following on each MDM:

```
echo cli_use_secure_communication=0 >> ~/.scli/conf.txt
```

NOTE: Changing from secure mode to non-secure mode after renewing certificates is not supported.

Logging in

To access the CLI, you must first log in to the management system using a terminal application.

If the CLI and the MDM do not reside on the same server, add the `--mdm_ip` parameter to all CLI commands.

In a non-clustered environment, use the MDM IP address. In a clustered environment, use the IP addresses of the primary and secondary MDMs, separated by a comma. For example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --login --username supervisor1 --password password1
--management_system_ip 10.11.12.13
```

If you do not enter the password, you will be prompted to enter the password.

login

Log in to the system by either specifying management credentials or using a pre-generated p12 certificate. It is recommended to avoid the password options. If not used, the user is prompted to enter the passwords in a secure manner.

Once you are authenticated by the system, all commands can be run with the respective role until a logout is performed, or until the session expires by reaching one of the following timeouts:

- Maximum session length (default: 8 hours)
- Session idle time (default: 10 minutes)

Before you log in for the first time, run the following commands:

`cp/opt/emc/scaleio/mdm/cfg/mgmt_ca.pem/`—Copy the management certificate to the root location.

`scli --add_certificate --certificate_file /opt/emc/scaleio/mdm/cfg/mgmt_ca.pem`—Add the management certificate to the trusted certificate.

Syntax

```
scli --login ( (--management_system_ip <MNO_IP> --username <NAME> [--password
<PASSWORD>] ) | ( [--p12_path <P12_PATH>] [--p12_password <P12_PASSWORD>] ) ) [--
accept_banner_by_scripts_only]
```

Parameters

`--management_system_ip <MNO_IP>`

IP address of the management system.

`--username <NAME>`

User name to log in

`--password <PASSWORD>`

User password. If the password is missing, the user is prompted to enter the password.

NOTE: In Linux, to prevent the password from being recorded in the history log, type the command without password parameter and then enter the password in the new line when it is prompted.

`--p12_path <P12_PATH>`

Path to p12 certificate file. If not specified, uses same default path as in generate_login_certificate

`p12_password <P12_PASSWORD>`

Password for p12 certificate. If not specified, the user is prompted to enter the password.

`--accept_banner_by_scripts_only`

Preemptive approval of login banner

Examples

```
scli --login --management_system_ip dellpowerflex.com --username myuser --password 2@3#4A
```

If a login banner is configured and enabled on your system, press any key to display the banner. Enter "q" to quit the login banner, and then enter "y" to approve the banner.

generate_login_certificate

Generate a P12 certificate for CLI.

Syntax

```
scli --generate_login_certificate --management_system_ip <MNO_IP> --username <NAME> [--
password <PASSWORD>] [--p12_password <P12_PASSWORD>] [--p12_path <P12_CERT_PATH>]
```

Parameters

--management_system_ip <MNO_IP>

IP address of the management system.

--username <NAME>

User name to log in

--password <PASSWORD>

User password. If the password is missing, the you are prompted to enter the password.

NOTE: In Linux, to prevent the password from being recorded in the history log, leave out the `password` flag and enter the password interactively.

--p12_password <P12_PASSWORD> (Optional)

The P12 password is used to encrypt the certificate. If missing, you are prompted to enter the password.

--p12_path <P12_CERT_PATH> (Optional)

The path where the p12 certificate is stored. If not specified, use the same default path as in login.

Examples

```
scli --login --management_system_ip dellpowerflex.com --username myuser --password
2@3#4A --p12_path /home/myuser/my_certificate.p12 --p12_password "password"
```

logout

Log out the current user of the system.

Syntax

```
scli --logout
```

Example

```
scli --logout
```

Getting help

The PowerFlex CLI supports auto-completion. To complete a command or parameters, press the Tab key while typing CLI commands.

NOTE:

In Windows, PowerFlex does not support auto-completion.

Alternately, you can run the `help` command.

Help

Display help for the CLI commands.

Syntax

```
scli --help
  [--mdm]
  [--sds]
  [--vol]
  [--sdc]
  [--general]
  [--all]
  [--user]
```

NOTE: To view all help options, run the command without any additional options:

```
scli --help
```

Options

--mdm
: Displays help for MDM commands

--sds
: Displays help for SDS commands

--vol
: Displays help for volume commands

--sdc
: Displays help for SDC commands

--general
: Displays help for general commands

--all
: Displays help for all commands

--user
: Displays help for commands related to user management

Example

```
scli --help --mdm
```

CLI command help syntax

- `Text`—Required
- `<>`—Argument
- `()`—Required element
- `[]`—Optional element
- `|`—Select from options A | B | C

Example command:

```
scli --cmd_example --req1 (--req2 | --req3 <VAL1>) [opt1 <VAL2> | opt2] Options
```

Some description of what `cmd_example` does.

Parameters:

`--req1`
req1 description

`--req2`
req2 description

`--req3 <V1>`
req3 description with possible VAL1 input values

`--opt1 <VAL2>`
opt1 description with possible VAL2 input values

`--opt2`
opt2 description

Options:
CHOOSE ONE/SEVERAL

`--spec_opt1 <VAL3>`
spec_opt1 description with possible VAL3 input values

`--spec_opt2`
spec_opt2 description

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Acceleration pool commands

This section contains commands for managing acceleration pools.

add_acceleration_pool

Add an acceleration pool to a protection domain.

Syntax

```
scli --add_acceleration_pool (--protection_domain_id <ID> | --protection_domain_name
<NAME>) (--media_type <TYPE> {SSD | NVDIMM})
  [--acceleration_pool_name <NAME>]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--media_type {SSD | NVDIMM}
Type of devices in the acceleration pool: SSD or NVDIMM

--acceleration_pool_name *<NAME>*
Name to assign the acceleration pool you are creating

Example

```
scli --add_acceleration_pool --protection_domain_name pd1 --media_type SSD --
acceleration_pool_name RfcAccPool
```

modify_metadata_acceleration

Modify the acceleration pool that holds the metadata for the specified storage pool.

Syntax

```
scli --modify_metadata_acceleration (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)
  [--on_device_metadata_on_missing_capacity]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--acceleration_pool_name *<NAME>*

Acceleration pool name

--acceleration_pool_id *<ID>*

Acceleration pool ID

--on_device_metadata_on_missing_capacity

When there is insufficient capacity in the acceleration pool, the metadata of some of the devices resides on the devices themselves.

Example

```
scli --modify_metadata_acceleration --protection_domain_name pd2 --
storage_pool_name sp4 --protection_domain_name pd2 --acceleration_name ap4 --
on_device_metadata_missing_capacity
```

query_acceleration_pool

Retrieve information about a protection domain's acceleration pool.

Syntax

```
scli --query_acceleration_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--acceleration_pool_name *<NAME>*

Acceleration pool name

--acceleration_pool_id *ID>*

Acceleration pool ID

Example

```
scli --query_acceleration_pool --protection_domain_name pd2 --acceleration_pool_name
NVDIMM_pool
```

Sample output

```
Acceleration Pool RFcache_Pool (Id: 6794008e00000000)
        Media Type: SSD
        Used for RFcache: yes
```

remove_acceleration_pool

Remove an acceleration pool from a protection domain.

Syntax

```
scli --remove_acceleration_pool (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id
<ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--acceleration_pool_name *<NAME>*
Acceleration pool name

--acceleration_pool_id *ID>*
Acceleration pool ID

Example

```
scli --remove_acceleration_pool --protection_domain_name pd1 --acceleration_pool_name
ap_flash1
```

rename_acceleration_pool

Name or rename an acceleration pool.

Syntax

```
scli --rename_acceleration_pool (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id
```

```
<ID>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--acceleration_pool_name *<NAME>*
Acceleration pool name

--acceleration_pool_id *ID>*
Acceleration pool ID

--new_name *<NAME>*
New name to be assigned to the acceleration pool

Example

```
scli --rename_accleration_pool --protection_domain_name pd1 --acceleration_pool_name
ap_1 --new_name ap_flash1
```

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Device commands

This section contains commands for managing devices.

abort_remove_sds_device

Stop the process of removing a storage data server (SDS) device from an SDS. If it is performed at too late a stage in the removal process, the command may fail and the device is removed.

NOTE: Removal of acceleration devices cannot be aborted.

Syntax

```
scli --abort_remove_sds_device (--device_id <ID> ((--sds_id <ID> | sds_name <NAME> |
sds_ip [sds_port <PORT>]) (device_name <NAME>  | device_path <PATH>)))
```

Parameters

--device_id *<ID>*

The ID of the storage device that was requested to be removed from the SDS

--sds_id *<ID>*

The ID of the SDS from which the device was to be removed

--sds_name *<NAME>*

The name of the SDS from which the device was to be removed

--sds_ip *<IP>*

The IP address of the SDS from which the device was to be removed

--sds_port *<PORT>*

The port associated with the SDS

--device_name *<NAME>*

The name of the storage device or file that was requested to be removed from the SDS

--device_path *<PATH>*

The path of the device that was requested to be removed

Example

```
scli --abort_remove_sds_device --sds_ip 192.168.1.6 --device_name /dev/sdb
```

activate_sds_device

Activate SDS devices that have completed their tests.

The system does not use SDS devices that are added using the `add_sds` or `add_sds_device` command, with the `test_only` parameter, until activated by this command.

This command does not apply to acceleration devices which are activated automatically.

NOTE: You can activate one or all SDS devices on an SDS with a single command.

Syntax

```
scli --activate_sds_device (--sds_id <ID> |
--sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_id <ID> | --device_name
<NAME>  | --device_path <PATH> | --activate_all_devices)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_id *<ID>*
ID of the SDS device to activate

--device_name *<NAME>*
Storage device name or file name

--device_path *<PATH>*
Path of the SDS device to activate

--activate_all_devices
Activate all of the specified SDS devices. If all the devices cannot be activated, this command fails.

Example

```
scli --mdm_ip 192.168.1.200 --activate_sds_device
--sds_ip 192.168.1.5 --activate_all_devices
```

add_sds_device

Add a storage or acceleration device to an SDS.

CAUTION: During execution of the command, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is returned, and the command fails for that device. If you want to overwrite existing data on the device by forcing the command, use the `force_device_takeover` flag. This flag destroys all existing data on the device.

Syntax

```
scli --add_sds_device (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --device_path <PATH> --device_name <NAME>] (--storage_pool_name <NAME> | --
storage_pool_id <ID> | --acceleration_pool_name <NAME> | --acceleration_pool_id <ID>)
  [--force_device_takeover]
  [--media_type {HDD | SSD}]
  [--external_acceleration {none | read | write | read_and_write}]
```

```
   [--test_time <TIME>]
   [--test_only | --no_test | <BLANK> ]
```

Parameters

--sds_id *<ID>*

ID of the SDS

--sds_name *<NAME>*

Name of the SDS

--sds_ip *<IP>*

SDS IP address

-sds_port *<PORT>*

Port associated with the SDS

--device_path *<PATH>*

Full path of the device to be added. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device.

--device_name *<NAME>*

Name to assign to the device

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--acceleration_pool_name *<NAME>*

Acceleration pool name

--acceleration_pool_id *<ID>*

Acceleration pool ID

--force_device_takeover

Add the device, ignoring data from other SDSs on the device.

CAUTION: Use this flag with caution, because all data on the device is destroyed.

--media_type {HDD | SSD}

Media type of the device, where the device will be HDD or SSD.

--external_acceleration *<TYPE>*

External acceleration for the device where *<TYPE>* can be one of the following:

none

The device is not accelerated by non-PowerFlex cache.

read

The device is accelerated by a non-PowerFlex read cache.

write

The device is accelerated by a non-PowerFlex write cache.

read_and_write

The device is accelerated by both non-PowerFlex read and write cache.

--test_time *<TIME>*

The maximum test run-time in seconds. Default: 10. The test will stop when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first. When `no_test` is selected, this switch is ignored.

Test options (choose one):

By default, PowerFlex tests the performance of the device being added before its capacity can be used, and saves the results. Two tests are performed: random writes and random reads. When the tests are

complete, the device capacity is added automatically to the storage pool used by the MDM. To modify this behavior, specify one of the test options:

--test_only

Devices will be tested, but not used. To start using their capacity, run the `--activate_sds_device` command.

--no_test

The device capacity will be used without any device testing.

<BLANK>

Read and write test will be run on devices before their capacity is used.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds_device
--sds_ip 192.168.1.6 --device_path /dev/sdb --device_name sd02
```

Devices

The following devices are considered to be not "clear," and cannot be added to an SDS:

- Linux: A complete device with either a file system or partition, or a partitioned device with a file system.
- Windows: A complete device with a partition, or a partitioned device with a file system.
- ESX: The same as the above, depending on the operating system of the SVM where the SDS is installed.

Limitations:

- SAN devices are not prevented from being added to an SDS.
- Devices in an LVM group cannot be added to an SDS.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

SDS device names

Assigning each SDS device a meaningful name facilitates future object identification, because the defined name remains constant even if the path changes. When a name has not been defined, the system may display default system-defined names that use the first IP address of the SDS.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

clear_sds_device_error

Clear an SDS device error.

NOTE: The command does not check whether the device error was fixed. If the error persists, the device returns to an error state when it is accessed. You can verify that the device error is fixed by running an SDS query.

Syntax

```
scli --clear_sds_device_error (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) (--device_id <ID> | --device_name <NAME> | --device_path <PATH> | --
clear_all)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_id *<ID>*
ID of the device

--device_name *<NAME>*
Storage device name or file name

--device_path *<PATH>*
Path of the device

--clear_all
Clear errors on all SDS devices.

Example (single device)

```
scli --mdm_ip 192.168.1.200 --clear_sds_device_error
--device_name /dev/sdb --sds_ip 192.168.1.7
```

Example (all devices)

```
scli --mdm_ip 192.168.1.200 --clear_sds_device_error
--clear_all --sds_ip 192.168.1.7
```

modify_sds_device_capacity

Modify the amount of an SDS device's storage capacity that is made available to PowerFlex.

Use this command to reduce (or increase) the device capacity that is available to PowerFlex, in order to assist in balancing storage, without affecting the physical disk size. Ensure that the modified capacity is less than the total physical size of the device. If you reduce capacity, data will be moved to other devices in an asynchronous fashion.

When reducing capacity, the excess device capacity is not accessible to the user.

Syntax

```
scli --modify_sds_device_capacity (--device_id <ID> | ((--sds_id <ID> | --sds_name
<NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path
<PATH>))) --size_gb <SIZE>
```

Parameters

--device_id *<ID>*
ID of the SDS device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

SDS storage device name or file

--device_path *<PATH>*

Path of the device whose capacity is to be modified

--size_gb *<SIZE>*

Size in GB of the device to use. The size must be within the device size limits and slightly less than the full device size.

Example

```
scli --mdm_ip 192.168.1.200 --modify_sds_device_capacity --sds_ip 192.168.1.6 --
device_name /dev/sdb --size_gb 150
```

query_all_device_latency_meters

Retrieve device latency meters for all the devices, or for devices in the specified protection domain.

NOTE: To query the devices in a specific SDS, use the `query_device_latency_meters` command.

Syntax

```
scli --query_all_device_latency_meters
  [(--protection_domain_id <ID> | --protection_domain_name <NAME>)]
```

NOTE: If the background device scanner is enabled, several device read statistics are dramatically affected.

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --query_all_device_latency_meters --protection_domain_name pd10
```

Sample output

```
Protection Domain: fcc9af5500000000, Name: pd1 contains 4 SDS
Storage Pool: 3df1cd9600000000, Name: sp1 contains 8 devices
  SDS sds_idd_1:
    Device: /dev/sdb Average latency read: 836 write: 921 (microseconds)
    Device: /dev/sdc Average latency read: 755 write: 843 (microseconds)

  SDS sds_idd_2:
    Device: /dev/sdb Average latency read: 658 write: 815 (microseconds)
    Device: /dev/sdc Average latency read: 689 write: 732 (microseconds)

  SDS sds_idd_4:
    Device: /dev/sdb Average latency read: 842 write: 811 (microseconds)
    Device: /dev/sdc Average latency read: 852 write: 918 (microseconds)

  SDS sds_idd_5:
    Device: /dev/sdb Average latency read: 844 write: 698 (microseconds)
    Device: /dev/sdc Average latency read: 795 write: 801 (microseconds)

  Devices with maximal read latency:
    SDS: sds_idd_4 Device: /dev/sdc  Latency: 852 (microseconds)
    SDS: sds_idd_5 Device: /dev/sdb  Latency: 844 (microseconds)
    SDS: sds_idd_4 Device: /dev/sdb  Latency: 842 (microseconds)
  Devices with maximal write latency:
    SDS: sds_idd_1 Device: /dev/sdb  Latency: 921 (microseconds)
    SDS: sds_idd_4 Device: /dev/sdc  Latency: 918 (microseconds)
    SDS: sds_idd_1 Device: /dev/sdc  Latency: 843 (microseconds)

Protection Domain: fcc9af5600000001, Name: pd2 contains 1 SDS
Storage Pool: 3df1cd9700000001, Name: sp2 contains 1 devices
  SDS sds_idd_3:
    Device: /dev/sdb Average latency read: 0 write: 0 (microseconds)


Protection Domain: fcc9d66500000002, Name: rrr contains 0 SDS
```

query_device_latency_meters

Retrieve the latency meters for all the devices of the specified SDS.

The SDS maintains statistics about reads and writes to its devices. This command returns the average I/O size and latency for both reads and writes.

NOTE: To query all devices in the system, or all devices in a specific protection domain, use the `query_all_device_latency_meters` command.

Syntax

```
scli --query_device_latency_meters (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) [(--storage_pool_name <NAME> | --storage_pool_id <ID>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

NOTE: If the background device scanner is enabled, several device read statistics are dramatically affected.

Example

```
scli --mdm_ip 192.168.1.200 --query_device_latency_meters --sds_name sds_idd_2 --
storage_pool_name sp1
```

Sample output

```
SDS sds_idd_2 returned information on 2 devices
Device: /dev/sdc
  Average read size: 277.1 KB (283763 Bytes)
  Average read latency: 873 (microseconds)
  Average write size: 355.6 KB (364140 Bytes)
  Average write latency: 1144 (microseconds)
Device: /dev/sdb
  Average read size: 292.8 KB (299816 Bytes)
  Average read latency: 928 (microseconds)
  Average write size: 362.0 KB (370731 Bytes)
  Average write latency: 1157 (microseconds)
```

query_device_test

Retrieve the SDS device test results. Every SDS saves results for the last two tests per device.

Syntax

```
scli --query_device_test ((--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
| (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
| --storage_pool_id <ID>) | (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>))
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_id *<ID>*

Device ID

--device_name *<NAME>*

SDS storage device name or file name

--device_path *<PATH>*

File path to storage device

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--acceleration_pool_name *<NAME>*

Storage pool name

--acceleration_pool_id *<ID>*

Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --query_device_test
--sds_ip 192.168.1.6 --device_name /dev/sdb
```

query_sds_device_info

Retrieve detailed information for the specified SDS device, or for all devices in the specified SDS.

Syntax

```
scli --query_sds_device_info (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
  [--all_devices]
```

Parameters

--device_id *<ID>*

ID assigned to the storage device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name assigned to the storage device

--device_path *<PATH>*

Device path or file path of the storage device

--all_devices

Return detailed information for all devices in the specified SDS.

Example

```
scli --query_sds_device_info --sds_id 06b108a400000004 --all_devices
```

Sample output

```
Storage Device Information (total 2 devices):
        Device ID: c8bea95800040000 Name: N/A Path: /dev/sdb
                ScaleIO Device Configuration:
                        Original Path: /dev/sdb
                        Storage Pool Name: sp1
                        Capacity Limit: 148.9 GB (152500 MB)
                        Device State: Normal
                        Data Layout: Medium Granularity
                        MG Persistent checksum state: PROTECTED
                Physical Device Information:
                        Device Type: UNKNOWN
                        Media Type: HDD
                        Auto Detected Media Type: SSD
                        External Acceleration Type: NONE
                        Queue Depth: 16
                        Preset Queue Depth: HDD
                        Vendor Name: N/A
                        Model Name: N/A
                        Serial Number: N/A
                        Slot Number: N/A
                        Firmware Version: N/A
                        Cache Look-ahead: not Active
                        Write Cache: not Active
                        ATA Security: not Active
                        Logical Sector Size: 0 B
                        Physical Sector Size: 0 B
                        Capacity: 0 GB
                        LED Setting: OFF
                Background Device Scanner Information:
                        Scanned: 1873793 MB
                        Unresolved Read Errors: 0/0
                        Unresolved Compare Errors: 0/0
                SMART Information:
                        Aggregated State: NEVER_FAILED
                        Temperature State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                        Media Wearout Indicator State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                RAID Controller Information:
                        Serial Number: N/A
                        RAID vDisk Status: N/A
                        RAID vDisk Type: N/A
                        RAID vDisk Cache: N/A

        Device ID: c8bea95900040001 Name: N/A Path: /dev/sdc
                ScaleIO Device Configuration:
                        Original Path: /dev/sdc
                        Storage Pool Name: sp1
                        Capacity Limit: 148.9 GB (152500 MB)
```

```
                        Device State: Normal
                        Data Layout: Medium Granularity
                        MG Persistent checksum state: PROTECTED
                Physical Device Information:
                        Device Type: UNKNOWN
                        Media Type: HDD
                        Auto Detected Media Type: SSD
                        External Acceleration Type: NONE
                        Queue Depth: 16
                        Preset Queue Depth: HDD
                        Vendor Name: N/A
                        Model Name: N/A
                        Serial Number: N/A
                        Slot Number: N/A
                        Firmware Version: N/A
                        Cache Look-ahead: not Active
                        Write Cache: not Active
                        ATA Security: not Active
                        Logical Sector Size: 0 B
                        Physical Sector Size: 0 B
                        Capacity: 0 GB
                        LED Setting: OFF
                Background Device Scanner Information:
                        Scanned: 1873886 MB
                        Unresolved Read Errors: 0/0
                        Unresolved Compare Errors: 0/0
                SMART Information:
                        Aggregated State: NEVER_FAILED
                        Temperature State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                        Media Wearout Indicator State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                RAID Controller Information:
                        Serial Number: N/A
                        RAID vDisk Status: N/A
                        RAID vDisk Type: N/A
                        RAID vDisk Cache: N/A


Acceleration Device Information (total 0 devices):

RAID Controllers Information (total 0 controllers):
```

remove_sds_device

Initiate removal of a storage device from an SDS. Note that this process is asynchronous, and will run in the background.

An SDS device can be removed at any time, and no downtime is required. The removal process is asynchronous and runs in the background.

During execution of this command, the associated data is replicated to different nodes. Therefore, the process may take a long time.

When removing an SDS device on a VMware server, use the vSphere client.

NOTE: If the capacity of this SDS is still used by volumes, and cannot be replaced, the command will fail.

Syntax

```
scli --remove_sds_device (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --
sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
```

Parameters

--device_id *<ID>*

ID of the device to be removed

--sds_id *<ID>*

ID of the SDS from which the device should be removed

--sds_name *<NAME>*

Name of the SDS from which the device should be removed

--sds_ip *<IP>*

IP address of the SDS from which the device should be removed

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name of the storage device or file to be removed from the SDS

--device_path *<PATH>*

Path of the storage device to be removed from the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds_device
--sds_ip 192.168.1.6 --device_name /dev/sdb
```

Related operations

To abort the removal, run the command `scli --abort_remove_sds_device`.

To track progress of the removal operation, use either of the following queries:

- `scli --query_sds_device_info`
- `scli --query_sds`

rename_device

Assign a name to, or rename, an SDS device.

Syntax

```
scli --rename_device (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --new_name
<NAME>
```

Parameters

--device_id *<ID>*

ID of the SDS device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name of the SDS device

--device_path *<PATH>*

Path of the SDS device

--new_name *<NAME>*

New name to be assigned to the SDS device

Example

```
scli --rename_device --sds_ip 8.8.8.100 --device_name flash_18 --new_name flash_10018
```

set_sds_device_external_acceleration_type

Set the external acceleration type of an SDS device.

Syntax

```
scli --set_sds_device_external_acceleration_type (--device_id <ID> | ((--sds_id <ID>
| --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --
device_path <PATH>))) --external_acceleration <TYPE>
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Device name

--device_path *<PATH>*

SDS storage device path or file path

--external_acceleration *<TYPE>*

External acceleration for the device, where TYPE is one of the following:

none

The device is not accelerated by a non-PowerFlex read or write cache.

read

The device is accelerated by a non-PowerFlex read cache.

write

The device is accelerated by a non-PowerFlex write cache.

read_and_write

The device is accelerated by both non-PowerFlex read and write cache.

Example

```
scli --set_sds_device_external_acceleration_type --sds_name sds_23 --device_name
storage_2310 --external_acceleration read
```

set_sds_device_led

Turn the LED on a storage or cache device on or off.

Syntax

```
scli --set_sds_device_led (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) (--
led_on | --led_off)
```

Parameters

--device_id *<ID>*
: ID assigned to the storage device

--sds_id *<ID>*
: SDS ID

--sds_name *<NAME>*
: SDS name

--sds_ip *<IP>*
: SDS IP address

--sds_port *<PORT>*
: Port associated with the SDS

--device_id *<ID>*
: Device ID.

--device_name *<NAME>*
: Name assigned to the storage device

--device_path *<PATH>*
: Device path or file path of a storage device

--led_on
: Turn the LED on.

--led_off
: Turn the LED off.

Example

```
scli --set_sds_device_led --sds_ip  192.168.1.5 --sds_port 7072 --device_name slot5 --
led_on
```

set_sds_device_media_type

Set the media type of an SDS device.

Syntax

```
scli --set_sds_device_media_type (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME>
| --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
media_type <TYPE>
```

Parameters

--device_id *<ID>*
Device ID

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path

--media_type *<TYPE>*
Media type of the device where TYPE is one of the following:

HDD
The device is an HDD device.

SSD
The device is an SSD device.

Example

```
scli --set_sds_device_media_type --sds_name sds22 --device_name storage_210 --media_type
HDD
```

start_device_test

Initiate a device test for one or all SDS devices.

Syntax

```
cli --start_device_test ((--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
| (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
|--storage_pool_id <ID>)) | (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)) (Options)
(Control)
  [--i_am_sure]
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

IP address associated with the SDS

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

SDS storage device name or file

--device_path *<PATH>*

SDS storage device path or file path

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--acceleration_pool_name *<NAME>*

Acceleration pool name

--acceleration_pool_id *<ID>*

Acceleration pool ID

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Options

Choose several:

--io_type_write

Use write I/O type (default is read). If the storage device is in use, only read testing can be performed.

--io_pattern_squential

Use sequential I/O pattern (default is random).

--io_size_kb *<SIZE>*

The read or write I/O size in KB. Range is 4-512 KB. Default: 8 KB.

Control

Choose several:

--test_time *<TIME>*

Maximum test run time in seconds. If not specified, the test will be limited by I/O size only.

--io_total_size_mb *<SIZE>*

The total size of the test reads or writes in MB. Range is 1-32768 MB, which specifies the total I/O size in MB. Default: 128 MB

<blank>

Use defaults

Example

```
scli --mdm_ip 192.168.1.200 --start_device_test
--sds_ip 192.168.1.6 --device_path /dev/sdb --io_type_write --io_size_kb 8
```

set_sds_device_path

Configure the path of the given storage device.

Syntax

```
scli --set_sds_device_path (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
new_device_path <PATH>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path. The file path is used only when the device is available at this path

--new_device_path *<PATH>*
The new path to configure

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Fault set commands

This section contains commands for managing fault sets.

add_fault_set

Add an empty fault set to a protection domain.

Fault sets are subsets of a given protection domain. They provide additional safeguards for protecting your data against hardware failure. When you define SDSs later on, you can add them to a fault set.

Fault sets are enabled by default.

Syntax

```
scli --add_fault_set (--protection_domain_id <ID> | --protection_domain_name <NAME>)
  [--fault_set_name <NAME>]
```

Parameters

--protection_domain_id *<ID>*

ID of the protection domain to which the fault set will be added

--protection_domain_name *<NAME>*

Name of the protection domain to which the fault set will be added

--fault_set_name *<NAME>*

Name of the new fault set

Example

```
scli --add_fault_set --protection_domain_name pd1 --fault_set_name fs1-mg
```

Fault set names

Assign each fault set a meaningful name. When a name has not been defined, the system may display default system-defined names that use the ID of the fault set.

Each fault set name must conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

clear_fault_set

Remove all SDSs from the specified fault set.

Syntax

```
scli --clear_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
  [--i_am_sure]
```

Parameters

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --clear_fault_set --protection_domain_name pd18 --fault_set_name sio-fs1
```

query_all_fault_sets

Retrieve information for all the fault sets in the specified protection domain.

Syntax

```
scli --query_all_fault_sets (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

Example

```
scli --query_all_fault_sets --protection_domain_name pd1
```

Sample output

```
Protection Domain has 1 Fault Sets:
Protection Domain ID: fcc9af5500000000 Fault Set: ID: 36c9d76800000000 Name: fs1_mg SDS
Count: 2
  SDS ID: 06b108a400000004 Name: sds_idd_1
  SDS ID: 06b108a300000003 Name: sds_idd_2
```

query_fault_set

Retrieve information about the specified fault set.

Syntax

```
scli --query_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
```

Parameters

--fault_set_id *<ID>*
: Fault set ID

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--fault_set_name *<NAME>*
: Name of the new fault set

Example

```
scli --query_fault_set --protection_domain_name pd18 --fault_set_name fs1_mg
```

Sample output

```
Protection Domain ID: fcc9af5500000000 Fault Set: ID: 36c9d76800000000 Name: fs1_mg SDS
Count: 2
  SDS ID: 06b108a400000004 Name: sds_idd_1
  SDS ID: 06b108a300000003 Name: sds_idd_2
```

remove_fault_set

Remove the specified fault set.

Ensure that the fault set is empty before executing this command. You can remove SDSs from the fault set using the `clear_fault_set` command.

Syntax

```
scli --remove_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
```

Parameters

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Name of the new fault set

Example

```
scli --remove_fault_set --protection_domain_name pd18 --fault_set_name sio-fs1
```

rename_fault_set

Rename the specified fault set.

Syntax

```
scli --rename_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)) --new_name <NAME>
```

Parameters

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--new_name *<NAME>*

New name to assign to the fault set

Example

```
scli --rename_fault_set --protection_domain_name pd18 --fault_set_name sio-fs-test --
new_name sio-fs1
```

set_sdt_fault_set

Set the fault set of an SDT.

Syntax

```
scli --set_sdt_fault_set (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) (--fault_set_id <ID> | ((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --fault_set_name <NAME>))
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_id *<ID>*
Fault set ID

--fault_set_name *<NAME>*
Fault set name

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General CLI commands

This section contains general CLI commands, including commands relating to licenses, certificates, login banner, and syslog.

acknowledge_scanner_error_counters

Acknowledge read and compare errors so that alerts are no longer be shown for the current counter value. Acknowledgment can be performed at storage pool level or at device level, and for counter type.

Syntax

```
scli --acknowledge_scanner_error_counters (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) | (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) ) [--
acknowledge_compare_errors] [--acknowledge_read_errors]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path

--acknowledge_read_errors
Acknowledge read error counter

--acknowledge_compare_errors

Acknowledge compare error counter

Example

```
scli --acknowledge_scanner_error_counters --sds_name sds25 --device_name abc123 --
acknowledge_compare_errors
```

add_trusted_ca

Upload the certificate from the specified file as a trusted certificate.

Syntax

```
scli --add_trusted_ca --certificate_file <FILE> --comment <COMMENT>
```

Parameters

--certificate_file *<FILE>*

The certificate file to upload as trusted

--comment *<COMMENT>*

Optional free text to describe the certificate

Example

```
scli --mdm_ip 192.168.1.200 --add_trusted_ca --certificate_file system2_ca.cer --comment
"Certificate for system 2"
```

extract_root_ca

Extract the MDM root certificate to the specified file. This is the first step in the process of exchanging certificates between replication peer systems.

Syntax

```
scli --extract_root_ca --certificate_file <FILE>
```

Parameters

--certificate_file *<FILE>*

The file to write the certificate to

Example

```
scli --mdm_ip 192.168.1.200 --extract_root_ca --certificate_file system1_ca.cer
```

list_trusted_ca

Display a list of the currently trusted certificates in the system.

Syntax

```
scli --list_trusted_ca
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --list_trusted_ca
```

set_license

Set or update the Dell Technologies PowerFlex license.

Syntax

```
scli --set_license --license_file <LICENSE_FILE>
```

Parameters

--license_file *<LICENSE_FILE>*
Path to a file containing the license key value provided by Dell Technologies

query_license

Retrieve information about the current license and restrictions.

Syntax

```
scli --query_license
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_license
```

query_disconnections

Query various connection statuses.

Syntax

```
scli --query_disconnections [--disconnection_type <TYPE>] [--print_all] [--print_proxy]
```

Parameters

--disconnection_type <TYPE>

Print disconnections of the given type only. The following options are available:

sdc_sds

SDC-SDS connections

sdc_sdr

SDC-SDR connections

sdr_sds

SDR-SDS connections

sdr_sdr

SDR-SDR connections

--print_all

Use this flag to print all disconnections

--print_proxy

When proxy access mode is active, print SDSs that can not serve as a proxy.

Example

```
scli --mdm_ip 192.168.1.200 --query_disconnections --disconnection_type sdr_sdr
```

query_remote_syslog

Retrieve the remote syslog module state and configuration.

Syntax

```
scli --query_remote_syslog
```

Parameters

None.

Example

```
scli --query_remote_syslog
```

query_oscillating_failure_counter_parameters

Retrieve oscillating failure counter parameters.

Syntax

```
scli --query_oscillating_failure_counter_parameters --failure_counter (<COUNTER FROM
GROUP A> (--protection_domain_id <ID> | --protection_domain_name <NAME>) | <COUNTER FROM
GROUP B> | <COUNTER FROM GROUP C> | --all_counters(((----protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>))
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be queried. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- mdm_sds_network_disconnections
- sds_sds_network_disconnections
- sds_decoupled
- sds_configuration_failures
- sds_receive_buffer_allocation_failures

Counter Group B

SDC-related counters:

- sdc_mdm_network_disconnections
- sdc_sds_network_disconnections
- sdc_long_operations
- sdc_memory_allocation_failures
- sdc_socket_allocation_failures

Counter Group C

SDS device-related counters:

- sds_device_long_successful_ios

--all_counters

Query all oscillating failure counters parameters of the selected group

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

Example

```
scli --query_oscillating_failure_counter_parameters --failure_counter
sds_device_long_successful_ios --protection_domain_name pd10 --storage_pool_name sp18
```

remove_trusted_ca

Remove the certificate with the given fingerprint.

Syntax

```
scli --remove_trusted_ca --fingerprint <FINGERPRINT>
```

Parameters

--fingerprint <FINGERPRINT>

The fingerprint of the certificate to be removed

Example

```
scli --mdm_ip 192.168.1.200 --remove_trusted_ca --fingerprint
B5:F0:43:1E:9B:7B:55:7F:05:98:B7:D9:7E:50:D4:9F:6D:1B:A7:5F
```

reset_oscillating_failure_counters

Reset the specified oscillating failure counters to zero. This command is useful when you have fixed a problem and want to ensure that an alert is no longer active in the system.

Syntax

```
scli --reset_oscillating_failure_counters (--failure_counter (<COUNTER FROM GROUP A> ((--
protection_domain_id <ID> | --protection_domain_name <NAME>) | --all_sds) | <COUNTER
FROM GROUP B> --all_sdc | <COUNTER FROM GROUP C> (((----protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) |
--all_devices) | --all_counters)
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be reset to zero. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- mdm_sds_network_disconnections
- sds_sds_network_disconnections
- sds_decoupled
- sds_configuration_failures
- sds_receive_buffer_allocation_failures

Counter Group B

SDC-related counters:

- `sdc_mdm_network_disconnections`
- `sdc_sds_network_disconnections`
- `sdc_long_operations`
- `sdc_memory_allocation_failures`
- `sdc_socket_allocation_failures`

Counter Group C

SDS device-related counters:

- `sds_device_long_successful_ios`

`--protection_domain_id <ID>`

Protection domain ID

`--protection_domain_name <NAME>`

Protection domain name

`--storage_pool_name <NAME>`

Storage pool name

`--storage_pool_id <ID>`

Storage pool ID

`--all_sds`

Reset the counter parameters on all SDSs.

`--all_sdc`

Reset the counter parameters on all SDCs.

`--all_devices`

Reset the counter parameters on all SDS devices.

`--all_counters`

Reset all counters to zero.

Example

```
scli --reset_oscillating_failure_counters --failure_counter
mdm_sds_network_disconnections --all_sds
```

set_cli_login_banner_acceptance_mode

Set the login banner acceptance mode.

Syntax

```
scli --set_cli_login_banner_acceptance_mode (--password_only | --accept_banner)
```

Parameters

`--password_only`

Allow CLI users to accept the login banner by entering a valid password.

`--accept_banner`

Require CLI users to accept the login banner by pressing 'y'.

Example

```
scli --set_cli_login_banner_acceptance_mode --password_only
```

set_cli_login_banner_preemptive_acceptance

Enable or prevent CLI users from preemptively accepting the login banner.

Syntax

```
scli --set_cli_login_banner_preemptive_acceptance (--enable | --disable)
```

Parameters

--enable

Allow CLI users to preemptively accept the login banner

--disable

Prevent CLI users from preemptively accepting the login banner

Example

```
scli --set_cli_login_banner_preemptive_acceptance --enable
```

set_login_banner

Set the login banner that is displayed at every login to the system, or remove an existing login banner.

The maximum file size for the banner is 16 KB. Only ASCII text files are supported.

Syntax

```
--set_login_banner (--filename <FILE> | --remove_banner)
```

Parameters

--filename *<FILE>*

File containing the login banner to set

--remove_banner

Remove the login banner.

Example

```
scli --set_login_banner --filename login_banner.txt
```

set_oscillating_failure_counter_parameters

Modify oscillating failure counter parameters. You can use this command to configure the time interval associated with each window type and the number of failures allowed before reporting commences for each window type, per counter.

Syntax

```
scli --set_oscillating_failure_counter_parameters --failure_counter (<COUNTER FROM GROUP
A> ((--protection_domain_id <ID> | --protection_domain_name <NAME>) | --all_sds) |
<COUNTER FROM GROUP B> --all_sdc | <COUNTER FROM GROUP C> (((----protection_domain_id
<ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) | --all_devices) --window_type {SHORT | MEDIUM | LONG} --window_interval
<INTERVAL> --failures_threshold<ID> <NUM OF FAILURES IN INTERVAL>
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be reset to zero. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- mdm_sds_network_disconnections
- sds_sds_network_disconnections
- sds_decoupled
- sds_configuration_failures
- sds_receive_buffer_allocation_failures

Counter Group B

SDC-related counters:

- sdc_mdm_network_disconnections
- sdc_sds_network_disconnections
- sdc_long_operations
- sdc_memory_allocation_failures
- sdc_socket_allocation_failures

Counter Group C

SDS device-related counters:

- sds_device_long_successful_ios

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--all_sds

Apply the counter parameters to all SDSs.

--all_sdc

Apply the counter parameters to all SDCs.

--all_devices

Apply the counter parameters to all SDS devices.

--window_type {SHORT | MEDIUM | LONG}

Window type to be modified: short, medium, or long

--window_interval *<INTERVAL>*

New window time interval in seconds. The valid interval range is 1-86400. If the interval is set to 0, the window type is disabled.

--failures_threshold*<ID>* *<NUM OF FAILURES IN INTERVAL>*

Maximum number of failures per window type to be ignored before error reporting begins

Example

```
scli --set_oscillating_failure_counter_parameters --failure_counter
mdm_sds_network_disconnections --window_interval 120 --failures_threshold 10 --
window_type short --all_sds
```

set_remote_read_only_limit_state

Configure the PowerFlex system to restrict access from remote clients to read-only operations on the MDM. When this restriction is enabled, remote client users cannot change any configurations on the MDM. However, users on a local host may still make configuration changes while the restriction is enforced.

NOTE: A local user is one that successfully communicates with the MDM using the IP address 127.0.0.1 (the default IP address used by SCLI). If you try to issue commands from the local machine using other local IP addresses, you will not be able configure the system.

Syntax

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state {enabled |
disabled}
```

Parameters

--remote_read_only_limit_state {enabled | disabled}

Enable or disable restricted state. Default: disabled.

Example

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state enabled
```

set_syslog_facility

Set the facility field of the syslog events. Legal values are 0 to 23.

Syntax

```
scli --set_syslog_facility --remote_syslog_server_ip <IP> --syslog_facility <FACILITY>
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

Example

```
scli --set_syslog_facility --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

start_remote_syslog

Start posting events to a remote syslog server.

Syntax

```
scli --start_remote_syslog --remote_syslog_server_ip <IP>
  [--remote_syslog_server_port <PORT>]
  [--syslog_facility <FACILITY>]
  [--attach_event_code]
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--remote_syslog_server_port *<PORT>*

Syslog server port. Default is 1468.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

--attach_event_code

Add the posted event code to the event message. This parameter is disabled by default.

Example

```
scli --start_remote_syslog --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

stop_remote_syslog

Stop posting events to a remote syslog server.

Syntax

```
scli --stop_remote_syslog (--remote_syslog_server_ip <IP> | --all_remote_syslog_servers)
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--all_remote_syslog_servers

Stop posting events to all remote syslog servers.

Example

```
scli --stop_remote_syslog --remote_syslog_server_ip 192.168.1.201
```

test_remote_system_connectivity

Perform a connectivity test between the specified SDTs of the protection domain and remote system.

Syntax

```
scli --test_remote_system_connectivity (--remote_system_id <ID> | --remote_system_name
<NAME>) (--protection_domain_id <ID> | --protection_domain_name <NAME>) [--
destination_volume_nsid <NSID>] [--full_output]
```

Parameters

--remote_system_id *<ID>*

The ID of the remote system

--remote_system_name *<NAME>*

The name of the remote system

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--destination_volume_nsid *<NSID>*

The destination volume NSID. If not specified, only an indication of the success of discovery information retrieval is returned.

--full_output

Print a list of IO connections to the remote system, If not specified, only the number of healthy connections is returned.

add_network_set

Add a network set to the PowerFlex system.

Syntax

```
scli --add_network_set [--network_set_name <NAME>] [--network_id <ID> | --network_name
<NAME>]
```

Parameters

--network_set_name *<NAME>*

A name to be assigned to the network set

--network_id *<ID>*

A comma separated list of system network IDs to be associated with the network set

--network_name *<NAME>*

A comma separated list of system network names to be associated with the network set

modify_network_set

Modify the list of system networks of an existing network set.

Syntax

```
scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [--
network_id <ID> | --network_name <NAME>]
```

Parameters

--network_set_id *<ID>*

Network set ID

--network_set_name *<NAME>*

Network set name

--network_id *<ID>*

A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*

A comma separated list of system network names to be replaced with the old one

query_network_set

Retrieve information about a network set.

Syntax

```
scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*

Network set ID

--network_set_name *<NAME>*

Network set name

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Protection domain commands

This section contains commands for managing protection domains.

activate_protection_domain

Activate a protection domain.

Protection domains are activated by default. Use this command to activate a protection domain that was inactivated.

Syntax

```
scli --activate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_activate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--force_activate

Activates the protection domain, ignoring current system errors

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --activate_protection_domain --protection_domain_name pd1
```

add_protection_domain

Add a protection domain.

Syntax

```
scli --add_protection_domain
  [--protection_domain_name <NAME>]
```

Parameters

--protection_domain_name *<NAME>*

Unique name to assign the protection domain

Example

```
Example
scli --mdm_ip 192.168.1.200 --add_protection_domain
--protection_domain_name rack_1.1
```

Protection domain names

Assign each protection domain a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the ID of the volume.

Each protection domain name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

disable_fgl_metadata_cache

Disable fine granularity metadata caching for the specified protection domain.

Syntax

```
scli --disable_fgl_metadata_cache (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --disable_fgl_metadata_cache --protection_domain_name PD1
```

enable_fgl_metadata_cache

Enable fine granularity metadata caching for the specified protection domain. Fine granularity metadata caching helps improve the performance of the system when using the fine granularity layout.

Syntax

```
scli --enable_fgl_metadata_cache (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --enable_fgl_metadata_cache --protection_domain_name PD1
```

inactivate_protection_domain

Inactivate the specified protection domain. Using this command is an effective way to shut down nodes and is preferable to shutting them down manually.

NOTE: When you inactivate a protection domain, the data remains on the SDSs. It is preferable to remove a protection domain if you no longer need it.

When this feature is in effect, the system may perform the following activities in the background:

- Determine whether there are any current rebuild/rebalance activities taking place. If so, the shutdown is delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Pause application I/O and disable access to volumes.
- Move the DRL mode of all SDSs to harden in preparation for rebooting the server.
- Reload all SDSs before re-enabling data access.

Syntax

```
scli --inactivate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_inactivate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--force_inactivate

Inactivate the protection domain, ignoring current system errors.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --inactivate_protection_domain --protection_domain_name pd1
```

query_port_flapping_status

Query the system port flapping status.

Syntax

```
scli --query_port_flapping_status [--weak] (--all | --protection_domain_id <ID> | --
protection_domain_name <NAME> | --sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>] | --sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --
sdc_ip<IP> | --sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] |--
apply_to_mdm | --peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP>
[--peer_system_port <PORT>] | --peer_system_object_id <ID> | --sdt_id <ID> | --sdt_name
<NAME> | --sdt_ip <IP> [--sdt_port <PORT>])
```

Parameters

--weak
Include reported paths with weak ERT violations.

--all
Query all protection domains in the system.

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--sds_id *<ID>*
SDS ID

--sds_ip *<IP>*
SDS IP address

--sds_name *<NAME>*
SDS name

--sds_port *<PORT>*
SDS port

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC global unique identifier

--sdc_ip *<IP>*
SDC IP address

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

--apply_to_mdm
Display MDM only

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_ip *<IP>*
Replication peer system IP

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

query_port_flapping_unsupported_components

Query components that do not support the port flapping feature, and/or have it disabled.

Syntax

```
scli --query_port_flapping_unsupported_components [(--full | --protection_domain_id <ID>
| --protection_domain_name <NAME>) [--raw]] [(--only_disabled | --only_unsupported)]
```

Parameters

--full
Display a full unsummarized report

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--raw
Outputs results in raw format with one component displayed per line

--only_disabled
Query only the the components that have the port flapping feature disabled

--only_unsupported

Query only the the components that do not support the port flapping feature. Omit components that have the feature disabled.

Example

```
scli --mdm_ip 192.168.1.200 --query_port_flapping_unsupported_components --full
```

query_protection_domain

Retrieve aggregated information about all objects in a single protection domain.

Syntax

```
scli --query_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --query_protection_domain --protection_domain_name pd
```

Sample output

```
Protection Domain pd (Id: bbf165fb00000000) has 1 storage pools, 1 acceleration pools, 0
Fault Sets, 5 SDS nodes and 24 volumes
        Free net capacity: 20.5 TB (20970 GB), Remaining Volume allocation limit: 365.9
TB (374688 GB), Total Volume allocation limit: 413.9 TB (423840 GB)
        Operational state is Active
        RFcache enabled, Mode: Write miss, Page Size 64 KB, Max IO size 128 KB,
Acceleration Pool ID: N/A
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Max Replication remote capacity ratio: 50%
        Max Replication remote capacity: 0 Bytes

Storage Pool sp_822jIk (Id: 2a2a65ce00000000) has 24 volumes and 20.5 TB (20970 GB) free
net capacity, 365.9 TB (374688 GB) Remaining volume allocation limit, 413.9 TB (423840
GB) Total volume allocation limit
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using No-Limit policy
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
```

```
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: none, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 21% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Doesn't use Read Flash Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Fine granularity
        Compression method: Normal
        Fine granularity acceleration pool: 68a806e600000000
        Fine granularity advanced configuration:
                Over provisioning factor: 20
                Write atomicity size: 64KB
                NVDIMM write cache size: 16MB
                NVDIMM metadata amortization 1.50
                Max compression ratio: 10
                Performance profile: high_performance
        Replication Capacity Max Allowed Ratio: 0%
        Replication Capacity Max Allowed: 0 Bytes
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0

Acceleration Pool ap_ORFKOk (Id: 68a806e600000000)
        Media Type: NVDIMM
        Used for RFcache: no

SDR-SDS connectivity status: All connected

SDS Summary:
        Total 5 SDS Nodes
        5 SDS nodes have membership state 'Joined'
        5 SDS nodes have connection state 'Connected'
        1 SDS nodes have maintenance state 'Exiting-Maintenance-Mode'

Physical data:
        52.4 TB (53651 GB) total capacity
                507.9 GB (520139 MB) in-use capacity
                        373.6 GB (382539 MB) user data
                        41.0 GB (41965 MB) metadata
                        93.4 GB (95634 MB) fine granularity spares
                        64.5 GB (66061 MB) capacity in reclamation
                41.0 TB (41940 GB) unused capacity
                10.9 TB (11202 GB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                507.9 GB (520139 MB) protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity

User data:
        365.2 GB (373913 MB) total user data
                0 Bytes medium granularity user data
                365.2 GB (373913 MB) fine granularity user data
                        365.2 GB (373913 MB) compressed data (2.0:1 compression ratio)
                        0 Bytes uncompressed data
        0 Bytes trimmed capacity
        365.2 GB (373913 MB) user data without trimmed capacity
        2.0:1 total compression ratio
        134.6:1 thin and snapshot saving
```

```
        263.1:1 overall saving
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                365.2 GB (373913 MB) at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                           7397 IOPS 115.6 MB (118352 KB)
per-second
        Primary-writes                          7444 IOPS 116.3 MB (119100 KB)
per-second
        Secondary-reads                         0 IOPS 0 Bytes per second
        Secondary-writes                        7317 IOPS 114.3 MB (117068 KB)
per-second
        Backward-rebuild-reads                  0 IOPS 0 Bytes per second
        Backward-rebuild-writes                 0 IOPS 0 Bytes per second
        Forward-rebuild-reads                   0 IOPS 0 Bytes per second
        Forward-rebuild-writes                  0 IOPS 0 Bytes per second
        Rebalance-reads                         0 IOPS 0 Bytes per second
        Rebalance-writes                        0 IOPS 0 Bytes per second
        Volume-migration-reads                  0 IOPS 0 Bytes per second
        Volume-migration-writes                 0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads  0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes 0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads   0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes  0 IOPS 0 Bytes per second

Volumes summary:
        24 thin-provisioned volumes. Total user data size: 365.2 GB (373913 MB)
        48.0 TB (49152 GB) all volumes size
        48.0 TB (49152 GB) base volumes size
        24 user volumes
```

remove_protection_domain

Remove a protection domain from the PowerFlex system. You can only remove a protection domain if no SDS and no storage pool are associated with it.

Syntax

```
scli --remove_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id <*ID*>

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --remove_protection_domain
--protection_domain_name my_protection_domain
```

rename_protection_domain

Assign a name to, or rename, a protection domain.

Syntax

```
scli --rename_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--new_name *<NAME>*

Example

```
scli --mdm_ip 192.168.1.200 --rename_protection_domain --protection_domain_name pd1 --
new_name protection_domain_1.
```

Protection domain names

Assign the protection domain a meaningful name associated with its operational role. The protection domain name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

set_default_fgl_metadata_cache_size

Configure the default fine granularity metadata cache size for a protection domain. New SDSs in the protection domain inherit this value when created.

Syntax

```
scli --set_default_fgl_metadata_cache_size (--protection_domain_id <ID> | --
protection_domain_name <NAME>) --metadata_cache_size_mb <SIZE>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--metadata_cache_size_mb *<SIZE>*

Define the default size of metadata cache for the protection domain in MB

Example

```
scli --mdm_ip 192.168.1.200 --set_default_fgl_metadata_cache_size --
protection_domain_name PD1 --metadata_cache_size_mb 10248
```

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Remote system commands

This section contains commands for managing remote systems.

add_remote_system

Add a remote system that can be used for data mobility.

Syntax

```
scli --add_remote_system --discovery_ip <IPs> [--remote_system_name <NAME>] [--
remote_system_type <SYSTEM_TYPE>] [--discovery_port <PORT>] [--remote_system_description
<DESCRIPTION>] [--transport <TRANSPORT_TYPE>] [--bw_limit_mbps <Mbps>]
```

Parameters

--discovery_ip *<IP>*
: A list of IP addresses seperated by commas

--remote_system_name *<NAME>*
: The remote system name

--remote_system_type *<SYSTEM_TYPE>*
: The remote system type. Default is NVMe

--discovery_port *<PORT>*
: NVMe discovery port. Default port is 8009

--remote_system_description *<DESCRIPTION>*
: Description of the remote system

--transport *<TRANSPORT_TYPE>*
: Currently, only TCP is supported

--bw_limit_mbps *<Mbps>*
: The remote system bandwidth limit

query_remote_system

Retrieve information about the remote system.

Syntax

```
scli --query_remote_system (--remote_system_id <ID> | --remote_system_name <NAME>)
```

Parameters

--remote_system_id *<ID>*
: The ID of the remote system

--remote_system_name *<NAME>*

The name of the remote system

query_all_remote_systems

Retrieve information about all the snapshot copy tasks.

rename_remote_system

Modify the remote system name.

Syntax

```
scli --rename_remote_system (--remote_system_id <ID> | --remote_system_name <NAME>) --
new_remote_system_name <NAME>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

--remote_system_name *<NAME>*

The name of the remote target system

--new_remote_system_name *<NAME>*

The new name for the remote system

modify_remote_system_bw_limit

Modify a remote system bandwidth limit.

Syntax

```
scli --modify_remote_system_bw_limit (--remote_system_id <ID> | --remote_system_name
<NAME>) --new_bw_limit_mbps <Mbps>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

The ID of the remote target system *<NAME>*

The name of the remote target system

--new_bw_limit_mbps *<Mbps>*

New bandwidth limit of the system

modify_remote_system_connectivity

Modify the IP list or port for the remote system.

Syntax

```
scli --modify_remote_system_connectivity (--remote_system_id <ID> | --remote_system_name
<NAME>) --discovery_ip <IP> --discovery_port <PORT>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

--remote_system_name *<NAME>*

The name of the remote target system.

Select one of the following options:

--discovery_ip *<IPs>*

A list of IP addresses seperated by commas

--discovery_port *<PORT>*

The new port of the system

modify_remote_system_description

Modify a remote system description.

Syntax

```
scli --modify_remote_system_description (--remote_system_id <ID> | --remote_system_name
<NAME>) --remote_system_description <DESCRIPTION>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

--remote_system_name *<NAME>*

The name of the remote target system

--remote_system_description *<DESCRIPTION>*

The description of the new system

remove_remote_system

Remove the remote system.

Syntax

```
scli --remove_remote_system (--remote_system_id <ID> | --remote_system_name <NAME>)
```

Parameters

--remote_system_id *<ID>*
The ID of the remote system

--remote_system_name *<NAME>*
The name of the remote system

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Replication commands

This section contains commands for managing replication.

abort_replication_consistency_group_snapshots_create

Cancel a previous request to create a snapshot of a replication consistency group (RCG).

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --abort_replication_consistency_group_snapshots_create (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --abort_replication_consistency_group_snapshots_create --
replication_consistency_group_name rcg27
```

activate_replication_consistency_group

Activate the specified replication consistency group (RCG).

Notes

Replication is enabled while activating the RCG. The initial copy process is started when there are sufficient system resources.

Syntax

```
scli --activate_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --activate_replication_consistency_group --
replication_consistency_group_name rcg27
```

activate_replication_consistency_group_mappings

Activate replication consistency group mappings on the peer system that is addressed by the command.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including activating mappings. This command is useful in situations when you choose to perform each of the individual operations directly. You can only activate mappings when the replication consistency group (RCG) role is neutral or source. You cannot activate mappings if the RCG role is target.

Syntax

```
scli --activate_replication_consistency_group_mappings (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --activate_replication_consistency_group_mappings --
replication_consistency_group_name rcg27
```

add_replication_consistency_group

Add a replication consistency group (RCG).

Notes

You may only add an RCG after the replication peer system has been added using `add_replication_peer_system`.

Syntax

```
scli --add_replication_consistency_group (--destination_system_id <ID> |
--destination_system_name <NAME> | --destination_system_ip <IP> [--
destination_system_port <PORT>] | --destination_system_object_id <ID>) --rpo
<PERIOD> [--replication_consistency_group_name <NAME>] (--protection_domain_id
<ID> | --protection_domain_name <NAME>) (--remote_protection_domain_id
<ID> | --remote_protection_domain_name <NAME>) [--target_volume_access_mode
<TARGET_VOLUME_ACCESS_MODE>] [--replication_consistency_group_activity_mode
<ACTIVITY_MODE>]
```

Parameters

--destination_system_id *<ID>*

Destination replication system ID

--destination_system_name *<NAME>*

Destination replication system Name

--destination_system_ip *<IP>*

Destination replication system IP address

--destination_system_port *<PORT>*

Port assigned to the destination replication system

--destination_system_object_id *<ID>*

Destination replication system internal object ID

--rpo *<PERIOD>*

Desired RPO in units of time. To specify the time period, use s|m|h respectively indicating seconds, minutes, or hours. Examples: 30s (30 seconds), 15m (15 minutes), or 1h (one hour).

--replication_consistency_group_name *<NAME>*

A unique name to be assigned to the replication consistency group

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--remote_protection_domain_id *<ID>*

Remote protection domain ID

--remote_protection_domain_name *<NAME>*

Remote protection domain name

--target_volume_access_mode *<TARGET_VOLUME_ACCESS_MODE>*

Target volume access mode. Can be one of the following: `no_access` (default) or `read_only`

--replication_consistency_group_activity_mode *<ACTIVITY_MODE>*

Replication consistency group activity mode. Can be one of the following:

active

(Default). When the RCG is active, replication is enabled and initial copy begins as soon as there are sufficient system resources.

inactive

The RCG is created in inactive state and data is not copied to the target. The RCG can be activated later using the `activate_replication_consistency_group` command.

Example

```
scli --mdm_ip 192.168.1.200 --add_replication_consistency_group --destination_system_id
459865f0774b81e6 --rpo 15m --protection_domain_id 1caf743100000000 --
remote_protection_domain_id dc65bd9900000000 --target_volume_access_mode read_only
```

add_replication_pair

Add a replication pair consisting of a volume from the source system and a volume from the target system.

A volume can be a member of at most one replication pair. When the replication pair is created, both the source volume and the target volume must be the same size. It is recommended, but not mandatory, that the volumes in the volume pair have the same attributes (including zero padding and granularity). Not doing so can impact performance and capacity.

Syntax

Syntax for adding pair without destination volume auto-provisioning (destination volume must be pre-created):

```
scli --add_replication_pair (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>) (--source_volume_id <ID> | --
source_volume_name <NAME>) (--destination_volume_id <ID> | --destination_volume_name
<NAME>) [--replication_pair_name <NAME>]
```

Syntax for adding pair with destination volume auto-provisioning:

```
scli --add_replication_pair  [--replication_pair_name <NAME>]
(--replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
(--source_volume_id <ID> | --source_volume_name <NAME>) (--destination_storage_pool_name
<NAME> | --destination_storage_pool_id <ID>) [--destination_sdc_id <ID> | --
destination_sdc_name <NAME> | --destination_sdc_guid <GUID> | --destination_sdc_ip <IP>]
 [--compression_method <METHOD>] [Provisioning options] [Use RAM Read Cache options]
[--access_mode <ACCESS_MODE>] [--allow_multi_map]
```

Parameters

--replication_pair_name *<NAME>*

Replication pair name

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--source_volume_id *<ID>*

Source volume ID

--source_volume_name *<NAME>*

Source volume name

--destination_sdc_ip *<IP>*

Destination SDC IP

Provisioning options:

If auto provision option is selected, provide the following:

Compression (optional):

--compression_method *<METHOD>*

Indicates the compression method to be used, relevant only for fine granularity data layouts.

`none`— No compression applied

`normal`—Normal compression applied

Destination storage pool option: Select any one of the following:

--destination_storage_pool_name *<NAME>*

Destination storage pool name

--destination_storage_pool_id *<ID>*

Destination storage pool ID

Provisioning options: Select any one of the following:

--thin_provisioned

This volume will be thin-provisioned

--thick_provisioned

This volume will be thick-provisioned

<BLANK>

Use storage pool default (thick for medium granularity, thin for fine granularity)

Use RAM read cache: Select any one of the following:

--use_rmcache

The specified volume uses caching

--dont_use_rmcache

The specified volume does not use caching

<BLANK>

Use storage pool default value

Indicates what dst (destination) SDC exposes the dst volume in case of disaster recovery:

--destination_sdc_id *<ID>*

SDC ID

--destination_sdc_name *<NAME>*

SDC name

--destination_sdc_guid *<GUID>*

SDC global unique identifier

--destination_sdc_ip *<IP>*

SDC IP address

Destination volume mapping to SDC (optional):

--access_mode <ACCESS_MODE>

Mapping access mode when not in disaster recovery. Can be one of the following: no_access (default), read_only

--allow_multi_map

Allow this volume to be mapped to more than one SDC

add_replication_peer_system

The peer system must be added to each of the systems involved in replication. There must be network connectivity between the two systems.

Syntax

```
scli --add_replication_peer_system --peer_system_ip <IP> [--peer_system_port <PORT>] --
peer_system_id <ID> [--peer_system_name <NAME>]
```

Parameters

--peer_system_ip *<IP>*

A comma-separated list of IP addresses assigned to the peer system. These are necessary for the replication to survive failures.

--peer_system_port *<PORT>*

Port assigned to the peer system. If this is omitted, the port used is the default peer listening port.

--peer_system_id *<ID>*

Peer system ID

--peer_system_name *<NAME>*

A name to be assigned to the peer system. The name should be unique.

Example

```
scli --mdm_ip 192.168.1.200 --add_replication_peer_system --peer_system_ip 10.55.105.21
--peer_system_id 7d897dd0269f5b0f --peer_system_name Durham1
```

add_sdr

Add an SDR to the specified protection domain. The SDR is one of the main components that must be installed on each of the peer protection domains for replication. It transfers data from the source volume to the target volume.

Syntax

```
scli --add_sdr --sdr_ip <IP> [--sdr_ip_role <ROLE>] [--sdr_name <NAME>] (--
protection_domain_id <ID> | --protection_domain_name <NAME>) [--sdr_port <PORT>] [--
profile <PROFILE>] [--force_clean [--i_am_sure]]
```

Parameters

--sdr_ip *<IP>*

A comma-separated list of IP addresses assigned to the SDR

--sdr_ip_role *<ROLE>*

The role assigned to each SDR IP address where *<ROLE>* can be one of the following:

application_only

Application only, for communicating with SDC

storage_only

Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

--sdr_name *<NAME>*

A name to be assigned to the SDR

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--sdr_port *<PORT>*

Port assigned to the SDR

--profile {compact | high_performance}

Set the performance profile as compact or high performance. The high performance profile configures a predefined set of parameters for very high performance use cases.

--force_clean

Clean a previous SDR configuration. Use this if the SDR was previously part of a PowerFlex system.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --add_sdr --sdr_ip 10.55.105.41,10.76.30.14 --sdr_name SDR2
--protection_domain_id 2c3071a200000001
```

SDR names

Assign each SDR a meaningful name. When a name has not been defined, the system may display default system-defined names that use IDs of the SDRs.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object’s property sheet in the PowerFlex Manager.

add_sdr_ip

Add an IP address to the specified SDR.

Syntax

```
scli --add_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_sdr_ip <IP> [--sdr_ip_role <ROLE>]
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--new_sdr_ip *<IP>*

New IP address to be assigned to the SDR

--sdr_ip_role *<ROLE>*

The role assigned to each SDR IP address where *<ROLE>* can be one of the following:

application_only

Application only, for communicating with SDC

storage_only

Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

Example

```
scli --mdm_ip 192.168.1.200 --add_sdr_ip --sdr_name SDR27 --new_sdr_ip 10.76.30.14 --
sdr_ip_role application_only
```

clear_broken_replication_consistency_group_alerts

Clear broken replication consistency group (RCG) indications for all volumes associated with the specified protection domain. The clearing process is useful in situations when the SDRs run out of journal capacity and RCG autotermination cannot be applied (such as when the remote MDM does not support it). At that point, an alert is produced for every replicated volume in the affected Protection Domain. These alerts can be cleared using this command.

Syntax

```
scli --clear_broken_replication_consistency_group_alerts (--protection_domain_id <ID> |
--protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --clear_broken_replication_consistency_group_alerts --
protection_domain_name pd1
```

create_replication_consistency_group_snapshots

Create replication consistency group snapshots on the target.

Notes

The resulting snapshots contain the latest consistent image. If a consistent image cannot be created, the command fails. The `--force` option can be used to create a snapshot even if a consistent image is not available. Use this option carefully as the resulting snapshot may be crash inconsistent and may not be usable by applications.

Syntax

```
scli --create_replication_consistency_group_snapshots (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) [--
force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Run the command even if some volumes would be excluded because of the uncompleted initial copy.

enter_sdr_maintenance_mode

Enter the SDR into maintenance mode to allow for maintenance of the SDR. Note that there must be at least two other active SDRs joined to the system for replication to continue. When maintenance mode is activated, the MDM changes the mapping and distributes the I/Os to the remaining SDRs.

Syntax

```
scli --enter_sdr_maintenance_mode (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--
sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --enter_sdr_maintenance_mode --sdr_name SDR27
```

exit_sdr_maintenance_mode

Move the SDR out of maintenance mode.

Notes

An SDR exiting maintenance mode requires free journal capacity in order to become active in the replication.

Syntax

```
scli --exit_sdr_maintenance_mode (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--
sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --exit_sdr_maintenance_mode --sdr_name SDR27
```

failover_replication_consistency_group

Failover the specified replication consistency group. During failover, the I/O applications are stopped at the source and the access mode of the source volumes is changed to inaccessible. The target volumes are moved to the newest consistent image available on the destination and their access mode is changed to read-write. Recovery from a failover state can be achieved by either restoring the replication or reversing it.

Syntax

```
scli --failover_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>) [--force_during_partial_consistency] [--
i_am_sure]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID.

--replication_consistency_group_name *<NAME>*

Replication consistency group name.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

--force_during_partial_consistency

Run the command even if some volumes would be excluded because of the uncompleted initial copy.

failover_test_start_replication_consistency_group

Start a test failover of the specified replication consistency group (RCG). This test allows you to demonstrate failover mode without interrupting data synchronization.

Syntax

```
scli --failover_test_start_replication_consistency_group (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) [--
force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Perform the command despite some volumes would be excluded due to uncompleted initial copy.

failover_test_stop_replication_consistency_group

Stop a failover test for the specified replication consistency group (RCG).

Syntax

```
scli --failover_test_stop_replication_consistency_group (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --failover_test_stop_replication_consistency_group --
replication_consistency_group_name rcg27
```

freeze_apply_replication_consistency_group

Apply the latest consistent image available at the destination to the target volumes and freeze the replication consistency group (RCG). While the RCG is frozen, replication remains active and I/O operations are replicated to the target journal. The I/Os are stored in the target journal and are not applied to the target volumes. Retaining in a frozen state for a long time runs the risk of running out of space in the target journal.

Syntax

```
scli --freeze_apply_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)[--force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Run the command even if some volumes would be excluded because of the uncompleted initial copy.

inactivate_replication_consistency_group_mappings

Inactivate replication consistency group mappings on the peer to which the command is applied.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including inactivating the replication consistency group mappings. This command is useful in situations when you choose to perform each of the individual operations directly.

Syntax

```
scli --inactivate_replication_consistency_group_mappings (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --inactivate_replication_consistency_group_mappings --
replication_consistency_group_name rcg27w
```

modify_replication_consistency_group_rpo

Modify the recovery point objective (RPO) of the replication consistency group.

The RPO is the maximum data loss (in units of time) that you are willing to lose. When you set the RPO, you are setting the goal of replication.

Syntax

```
scli --modify_replication_consistency_group_rpo (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>) --rpo <PERIOD>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--rpo *<PERIOD>*

Desired RPO in units of time, between 30 seconds and 1 hour. To specify the time period, use s|m|h respectively indicating seconds, minutes, or hours. Examples: 30s (30 seconds), 15m (15 minutes), or 1h (one hour).

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_consistency_group_rpo --
replication_consistency_group_name rcg27 --rpo 20m
```

modify_replication_consistency_group_target_volume_access_mode

Modify the target volume access mode of a Replication Consistency Group.

Syntax

```
scli --modify_replication_consistency_group_target_volume_access_mode (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) --
target_volume_access_mode <TARGET_VOLUME_ACCESS_MODE>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--target_volume_access_mode *<TARGET_VOLUME_ACCESS_MODE>*

Target volume access mode. Can be one of the following: `no_access` (default), `read_only`

Example

```
scli --mdm_ip 192.168.1.200
--modify_replication_consistency_group_target_volume_access_mode --
replication_consistency_group_name rcg27 --target_volume_access_mode read_only
```

modify_replication_peer_system_ip

Modify an IP address of a replication peer system.

Syntax

```
scli --modify_replication_peer_system_ip (--peer_system_id <ID> | --peer_system_name
<NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id
<ID>) --new_peer_system_ip <IP>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
Replication peer system IP address

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--new_peer_system_ip *<IP>*
A comma-separated list of new IP addresses assigned to the peer system

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_peer_system_ip --peer_system_name RPS1
--new_peer_system_ip 10.55.105.21
```

modify_replication_peer_system_port

Modify the port of the specified replication peer system.

Notes

Before running this SCLI command to modify port assigned to the replication peer system, you must first edit the `/opt/emc/scaleio/mdm/cfg/conf.txt` configuration file and set the `mdm_external_port` field to the new port. After you've saved the configuration file, restart the MDM service. Then run this command to modify the port assigned to the replication peer system.

Syntax

```
scli --modify_replication_peer_system_port (--peer_system_id <ID> | --peer_system_name
<NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id
<ID>) --new_peer_system_port <PORT>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
IP address of the replication peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--new_peer_system_port *<PORT>*

New port assigned to the peer system. When this parameter is omitted, the default listening port is used.

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_peer_system_port --peer_system_name RP1
--new_peer_system_port 7611
```

modify_sdr_ip_role

Modify the role of the specified SDR's IP address.

Syntax

```
scli --modify_sdr_ip_role (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP>  [--
sdr_port <PORT>]) [--sdr_ip_to_modify <IP>] --new_sdr_ip_role <ROLE>
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--sdr_ip_to_modify *<IP>*

(Optional) The IP address of the SDR that you wish to modify

--new_sdr_ip_role *<ROLE>*

The new role assigned to the modified SDR IP address where *<ROLE>* can be one of the following:

application_only

Application only, for communicating with SDC

storage_only

Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

Example

```
scli --mdm_ip 192.168.1.200 --modify_sdr_ip_role --sdr_name SDR27  --sdr_ip_to_modify
10.76.30.14 --new_sdr_ip_role application_and_storage
```

modify_sdr_port

Modify the port assigned to the specified SDR.

Syntax

```
scli --modify_sdr_port (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_sdr_port <PORT>
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address of the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

--new_sdr_port *<PORT>*
New port to be assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --modify_sdr_port --sdr_id ffa0be3900000000 --new_sdr_port
11088
```

pause_pair_initial_copy

Pause the initial copy process of the specified replication pair.

Syntax

```
scli --pause_pair_initial_copy (--replication_pair_id <ID> | --replication_pair_name
<NAME>)
```

Parameters

--replication_pair_id *<ID>*
Replication pair ID

--replication_pair_name *<NAME>*

Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --pause_pair_initial_copy --replication_pair_id
ca77eb5e00000000
```

pause_replication_consistency_group

Pause the replication data transfer of a replication consistency group.

Syntax

```
scli --pause_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --pause_replication_consistency_group --
replication_consistency_group_name rcg27
```

print_recreate_replication_consistency_group_script

Display a list of SCLI commands required to recreate the specified replication consistency group.

Syntax

```
scli --print_recreate_replication_consistency_group_script (--
replication_consistency_group_name <NAME>| --replication_consistency_group_id <ID>)
```

Parameters

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--replication_consistency_group_id *<ID>*

Replication consistency group ID

Example

```
scli --mdm_ip 192.168.1.200 --print_recreate_replication_consistency_group_script --
replication_consistency_group_name rcg27
```

Sample output

```
scli --add_replication_consistency_group --replication_consistency_group_name RCG1 --
destination_system_object_id c876517d00000000 --protection_domain_id 036a797000000000
--remote_protection_domain_id 073b95a000000000 --rpo 3600 --target_volume_access_mode
no_access
scli --add_replication_pair --replication_consistency_group_name RCG1 --source_volume_id
6730b3a800000003 --destination_volume_id 2d3d6cd500000003 --copy_type full
```

query_all_replicated_volumes

Retrieve information about all replicated volumes. The query result can be optionally filtered by protection domain or by storage pool.

Syntax

```
scli --query_all_replicated_volumes [(((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replicated_volumes
```

Sample output

```
Protection Domain f11d784d00000000 Name: pd1
Storage Pool bf83a76700000000 Name: sp1
 Volume ID: c3a62eb200000003 Name: vol1 Size: 16.0 GB (16384 MB)
   Pair ID: ef199ea900000000 (CG ID: 4607a08e00000000)

Protection Domain f11d784e00000001 Name: pd2
Storage Pool bf83a76800000001 Name: sp2
 <No volumes defined>
```

```
Protection Domain f11d9f5d00000002 Name: dfg
 <No volumes defined>
Query-all-replicated-volumes returned 1 volumes
```

query_all_replication_consistency_groups

Retrieve information about all the replication consistency groups (RCG).

Syntax

```
scli --query_all_replication_consistency_groups
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_consistency_groups
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
 ID: 399c609c00000000 Name: rcg1 Activity: INACTIVE Peer System: peer_to_118 RPO: 1m
Compliance: 0% Status: Stopped by User

Protection Domain fcc9af5600000001 Name: pd2

Protection Domain fcc9d66500000002 Name: rrr

query_all_replication_consistency_groups returned 1 Replication Consistency Groups.
```

query_all_replication_journal_capacity

Retrieve information about all replication journal volumes. The results can be filtered by protection domain or by storage pool.

Syntax

```
scli --query_all_replication_journal_capacity [(((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_journal_capacity --storage_pool_name
SP1
```

Sample output

```
Protection Domain a86cde9d00000000 Name: pd1
Storage Pool 6d1751fe00000000 Name: sp1
 Volume ID: f7d341ef00000000 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35600000000
Thin-provisioned  SdrSessionIndex 0
 Volume ID: f7d341f100000002 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35800000002
Thin-provisioned  SdrSessionIndex 2
 Volume ID: f7d341f400000001 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35a00000001
Thin-provisioned  SdrSessionIndex 1

Journal Capacity - Allocated: 24.0 GB (24576 MB) Max Allowed: 26.0 GB (26624 MB) (10% of
total)

Query-all-volumes returned 3 volumes
```

query_all_replication_pairs

Retrieve information about all the replication pairs in the system.

Syntax

```
scli --query_all_replication_pairs
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_pairs
```

Sample output

```
CG Id: 399c609c00000000 CG Name: rcg1
Peer System Name: peer_to_118 Direction: LOCAL_TO_REMOTE
    Local ID: bee0361f00000000 Remote ID: c557f02500000000 Copy Type: IDENTICAL Initial
Copy State: DONE Name: (null)
    Local Volume ID f6351e3700000003 Remote Volume ID: dbf2341e00000003 Local Volume
Name: vol1_114 Remote Volume Name: vol1_118
```

```
    Initial Copy progress: 100% Local Activity State: INACTIVE
query_all_replication_pairs returned 1 Replication Pairs.
```

query_all_replication_peer_system

Retrieve information about all the replication peer systems.

Syntax

```
scli --query_all_replication_peer_system
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_peer_system
```

Sample output

```
Query-all-Replication Peer System returned 1 Replication Peer System nodes.

Replication Peer System ID: 7c3e2613207b310f Replication Peer System
internal ID: 3ea1845400000000 Name: peer_to_118 State: Joined, SUCCESS IP:
10.55.137.118,10.55.137.120 Port: 7611 Version: 3.6.0

SDR-SDR connectivity status: All connected
```

query_all_sdr

Retrieve information about all SDRs.

Syntax

```
scli --query_all_sdr
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sdr
```

Sample output

```
Query-all-SDR returned 3 SDR nodes.

Protection Domain ID: 036a797000000000
SDR ID: 5747c3f800000000 Name: SDR2 State: Connected, Joined IP:
10.55.105.40,10.76.30.11,192.168.177.28 Port: 11088 Version: 3.5.0
SDR ID: 5747c3f900000001 Name: SDR3 State: Connected, Joined IP:
10.55.105.41,10.76.30.14 Port: 11088 Version: 3.5.0
SDR ID: 5747c3fa00000002 Name: SDR1 State: Connected, Joined IP:
10.55.105.39,10.76.30.10 Port: 11088 Version: 3.5.0
```

query_replication_consistency_group

Retrieve information about the specified Replication Consistency Group.

Syntax

```
scli --query_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_consistency_group --
replication_consistency_group_id 399c609c00000000
```

Sample output

```
Replication Consistency Group ID: 399c609c00000000 Name: rcg1
  Protection Domain ID: fcc9af5500000000 Name: pd1

Configuration:
  RPO: 1m
  Target volume access mode: NO_ACCESS
  Number of volume pairs: 1, Capacity: 16.0 GB (16384 MB)

Remote info:
  Peer System ID: 7c3e2613207b310f Name: peer_to_118
  Protection Domain ID: e6bce87900000000 Name: pd1
  Remote CG Id: 2215ba5800000000

State:
  Lifetime State: NORMAL
  Transmit State: NOT_PAUSED
  Freeze State: UNFROZEN
  Consistency Mode: PARTIALLY_CONSISTENT
  Direction: LOCAL_TO_REMOTE
  Disaster recovery flow: NONE
```

```
  Disaster recovery state: None

Consistency state:
  RPO Compliance: 0%
  Lag Received: 0s (at 10:51:55)
  Lag Journalled: 0s (at 10:51:55)
  Lag Applied: 0s (at 10:51:55)
  Last Barrier: 0s (at 10:51:55)
  Next Planned Barrier: 0s (at 10:51:55)

Bandwidth:
  User Read: 479 IOPS 239.2 MB (244941 KB) per-second
  User Write: 514 IOPS 256.8 MB (262960 KB) per-second
  Transmit: 0 IOPS 0 Bytes per-second
  Apply: 0 IOPS 0 Bytes per-second

Latency:
  User Read: 240 IOPS 20.917 ms average latency
  User Write: 257 IOPS 221.345 ms average latency
  Transmit: 0 IOPS 0 us average latency
  Apply: 0 IOPS 0 us average latency
  Journaler Write: 0 IOPS 0 us average latency
  Journaler Read: 0 IOPS 0 us average latency

Journalled data:
  Journalled on Source: 0 Bytes
  Ready For Transmit: 0 Bytes
  Journalled on Target: 0 Bytes
  Ready For Apply: 0 Bytes

Activity State:
  Local Activity State: ACTIVE
  Remote Activity State: ACTIVE
```

query_replication_pair

Retrieve information about the specified replication pair.

Syntax

```
scli --query_replication_pair --replication_pair_id <ID> --replication_pair_name <NAME>
```

Parameters

--replication_pair_id *<ID>*
Replication pair ID

--replication_pair_name *<NAME>*
Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_pair --replication_pair_id
bee0361f00000000
```

Sample output

```
Local ID: bee0361f00000000 Remote ID: c557f02500000000 Name: N/A Initial Copy State:
DONE Initial Copy Priority: -1 Initial Copy paused by user: FALSE
Peer System Name: peer_to_118 Direction: LOCAL_TO_REMOTE CG ID: 399c609c00000000 CG
Name: rcg1
Local Volume ID: f6351e3700000003 Remote Volume ID: dbf2341e00000003 Local Volume Name:
vol1_114 Remote Volume Name: vol1_118 Remote Volume Capacity (MB): 16384 Copy Type:
IDENTICAL Lifetime State: NORMAL
Initial Copy progress: 100%
Local Activity State: INACTIVE Remote Activity State: INACTIVE Last EnableRpl Failure
Reason: INVALID
```

query_replication_peer_system

Retrieve information about the specified replication peer system.

Syntax

```
scli --query_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
: Replication peer system ID

--peer_system_name *<NAME>*
: Replication peer system name

--peer_system_ip *<IP>*
: Replication peer system IP address

--peer_system_port *<PORT>*
: Port assigned to the replication peer system port

--peer_system_object_id *<ID>*
: Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_peer_system --peer_system_id
7c3e2613207b310f
```

Sample output

```
Replication Peer System 7c3e2613207b310f Internal ID 3ea1845400000000 Name: peer_to_118
Version: 3.6.0
IP information (total 2 IP addresses):
         1: 10.55.137.118
         2: 10.55.137.120
         Port: 7611
Membership-state: Joined; Coupling_rc: SUCCESS;
```

query_replication_peer_system_destination_volume_candidates

Retrieve information about destination volume candidates in the replication peer system. This command returns available remote volumes in the specified protection domain of the peer system with the specified capacity.

Syntax

```
scli --query_replication_peer_system_destination_volume_candidates (--peer_system_id
<ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>]
| --peer_system_object_id <ID>) (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --size_gb <SIZE>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
Replication peer system IP address

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--size_gb *<SIZE>*
Size of volume in GBs

Example

```
scli --mdm_ip 192.168.1.200 --
query_replication_peer_system_destination_volume_candidates --peer_system_id
1820896a6b53a70f --protection_domain_name pd1 --size_gb 540
```

query_replication_peer_system_protection_domains

Retrieve information about the available protection domains in the target replication peer system.

Syntax

```
Usage: scli --query_replication_peer_system_protection_domains (--peer_system_id <ID>
| --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --
peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
IP address assigned to the replication peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_peer_system_protection_domains --
peer_system_id 1820896a6b53a70f
```

Sample output

```
Received 1 PDs
PD Id 382dca8200000000
PD Name pd1
```

query_sdr

Retrieve information about the specified SDR.

Syntax

```
scli --query_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --query_sdr --sdr_id 82dc5ccd00000000
```

Sample output

```
SDR 82dc5ccd00000000 Name: SDR3 Version: 3.6.0
 Protection Domain ID: fcc9af5500000000
IP information (total 1 IP addresses):
         1: 10.55.137.116    Role: all (application, storage, external)
         Port: 11088
Membership-state: Joined; Connection-state: Connected; Maintenance-state: Normal; SDR-
state: Normal;
```

query_sync_now_replication_consistency_group

Retrieve the status of a synchronization process for the specified replication consistency group (RCG).

Syntax

```
scli --query_sync_now_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>) --sync_now_key <SYNC_NOW_KEY>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--sync_now_key *<SYNC_NOW_KEY>*

The key returned from running the `--sync_now_replication_consistency_group` command.

Example

```
scli --mdm_ip 192.168.1.200 --query_sync_now_replication_consistency_group --
replication_consistency_group_name rcg27 --sync_now_key 667432
```

Sample output

```
Successfully queried SyncNowKey 190531 of Replication Consistency Group RCG1: The data
was applied to the destination volumes
```

Return codes

The following are the possible return codes for this command:

Table 4. Replication Consistency Group Return Codes

Return code	Message
MOS_RC__SUCCESS	Success.
MOS_RC__RPL_CG_INVALID_ID	Invalid replication consistency group ID.
MOS_RC__RPL_CG_INVALID_KEY	Invalid replication consistency group key.

Table 4. Replication Consistency Group Return Codes (continued)

Return code	Message
MOS_RC__RPL_CG_NOT_FOUND	Replication consistency group not found.
MOS_RC__RPL_CG_IS_INACTIVE	The RCG is not active due to deactivated protection domain.
MOS_RC__RPL_CG_COMMAND_IS_NOT_ALLOWED_ON_SYSTEM_RCG	A user operation is not allowed on a system RCG.
MOS_RC__RPL_CG_COMMAND_IS_NOT_ALLOWED_DURING_DISASTER_RECOVERY	The command is not allowed during disaster recovery.
MOS_RC__RPL_CG_IS_NOT_IN_NORMAL_STATE	Replication consistency group is not in normal state.

query_replication_destination_sdc_candidates

Query a replication peer-system for available SDCs. These SDCs can be used in destination volumes auto-provisioning.

Syntax

```
scli --query_replication_destination_sdc_candidates (--peer_system_id <ID> | --
peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --
peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
Replication peer system IP address

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

query_replication_destination_storage_pool_candidates

Query a replication peer-system for storage pools on which a volume of specified size and provisioning mode can be created. These storage pools can be used in destination volumes auto-provisioning.

Syntax

```
scli --query_replication_destination_storage_pool_candidates (--peer_system_id
<ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port
<PORT>] | --peer_system_object_id <ID>) (--remote_protection_domain_id <ID> | --
remote_protection_domain_name <NAME>) --size_gb <SIZE> [Provisioning options]
```

Parameters

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--remote_protection_domain_id *<ID>*

Remote protection domain ID

--remote_protection_domain_name *<NAME>*

Remote protection domain name

--size_gb *<SIZE>*

The required volume size. The query results show only storage pools on which a volume of this size can be created.

Select any one of the following provisoning options:

--thin_provisioned

Volume is thin-provisioned

--thick_provisioned

Volume is thick-provisioned

<BLANK>

Use storage pool default (thick for medium granularity, thin for fine granularity)

remove_replication_consistency_group

Initiate removal of a replication consistency group.

Notes

Removal of a replication consistency group is a lengthy process and it may complete some time after the command completion.

Syntax

```
Usage: scli --remove_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_consistency_group --
replication_consistency_group_name rcg27
```

remove_replication_pair

Remove the specified replication pair.

NOTE: The target volume image may not be consistent after the pair removal. To obtain a consistent image on the target volume: Run `freeze_apply_replication_consistency_group` on the the replication consistency group, remove the replication pair, and then run `unfreeze_apply_replication_consistency_group` on the replication consistency group.

Syntax

```
scli --remove_replication_pair (--replication_pair_id <ID> | --replication_pair_name
<NAME>) [--i_am_sure]
```

Parameters

--replication_pair_id *<ID>*
Replication pair ID

--replication_pair_name *<NAME>*
Replication pair name

--i_am_sure
Preemptive approval

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_pair --replication_pair_name rp1 --
i_am_sure
```

remove_replication_peer_system

Remove a replication peer system.

Syntax

```
scli --remove_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
Peer system ID

--peer_system_name *<NAME>*
Peer system name

--peer_system_ip *<IP>*
A comma-separated list of IP addresses assigned to the peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_peer_system --peer_system_name RPS1
```

remove_sdr

Initiate removal of the specified SDR.

Syntax

```
scli --remove_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --remove_sdr --sdr_name SDR2
```

remove_sdr_ip

Remove an IP address from an SDR.

Syntax

```
scli --remove_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --sdr_ip_to_remove <IP>
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--sdr_ip_to_remove *<IP>*

IP address to remove from the SDR

Example

```
scli --mdm_ip 192.168.1.200 --remove_sdr_ip --sdr_name SDR27 --sdr_ip_to_remove
10.76.30.14
```

rename_replication_consistency_group

Name or rename the specified replication consistency group.

Syntax

```
scli --rename_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>) (--new_name <NAME> | --remove_name)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--new_name *<NAME>*

New name to be assigned to the replication consistency group

--remove_name

Use this flag to delete the replication consistency group name without giving it a new name.

Example

```
scli --mdm_ip 192.168.1.200 --rename_replication_consistency_group --
replication_consistency_group_name rcg27 --new_name RCG27
```

rename_replication_peer_system

Name or rename the replication peer system. The name should be unique.

Syntax

```
scli --rename_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>) --
new_peer_system_name <NAME>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
IP address assigned to the replication peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--new_peer_system_name *<NAME>*
A new unique name to be assigned to the peer system.

Example

```
scli --mdm_ip 192.168.1.200 --rename_replication_peer_system --peer_system_id
1820896a6b53a70f --new_peer_system_name RPS1
```

rename_sdr

Name or rename an SDR.

Syntax

```
scli --rename_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_name <NAME>
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

--new_name *<NAME>*
New name to be assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --rename_sdr --sdr_name SDR2 --new_name SDR27
```

resume_pair_initial_copy

Resume the paused initial copy process of the specified replication pair.

Syntax

```
scli --resume_pair_initial_copy (--replication_pair_id <ID> | --replication_pair_name
<NAME>)
```

Parameters

--replication_pair_id <ID>

Replication pair ID

--replication_pair_name <NAME>

Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --resume_pair_initial_copy --replication_pair_name RP1
```

resume_replication_consistency_group

Resume a paused replication consistency group (RCG). Once the RCG is resumed, the source will resume transmission of the collected write data to the target.

Syntax

```
scli --resume_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --resume_replication_consistency_group --
replication_consistency_group_name rcg27
```

restore_replication_consistency_group

Once the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Restoring replication maintains the replication direction from the original source and overwrites all data at the target.

Notes

During a restore operation, data movement from the source to the target is carried out without maintaining consistency of the data during the process. Consistency is only achieved at the target once the source and target synchronized.

Prior to executing the restore operation, you consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Syntax

```
scli --restore_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --restore_replication_consistency_group --
replication_consistency_group_name rcg27
```

reverse_replication_consistency_group

Once the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Reversing replication changes the direction so that the original target becomes the source. All data at the original source is overwritten by the data at the target.

Notes

During a reverse operation, data movement from the new source to the new target is carried out without maintaining consistency of the data during the process. Consistency is only achieved at the new target once the new source and the new target synchronized.

Prior to executing the reverse operation, you consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Syntax

```
scli --reverse_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --reverse_replication_consistency_group --
replication_consistency_group_name rcg27
```

set_pair_initial_copy_queue_position

Set the priority of the initial copy process of the specified replication pair to be the highest or lowest priority in the initial copy queue.

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --set_pair_initial_copy_queue_position (--replication_pair_id <ID> | --
replication_pair_name <NAME>) (--initial_copy_priority <TYPE>)
```

Parameters

--replication_pair_id *<ID>*

Replication pair ID

--replication_pair_name *<NAME>*

Replication pair name

--initial_copy_priority *<TYPE>*

Desired initial copy priority. Can be one of the following:

highest

Move the initial copy process to the front of the queue

lowest

Move the initial copy process to the end of the queue

Example

```
scli --mdm_ip 192.168.1.200 --set_pair_initial_copy_queue_position --
replication_pair_name RP1 --initial_copy_priority highest
```

set_replication_consistency_group_consistent

Set the specified replication consistency group (RCG) as consistent. While in consistent mode, the journal at the target applies its content to the target volumes in a way that guarantees that a consistent image of the target volumes can be achieved.

NOTE: This command cannot be performed when the specified RCG is inactive. It cannot be performed when in failover or switchover mode. Similarly, it cannot be performed on a disabled protection domain or during freeze apply.

Syntax

```
scli --set_replication_consistency_group_consistent (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_consistency_group_consistent --
replication_consistency_group_name rcg27
```

set_replication_consistency_group_inconsistent

Set the specified replication consistency group (RCG) as inconsistent. While in inconsistent mode, the contents of the target journal are applied to the target volumes without guarantee of consistency. This means that the target volume image may not be consistent when the RCG is inconsistent. The RCG automatically regains consistency when possible.

It is recommended to create a snapshot of the application volumes at the target before running this command. The snapshot captures the last consistent image before the RCG is inconsistent.

NOTE: This command cannot be performed when the specified RCG is inactive or while in failover or switchover mode. It also cannot be performed on a disabled protection domain, during freeze apply, while creating snapshots, or during failover test.

Syntax

```
scli --set_replication_consistency_group_inconsistent (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_consistency_group_inconsistent --
replication_consistency_group_name rcg27
```

set_replication_journal_capacity

Set the replication journal capacity ratio for the specified storage pool. This defines the maximum ratio of a storage pool capacity that can be used by the replication system for journaling.

Notes

Journal capacity is defined as a percentage of the total storage capacity in the storage pool and must equal at least 28 GB per SDR. In general, journal capacity should be at least 5% of replicated usable capacity in the protection domain, including volumes used as source and targets. It is important to assign enough storage capacity for the replication journal.

Syntax

```
scli --set_replication_journal_capacity (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --
replication_journal_capacity_max_ratio <RATIO>)
```

Parameters

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--storage_pool_name *<NAME>*
: Storage pool name

--storage_pool_id *<ID>*
: Storage pool ID

--replication_journal_capacity_max_ratio *<RATIO>*
: Maximum capacity percentage to be allocated for journal capacity (from 0 to 100).

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_journal_capacity --protection_domain_name
PD1 --storage_pool_name SP1 --replication_journal_capacity_max_ratio 25
```

set_remote_replication_journal_capacity

Define the maximum capacity allowed for the remote system to use for replication consistency groups (RCG) sending to this protection domain.

Syntax

```
scli --set_remote_replication_journal_capacity (--protection_domain_id <ID>
| --protection_domain_name <NAME>) (--remote_capacity_max_ratio <RATIO>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--remote_capacity_max_ratio *<RATIO>*

New capacity ratio to be assigned to the remote Protection domain.

Valid range: 10-100

Example

```
scli --mdm_ip 192.168.1.200 --set_remote_replication_journal_capacity --
protection_domain_id fcc9af5500000000 --remote_capacity_max_ratio 50
```

switchover_replication_consistency_group

Switch over the specified replication consistency group (RCG). Switchover is a planned failover. The application I/Os are stopped at the source, and the data is synchronized so that the target volume is consistent with the source volume. The access mode of the original source volumes is changed to unavailable or read-only. The access mode of the original target volumes is changed so that the hosts can read and write to the volumes.

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --switchover_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>)[--i_am_sure]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --switchover_replication_consistency_group --
replication_consistency_group_name rcg27 --i_am_sure
```

sync_now_replication_consistency_group

Synchronize the source and target to ensure that all new data that has been written to the source is also written to the target volumes. A sync key is returned that can be used to query the status of the synchronization process.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including synchronizing the source and target volumes. This command is useful in situations when you choose to perform each of the individual operations directly.

Syntax

```
scli --sync_now_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>) [--force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Perform the command despite some volumes would be excluded due to uncompleted initial copy.

terminate_replication_consistency_group

Terminate the specified replication consistency group (RCG).

Notes

Terminating the RCG moves it into an inactive state. Replication is stopped, and the journal capacity is released. From this point on, the I/Os are handled by the SDS and not the SDR and no journal capacity is consumed. All RCG configurations are maintained, including source and target roles, pair configuration, and volume access modes.

Terminating an RCG does not require connectivity between the sites. If the RCG is terminated on a peer when there is no connectivity, then the other peer terminates once connectivity is restored.

NOTE: Activating a terminated RCG requires full initialization. The initial copy process may take a long time.

Syntax

```
scli --terminate_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --terminate_replication_consistency_group --
replication_consistency_group_name rcg27
```

unfreeze_apply_replication_consistency_group

Unfreeze the replication consistency group (RCG) and apply the I/Os stored in the target journal to the target volumes.

Syntax

```
scli --unfreeze_apply_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --unfreeze_apply_replication_consistency_group --
replication_consistency_group_name rcg27
```

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Return messages

This section describes the contents and the types of return messages in the PowerFlex CLI.

Each return message contains the following:

- Name of the return message (return code)
- Message presented with the return code
- Commands that may return the relevant code
- Action or actions that may resolve the issue (if relevant)

Return messages overview

This section describes the contents and the types of return messages in the PowerFlex CLI.

Each return message contains the following:

- Name of the return message (return code)
- Message presented with the return code
- Commands that may return the relevant code
- Action or actions that may resolve the issue (if relevant)

SDS device name changed

Table 5. SDS_DEVICE_NAME_CHANGED parameters

Parameter	Description
Name	SDS_DEVICE_NAME_CHANGED
Message	The SDS detected that the device name has changed. Re-run the command using the device ID.
Commands	`remove_sds_device, abort_remove_sds_device`
Description	In some situations (usually following a reboot), the operating system changes the device name. The SDS detects this change. `Remove_sds_device` is a time consuming operation. To ensure that you have chosen the correct device, PowerFlex alerts you to the name change.
Action	Identify the device in the output of `query_sds` and use the device ID to run the command again. You can also resynchronize the device name.

Success

Table 6. SUCCESS parameters

Parameter	Description
Name	SUCCESS

Table 6. SUCCESS parameters (continued)

Parameter	Description
Message	Success
Commands	All
Description	Command was executed successfully
Action	None

No resources

Table 7. NO_RESOURCES parameters

Parameter	Description
Name	NO_RESOURCES
Message	Insufficient resources
Commands	`add_volume`
Description	There is insufficient space to create a volume of the requested size
Action	Add capacity to the system by adding another SDS/device, or create a smaller volume

Timeout

Table 8. TIMEOUT parameters

Parameter	Description
Name	TIMEOUT
Message	A timeout occurred
Commands	All
Description	No reply received from the MDM within the specified timeout period
Action	Validate the requested operational state using queries

Already exists

Table 9. ALREADY_EXISTS error parameters

Parameter	Description
Name	ALREADY_EXISTS
Message	Already Exists
Commands	All object addition commands
Description	The added object already exists
Action	Check command parameters, and use queries to view existing objects' properties

Communication error

Table 10. COMMUNICATION_ERROR parameters

Parameter	Description
Name	COMMUNICATION_ERROR
Message	Communication error
Commands	SDS related commands
Description	SDS has communication errors
Action	Use `query_all` to determine which SDS is disconnected

Wrong reconfiguration mode

Table 11. WRONG_RECONF_MODE error information

Parameter	Description
Name	WRONG_RECONF_MODE
Message	SDS being configured, retry command
Commands	All SDS related commands
Description	SDS is being configured by the MDM
Action	Retry the command

SDS add device “open” error

Table 12. TGT_ADD_DEV_OPEN_ERROR parameters

Parameter	Description
Name	TGT_ADD_DEV_OPEN_ERROR
Message	Could not open SDS device
Commands	`add_sds, add_sds_device`
Description	SDS cannot open one of the specified devices
Action	Validate the command parameters. If correct, validate that the device can be accessed on the SDS.

SDS add device IO error

Table 13. TGT_ADD_DEV_IO_ERROR parameters

Parameter	Description
Name	TGT_ADD_DEV_IO_ERROR
Message	Add SDS device IO error
Commands	`add_sds, add_sds_device`
Description	The SDS was unable to write to one of the specified devices

Table 13. TGT_ADD_DEV_IO_ERROR parameters (continued)

Parameter	Description
Action	Validate the command parameters, and check the specified devices

SDS add device size problem

Table 14. SDS add device size problem parameters

Parameter	Description
Name	TGT_ADD_DEV_SIZE_PROBLEM
Message	SDS device size error
Commands	`add_sds, add_sds_device`
Description	The device size is smaller or larger than the allowed size.
Action	Validate the command parameters, and set the device size appropriately

SDS actively attach to this MDM

Table 15. TGT_ACTIVE_ATTACH_TO_THIS_MDM parameters

Parameter	Description
Name	TGT_ACTIVE_ATTACH_TO_THIS_MDM
Message	SDS is actively attached to this MDM
Commands	`add_sds`
Description	SDS was already configured by this MDM. This usually implies that the provided IP address resolves to an already configured SDS that is currently connected. “Actively” implies that the SDS is configured and communicating with the MDM.
Action	Check the provided IP address. Contact Dell Technologies support.

SDS already attached to this MDM

Table 16. TGT_ALREADY_ATTACH_TO_THIS_MDM parameters

Parameter	Description
Name	TGT_ALREADY_ATTACH_TO_THIS_MDM
Message	SDS is already attached to this MDM
Commands	`add_sds`
Description	SDS was previously configured by this MDM. This usually indicates that the SDS was previously removed using `remove_sds`, but was not cleaned up properly.
Action	Validate the SDS IP address and port, and check the MDM configuration to validate that the SDS is not already configured under a different IP address. If the SDS is not found in the MDM configuration, this might indicate that it was not cleaned up properly when it was removed. To add the

Table 16. TGT_ALREADY_ATTACH_TO_THIS_MDM parameters (continued)

Parameter	Description
	SDS forcefully, thus overriding its configuration and data, use the `force_clean` flag.

SDS active attach to other MDM

Table 17. TGT_ACTIVE_ATTACH_TO_OTHER_MDM parameters

Parameter	Description
Name	TGT_ACTIVE_ATTACH_TO_OTHER_MDM
Message	SDS is actively attached to another MDM
Commands	`add_sds`
Description	SDS is already managed by another MDM. "Actively" implies that the SDS is configured and communicating with the MDM.
Action	Check the command parameters to validate the SDS IP address and port. Check the configuration of any other MDMs installed that might be connected to this SDS. Contact Dell Technologies Support.

SDS already attached to other MDM

Table 18. TGT_ALREADY_ATTACH_TO_OTHER_MDM parameters

Parameter	Description
Name	TGT_ALREADY_ATTACH_TO_OTHER_MDM
Message	SDS is already attached to another MDM
Commands	`add_sds`
Description	SDS was previously configured by another MDM
Action	Check your command parameters to validate the SDS IP address and port. Check for other PowerFlex installations of MDMs. If the SDS is not found in any MDM configuration, this might indicate that it was not cleaned up properly when it was removed. To add the SDS, use the `force_clean` flag. **NOTE:** This command erases the existing SDS configuration and data.

SDS not found

Table 19. TGT_NOT_FOUND parameters

Parameter	Description
Name	TGT_NOT_FOUND
Message	Could not find SDS
Commands	All SDS related commands
Description	The SDS was not found

Table 19. TGT_NOT_FOUND parameters (continued)

Parameter	Description
Action	Check the provided parameters. Use `query_all_sds` to view the configured SDSs.

Volume not found

Table 20. VOL_NOT_FOUND parameters

Parameter	Description
Name	VOL_NOT_FOUND
Message	Could not find the volume
Commands	All volume related commands
Description	The volume was not found
Action	Check the provided parameters. Use `query_all_volumes` to view the configured volumes.

SDC not found

Table 21. INI_NOT_FOUND parameters

Parameter	Description
Name	INI_NOT_FOUND
Message	Could not find the SDC
Commands	All SDC related commands
Description	SDC was not found
Action	Use `query_all_sdc` to view the configured SDCs

Volume already mapped to SDC

Table 22. VOL_ALREADY_MAPPED_TO_THIS_INI parameters

Parameter	Description
Name	VOL_ALREADY_MAPPED_TO_THIS_INI
Message	The volume is already mapped to this SDC
Commands	`map_volume_to_sdc` (when mapping to a single SDC)
Description	The specified volume is already mapped to the specified SDC
Action	Check the command parameters, and use `query_all_volumes` to validate the current mapping

Volume already mapped to all SDCs

Table 23. VOL_ALREADY_MAPPED_TO_ALL_INIS parameters

Parameter	Description
Name	VOL_ALREADY_MAPPED_TO_ALL_INIS

Table 23. VOL_ALREADY_MAPPED_TO_ALL_INIS parameters (continued)

Parameter	Description
Message	The volume is already mapped to all SDCs
Commands	`map_volume_to_sdc` (when mapping to all SDCs)
Description	The specified volume is already mapped to all SDCs
Action	Check the command parameters, and use `query_all_volumes` to validate the current mapping

Volume not mapped to SDC

Table 24. VOL_NOT_MAPPED_TO_INI parameters

Parameter	Description
Name	VOL_NOT_MAPPED_TO_INI
Message	The volume is not mapped to SDC
Commands	`unmap_volume_from_sdc`, `query_sdc_volume_limits`, `set_sdc_volume_limits`
Description	The specified volume is not mapped to the specified SDC
Action	Check the command parameters, and use `query_all_sdc` to validate the current mapping

Volume mapped to all SDCs

Table 25. VOL_MAPPED_TO_ALL_INIS parameters

Parameter	Description
Name	VOL_MAPPED_TO_ALL_INIS
Message	The volume is mapped to all SDCs
Commands	`map_volume_to_sdc`, `unmap_volume_from_sdc`, `set_sdc_volume_limits`
Description	The volume is mapped to all SDCs
Action	Check the command parameters, and use `query_all_volumes` to validate the current mapping

Too many mappings

Table 26. TOO_MANY_MAPPINGS parameters

Parameter	Description
Name	TOO_MANY_MAPPINGS
Message	There are too many mappings
Commands	`map_volume_to_sdc`
Description	Mapping of an additional volume will exceed the system limit
Action	Validate if all existing volumes that are mapped are relevant. Contact Dell Technologies Support.

Invalid volume ID

Table 27. Invalid volume ID parameters

Parameter	Description
Name	INVALID_VOL_ID.
Message	Invalid volume ID. Try again with a valid ID.
Commands	All volume-related commands.
Description	The specified volume ID is invalid.
Action	Check the specified volume ID. Use `query_all_volumes` to find the correct ID, or use the volume name.

Invalid SDS ID

Table 28. INVALID_TGT_ID parameters

Parameter	Description
Name	INVALID_TGT_ID.
Message	Invalid SDS ID. Try again with a valid ID.
Commands	All SDS commands.
Description	The specified SDS ID is invalid.
Action	Check the specified SDS ID. Use `query_all_sds` to find the correct ID, or use the SDS name or IP address.

Invalid SDC ID

Table 29. INVALID_INI_ID parameters

Parameter	Description
Name	INVALID_INI_ID.
Message	Invalid SDC ID. Try again with a valid ID.
Commands	All SDC commands.
Description	The specified SDC ID is invalid.
Action	Check the specified SDC ID. Use `query_all_sdc` to find the correct ID, or use the SDC IP address.

Host not found

Table 30. HOST_NOT_FOUND parameters

Parameter	Description
Name	HOST_NOT_FOUND
Message	Could not find the host
Commands	All SDS commands
Description	The specified host name was not found
Action	Check the specified host name

Duplicate host name

Table 31. DUP_HOST_NAME parameters

Parameter	Description
Name	DUP_HOST_NAME
Message	Duplicate host name
Commands	`add_sds`
Description	The list of host names includes two identical host names or IP addresses, or two host names are resolved to the same IP address
Action	Check the specified host name list

Address and port in use

Table 32. TGT_ADDR_AND_PORT_IN_USE parameters

Parameter	Description
Name	TGT_ADDR_AND_PORT_IN_USE
Message	The SDS IP address and port already in use. Try with different ones.
Commands	`add_sds`
Description	The specified IP address and port are already in use by another defined SDS.
Action	Check the specified IP address and port.

Too many SDSs in Protection Domain

Table 33. TOO_MANY_TGTS_IN_FDOM parameters

Parameter	Description
Name	TOO_MANY_TGTS_IN_FDOM
Message	There are too many SDSs in a protection domain.
Commands	`add_sds`
Description	Adding the SDS exceeds the system limit of allowed SDSs in a protection domain.
Action	Check the configuration.

Too many SDSs

Table 34. TOO_MANY_TGTS parameters

Parameter	Description
Name	TOO_MANY_TGTS
Message	There are too many SDSs
Commands	`add_sds`

Table 34. TOO_MANY_TGTS parameters (continued)

Parameter	Description
Description	Adding the SDS will exceed the system limit of allowed SDSs in the system
Action	Check configuration

Too many volumes in protection domain

Table 35. Too many volumes in protection domain parameters

Parameter	Description
Name	TOO_MANY_VOLUMES_IN_FD
Message	Too many volumes in a protection domain.
Commands	`add_volume`
Description	Adding a volume exceeds the limit for the number of volumes allowed in a protection domain.
Action	Check the configuration.

Too many volumes

Table 36. Too many volumes parameters

Parameter	Description
Name	TOO_MANY_VOLS
Message	There are too many volumes
Commands	`add_volume`
Description	Adding a volume will exceed the limit for the number of volumes allowed in a system
Action	Check configuration

SDS name in use

Table 37. TGT_NAME_IN_USE parameters

Parameter	Description
Name	TGT_NAME_IN_USE
Message	SDS name already in use. Use a different name.
Commands	`add_sds, rename_sds`
Description	The specified SDS name is already in use by another SDS.
Action	Use `query_all_sds` to validate existing names, and issue the command with another name.

Volume name in use

Table 38. VOL_NAME_IN_USE parameters

Parameter	Description
Name	VOL_NAME_IN_USE
Message	Volume name already in use. Use a different name.
Commands	`add_volume, rename_volume`
Description	The specified volume name is already in use by another volume.
Action	Use `query_all_volumes` to validate existing names, and issue the command with another name.

Duplicate SDS device name

Table 39. DUPLICATE_TGT_DEVICE_NAME parameters

Parameter	Description
Name	DUPLICATE_TGT_DEVICE_NAME
Message	Duplicate SDS device name
Commands	`add_sds`
Description	A device appears more than once in the `add_sds` command
Action	Validate command parameters

Duplicate SDC IP address

Table 40. DUPLICATE_SDC_IP parameters

Parameter	Description
Name	DUPLICATE_SDC_IP
Message	More than one SDC exists for the same IP address. Use name or ID instead, or remove the duplicate SDC.
Commands	`map_volume_to_sdc, unmap_volume_from_sdc, set_sdc_volume_limits, query_sdc_volume_limits`
Description	The specified SDC IP address is not unique. It is associated with more than one SDC.
Action	Use `query_all_sdc` to find the relevant ID. Use the SDC ID in the command instead of the SDC IP address.

SDS not under removal

Table 41. TGT_NOT_UNDER_REMOVAL parameters

Parameter	Description
Name	TGT_NOT_UNDER_REMOVAL

Table 41. TGT_NOT_UNDER_REMOVAL parameters (continued)

Parameter	Description
Message	SDS is not being removed
Commands	`abort_remove_sds`
Description	The SDS is not being removed
Action	Check command parameters. If the parameters are correct, it is still possible that the SDS has already been removed.

SDS being removed

Table 42. TGT_BEING_REMOVED parameters

Parameter	Description
Name	TGT_BEING_REMOVED
Message	SDS is being removed
Commands	SDS related commands
Description	The SDS is currently being removed
Action	None

Volume mapped

Table 43. VOL_MAPPED parameters

Parameter	Description
Name	VOL_MAPPED
Message	Volume is mapped
Commands	`remove_volume, remove_consistency_group`
Description	The volume (for `remove_volume`) or any volume in the group (`remove_consistency_group`) is mapped to an SDC
Action	Unmap before removing the volume or consistency group. Use `query_all_volumes` to check current mapping.

Use MDM IP address

Table 44. USE_MDM_IP parameters

Parameter	Description
Name	USE_MDM_IP
Message	Use cluster MDM physical IP to access the MDM.
Commands	All cluster commands.
Description	The command reached the secondary MDM. Use the physical IP address to access the MDM.
Action	Add `--mdm_ip` to the commands.

Command only in cluster

Table 45. CMD_ONLY_IN_CLUSTER parameters

Parameter	Description
Name	CMD_ONLY_IN_CLUSTER
Message	Command can only be used in cluster mode
Commands	`switch_mdm_ownership,`
Description	This command can be issued only when in cluster mode
Action	Use `query_cluster` to check the current configuration

Command only in single mode

Table 46. CMD_ONLY_IN_SINGLE parameters

Parameter	Description
Name	CMD_ONLY_IN_SINGLE
Message	Command can only be used in a single mode.
Commands	MDM cluster commands
Description	This command can be issued in a single mode only.
Action	Use `switch_mdm_ownership` to switch to single mode before issuing this command.

Command error when cluster degraded

Table 47. CMD_ERROR_CLUSTER_DEGRADED parameters

Parameter	Description
Name	CMD_ERROR_CLUSTER_DEGRADED
Message	Command cannot be used when a cluster is degraded
Commands	`switch_mdm_ownership`
Description	This command cannot be issued when the cluster is degraded
Action	Check the cluster state and hardware configuration to determine the reason for cluster degradation

Too many devices

Table 48. TOO_MANY_DEVICES parameters

Parameter	Description
Name	TOO_MANY_DEVICES
Message	There are too many devices for SDS
Commands	`add_sds, add_sds_device`

Table 48. TOO_MANY_DEVICES parameters (continued)

Parameter	Description
Description	Too many devices specified (`add_sds`), or the SDS already has the maximum number of allowed devices
Action	None

SDS device not found

Table 49. TGT_DEVICE_NOT_FOUND parameters

Parameter	Description
Name	TGT_DEVICE_NOT_FOUND
Message	Could not find the SDS device
Commands	SDS device related commands
Description	SDS device was not found
Action	Check the command parameters. Use `query_sds` to view the device details.

License too long

Table 50. LICENSE_TOO_LONG parameters

Parameter	Description
Name	LICENSE_TOO_LONG
Message	The license key is too long. Check your key and enter it again.
Commands	`set_license`
Description	The specified license key is too long.
Action	Validate the specified license key with the one received from Dell.

License error

Table 51. License error parameters

Parameter	Description
Name	LICENSE_ERROR
Message	License key is corrupted.
Commands	`set_license`
Description	The specified license key is invalid or does not match this version. Contact Dell Technologies Support.
Action	Validate the specified license key with the one received from Dell.

Unknown license version

Table 52. LICENSE_UNKNOWN_VERSION parameters

Parameter	Description
Name	LICENSE_UNKNOWN_VERSION
Message	Unknown license version
Commands	`set_license`
Description	The specified license key is corrupted
Action	Validate the specified license key with the one received from Dell.

License not set

Table 53. LICENSE_NOT_SET parameters

Parameter	Description
Name	LICENSE_NOT_SET
Message	License has not been set
Commands	`query_license, add_sds, add_sds_device, set_data_copy_limit`
Description	License was not set. Product cannot be used.
Action	Set the license before continuing to work

Configuration exceeds new license

Table 54. CONFIG_EXCEEDS_NEW_LICENSE parameters

Parameter	Description
Name	CONFIG_EXCEEDS_NEW_LICENSE
Message	The current system configuration exceeds the license entitlements
Commands	`set_license`
Description	The specified license decreases the amount of licensed capacity. The current system capacity already exceeds the specified license.
Action	Contact Dell Technologies Support to receive a license for a larger amount of capacity

License capacity exceeded

Table 55. CAPACITY_LICENSE_EXCEEDED parameters

Parameter	Description
Name	CAPACITY_LICENSE_EXCEEDED
Message	The operation could not be completed. The license capacity has been exceeded.
Commands	`add_sds, add_sds_device`
Description	Adding the SDS or device exceeds the current licensed capacity
Action	Contact Dell Technologies Support to receive a license for a larger amount of capacity

License expired

Table 56. LICENSE_TIME_EXCEEDED parameters

Parameter	Description
Name	LICENSE_TIME_EXCEEDED
Message	The license has expired
Commands	Most commands
Description	The system time-based license has expired
Action	Contact Dell Technologies Support to receive a new license

Installation ID mismatch

Table 57. INSTALL_ID_MISMATCH parameters

Parameter	Description
Name	INSTALL_ID_MISMATCH
Message	The license installation ID does not match the ID of this system
Commands	`set_license`
Description	The provided license key does not match the current installation
Action	Contact Dell Technologies Support

Customer ID mismatch

Table 58. CUSTOMER_ID_MISMATCH parameters

Parameter	Description
Name	CUSTOMER_ID_MISMATCH
Message	License customer ID does not match this system

Table 58. CUSTOMER_ID_MISMATCH parameters (continued)

Parameter	Description
Commands	`set_license`
Description	The provided license key does not match the current installation
Action	Contact Dell Technologies Support

Unsupported license version

Table 59. LICENSE_UNSUPPORTED_VERSION parameters

Parameter	Description
Name	LICENSE_UNSUPPORTED_VERSION
Message	Unsupported license version
Commands	`set_license`
Description	The provided license key does not match the current installation
Action	Contact Dell Technologies Support

Volume decrease not supported

Table 60. VOL_DECREASE_NOT_SUPPORTED parameters

Parameter	Description
Name	VOL_DECREASE_NOT_SUPPORTED
Message	Volume capacity can only be increased
Commands	`modify_volume_capacity`
Description	The specified volume capacity is smaller than the existing volume capacity. Decreasing volume capacity is not supported.
Action	Use `query_all_volumes` to validate your current configuration

SDS device being removed

Table 61. TGT_DEV_BEING_REMOVED parameters

Parameter	Description
Name	TGT_DEV_BEING_REMOVED
Message	The SDS device is being removed
Commands	`remove_sds, remove_sds_device, activate_sds_device, start_sds_device_test`

Table 61. TGT_DEV_BEING_REMOVED parameters (continued)

Parameter	Description
Description	The specified SDS device is currently being removed. The command cannot be executed.
Action	Wait for the device to be removed

Cannot remove last SDS device

Table 62. CANNOT_REMOVE_LAST_SDS_DEVICE parameters

Parameter	Description
Name	CANNOT_REMOVE_LAST_SDS_DEVICE
Message	Last remaining SDS device cannot be removed
Commands	`remove_sds_device`
Description	The command attempts to remove the last remaining SDS device. This is an illegal operation.
Action	To remove the last remaining device, remove the SDS

Device not under removal

Table 63. DEV_NOT_UNDER_REMOVAL parameters

Parameter	Description
Name	DEV_NOT_UNDER_REMOVAL
Message	The SDS device is not being removed
Commands	`abort_remove_sds`
Description	The specified device is not in the process of being removed
Action	Check the command parameters

SDS already adding device

Table 64. TGT_ALREADY_ADDING_DEV parameter details

Parameter	Description
Name	TGT_ALREADY_ADDING_DEV
Message	A device is being added to this SDS
Commands	`add_sds_device, remove_sds`
Description	This operation cannot be performed while a device is being added to the SDS
Action	Wait for the device to be added

Invalid device ID

Table 65. INVALID_DEV_ID parameters

Parameter	Description
Name	INVALID_DEV_ID
Message	The device ID is invalid. Use a valid device ID.
Commands	SDS device-related commands.
Description	The specified SDS device ID is invalid.
Action	Use `query_sds` to determine the correct ID.

Too many protection domains

Table 66. TOO_MANY_FDs parameters

Parameter	Description
Name	TOO_MANY_FDs
Message	There are too many protection domains.
Commands	`add_protection_domain`
Description	Adding a protection domain exceeds the system limit of the allowed protection domains.
Action	None

Protection domain name in use

Table 67. FD_NAME_IN_USE parameters

Parameter	Description
Name	FD_NAME_IN_USE
Message	Protection domain name already in use. Try a different name.
Commands	`add_protection_domain, rename_protection_domain`
Description	The specified protection domain name is already in use by another protection domain.
Action	Use `query_all` to validate the existing names, and issue the command with another name.

Protection domain not found

Table 68. FD_NOT_FOUND parameters

Parameter	Description
Name	FD_NOT_FOUND.
Message	Could not find protection domain.
Commands	All protection domain-related commands.

Table 68. FD_NOT_FOUND parameters (continued)

Parameter	Description
Description	The specified protection domain name or ID do not match any existing protection domains.
Action	Check the command parameters. Use `query_all` to validate the existing protection domain names.

Protection domain has SDSs

Table 69. Protection domain has SDSs parameters

Parameter	Description
Name	FD_HAS_TGTS
Message	Protection domain has SDSs
Commands	`remove_protection_domain`
Description	The protection domain contains SDSs and cannot be removed.
Action	Remove the SDSs before attempting to remove the protection domain.

Not enough SDSs in storage pool

Table 70. NOT_ENOUGH_TGTS_IN_STORAGE_POOL parameters

Parameter	Description
Name	NOT_ENOUGH_TGTS_IN_STORAGE_POOL
Message	Fewer than three devices from different SDSs are defined in this storage pool. Add devices to the storage pool from additional SDSs.
Commands	`add_volume`
Description	The storage pool must contain devices from at least three different SDSs in order to create a volume.
Action	Add more devices to the storage pool before creating a volume.

Remove in progress

Table 71. REMOVE_IN_PROGRESS parameters

Parameter	Description
Name	REMOVE_IN_PROGRESS
Message	Removal of the object is in progress. Please wait.
Commands	Multiple commands
Description	The operation cannot be performed, because the volume is being removed
Action	Wait for the volume to be removed

No network test results

Table 72. NET_TEST_NOT_FOUND parameters

Parameter	Description
Name	NET_TEST_NOT_FOUND
Message	Could not find network test results
Commands	`query_sds_network_test_results`
Description	The specified SDS has no network test results
Action	Check the command parameters to make sure that a `start_sds_network_test` was sent to the specified SDS. If this is the correct SDS, resend the `start_sds_network_test` command.

Network test in progress

Table 73. NET_TEST_IN_PROGRESS parameters

Parameter	Description
Name	NET_TEST_IN_PROGRESS
Message	Network test in progress
Commands	`query_sds_network_test_results`
Description	The SDS network test has not been completed
Action	Wait for the test to be completed. Use the same query periodically until results are displayed.

SDS device already active

Table 74. TGT_DEV_ALREADY_ACTIVE parameters

Parameter	Description
Name	TGT_DEV_ALREADY_ACTIVE
Message	The SDS device is already active
Commands	`activate_sds_device`
Description	At least one specified device is already active
Action	Use `query_sds` to view the device status

SDS device wrong test mode

Table 75. TGT_DEV_WRONG_TEST_MODE parameters

Parameter	Description
Name	TGT_DEV_WRONG_TEST_MODE
Message	The SDS device is in incorrect test mode

Table 75. TGT_DEV_WRONG_TEST_MODE parameters (continued)

Parameter	Description
Commands	`activate_sds_device`
Description	At least one specified device will automatically be activated when its test is completed
Action	Use `query_sds` to view the device status

SDS device wrong test state

Table 76. TGT_DEV_WRONG_TEST_STATE parameters

Parameter	Description
Name	TGT_DEV_WRONG_TEST_STATE
Message	The SDS device test is not complete
Commands	`activate_sds_device`
Description	At least one specified device test is not yet complete. This device cannot be activated.
Action	Use `query_sds` to view the devices' status

SDS device test in progress

Table 77. TGT_DEV_TEST_IN_PROGRESS parameters

Parameter	Description
Name	TGT_DEV_TEST_IN_PROGRESS
Message	The SDS device test is in progress
Commands	`start_sds_device_test`
Description	At least one specified device test is in-progress. A new test for this device cannot be started now.
Action	Use `query_sds` to view the devices' test status

Volume allocation "busy" error

Table 78. VOL_ALLOC_ERROR_BUSY parameters

Parameter	Description
Name	VOL_ALLOC_ERROR_BUSY
Message	The system is busy. Retry later.
Commands	`add_volume`
Description	The MDM is attempting to recover a connection with an SDS. You cannot create a volume now.

Table 78. VOL_ALLOC_ERROR_BUSY parameters (continued)

Parameter	Description
Action	Use `query_all` to view the system status. Retry the command when the situation has returned to normal.

Capacity highly unbalanced

Table 79. CAPACITY_HIGHLY_UNBALANCED parameters

Parameter	Description
Name	CAPACITY_HIGHLY_UNBALANCED
Message	System capacity is unbalanced
Commands	`add_volume`
Description	System capacity is unbalanced. This can be a temporary or permanent state. • Temporary Due to adding or removing capacity, or due to disconnections. In this case, the system will automatically work in the background to correct the situation. • Permanent The capacity of the devices added to PowerFlex is highly unbalanced. This requires user intervention to correct the situation.
Action	If a rebalance is in progress, wait for it to finish and resubmit the command. If the situation persists, contact support.

Volume creation failed

Table 80. VOL_CREATE_FAIL parameters

Parameter	Description
Name	VOL_CREATE_FAIL
Message	Could not create the volume
Commands	`add_volume`
Description	The volume creation failed
Action	Use `query_all` to validate the system state, and retry the command

Password authentication failure

Table 81. Invalid password error parameters

Parameter	Description
Name	INVALID_PASSWORD
Message	Invalid password. Try again.
Commands	Most commands.
Description	Indicates that the password specified in the command is not valid for the current operation.

Table 81. Invalid password error parameters (continued)

Parameter	Description
Action	Check that you are using the correct password in the command.

No permissions

Table 82. NO_PERMISSIONS parameter details

Parameter	Description
Name	NO_PERMISSIONS
Message	You do not have permission to perform this operation. Contact your system administrator for assistance.
Commands	Most commands.
Description	The specified user does not have permission to issue this command.
Action	Contact the system administrator.

Too many storage pools in protection domain

Table 83. Too many storage pools in protection domain parameter details

Parameter	Description
Name	TOO_MANY_STORAGE_POOLS_IN_PROTECTION_DOMAIN
Message	There are too many storage pools in protection domain.
Commands	`add_storage_pool`
Description	Addition of another storage pool exceeds the system limit of allowed storage pools in a protection domain.
Action	Use `query_protection_domain` to view the storage pools in the protection domain

Storage pool already exists

Table 84. STORAGE_POOL_ALREADY_EXISTS parameter details

Parameter	Description
Name	STORAGE_POOL_ALREADY_EXISTS
Message	Storage pool exists.
Commands	`add_storage_pool`
Description	A storage pool with the specified name exists in the specified protection domain.
Action	Use `query_protection_domain` to view the storage pools in the protection domain, and choose a different name.

Storage pool not found

Table 85. STORAGE_POOL_NOT_FOUND parameter details

Parameter	Description
Name	STORAGE_POOL_NOT_FOUND
Message	Could not find a storage pool.
Commands	All storage pool–related commands.
Description	The specified storage pool was not found.
Action	Use `query_protection_domain` to view the storage pools in the protection domain, and check the issued parameters.

Storage pool has devices

Table 86. STORAGE_POOL_HAS_DEVS parameter details

Parameter	Description
Name	STORAGE_POOL_HAS_DEVS
Message	The storage pool has SDS devices.
Commands	`remove_storage_pool`
Description	The specified storage pool has SDS devices that are associated with it.
Action	Remove the SDS devices before removing the storage pool.

Storage pool name already exists

Table 87. STORAGE_POOL_NAME_ALREADY_EXISTS parameter details

Parameter	Description
Name	STORAGE_POOL_NAME_ALREADY_EXISTS
Message	Storage pool name exists.
Commands	`rename_storage_pool`
Description	A storage pool with the specified new name exists in the protection domain.
Action	Use `query_protection_domain` to view the storage pools in the protection domain. Choose another name for the storage pool.

Invalid storage pool ID

Table 88. INVALID_STORAGE_POOL_ID parameter details

Parameter	Description
Name	INVALID_STORAGE_POOL_ID
Message	Invalid storage pool ID.

Table 88. INVALID_STORAGE_POOL_ID parameter details (continued)

Parameter	Description
Commands	All storage pool-related commands
Description	The specified storage pool ID is invalid
Action	Check the specified storage pool ID. Use `query_protection_domain` to find the correct ID, or use the storage pool name.

Protection domain has storage pools

Table 89. FD_HAS_STORAGE_POOLS parameter details

Parameter	Description
Name	FD_HAS_STORAGE_POOLS
Message	Protection domain has storage pools.
Commands	`remove_protection_domain`
Description	The protection domain has more than one storage pool associated with it.
Action	Remove the storage pools before removing the protection domain.

SDC has mappings

Table 90. INI_HAS_MAPPINGS parameter details

Parameter	Description
Name	INI_HAS_MAPPINGS
Message	SDC has mapped volume(s)
Commands	`remove_sdc`
Description	At least one volume is mapped to the specified SDC
Action	Remove the mapping before removing the SDC

Too many volumes in V-Tree

Table 91. TOO_MANY_VOLS_IN_VTREE parameter details

Parameter	Description
Name	TOO_MANY_VOLS_IN_VTREE
Message	There are too many volumes in the V-Tree
Commands	`snapshot_volume`
Description	Taking a snapshot of the specified volume will exceed the allowed system limit of snapshots per volume

Table 91. TOO_MANY_VOLS_IN_VTREE parameter details (continued)

Parameter	Description
Action	Use `query_volume` to view the current snapshots. Consider deleting some snapshots.

Too many snapshots in V-Tree

Table 92. TOO_MANY_SNAPS_IN_VTREE parameter details

Parameter	Description
Name	TOO_MANY_SNAPS_IN_VTREE
Message	There are too many snapshots in the V-Tree
Commands	`snapshot_volume`
Description	Taking a snapshot of the specified volume will exceed the allowed system limit of snapshots per volume
Action	Contact Dell Technologies Support

No volume to delete

Table 93. NOTHING_TO_DO parameter details

Parameter	Description
Name	NOTHING_TO_DO
Message	No volume to delete
Commands	`remove_volume`
Description	The combination of parameters resulted in an empty set of volumes to delete
Action	Use `query_volume` to view the current state, and review the command parameters accordingly

Too many devices in storage pool

Table 94. TOO_MANY_DEVICES_IN_STORAGE_POOL parameter details

Parameter	Description
Name	TOO_MANY_DEVICES_IN_STORAGE_POOL
Message	There are too many devices in storage pool.
Commands	`add_sds, add_sds_device`
Description	Adding a device to the specified storage pool exceeds the allowed system limit of SDS devices per storage pool.
Action	Use `query_sds` to find the association of devices to storage pools.

Volume not found for consistency group

Table 95. VOL_NOT_FOUND_FOR_SNAP_GROUP parameter details

Parameter	Description
Name	VOL_NOT_FOUND_FOR_SNAP_GROUP
Message	No volumes were found for the given consistency group
Commands	`remove_consistency_group`
Description	No volume found for the specified consistency group
Action	Use `query_all_volumes` to view the existing consistency groups, and review command parameters accordingly

Too many IP addresses

Table 96. TOO_MANY_IPS parameter details

Parameter	Description
Name	TOO_MANY_IPS
Message	There are too many IP addresses for SDS
Commands	`add_sds_ip`
Description	Addition of an IP address to the SDS exceeds the allowed system limit of IP addresses per SDS
Action	Use `query_sds` to view the SDS IP addresses

Cannot remove last SDS IP address

Table 97. LAST_IP parameter details

Parameter	Description
Name	LAST_IP
Message	SDS must have at least one IP address
Commands	`remove_sds_ip`
Description	An SDS must have at least one IP address. Cannot remove the last remaining IP address.
Action	Use `query_sds` to view the SDS IP addresses.

SDS IP address not found

Table 98. SDS_IP_NOT_FOUND parameter details

Parameter	Description
Name	SDS_IP_NOT_FOUND
Message	SDS IP not found

Table 98. SDS_IP_NOT_FOUND parameter details (continued)

Parameter	Description
Commands	`remove_sds_ip`
Description	The specified SDS IP address was not found
Action	Use `query_sds` to view the SDS IP addresses

Volume size too large

Table 99. VOL_SIZE_TOO_LARGE parameter details

Parameter	Description
Name	VOL_SIZE_TOO_LARGE
Message	The volume size too large
Commands	`add_volume, modify_volume_capacity`
Description	The specified volume size is too large
Action	Use a smaller volume size

Capacity too low for snapshot

Table 100. CAPACITY_TOO_LOW_FOR_SNAP parameter details

Parameter	Description
Name	CAPACITY_TOO_LOW_FOR_SNAP
Message	The capacity of the storage pool is too small to create new snapshots.
Commands	`snapshot_volume`
Description	Snapshots are disabled when the free capacity in the storage pool is very low. The threshold is defined using the `set_capacity_alerts_threshold` command. The default threshold is 10% below the predefined spare.
Action	Add storage capacity, or free up space by removing some snapshots.

IP address belongs to other SDS

Table 101. IP_BELONGS_TO_OTHER_TGT parameter details

Parameter	Description
Name	IP_BELONGS_TO_OTHER_TGT
Message	The specified IP address belongs to another SDS. Use the correct IP address.
Commands	`add_sds, add_sds_ip`
Description	The specified IP address is already assigned to another SDS.

Table 101. IP_BELONGS_TO_OTHER_TGT parameter details (continued)

Parameter	Description
Action	Check command parameters.

Invalid port

Table 102. INVALID_PORT parameter details

Parameter	Description
Name	INVALID_PORT
Message	The specified SDS port is not in the valid range. Try again with a valid port number.
Commands	`add_sds, add_sds_ip`
Description	The specified port must be greater than 1024.
Action	Check command parameters.

LIA package not installed

Table 103. LIA_PACKAGE_NOT_INSTALLED parameter details

Parameter	Description
Name	LIA_PACKAGE_NOT_INSTALLED
Message	The package is not installed.
Description	You are trying to uninstall the LIA package, but the package is not installed.
Action	Check why you are trying to uninstall a component that is not installed. Verify that you have the correct package name for the package that you want to uninstall.

LIA multiple packages installed

Table 104. LIA_MULTIPLE_PACKAGES_INSTALLED parameter details

Parameter	Description
Name	LIA_MULTIPLE_PACKAGES_INSTALLED
Message	More than one package is installed.
Description	More than one package of the specified component is installed.

Action	Uninstall the packages manually: Linux:
	1. Find the package name:
	`rpm -qa | grep -i lia`
	The LIA package name is displayed. For example:
	`EMC-ScaleIO-lia-2.50-0.30.el6.x86_64`

Table 104. LIA_MULTIPLE_PACKAGES_INSTALLED parameter details (continued)

Parameter	Description
	2. Uninstall the package: `rpm -e <PACKAGE_NAME>` Windows: • **Windows Control Panel > Add/Remove programs** > **EMC-ScaleIO-lia**

LIA command failed

Table 105. LIA_COMMAND_FAILED parameter details

Parameter	Description
Name	LIA_COMMAND_FAILED
Message	The command failed. Try again.
Description	This is a general error message.
Action	Look for more information about the error in the LIA log: Linux: `/opt/emc/scaleio/lia/logs` Windows: `C:\Program Files\emc\scaleio\lia\logs`

LIA invalid password

Table 106. LIA_INVALID_PASSWORD Parameter Details

Parameter	Description
Name	LIA_INVALID_PASSWORD
Message	Token authentication was not successful
Description	The wrong LIA token was used, and authentication failed.
Action	Verify that you have the correct LIA password (token).

LIA operation not permitted

Table 107. LIA_OPERATION_NOT_PERMITTED parameter details

Parameter	Description
Name	LIA_OPERATION_NOT_PERMITTED
Message	The operation is not permitted by current configuration
Description	LIA was configured to disallow the operation that you tried to perform.
Action	Contact your system administrator; change the LIA configuration in your system.

LIA file is too large

Table 108. LIA_FILE_IS_TOO_LARGE parameter details

Parameter	Description
Name	LIA_FILE_IS_TOO_LARGE
Message	The file is too large to send back to the user.
Description	The file returned by the `get_info` command is larger than 500 MB, and therefore cannot be sent back to you.
Action	Contact Dell Technologies Support.

LIA wrong ID

Table 109. LIA_WRONG_ID parameter details

Parameter	Description
Name	LIA_WRONG_ID
Message	LIA installation ID verification failed
Description	The LIA ID that was passed in the installation/uninstall package is different to the installation ID that LIA is currently using.
Action	Edit the LIA's configuration file with LIA token and restart the LIA. The configuration file is found at: • Linux: `/opt/emc/scaleio/lia/cfg/cont.txt` • Windows: `C:\Program Files\emc\scaleio\lia\cfg\conf.txt`

LIA disk does not exist

Table 110. LIA_DISK_DOES_NOT_EXIST parameter details

Parameter	Description
Name	LIA_DISK_DOES_NOT_EXIST
Message	The disk does not exist in the system
Description	The disk name that was passed with the `prepare_disk` command does not exist in the system
Action	Check that you are using the correct the disk name

SDS IP address not found

Table 111. TGT_IP_NOT_FOUND parameter details

Parameter	Description
Name	TGT_IP_NOT_FOUND
Message	The SDS IP address cannot be found.
Commands	`modify_sds_ip, remove_sds_ip`

Table 111. TGT_IP_NOT_FOUND parameter details (continued)

Parameter	Description
Description	You tried to remove or modify an SDS IP address that does not exist in the system.
Action	Verify that you are specifying the correct IP address in the command.

Bad threshold value

Table 112. BAD_THRESHOLD_VALUE parameter details

Parameter	Description
Name	BAD_THRESHOLD_VALUE
Message	The threshold value is out of range. Enter a valid percentage value.
Commands	`set_capacity_alert_threshold`
Description	The legal range for the capacity alert threshold is a percentage value between 0 and 100. The value that is specified for critical alerts must be less than the value for high alerts.
Action	Specify a value between 0 and 100. Ensure that the value for critical alerts is less than the value for high alerts.

Storage pool has volumes

Table 113. Storage pool has volumes parameter details

Parameter	Description
Name	STORAGE_POOL_HAS_VOLS
Message	The storage pool has volumes. Remove all the volumes before removing the storage pool.
Commands	`remove_storage_pool` `remove_protection_domain`
Description	Only empty storage pools can be removed.
Action	Remove all volumes before removing the storage pool or protection domain.

Protection Domain and Storage Pool mismatch

Table 114. FD_AND_SP_MISMATCH parameter details

Parameter	Description
Name	FD_AND_SP_MISMATCH
Message	The specified storage pool is not in the specified protection domain.
Commands	Any command which uses the combination of storage pool and protection domain.

Table 114. FD_AND_SP_MISMATCH parameter details (continued)

Parameter	Description
Description	The operation could not be performed because the combination of storage pool and protection domain is incorrect and the storage pool could not be found.
Action	Use the correct combination of storage pool and protection domain in the command. Use the command `query_all` to list all the storage pools to find this information.

SDS IP address already exists

Table 115. TGT_IP_ALREADY_EXISTS parameter details

Parameter	Description
Name	TGT_IP_ALREADY_EXISTS
Message	This SDS IP exists. Use a different IP address.
Commands	`add_sds_ip`
Description	The IP address that is used in the command is already in use in the system.
Action	Verify that you are using the correct IP address in the command, and that the address has not been assigned to any other SDSs.

IP role already set

Table 116. ROLE_ALREADY_SET parameter details

Parameter	Description
Name	ROLE_ALREADY_SET
Message	The IP role for this SDS has already been configured.
Commands	`modify_sds_ip`
Description	The IP role for the specified SDS has already been configured.
Action	Verify that you are configuring the intended SDS, and that you are specifying the correct IP address.

Cluster ID mismatch

Table 117. Cluster ID mismatch parameter details

Parameter	Description
Name	CLUSTER_ID_MISMATCH
Message	The system ID number that you provided does not match the actual system ID.
Commands	`query_properties`, `query_system`
Description	The system ID number has been modified, and does not match the ID specified in the command.

Table 117. Cluster ID mismatch parameter details (continued)

Parameter	Description
Action	No action required - this code is provided for information purposes for advanced users.

Device already exists

Table 118. Device already exists parameter details

Parameter	Description
Name	DEV_ALREADY_EXISTS
Message	A device with the given name already exists in the SDS.
Commands	`add_sds_device`
Description	You tried to add a device to the system, using a name that is already defined in the SDS.
Action	Verify that the specified device has not already been added to the system.

Volume size is illegal

Table 119. VOL_SIZE_ILLEGAL parameter details

Parameter	Description
Name	VOL_SIZE_ILLEGAL
Message	The specified volume size is illegal. Volume size must be in 8 GB blocks.
Commands	`add_volume, modify_volume_size`
Description	You tried to define a volume size that is not a multiple of 8 GB. Volume size can only be defined in multiples of 8 GB.
Action	Specify volume size in blocks of 8 GB.

SDC must be disconnected

Table 120. SDC must be disconnected parameter details

Parameter	Description
Name	INI_MUST_BE_DISCONNECTED
Message	The SDC must be disconnected.
Commands	`remove_sdc`
Description	Only SDCs that are currently inactive (disconnected) can be removed.
Action	This command is intended to remove old, disconnected SDCs. Do not remove active SDCs.

Invalid percentage

Table 121. INVALID_PERCENTAGE parameter details

Parameter	Description
Name	INVALID_PERCENTAGE
Message	The percentage value that you provided is invalid. Use a value between 0—100.
Commands	`set_thick_volume_percentage`
Description	The value that you entered was less than 0 or greater than 100.
Action	Use a value between 1 and 100 in the command.

Wrong receive group

Table 122. WRONG_RECEIVE_GROUP parameter details

Parameter	Description
Name	WRONG_RECEIVE_GROUP
Message	The command was sent to the wrong receive group due to component software mismatch.
Commands	All commands
Description	There are different software versions installed on some components in your system, and this is causing a software mismatch error.
Action	Verify that the same software version is installed on all components, and fix, if necessary.

Mismatched SW ID

Table 123. MISMATCHING_SWID parameter details

Parameter	Description
Name	MISMATCHING_SWID
Message	The license contains a mismatch of the SWID number. Contact Dell Technologies Support.
Commands	`set_license`
Description	The SWID number that you entered does not match your license.
Action	Contact Dell Technologies Support.

Invalid license issuer

Table 124. INVALID_LICENSE_ISSUER parameter details

Parameter	Description
Name	INVALID_LICENSE_ISSUER
Message	The issuer of the license you are attempting to add does not match that of the product.
Commands	`set_license`
Description	There is a problem with your license.
Action	Contact Dell Technologies Support

Number of parallel messages is too low

Table 125. NUM_PARALLEL_MSG_TOO_LOW parameter details

Parameter	Description
Name	NUM_PARALLEL_MSG_TOO_LOW
Message	The number of parallel messages is too low. Use a number in the range 1 - 6.
Commands	`start_sds_network_test`
Description	You used a number less than 1 in the command.
Action	Use a number in the range of 1 to 6 for parallel messages in the command.

Number of parallel messages is too high

Table 126. NUM_PARALLEL_MSG_TOO_HIGH parameters

Parameter	Description
Name	NUM_PARALLEL_MSG_TOO_HIGH
Message	The number of parallel messages is too high. Use a number in the range 1 - 6.
Commands	`start_sds_network_test`
Description	You used a number greater than 6 in the command.
Action	Use a number in the range of 1 to 6 for parallel messages in the command.

Network test size is too high

Table 127. NETWORK_TEST_SIZE_TOO_HIGH parameters

Parameter	Description
Name	NETWORK_TEST_SIZE_TOO_HIGH

Table 127. NETWORK_TEST_SIZE_TOO_HIGH parameters (continued)

Parameter	Description
Message	The network test size is too high. Use a test size less than 10.
Commands	`start_sds_network_test`
Description	You used a number greater than 10 in the command.
Action	Use a number less than 10 for the test size in the command.

Network test length too long

Table 128. Network test length too long parameters

Parameter	Description
Name	NETWORK_TEST_LENGTH_TOO_HIGH
Message	The network test length is too long. Use a test length less than 43200 seconds.
Commands	`start_sds_network_test`
Description	You used a number greater than 43200 for the test length in the command.
Action	Use a number less than 43200 for test length in the command.

Network test must limit either time or size

Table 129. NETWORK_TEST_MUST_LIMIT_EITHER_TIME_OR_SIZE parameters

Parameter	Description
Name	NETWORK_TEST_MUST_LIMIT_EITHER_TIME_OR_SIZE
Message	The network test must limit either time or size.
Commands	`start_sds_network_test`
Description	Your command did not contain a value for time or size.
Action	Include a value for either time or size in the command.

Not enough SDSs in protection domain for test

Table 130. NOT_ENOUGH_TGTS_IN_FD_FOR_TEST parameters

Parameter	Description
Name	NOT_ENOUGH_TGTS_IN_FD_FOR_TEST
Message	To start the test, there must be at least two SDSs in the protection domain.
Commands	`start_sds_network_test`

Table 130. NOT_ENOUGH_TGTS_IN_FD_FOR_TEST parameters (continued)

Parameter	Description
Description	There are fewer than two SDSs in the protection domain, and therefore the test operation cannot be performed.
Action	Ensure that there are at least two SDSs in the protection domain before attempting to test it.

Not allowed in Storage Pool with devices

Table 131. Not allowed in storage pool with devices parameters

Parameter	Description
Name	NOT_ALLOWED_IN_SP_WITH_DEVS
Message	This operation is only allowed when there are no devices in the storage pool. Remove all devices from the storage pool.
Commands	`modify_zero_padding_policy`
Description	You can only modify the zero padding policy of empty storage pools.
Action	Ensure that all devices are removed from the storage pool before you attempt to modify its zero padding policy.

License capacity mismatch

Table 132. LICENSE_CAPACITY_MISMATCH parameters

Parameter	Description
Name	LICENSE_CAPACITY_MISMATCH
Message	The license contains a mismatch of the capacity values for basic and advanced features. Contact Dell Technologies Support.
Commands	`set_license`
Description	The capacity allowed for basic and advanced features in your license do not match your installation.
Action	Contact Dell Technologies Support.

Exceeds system capacity limitations

Table 133. Exceeds system capacity limitations parameters

Parameter	Description
Name	EXCEED_SYSTEM_CAPACITY_LIMITATIONS
Message	The request exceeds system limitations
Commands	`add_volume, modify_volume_capacity`
Description	The command specifies a volume larger in size than the maximum system capacity.

Table 133. Exceeds system capacity limitations parameters (continued)

Parameter	Description
Action	Remove one or more volumes in the system, to make room for the new one.

SDC mapped to too many volumes

Table 134. INI_MAPPED_TO_TOO_MANY_VOLS parameters

Parameter	Description
Name	INI_MAPPED_TO_TOO_MANY_VOLS
Message	The SDC is mapped to too many volumes. Remove some mapping.
Commands	`map_volume_to_sdc`
Description	You attempted to map more than the maximum allowed number of volumes (8192) to one or more SDCs.
Action	Avoid mapping more than the maximum allowed volumes to a single SDC.

Not enough devices in storage pool for removal

Table 135. NOT_ENOUGH_DEVICES_IN_STORAGE_POOL_FOR_REMOVAL parameters

Parameter	Description
Name	NOT_ENOUGH_DEVICES_IN_STORAGE_POOL_FOR_REMOVAL
Message	Not enough devices in the storage pool.
Commands	`remove_sds, remove_sds_device`
Description	Each storage pool must contain at least two devices. If there are two or less devices in the storage pool and you try to remove one, the command fails.
Action	Add devices to the storage pool before trying to remove this one.

Invalid network limits

Table 136. INVALID_TGT_NETWORK_LIMITS parameters

Parameter	Description
Name	INVALID_TGT_NETWORK_LIMITS
Message	Bad limits given. The overall must be larger than both rebuild and rebalance and all limits must be over 5MB
Commands	`sds_network_limits`

Table 136. INVALID_TGT_NETWORK_LIMITS parameters (continued)

Parameter	Description
Description	The total (overall) limit must be greater than the sum of the rebuild limit and the rebalance limit. All limits must be greater than 5 MB.
Action	Ensure that limit values are greater than 5 MB, and that the overall limit value is greater than rebuild limit+rebalance limit.

A device is currently being removed

Table 137. Device currently being removed parameters

Parameter	Description
Name	ONE_TGT_DEV_BEING_REMOVED
Message	This command cannot be completed; An SDS device is currently being removed.
Commands	`activate_sds_device` with the parameter `activate_all_devices`
Description	If a device is in the process of being removed, you cannot use the `activate_all_devices` parameter to activate the remaining ones.
Action	Wait until the device removal is complete, and try again.

A device is already active

Table 138. Device already active parameters

Parameter	Description
Name	ONE_TGT_DEV_ALREADY_ACTIVE
Message	This command cannot be completed; An SDS device is currently active.
Commands	`activate_sds_device` with the parameter `activate_all_devices`
Description	A device is already active.
Action	Do not use the `activate_all_devices` option in the command. Instead, specify each device individually in the command.

RAM cache wrong state

Table 139. RMCACHE_WRONG_STATE parameters

Parameter	Description
Name	RMCACHE_WRONG_STATE
Message	Wrong RAM cache state
Commands	`enable_sds_rmcache`

Table 139. RMCACHE_WRONG_STATE parameters (continued)

Parameter	Description
	`disable_sds_rmcache` `set_sds_rmcache_size`
Description	The cache is in transient state, so the requested operation cannot be executed.
Action	Wait for a few seconds and try again.

More than one mapping not allowed

Table 140. NOT_ALLOWED_MORE_THAN_ONE_MAPPING parameters

Parameter	Description
Name	NOT_ALLOWED_MORE_THAN_ONE_MAPPING
Message	Only a single SDC may be mapped to this volume at a time
Commands	`map_volume_to_sdc`
Description	The volume is already mapped, and the `allow_multi_map` flag was not used in the command
Action	The volume is already mapped. Use the `allow_multi_map` flag to map the volume to additional SDCs.

At least one SDS failed

Table 141. AT_LEAST_ONE_TGT_FAILED parameters

Parameter	Description
Name	AT_LEAST_ONE_TGT_FAILED
Message	At least one SDS failed
Commands	`enable_sds_rmcache` `disable_sds_rmcache` `set_sds_rmcache_size` for the entire protection domain
Description	You tried to enable or disable, or change the size of cache for all SDSs in the protection domain, and this failed for at least one of the SDSs
Action	Check the state of the cache in all the SDSs, and try again for the specific SDS with the problem.

A device is in incorrect test state

Table 142. ONE_TGT_DEV_WRONG_TEST_STATE parameters

Parameter	Description
Name	ONE_TGT_DEV_WRONG_TEST_STATE
Message	This command cannot be completed; An SDS device is currently in incorrect test mode
Commands	`activate_sds_device` with `activate_all_devices` parameter

Table 142. ONE_TGT_DEV_WRONG_TEST_STATE parameters (continued)

Parameter	Description
Description	A device was not added with the `test_only` option, so it cannot be activated with this command.
Action	It will activate itself on its own. No action required.

A device test is in progress

Table 143. ONE_TGT_DEV_WRONG_TEST_MODE parameters

Parameter	Description
Name	ONE_TGT_DEV_WRONG_TEST_MODE
Message	This command cannot be completed; An SDS device test is currently in progress
Commands	`activate_sds_device` with `activate_all_devices` parameter
Description	A device test is currently in progress, and the device cannot be activated until the test is finished.
Action	Wait for the device test to finish, and then try again.

Enterprise features not enabled

Table 144. ENTERPRISE_FEATURES_NOT_ENABLED parameters

Parameter	Description
Name	ENTERPRISE_FEATURES_NOT_ENABLED
Message	Enterprise features are not enabled for this system.
Commands	Any command that requires a special license
Description	The command that you tried to use cannot be used with the type of license that you have activated in your system.
Action	Verify that you are using a license with enterprise features enabled. If problems persist, contact Dell Technologies Support.

Volume already mapped to an SDC

Table 145. VOL_ALREADY_MAPPED_TO_AN_INI parameters

Parameter	Description
Name	VOL_ALREADY_MAPPED_TO_AN_INI
Message	The volume is already mapped to a SDC
Commands	`map_volume_to_sdc` with the `all_sdcs` option
Description	If the volume is already mapped to one or more SDCs, you cannot use the `all_sdcs` option.
Action	Remove the volume's mapping to the SDCs, and then try again.

SDS already exists and is being removed

Table 146. TGT_ALREADY_EXISTS_AND_BEING_REMOVED parameters

Parameter	Description
Name	TGT_ALREADY_EXISTS_AND_BEING_REMOVED
Message	The SDS exists and is being removed. Wait for the operation to finish.
Commands	`add_sds`
Description	If you try to add an SDS that exists, and is in the process of being removed, this operation fails. **NOTE:** If an `add_sds` command failed, the system tries to remove it automatically, without user intervention. Therefore, if you issue the same `add_sds` command twice in a row, and the first time failed, this return code may be generated (depending on why adding the SDS failed).
Action	Verify that you are adding the correct SDS. If you are adding an SDS which is in the process of being removed, wait for the removal process to finish before trying to add the SDS back to the system.

Invalid spare percentage

Table 147. INVALID_SPARE_PERCENTAGE parameters

Parameter	Description
Name	INVALID_SPARE_PERCENTAGE
Message	The spare percentage value that you provided is invalid. Use a value between 0 - 99.
Commands	`modify_spare_policy`
Description	The spare percentage that is required in the command is invalid.
Action	Verify that the correct input parameters (a value between 0-99) are used with the command.

Invalid background device scanner bandwidth limit

Table 148. INVALID_SCANNER_BW_LIMIT parameters

Parameter	Description
Name	INVALID_SCANNER_BW_LIMIT
Message	The scanner bandwidth limit is invalid. Please use a value between 10KB - 1GB
Commands	`enable_background_device_scanner`
Description	The bandwidth value that you provided for the bandwidth limit parameter is invalid.

Table 148. INVALID_SCANNER_BW_LIMIT parameters (continued)

Parameter	Description
Action	Verify that correct input parameters (a value between 10KB–1GB) are used with the command

Zero padding with background device scanner data comparison mode

Table 149. ZEROPAD_AND_SCANNER parameters

Parameter	Description
Name	ZEROPAD_AND_SCANNER
Message	Zero padding must be enabled in order to set scanner to data comparison mode.
Commands	`enable_background_device_scanner`
Description	The background device scanner has two modes: device only, and data comparison mode. The storage pool must have zero padding enabled before you attempt to enable background device scanning in data comparison mode.
Action	Enable zero padding in the storage pool (you must do this before adding devices to the storage pool)

IP address already assigned to another SDS

Table 150. ADDRESS_BELONGS_TO_DIFFERENT_TGT parameters

Parameter	Description
Name	ADDRESS_BELONGS_TO_DIFFERENT_TGT
Message	The given address belongs to a different SDS. Check the system configuration.
Commands	`add_sds, add_sds_ip`
Description	This message is returned if the specified IP address has already been assigned to another SDS in the system.
Action	Check SDS network configuration and IP address assignments.

Capacity too low for thin volumes

Table 151. CAPACITY_TOO_LOW_FOR_THIN_VOL parameters

Parameter	Description
Name	CAPACITY_TOO_LOW_FOR_THIN_VOL
Message	Storage pool reached critical capacity utilization. Unable to create new thin volumes.
Commands	`add_volume, modify_volume_capacity`

Table 151. CAPACITY_TOO_LOW_FOR_THIN_VOL parameters (continued)

Parameter	Description
Description	Thin volumes cannot be allocated if the specified storage pool has reached the critical capacity level.
Action	Add more devices to the storage pool.

Too many Storage Pools in the system

Table 152. Too many storage pools in the system parameters

Parameter	Description
Name	TOO_MANY_STORAGE_POOLS
Message	There are too many storage pools in the system.
Commands	`add_storage_pool`
Description	Addition of another storage pool exceeds the system limit of allowed storage pools.
Action	Remove unused storage pools. If the problem is not solved, install a new system.

Remote access to the MDM is blocked

Table 153. REMOTE_PERMISSION_DENIED parameters

Parameter	Description
Name	REMOTE_PERMISSION_DENIED
Message	Permission denied. Remote read-only limit state is enabled and is blocking this operation from a remote location. Run this command from the primary MDM host.
Commands	All
Description	Remote read-only limit state is enabled and is blocking this operation from a remote location.
Action	Run the command from the primary MDM host.

Cannot add a partitioned device

Table 154. TGT_ADD_DEV_IS_PARTITIONED parameters

Parameter	Description
Name	TGT_ADD_DEV_IS_PARTITIONED
Message	Unable to add a device that was already partitioned
Commands	`add_sds, add_sds_device`
Description	The added device is partitioned and cannot be used by the system
Action	Remove partitions from the device, or add the partitioned device to the system

Cannot add a mounted device

Table 155. Cannot add a mounted device parameters

Parameter	Description
Name	TGT_ADD_DEV_IS_MOUNTED
Message	Unable to add a device that was already mounted
Commands	`add_sds, add_sds_device`
Description	The added device is mounted and cannot be used by the system
Action	Unmount the device

Device header mismatch

Table 156. INVALID_DEVICE_HEADER_SIGNATURE parameters

Parameter	Description
Name	INVALID_DEVICE_HEADER_SIGNATURE
Message	Found an invalid device header signature
Commands	`add_sds, add_sds_device`
Description	The device was previously added to the system, or after SDS reboot the signature of the header does not match the expected signature
Action	Check the specific SDS devices configuration

Device has user content

Table 157. USER_CONTENT_DETECTED parameters

Parameter	Description
Name	USER_CONTENT_DETECTED
Message	User content is present on device
Commands	`add_sds, add_sds_device`
Action	Clear the device from all content/partitions before adding the device

Missing storage device media type

Table 158. Missing media type parameters

Parameter	Description
Name	MISSING_MEDIA_TYPE
Message	The media type is not configured.
Commands	`add_storage_pool`, `add_sds`, `add_sds_device`, `set_storage_pool_media_type`

Table 158. Missing media type parameters (continued)

Parameter	Description
Action	Specify the media type of the storage devices when issuing the command. Storage pools must contain devices of the same media type.

Invalid media type

Table 159. INVALID_MEDIA_TYPE parameters

Parameter	Description
Name	INVALID_MEDIA_TYPE
Message	The media type is invalid
Commands	`add_storage_pool`, `add_sds`, `add_sds_device`, `set_storage_pool_media_type`, `set_sds_device_media_type`
Action	Issue the command with an appropriate media type

Media type mismatch

Table 160. MEDIA_TYPE_MISMATCH parameters

Parameter	Description
Name	MEDIA_TYPE_MISMATCH
Message	The device media type is not compatible with the storage pool media type
Commands	`add_sds`, `add_sds_device`, `set_storage_pool_media_type`, `set_sds_device_media_type`
Action	Ensure that the storage device and the storage pool are the same media type. For example, storage pool defined for HDD can only contain HDD devices.

Invalid external acceleration type

Table 161. INVALID_EXTERNAL_ACCELERATION_TYPE parameters

Parameter	Description
Name	INVALID_EXTERNAL_ACCELERATION_TYPE
Message	The external acceleration type is invalid
Commands	`add_sds_device`, `set_storage_pool_external_acceleration_type`, `set_sds_device_external_acceleration_type`
Action	Issue the command with an appropriate external acceleration type

External acceleration type mismatch

Table 162. EXTERNAL_ACCELERATION_TYPE_MISMATCH parameters

Parameter	Description
Name	EXTERNAL_ACCELERATION_TYPE_MISMATCH
Message	The device external acceleration type is not compatible with the storage pool external acceleration type.
Commands	`set_storage_pool_external_acceleration_type`
Action	Ensure that the external acceleration type assigned to the storage pool and its storage devices are set to the same type.

Metadata feature cannot be configured on the acceleration device

Table 163. TGT_DEV_METADATA_NOT_CONFIGURABLE parameters

Parameter	Description
Name	TGT_DEV_METADATA_NOT_CONFIGURABLE
Message	The SDS device metadata feature cannot be configured for the selected devices because they are still undergoing previous configuration changes
Commands	`enable_metadata_acceleration`, `disable_metadata_acceleration`, `modify_metadata_acceleration`, `enable_metadata_acceleration_per_device`, `disable_metadata_acceleration_per_device`, `modify_metadata_acceleration_per_device`
Action	Wait until the current configuration changes have been applied, and then try again.

Metadata feature is not enabled on the storage pool

Table 164. Metadata not enabled on storage pool parameters

Parameter	Description
Name	TGT_DEV_METADATA_NOT_ENABLED_ON_SP
Message	The SDS device metadata feature is not enabled on the selected storage pool.
Commands	`modify_metadata_acceleration`
Action	Enable the feature at storage pool level and then issue the command again.

Metadata feature is already enabled on the storage pool

Table 165. TGT_DEV_METADATA_ALREADY_ENABLED_ON_SP parameters

Parameter	Description
Name	TGT_DEV_METADATA_ALREADY_ENABLED_ON_SP

Table 165. TGT_DEV_METADATA_ALREADY_ENABLED_ON_SP parameters (continued)

Parameter	Description
Message	The SDS device metadata feature is already enabled on the selected storage pool. To change the feature settings, use the modify command.
Commands	`enable_metadata_acceration`
Action	None required

Metadata feature is not enabled on the devices

Table 166. TGT_DEV_METADATA_NOT_ENABLED_ON_DEVICES parameters

Parameter	Description
Name	TGT_DEV_METADATA_NOT_ENABLED_ON_DEVICES
Message	The device metadata feature is not enabled on some or all of the selected devices
Commands	`modify_metadata_acceleration_per_device`
Action	Enable the metadata feature on all of the selected devices

Metadata is already enabled on devices

Table 167. TGT_DEV_METADATA_ALREADY_ENABLED_ON_DEVICES parameters

Parameter	Description
Name	TGT_DEV_METADATA_ALREADY_ENABLED_ON_DEVICES
Message	The SDS device metadata feature is already enabled on some or all of the selected devices. To change the feature settings for those devices, use the modify command.
Commands	`modify_metadata_acceleration_per_device`
Action	None required

Insufficient capacity for accelerating metadata

Table 168. INSUFFICIENT_ACC_METADATA_CAPACITY parameters

Parameter	Description
Name	INSUFFICIENT_ACC_METADATA_CAPACITY
Message	The required acceleration capacity for metadata exceeds the amount of available capacity on the acceleration device(s)
Commands	`modify_metadata_acceleration_per_device` , `add_sds`, `add_sds_device`, `enable_metadata_acceleration_per_device`
Action	Free capacity in the acceleration pool, or add one or more acceleration devices to the pool

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NVMe host and SDC commands

This section contains commands for managing NVMe host and SDC.

Understanding the effect of network latency on NVMe over TCP latency counters

This section describes the possible effects of network latency between the host and storage data target (SDT) in an NVMe over TCP environment.

To access the performance counters, run the following commands:

- For storage data target

```
scli --query_properties --object_type SDT --all_objects --properties
SDT_DATA_READ_LATENCY,SDT_DATA_WRITE_LATENCY,SDT_HOST_READ_LATENCY,SDT_HOST_WRITE_LATE
NCY
```

  Output sample:

```
754135e800000000:
          SDT_DATA_READ_LATENCY             3595 IOPS 0.823 ms average latency
          SDT_DATA_WRITE_LATENCY          0 IOPS 0 us average latency
          SDT_HOST_READ_LATENCY             899 IOPS 82 microseconds average latency
          SDT_HOST_WRITE_LATENCY          899 IOPS 12 microseconds average latency
```

- For host:

```
scli --query_properties --object_type HOST --all_objects --properties
USER_DATA_SDC_READ_LATENCY,USER_DATA_SDC_WRITE_LATENCY
```

  Output sample:

```
f3e742f300010000:
          USER_DATA_SDC_READ_LATENCY        930 IOPS 69 microseconds average latency
          USER_DATA_SDC_WRITE_LATENCY       930 IOPS 12 microseconds average latency
```

Table 169. I/O latency measurement by counters

PowerFlex object	Performance counter	Measures I/O latency		
		Type	From	To
storage data target	SDT_DATA_**READ**_LATENCY	Read	SDS	storage data target
	SDT_DATA_**WRITE**_LATENCY	Write	storage data target	SDS
	SDT_HOST_**READ**_LATENCY	Read	storage data target	Host
	SDT_HOST_**WRITE**_LATENCY	Write	Host	storage data target
HOST	USER_DATA_SDC_**READ**_LATENCY	Read	SDS	storage data target
	USER_DATA_SDC_**WRITE**_LATENCY	Write	storage data target	SDS

NOTE: The term storage data client (SDC) used in the counter name is for backward compatibility purposes only.

Limitations of measuring latency between the host and SDT

In an NVMe over TCP setup, PowerFlex does not have a component that is installed on the host. As a result of the above fact there is no way of knowing when the host initiates the I/O request or when it receives requested data or a response about data it sent. Therefore, it is impossible to measure latency between the host and storage data target. The effect this has on the host-related counters are described below. Counters for traffic between storage data server (SDS) and storage data target (SDT) are reliable and reflect any network latency between those nodes.

Exception

Latency can be measured in one scenario. The following table describes how network latency affects the counters that measure latency between the host and the storage data target.

Table 170. Network latency effect on counters measuring latency between host and storage data target

I/O type	I/O size	I/O direction	Result
READ	All	Host to and from the storage data target	Values are estimates based on the I/O size. Network latency does not affect the values, and the values cannot provide any information about such issues.
WRITE	Less than or equal to 4 KB		
	More than 4 KB	Storage data target to the host	PowerFlex can accurately measure the network latency between the host and the storage data target. The network latency affects the counter value **twice**. That is, if network latency is increased by 100 microseconds then the counter value increases by 200 microseconds.
		Host to the storage data target	

Significance of large write requests

In all the host to storage data target flow cases, the host sends a read or write request to the storage data target.

The flow is complicated while writing large IOs in the host to the storage data target:

1. Host sends a write request to the storage data target.
2. Storage data target sends the host a request to send the write data.
3. Host sends the write data to the storage data target.

Since the system knows when it initiated step 2 and it knows when step 3 is completed it can measure the time of these steps. Any network latency affects both steps 2 and 3, and this is the reason that the latency affects twice as described in the table above.

Summary

To check for possible effects of network latency between the host and the storage data target in an NVMe over TCP setup, complete the following:

1. Run a write I/O load with I/O size larger than 4 KB and examine the **SDT_HOST_WRITE_LATENCY** counter.
2. If a baseline value is available to compare with, divide the delta of the two counter values by two for an estimate of the network latency.

add_sdc

Approve an SDC to map volumes and allow an SDC to access the MDM.

Syntax

```
scli --add_sdc --sdc_ip <ip> | --sdc_guid <guid>
  [--sdc_name <name>]
```

Parameters

--sdc_ip *<ip>*
SDC IP address

--sdc_guid *<guid>*
SDC GUID

--sdc_name *<name>*
Name to assign to the SDC

NOTE: You can find the GUID of the SDC using any of these methods:

- From the command line, run:

  ```
  scli --query_all_sdc
  ```

- View the properties of the SDC in PowerFlex Manager.
- Locally in `/opt/emc/scaleio/sdc/bin/drv_cfg`, with root access permission, run:

  ```
  drv_cfg --query_guid
  ```

Example

```
scli --add_sdc --sdc_ip 10.125.120.53 --sdc_name sdc53
```

remove_host

Remove a host from the system.

Syntax

```
scli --remove_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --remove_host --host_name host1
```

rename_host

Name or rename an existing host.

Syntax

```
scli --rename_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>) --new_name <NAME>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

---sdc_ip *<IP>*

SDC IP address

- --new_name *<NAME>*

New name assigned to the host.

Example

```
scli --rename_host --host_name host1 --new_name new_host_name
```

check_sdc_authentication_status

Print the SDC authentication password status for all SDCs.

Syntax

```
scli --check_sdc_authentication_status [--run_test] [--filename <FILENAME>]
```

Parameters

--run_test

Run a connectivity test to check whether the SDCs can successfully authenticate using CHAP.

--filename *<FILENAME>*

The filename and filepath where detailed SDC authentication information will be stored. This parameter is mandatory when using the `--run_test` option.

Example

```
scli --mdm_ip 192.168.1.200 --check_sdc_authentication_status --run_test --filename
report.txt
```

generate_sdc_password

Generate a new SDC authentication password for the specified SDC.

Syntax

```
scli --generate_sdc_password (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> |
--sdc_ip <IP>) [--reason <REASON>]
```

Parameters

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC GUID

--sdc_ip *<IP>*

SDC IP address

--reason *<REASON>*

Free text to describe the reason for the SDC password reset

Example

```
scli --mdm_ip 192.168.1.200 --generate_sdc_password --sdc_name SDC1 --reason
MonthlyUpdate
```

query_all_approved_sdc

Query the system when it is operating in restricted SDC mode for all SDCs that have been approved by the system.

NOTE: This query is useful when restricted SDC mode is enabled and you want to determine which SDCs can be used for volume mapping.

Syntax

```
scli  --query_all_approved_sdc
```

Parameters

None.

Example

```
scli --query_all_approved_sdc
```

query_host

Retrieve information about a host.

Syntax

```
scli --query_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe Host NQN

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

query_all_hosts

Retrieve information about all SDCs and NVMe hosts.

Syntax

```
scli --query_all_hosts
```

query_potential_nvme_hosts

Retrieve information about all potential NVMe hosts. The query result can be optionally filtered by protection domain.

Syntax

```
scli --query_potential_nvme_hosts (--all | --protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--all

Retrieve information about all potential hosts.

--protection_domain_id *<ID>*

Protection domain ID

-protection_domain_name *<NAME>*

Protection domain name

query_all_sdc

Retrieve information about all SDCs in the system.

Syntax

```
scli --query_all_sdc [--agent_status] [--mdm_ip_addrs_current] [--mdm_ip_addrs_hardened]
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sdc
```

Sample output

```
MDM restricted SDC mode: NONE
Query all SDC returned 5 SDC nodes.
SDC ID: 30b5c18600000000 Name: N/A IP: 10.55.137.119 State: Connected GUID:
96CCE42A-6ADB-45EA-8E3F-B71C1F67808C  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
```

```
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18700000001 Name: N/A IP: 10.55.137.114 State: Connected GUID:
C08C87B5-7E28-421A-A82C-C9C5B3070FF2  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18800000002 Name: N/A IP: 10.55.137.115 State: Connected GUID:
89947FF3-6E84-4DF9-B869-481BCB8E3B3D  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18900000003 Name: N/A IP: 10.55.137.116 State: Connected GUID: 7F5BC40E-
A1FB-44EC-A8B0-76C63A7B1123  OS Type: LINUX Loaded Version: 3.6.0 Installed Version:
3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18a00000004 Name: N/A IP: 10.55.137.117 State: Connected GUID:
34EE9476-65D3-44C1-83DC-3CF6912CCB86  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
```

query_restricted_sdc_mode

Query whether restricted SDC mode is enabled or disabled in the PowerFlex system.

If restricted SDC mode is enabled, you must register to the system each SDC to which you want to map volumes using the `--add_sdc` command.

Syntax

```
scli --query_restricted_sdc_mode
```

Parameters

None.

Example

```
scli --query_restricted_sdc_mode
```

query_sdc

Retrieve information about the specified SDC.

Syntax

```
scli --query_sdc --sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>
  [--show_oscillating_failures]
```

Parameters

`--sdc_id <ID>`

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

--show_oscillating_failures

Show extended oscillating failure information if there are one or more failures.

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc --sdc_id c27ab52600000001
```

Sample output

```
SDC ID: c27ab52600000001 Name: N/A IP: 10.55.137.119 State: Connected GUID:
C6533636-224B-44E3-A9EB-3CD651116CF5  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  525 IOPS 262.4 MB (268697 KB) per-second
    Writes: 503 IOPS 251.2 MB (257222 KB) per-second

Volume information (total 2 volumes):
         1: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB) Access mode: READ_WRITE
         2: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB) Access mode: READ_WRITE
```

query_sdc_volume_limits

Retrieve the IOP and bandwidth limits that one SDC generates for the specified volume.

Syntax

```
scli --query_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc_volume_limits --volume_id f6351e3700000003 --
sdc_id c27ab52600000001 --mdm_port 6611
```

Sample output

```
Successfully queried limits from SDC with ID c27ab52600000001 to volume with ID
f6351e3700000003: IOPS limit is unlimited. Bandwidth limit is unlimited.
```

reset_sdc_sds_connections

Reset SDC-SDS connections.

Syntax

```
scli --reset_sdc_sds_connections --i_am_sure
```

Parameters

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --reset_sdc_sds_connections --i_am_sure
```

set_restricted_sdc_mode

Enable or disable restricted SDC mode. When enabled, SDCs must be approved (by GUID or IP address) before volumes can be mapped to them.

NOTE: In a system that has been upgraded and already has volumes mapped to SDCs, if you want to enable restricted SDC mode, you must first approve the SDCs and only then enable restricted SDC mode.

Syntax

```
scli --set_restricted_sdc_mode --restricted_sdc_mode {approved_id | guid | none}
```

Parameters

--restricted_sdc_mode {approved_ip | guid | none

Set the restricted SDC mode to one of the following:

- `approved_ip` — Volumes can be mapped only to approved SDCs that are also approved by IP address.
- `guid` — Volumes can be mapped only to approved SDCs.
- `none` — Restricted SDC mode is disabled and volumes can be mapped to any SDC.

Example

```
scli --set_restricted_sdc_mode --restricted_sdc_mode guid
```

set_sdc_authentication

Enable or disable SDC authentication. This feature ensures security by enforcing password-based authentication of the SDC to the MDM for access to mapped volumes. This prevents the SDC from accessing unauthorized volumes.

Syntax

```
scli --set_sdc_authentication (--enable | --disable) [--allow_old_sdc] [--
allow_no_sdc_password]
```

Parameters

--enable

Enable SDC authentication and authorization

--disable

Disable SDC authentication and authorization

--allow_old_sdc

Allow enabling SDC authentication and authorization when there is an old SDC. All old SDCs will be disconnected

--allow_no_sdc_password

Allow enabling SDC authentication and authorization when there is an SDC without a password. All SDCs without a password will be disconnected.

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_authentication --enable --allow_old_sdc
```

set_sdc_volume_limits

Set limits to the IOPS and bandwidth that one SDC generates for the specified volume. This enables you to control the quality of service (QoS).

Syntax

```
scli --set_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID> |
--sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>) (Options)
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

Options

Choose at least one:

--limit_iops *<NUMBER>*

Limit the volume IOPS. The number of IOPS must be larger than 10. 0 is unlimited.

--limit_bandwidth *<NUMBER>*

Limits the volume network bandwidth. The bandwidth is in MB/s. 0 is unlimited

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_volume_limit
--volume_name vol1 --sdc_ip 192.168.1.3 --limit_iops 100
```

set_approved_sdc_ips

Set a list of up to four IP addresses that are approved for the given SDC.

Syntax

```
scli --set_approved_sdc_ips (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --
sdc_ip <IP>) --sdc_ips <IPs>
```

Parameters

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC GUID

--sdc_ip *<IP>*
IP address of an SDC

--sdc_ips *<IPS>*
A comma-separated list of SDC IP addresses. Omit the space after each comma.

Example

```
scli --set_approved_sdc_ips --sdc_ips 10.76.60.10,10.76.60.11,10.76.60.12,10.76.60.13
```

add_nvme_host

Add an NVMe host to the system.

Syntax

```
scli --add_nvme_host --nvme_host_nqn <NQN> [--host_name <NAME>] [--max_number_paths
<VALUE>] [--max_number_sys_ports <VALUE>] [--host_os_full_type <OS>] [--host_group_name
<ID> | --host_group_id <NAME>] | --force
```

Parameters

--nvme_host_nqn *<NQN>*
NVMe host NQN

--host_name *<NAME>*
Host name

--max_number_paths *<VALUE>*
Maximal number of paths allowed per mapped volume

Valid range: 2 to 8

Dafault: 4

-max_number_sys_ports *<VALUE>*-

Maximal number of system ports allowed per protection domain

Valid range: 2 to 16

Dafault: 10

--host_os_full_type *<OS>*

OS type. Used for OS with known NVMe host issues

Options:

- generic (default)
- powerflex

--host_group_id *<ID>*

Host group ID.

--host_group_name *<NAME>*

Host group name.

--force

Force NQN, ignore all NQN formating rules.

Syntax

```
scli --add_nvme_host --host_name host1 --nvme_host_nqn
nqn.2014-08.org.nvmexpress:uuid:11111111-1111-2222-3333-444455556666
```

modify_nvme_host_max_paths

Modify the maximum number of paths per mapped volume for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_max_paths (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn
<NQN>) --max_number_paths <VALUE>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--volume_id *<ID>*

Volume ID

--max_number_paths *<VALUE>*

Maximal number of paths allowed per mapped volume

Valid range: 2-8

Default: 4

Example

```
scli --modify_nvme_host_max_paths --host_name host1 --max_number_paths 6
```

modify_nvme_host_max_sys_ports

Modify the maximum number of system ports per protection domain for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_max_sys_ports (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) --max_number_sys_ports <VALUE>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--max_number_sys_ports *<VALUE>*

Maximal number of system ports allowed per protection domain

Valid range: 2-128

Default: 10

Example

```
scli --modify_nvme_host_max_sys_ports --host_name host1 --max_number_sys_ports 6
```

modify_nvme_host_os_full_type

Modify the operating system (OS) type and version for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_os_full_type (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) --host_os_full_type <OS>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe Host NQN

--host_os_full_type *<OS>*

OS type. Used for OS with known NVMe host issues

Options:

- generic (default)
- powerflex

set_manual_nvme_host_connectivity

Set a manual connectivity goal for NVMe host to a protection domain.

Syntax

```
scli --set_manual_nvme_host_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--protection_domain_id <ID> | --protection_domain_name <NAME>) (--
system_port_ip <IP> | --system_port_id <ID>) --force
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--system_port_ip *<IP>*

A comma separated list of system port IP addresses.

--system_port_id *<ID>*

A comma separated list of system port IDs (4 hexadecimal digits).

--force

Force non-standard connectivity settings

reset_manual_nvme_host_connectivity

Reset manual connectivity goal for NVMe host to a protection domain.

Syntax

```
scli --reset_manual_nvme_host_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--protection_domain_id <ID> | --protection_domain_name <NAME>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

set_manual_nvme_mapping_connectivity

Set manual connectivity for mapping of a volume to an NVMe host.

Syntax

```
scli --set_manual_nvme_mapping_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--volume_id <ID> | --volume_name <NAME>) [--protection_domain_id
<ID> | --protection_domain_name <NAME>] (--system_port_ip <IP> | --system_port_id <ID>)
--force
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--system_port_ip *<IP>*
A comma separated list of system port IP addresses.

--system_port_id *<ID>*
A comma separated list of system port IDs (4 hexadecimal digits)

--force
Force non standard connectivity settings

reset_manual_nvme_mapping_connectivity

Reset manual connectivity of volume mapping to an NVMe host.

Syntax

```
scli --reset_manual_nvme_mapping_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--volume_id <ID> | --volume_name <NAME>) [--protection_domain_id
<ID> | --protection_domain_name <NAME>]
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

rebalance_nvme_host_connectivity

Set connectivity goal and system ports balancing on all system NVMe Hosts.

Syntax

```
scli --rebalance_nvme_host_connectivity
```

query_nvme_discovery_info

Get the information required to set up a discovery service on hosts.

Syntax

```
scli --query_nvme_discovery_info (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --service <TYPE> --transport <TYPE>
```

Parameters

--protection_domain_id *<ID>*

The failure domain ID

--protection_domain_name *<NAME>*

The failure domain name

--service *<TYPE>*

The service type

Option: discovery

--transport *<TYPE>*

The transport type

Option: TCP

--remote_system_description *<DESCRIPTION>*

Description of the remote system

--transport *<TRANSPORT_TYPE>*

Currently only TCP is supported

clear_nvme_reservation

Clear NVMe reservation and registration from the given volume. If there is a host that is a registrant and has an active controller, the command fails unless the `--force` flag is used.

Syntax

```
scli --clear_nvme_reservation (--volume_id <ID> | --volume_name <NAME>) [--wait] [--
force] [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--wait

Wait for the command to complete. If specified, the return code is the actual return code of the command

--force

Force clearing NVMe reservation

--i_am_sure

Preemptive approval

add_host_group

Add a host group to the system.

Syntax

```
scli --add_host_group [--host_group_name <NAME>] [--network_id <ID> | --network_name
<NAME>] [--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN>] [--force]
```

Parameters

--host_group_name *<NAME>*

A name to be assigned to the host group

--network_id *<ID>*

A comma separated list of system network IDs to be associated with the group

--network_name *<NAME>*

A comma separated list of system network names to be associated with the group

--host_id *<ID>*

A comma separated list of NVMe host IDs to be added to the group

--host_name *<NAME>*

A comma separated list of NVMe host names to be added to the group

--nvme_host_nqn *<NQN>*

A comma separated list of NVMe host NQNs to be added to the group

--force

Force assigning NVMe hosts to group regardless of system networks accessibility

assign_host_to_host_group

Assign a host to a host group.

Syntax

```
scli --assign_host_to_host_group (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn
<NQN>) (--host_group_id <ID> | --host_group_name <NAME>) [--force]
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--host_group_id *<ID>*

Host group ID

--host_group_name *<NAME>*

Host group name

--force *<NAME>*

Force assigning NVMe host to host group regardless of system networks accessibility

query_all_host_groups

Retrieve information about all host groups.

query_host_group

Retrieve information about a host group.

Syntax

```
scli --query_host_group (--host_group_id <ID> | --host_group_name <NAME>)
```

Parameters

--host_group_id *<ID>*
Host group ID

--host_group_name *<NAME>*
Host group name

modify_host_group

Modify the list of system networks of an existing host group.

Syntax

```
scli --modify_host_group (--host_group_id <ID> | --host_group_name <NAME>) [--network_id
<ID> | --network_name <NAME>]
```

Parameters

--host_group_id *<ID>*
Host group ID

--host_group_name *<NAME>*
Host group name

--network_id *<ID>*
A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*
A comma separated list of system network names to be replaced with the old one

rename_host_group

Name or rename a host group.

Syntax

```
scli --rename_host_group (--host_group_id <ID> | --host_group_name <NAME>) --new_name
<NAME>
```

Parameters

--host_group_id *<ID>*
Host group ID

--host_group_name *<NAME>*
Host group name

--new_name *<NAME>*
A new name to be assigned to the host group

remove_host_from_host_group

Remove a host from a host group.

Syntax

```
scli --remove_host_from_host_group (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

remove_host_group

Remove a host group from the system.

Syntax

```
scli --remove_host_group (--host_group_id <ID> | --host_group_name <NAME>)
```

Parameters

--host_group_id *<ID>*
Host group ID

--host_group_name *<NAME>*
Host group name

set_auto_update_sdc_with_mdm_ip_addresses

Enable or disable SDC automatic update with MDM IP addresses.

Syntax

```
scli --set_auto_update_sdc_with_mdm_ip_addresses (--enable | --disable)
```

Parameters

--enable
Enable SDC automatic update with MDM IP addresses.

--disable
Disable SDC automatic update with MDM IP addresses.

restart_connections_to_realign_net_path

Set a restart connection to realign the network path interval.

Syntax

```
scli --restart_connections_to_realign_net_path [--disable | --enable | --force | --
interval_hour <hours> | --enable --interval_hour <hours>]
```

Parameters

--enable
Restart connections to realign network path. The default interval is 24 hours.

--disable
Disable periodic connections realignment.

--interval_hour *<INTERVAL_HOUR>*
Set the restart connection interval, 1-168 hours.

--force
Force on time refresh, even if periodic connections realignment has been disabled.

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SDC operations

PowerFlex uses different methods to add external SDCs to an existing PowerFlex system, depending on the operation system of the server. For existing SDCs, many operations use the drv_cfg command line, which is a local utility that affects only the client on which the SDC is running.

Add an external SDC to an existing PowerFlex system

Add an external SDC to an existing PowerFlex system.

During manual installation of an SDC, you can install the SDC according to the operating system-specific instructions in the following section, and it will be connected to the existing PowerFlex system.

Install the SDC on an ESXi-based server and connect it to PowerFlex using esxcli

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Customer Support site)
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID needs to conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

The following procedure explains how to manually install an external SDC on an ESXi-based using esxcli in command line.

NOTE: This procedure requires two server reboots.

Steps

1. On the ESXi-based server on which you are installing the SDC, set the acceptance level:

   ```
   esxcli software acceptance set --level PartnerSupported
   ```

2. Install the SDC:

Option	Description
ESXi 6.x	`esxcli software vib install -d <full_path_to_offline_bundle>`

Option	Description
	where *<full_path_to_offline_bundle>* is the absolute path to the SDC VIB file. **NOTE:** The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.
ESXi 7	`esxcli software component apply -d <path_to_zip> -n scaleio-sdc-esx7` where *<path_to_zip>* is the path to the SDC zip file.

3. Get permission to reboot the ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.
   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.
4. Set the IP address of the MDM:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the "+" symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

       `10.20.30.40,50.60.70.80+11.22.33.44`
   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.
5. Back up the parameters:

   ```
   /bin/auto-backup.sh
   ```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

   ```
   vmkload_mod -l | grep scini
   ```

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

   The SDC component is installed on the ESXi-based server.

Results

The SDC is installed on the ESXi-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Linux-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).
- Secure Boot is not supported; ensure that Secure Boot is disabled in the BIOS.

About this task

The following procedure explains how to manually install an external SDC on a Linux-based server. External SDCs can be deployed by the PowerFlex Installer.

The steps in the procedure below are relevant for Linux RHEL/CentOS/Oracle Linux operating systems. Deployment on other operating systems requires the modifications described in the following table:

Table 171. Operating system modifications for SDC deployment

Operating system	Modifications required
Ubuntu/OL	Before installing PowerFlex, ensure that you have followed the required preparation procedures relating to various types of Linux operating systems. ● PowerFlex component packages are delivered as TAR files. Before installing, perform the following: 1. Untar all the packages: `tar -xvf <tar_file>` This yields SIOB files. 2. Extract the DEB from the SIOB files: `siob_extract <siob_file>` 3. The DEB file is in the following format: `EMC-ScaleIO-sdc-.<operating_system>.X.X.x86_64.deb` Use the extracted DEB files for the installation. ● Some commands are a bit different, noted where applicable.

Steps

1. Install the GPG key on every server on which SDC must be installed. From the PowerFlex installation folder, run the following command on every server:

```
rpm --import RPM-GPG-KEY-ScaleIO
```

2. In the command line, install the SDC:
   - RHEL/CentOS /Oracle Linux/SLES

```
MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
```

   - Ubuntu (These files must be extracted before use, as described above.)

```
MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.
3. To verify that the driver is loaded, the following command can be used:

- SLES 15:

```
service sdc status
```

- All other Linux flavors:

```
service scini status
```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that were added to the system.

Install SDC on an AIX-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. These instructions are relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).

About this task

The following procedure explains how to manually install an external SDC on an AIX-based server. The PowerFlex Installer cannot be used.

Steps

1. In the command line, install the SDC:

```
MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
```

   Where

   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.

     The SDC package is in a format similar to this: `EMC-ScaleIO-sdc-.aix7.aix7.2.ppc.rpm`

2. Optional: For SDC authentication to the MDM for access, in the command line, enter:

```
chdev -l scini -a "mdm_password=<ip>-<password>"
```

Results

The SDC is installed on the AIX-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the recommended post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Windows-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).

About this task

The following procedure explains how to manually install an external SDC on a Windows-based server. Alternatively, you can install the external SDC using the PowerFlex Installer. For more information about adding system components to an existing system, see the *Dell PowerFlex 4.5.x Install and Upgrade Guide*.

NOTE: This procedure requires a server reboot.

Steps

1. On the Windows-based server on which you are installing the SDC, run the following command in the command line:

   ```
   msiexec /i <SDC_PATH>.msi MDM_IP=<LIST_VIP_MDM_IPS>
   ```

   Where
   - *<SDC_PATH>* is the path where the SDC installation package is located.
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not add tiebreaker addresses here.
2. Get permission to reboot the Windows-based server, and perform a reboot to load the SDC driver on the server.

Results

The SDC is installed on the Windows-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Performing SDC operations

Many SDC operations use `drv_cfg`. The `drv_cfg` command line is a local CLI utility that affects only the client on which the SDC is running.

Possible SDC operations include updating the SDC driver with IP changes, detecting new volumes, querying volumes, loading a configuration file, adding an MDM, modifying an MDM IP address, and enabling support of PDL state.

Update the SDC driver with IP address changes

Use the drv_cfg to update the IP address.

Steps

1. Edit `drv_cfg.txt` and change the IP address in the last line to the new IP address.

Location of `drv_cfg.txt`:

- Linux: `/etc/emc/scaleio/`
- Windows: In the following registry key - `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\services\scini\Parameters\mdms`

NOTE:

On ESXi, GUID and MDM lists are stored as module parameters, and not in a `drv_cfg.txt` file. To modify these parameters, use `esxcli` commands.

2. Save and close the file.
3. Type the following command:

```
/etc/init.d/scini restart
```

NOTE: ESXi guidelines are described in the PowerFlex plug-in chapter.

Detecting new volumes

Volumes are always exposed to the operating system as devices with the prefix `scini` (such as `/dev/scinia`, `/dev/scinib` and so on). Unique names can be found under `/dev/disk/by-id/`.

PowerFlex periodically scans the system to detect new volumes. You can initiate a scan for the most up-to-date status on a particular SDC node. This command is unique because it is not a CLI command, but rather a command issued on the specific SDC.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: See the vSphere Client VMware guidelines on how to detect new storage. If troubleshooting is needed, contact Customer Support.

Command

`drv_cfg --rescan`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --rescan
```

Query volumes using drv_cfg

This utility retrieves information about all known active volume objects in kernel mode. You can use this utility to determine which volumes are mapped, and the ID of each volume in PowerFlex.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: Refer to "Update the SDC parameters in VMware based HCI or Compute node".

Command

drv_cfg --query_vols

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_vols
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_vols
```

Query tgt objects using drv_cfg

This utility retrieves information about all known active tgt objects (SDSs) in kernel mode.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: Refer to "Update the SDC parameters in VMware based HCI or Compute node".

Command

drv_cfg --query_tgts

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_tgts
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_tgts
```

Query GUID using drv_cfg

This utility retrieves the unique ID of the kernel module. The utility can be used to verify that all SDC GUIDs in the system are unique.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: Refer to "Update the SDC parameters in VMware based HCI or Compute node".

NOTE:

If the SDC was removed and reinstalled, the GUID of the SDC is different than its original GUID. In such a case, you may have to remove the SDC, if two SDCs now have the same GUID.

Command

drv_cfg --query_guid

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_guid
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_guid
```

Query MDMs using drv_cfg

This utility retrieves information about all known MDM objects in kernel mode. This utility is typically used to determine to which MDM an SDC is connected.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: Refer to "Update the SDC parameters in VMware based HCI or Compute node".

Command

drv_cfg --query_mdms

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_mdms
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_mdms
```

Loading a configuration file using drv_cfg

This utility reads a configuration file containing MDM IP addresses, and calls the kernel to connect to them.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`

- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: Refer to "Update the SDC parameters in VMware based HCI or Compute node".

The configuration file that is loaded when using the `drv_cfg --load_cfg_file` utility is not persistent; when you restart the SDC, the changes will be lost.

To make the changes persistent, perform either of the following:

- Install the SDC on every server that exposes volumes to the application running, by running the following command:

```
MDM_IP=<IP of the MDM> rpm -i <full rpm file path>
```

- Use the following `drv_cfg` command:

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip <EXISTING_MDM_IP_ADDRESS> --new_mdm_ip
<NEW_MDM_IP_ADDRESSES>
```

Command

drv_cfg --load_cfg_file

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

This command can not be used on ESXi servers. Instead, follow the steps described in "Modify parameters on ESXi servers".

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg
--load_cfg_file <FILE_NAME>
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg
--load_cfg_file /etc/emc/scaleio/drv_cfg.txt
```

Adding an MDM using drv_cfg

This utility calls the kernel module to connect to an MDM. This command is typically used in cases where an SDC is connected to more than one PowerFlex system.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: Refer to "Update the SDC parameters in VMware based HCI or Compute node".

NOTE:

Extending your PowerFlex system with another MDM requires that you update all SDCs in your system with the new MDM IP address. Run the `drv_cfg` utility with the `--add_mdm` option, and to make the change persistent, use the `--file` parameter. In addition, any additional objects or systems which interface with the MDM must also be updated. For more information, see Modifying an MDM IP address using drv_cfg.

Command

drv_cfg --add_mdm

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

This command cannot be used on ESXi servers.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
--ip <MDM_IP_ADDRESS_LIST>
```

Parameters

Table 172. drv_cfg utility parameters for MDM configuration

Parameter	Description
--ip <MDM_IP_ADDRESS_LIST>	List of IP addresses (comma delimited) for this primary or secondary MDM
Optional:	-
--file <CONFIG_FILE_NAME>	Name of the configuration file to which the MDM information should be written
--only_cfg	Do not call the kernel to actually connect

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
--ip 10.100.22.20,10.100.22.30
--file /etc/emc/scaleio/drv_cfg.txt
```

Modifying an MDM IP address using drv_cfg

This utility calls the kernel to modify the IP address list of an MDM. It is typically used in cases when an MDM IP address has changed, or when MDMs are added/removed from/to the system. The command must be run on every SDC in the system. To bring the changes into effect, a server restart is required.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: Refer to "Update the SDC parameters in VMware based HCI or Compute node".

NOTE:

Extending your PowerFlex system with another MDM requires that you update all SDCs in your system with the new MDM IP address. Run the `drv_cfg` utility with the `--add_mdm` option, and to make the change persistent, use the `--file` parameter. In addition, any additional objects or systems which interface with the MDM must also be updated.

Command

`drv_cfg --mod_mdm_ip`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server. This command can not be used on ESX servers. Instead, follow the steps described in "Modify parameters in ESXi servers".

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip <EXISTING_MDM_IP_ADDRESS>
--new_mdm_ip <NEW_MDM_IP_ADDRESSES> [--file <CONFIG_FILE_NAME>]["-""-"only_cfg]
```

Parameters

Table 173. drv_cfg utility parameters for modifying MDM IP addresses

Parameter	Description
--ip <EXISTING_MDM_IP_ADDRESS>	One of the existing MDM IP addresses
--new_mdm_ip <NEW_MDM_IP_ADDRESSES>	The new IP address list (comma delimited) for this MDM. If you want to retain the existing address(es), include them in this list.
Optional:	-
--file <CONFIG_FILE_NAME>	The name of the configuration file to which the MDM information should be written
--only_cfg	Do not call the kernel to actually connect

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip 10.100.20.20
--new_mdm_ip 10.100.20.20,10.100.20.30,10.100.20.40
```

Permanent device loss state

When the MDM has disconnected from the SDC and a volume that is mapped to this SDC has experienced an I/O error, the ESXi host continuously sends I/Os to the device to determine if the device has become accessible. This can subsequently cause a high I/O-error load that can lead to the host freezing. In cases where the device is disconnected long-term, such as when the entire MDM cluster is down during a network upgrade, the SDC can change the volume state to permanent device loss (PDL) to prevent more I/O errors coming from the ESXi.

PDL is an ESXi state, which, once enabled, is supported in PowerFlex. Once the ESXi host loses connectivity with a device, if a timeout value is reached, the ESXi is notified that the device is in a PDL state. The timeout value can be manually set. Once a device is in a PDL state, the ESXi host no longer attempts to re-establish connectivity or issue commands to the device.

Recovering a device from PDL state is described in the VMware documentation for your operating system version.

Use the instructions that match your environment.

Enable support of PDL state on the ESXi

Enable support of permanent device loss (PDL) state on the ESXi host.

Steps

1. On the ESXi host:

   ```
   esxcli system module parameters set -m scini "<<PREVIOUS_MODULE_PARAMS>>
   bBlkDevIsPdlActive=1 blkDevPdlTimeoutMillis=<TIMEOUT_VALUE>"
   ```

   where:

   - *<<PREVIOUS_MODULE_PARAMS>>* is any previous module parameters being used for this ESXi host.

- *<TIMEOUT_VALUE>* is the timeout time in milliseconds. Its value can be 1000-3600000 (default is 60000) and including it in the command is optional.

2. Reboot the host.

Disable support of PDL state on the ESXi

Disable support of permanent device loss (PDL) state on the ESXi host.

Steps

1. On the ESXi host:

```
esxcli system module parameters set -m scini "<<PREVIOUS_MODULE_PARAMS>>
bBlkDevIsPdlActive=0"
```

   Where *<<PREVIOUS_MODULE_PARAMS>>* is any previous module parameters being used for this ESXi host.

2. Reboot the host.

Run the SDC driver on RHEL\SLES with multipath enabled

Run the SDC driver on RHEL\SLES with multipath enabled

Steps

1. On the RHEL\SLES machine where the SDC is running, open `/etc/udev/rules.d/20-scini.rules` for editing
2. In the first line at the end add:

```
ENV{ID_WWN}="%c"
```

   The line should read like this:

```
KERNEL=="scini*[!0-9]", SUBSYSTEM=="block", PROGRAM="/bin/emc/scaleio/drv_cfg --
query_block_device_id --block_device $tempnode", SYMLINK+="disk/by-id/emc-vol-%c",
ENV{ID_BUS}="scsi", ENV{ID_SERIAL}="%c", ENV{ID_WWN}="%c"
```

3. Open `/etc/multipath.conf` and add the following additional settings to the configuration file:

```
defaults {
          retain_attached_hw_handler "no"
                find_multipaths no
}
```

```
blacklist_exceptions {
                    devnode "scini"
}
```

   NOTE: From the command line, run `dracut -f.` This includes the changes in initrd.
   `[retain_attached_hw_handler no]` is the default setting in RHEL 7.X\8.X.

4. Restart the PowerFlex service and multipathd service:

```
systemctl restart scini
```

```
systemctl restart multipathd.service
```

   NOTE: If LUNs are in use and a restart is not possible, restart the node.

SDC authentication

This feature ensures secured connectivity between the SDC and the storage cluster over the data path by applying CHAP (Challenge-Handshake Authentication Protocol) based authentication of the SDC to the MDM for access to the system in general and to mapped volumes in particular. This prevents the SDC from accessing the cluster it it is not admitted as well as unauthorized volumes. Once enabled, the SDC internally performs mutual CHAP authentication with the SDSs and the SDRs with no manual intervention.

Prerequisites

Enable SDC authentication according to the following rules:

- For each SDC, a CHAP authentication password is generated by the MDM
- All SDCs must be configured with their generated passwords
- Run the `--check_sdc_authentication_status` command, to check the status of the SDCs and whether they are ready to authenticate

About this task

This procedure describes how to enable SDC authentication.

NOTE: Using CHAP authentication with SDC also means that an SDC can only perform I/O operations on volumes explicitly mapped to it. The SDS will block SDC I/O operations on unmapped volumes. CHAP authentication is also used internally for I/O authentication to the SDS and SDR, however it is always enabled and not controlled by the user.

Steps

1. Get the shared generated password for SDC from the MDM using the command:

   ```
   scli --generate_sdc_password --(sdc_id <ID> | sdc_name <NAME) | sdc_guid <GUID> |
   sdc_ip <IP>) [--reason <REASON>]
   ```

   Use the reason parameter to verify that the SDC password is being reset and not changed by accident. The reason is stored in the MDM events log.

   NOTE: SDCs not configured with a password are disconnected after the feature is enabled in step 3.

   NOTE: You must comply with organization password protection policy rules when performing this operation.

   Copy the password that was generated in `<SDC_PASSWORD_STRING>`, used in the next step.

2. On the SDC, run the following command:
   - Linux:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --set_mdm_password --ip <MDM_IP> --password
     <SDC_PASSWORD_STRING> --file/etc/emc/scaleio/drv_cfg.txt
     ```

     NOTE: The file option is required for password persistency, for cases such as service scini restart or SDC reboot. Open the file to verify the `<SDC_PASSWORD_STRING>` is logged at the end of the MDM line.

   - ESXi:

     a. `cat /etc/vmware/esx.conf | grep scini | grep options`

     A string is returned representing all of the ESXi configuration parameters currently set. Copy the string with the enclosing quotation marks and paste in a text editor for editing.

     b. At the end of the string, add the following text, within the quotation marks:

     ```
     IoctlMdmPasswordStr=<MDM_IP>-<MDM_PASSWORD>
     ```

     where:
     - *<MDM_IP>* is the MDM IP address
     - *<MDM_PASSWORD>* is the MDM password

For example:

```
"IoctlIniGuidStr=cd069ce3-bf2a-5dea-b50a-1a5ebc8ef3de
IoctlMdmIPStr=192.169.217.165,172.17.217.165,192.169.217.166,172.17.217.166,192.1
69.217.167,172.17.217.167 IoctlMdmPasswordStr=192.169.217.165-AQAAAAAAAADu/
10fXW3BS1wPBDgnkR06tdneGoUK7VQ"
```

c. Run the following command with the string appended to the end:

```
esxcli system module parameters set -m scini -p <STRING>
```

For example:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=cd069ce3-
bf2a-5dea-b50a-1a5ebc8ef3de
IoctlMdmIPStr=192.169.217.165,172.17.217.165,192.169.217.166,172.17.217.166,192.1
69.217.167,172.17.217.167 IoctlMdmPasswordStr=192.169.217.165-AQAAAAAAAADu/
10fXW3BS1wPBDgnkR06tdneGoUK7VQ"
```

3. To check SDC readiness for all SDCs in the system, before enabling SDC authentication, run the following command:

NOTE: It is important to complete the previous steps for all SDCs before running the command.

```
scli --check_sdc_authentication_status [--run_test] [--file_name <FILENAME>]
```

Where:

- `--run_test` runs a connectivity test to check whether the SDCs can successfully authenticate using CHAP
- `--filename <FILENAME>` is the full file name and path for the generated report.

The command sends a report that includes the SDCs authentication password status.

NOTE: When running this command, the SDCs are disconnected for a very short period from the MDM. This does not interrupt running I/Os or have any impact on MDM/SDC activity. It is recommended to run the command when the system is in a healthy state and not during rebalancing or rebuilding operations.

4. To enable SDC authentication, run the following command:

```
scli --set_sdc_authentication --enable
```

5. To disable SDC authentication, run the following command:

```
scli --set_sdc_authentication --disable
```

6. Reboot the ESXi for the configuration to take effect. For more information, see the "Gracefully shut down or reboot a node" topic in *Dell PowerFlex 4.5.x Install and Upgrade Guide*.

Results

SDC authentication is enabled or disabled.

Manual SDC configuration/verification

If there are problems connecting to a PowerFlex system after a reboot, users can modify the `drv_cfg.txt` file or use the drv_cfg utility to ensure that the SDC reconnects to the MDM automatically upon node restart/reboot.

To do this, perform one of the following procedures:

- Edit `/etc/emc/scaleio/drv_cfg.txt` and change the #MDM line to include each IP address for any node(s) containing an MDM in your cluster.

  For example, change the line:

  From:

#mdm 10.20.30.40

To:

mdm 10.108.158.48,10.108.158.49

where .48 is the primary node, and .49 is the secondary node. The tiebreaker node should not be included.

- Use the drv_cfg binary or drv_cfg.exe on Windows to re-attach the MDM.
- Rescan all volumes using the drv_cfg utility.

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SDS commands

This section contains commands for managing storage data severs (SDS).

abort_remove_sds

Abort the removal of an SDS from a PowerFlex system. If it is performed at too late a stage in the removal process, the command may fail and the SDS will be removed.

Syntax

```
scli --abort_remove_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>])
```

Parameters

--sds_id *<ID>*
The ID of the SDS whose removal you want to stop

--sds_name *<NAME>*
The name of the SDS whose removal you want to stop

--sds_ip *<IP>*
The IP address of the SDS whose removal you want to stop

--sds_port *<PORT>*
The port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --abort_remove_sds --sds_ip 192.168.1.4
```

add_sds

Add an SDS to a protection domain.

This command creates an SDS component and can be used to apply many configuration options at the time of creation. It informs the SDS about which devices to use, and for what purpose to use them (storage or acceleration). By default, the command also performs tests on the devices it adds, and saves the results.

NOTE: During execution of the command, PowerFlex checks that the devices are clear before adding them. If a device is not clear, an error message is returned, and the command fails for that device. If you would like to overwrite existing data on the devices by forcing the command, use the `force_device_takeover` flag. Note that this flag destroys all existing data on the device!

Syntax

```
scli --add_sds --sds_ip <IP> [--sds_ip_role <ROLE> {sdc_only | sds_only
| all}] (--protection_domain_id <ID> | --protection_domain_name <NAME>) [(--
storage_pool_name <NAME> | --storage_pool_id <ID>) --device_path <PATHS> [--
device_name <NAMES>] [--media_type <TYPES> {HDD | SSD}]] [(--acceleration_pool_name
<NAME> | --acceleration_pool_id <ID>) --acceleration_device_path <PATHS> [--
acceleration_device_name <NAMES>]] [--sds_name <NAME>] [--fault_set_id <ID> | --
fault_set_name <NAME>]
   [--rmcache_size_mb <SIZE>]
   [{--enable_rmcache | --disable_rmcache}]
   [--sds_port <PORT>]
   [--force_clean]
   [--force_device_takeover]
   [--test_time <TIME>]
   [{--test_only | --no_test | <BLANK>}]
   [--profile <PROFILE>]
   [--i_am_sure]
```

Parameters

--sds_ip *<IP>*

Comma-separated list of one or more IP addresses associated with the SDS over which the data will be transferred. See "SDS IP addresses and roles" below. Omit the space after each comma.

--sds_ip_role {sdc_only | sds_only | all}

Comma-separated list of IP roles associated with the SDS IP addresses. Omit the space after each comma. See "SDS IP addresses and roles" below.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Comma-separated list of storage pool names to be associated with the storage devices, respectively. Omit the space after each comma. If only one storage pool is specified in the list, all devices are associated with it. Otherwise, each device is associated with the storage pool situated in the same position in the list.

--storage_pool_id *<ID>*

Comma-separated list of storage pool IDs to be associated with the storage devices, respectively. Omit the space after each comma. If only one storage pool is specified in the list, all devices are associated with it. Otherwise, each device is associated with the storage pool situated in the same position in the list.

--device_path *<PATHS>*

Comma-separated list of the full path to each devices to be added. Omit the space after each comma. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device. To force device take-over, use this command with the `force_device_takeover` flag, proceeding with caution.

--device_name *<NAMES>*

Comma-separated list of device names. Omit the space after each comma. When using this flag to assign names to the SDS devices. The number of names must equal the number of devices, and the names will be appropriated to the devices in the same order as they appear in the `device_path` list.

--media_type {HDD | SSD}

If one media type is specified in the list, all devices are assigned with this type. Otherwise, each device is assigned the media type located in the same position as the list.

--acceleration_pool_name *<NAME>*

Comma-separated list of acceleration pool names to be associated with the acceleration devices. Omit the space after each comma. If one acceleration pool is specified, all acceleration devices are assigned to it. Otherwise, each device is assigned to the acceleration pool located in the same position in the list.

--acceleration_pool_id *<ID>*

Comma-separated list of acceleration pool IDs to be associated with the acceleration devices. Omit the space after each comma. If one Acceleration Pool is specified, all acceleration devices are assigned to it. Otherwise, each device is assigned to the acceleration pool located in the same position in the list.

--acceleration_device_path *<PATHS>*

Comma-separated list of SDS acceleration device paths. Omit the space after each comma. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device. To force device take-over, use the command with the `force_device_takeover` flag, proceeding with caution.

--acceleration_device_name *<NAMES>*

Comma-separated list of acceleration device names. Omit the space after each comma.

--sds_name *<NAME>*

Name assigned to the SDS

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

--rmcache_size_mb *<SIZE>*

Size of the SDS RAM cache in MB. Range: 128 MB to 300GB

--enable_rmcache | --disable_rmcache

Enable/disable RAM cache on the SDS

--sds_port *<PORT>*

Port associated with the SDS. Default: 7072

--force_clean

Clean a previous SDS configuration. Use this if the SDS was previously part of a PowerFlex system.

--force_device_takeover

Force adding the devices, ignoring data from other SDSs that are stored on the devices.

CAUTION: Use this flag with caution. All data saved on the devices will be destroyed.

--test_time *<TIME>*

The maximum test run-time in seconds. Default: 10. The test will stop when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first.

Test options (choose one):

By default, PowerFlex tests the performance of each device being added before before its capacity can be used. Two tests are performed on the given devices: random writes and random reads. When the tests are complete, the device capacity is added automatically to the protection domain storage used by the MDM. To modify this behavior, specify one of the test options:

--test_only

Devices are tested, but not used. To start using their capacity, run the `--activate_sds_device` command.

--no_test

The device capacity is used without any device testing.

<BLANK>

Read and write test is run on devices before their capacity is used.

--profile {compact | high_performance}

Set the performance profile. The default is `high_performance`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds --sds_ip 192.168.1.9,192.168.1.29 -sds_ip_role
sds_only,sdc_only --protection_domain_name rack_1.1 --device_path /dev/cciss/c0d1p3,/dev/
cciss/c0d1p4 --device_name sds01,sds02 --sds_name sds_1
```

SDS names

Assign each SDS and SDS device a meaningful name in order to facilitate future object identification. This can be helpful for SDS devices, because the defined name remains constant even if the path changes. When a name has not been defined, the system may display default system-defined names that use the first IP address of the SDS.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

SDS IP addresses and roles

Configure up to eight IP addresses per SDS. Define each one with one of the following roles:

- sdc_only: Communications are enabled only between this SDS and SDCs. The MDM is not aware of this IP; therefore, internal error-checking is not performed. If an incorrect IP address is supplied, throughput may be affected.
- sds_only: Communications are enabled only between this SDS and other SDS and MDM components.
- all (default): Communications are enabled with all components:

You can define a role for every IP address, or use the default for all of them. A situation where all IP addresses are either sdc_only or sds_only is not valid. Each SDS must have one of the following:

- At least one IP address with the role all

  or

- Two IP addresses with the roles sdc_only and sds_only

IP roles can be configured when adding an SDS, by adding a new IP address to an SDS, or by editing a current IP role.

add_sds_ip

Add an IP address to an existing SDS.

Syntax

```
scli --add_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_sds_ip <IP>
  [--sds_ip_role {sdc_only | sds_only | all}] [--force_add_ip]
```

Parameters

`--sds_id <ID>`

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port number on the server to be associated with the SDS

--new_sds_ip *<IP>*

New IP address to be associated with the SDS

--sds_ip_role {sdc_only | sds_only | all}

The IP role assigned to each SDS IP address: SDC only, SDS only, or all (the default). When adding multiple IP addresses, use a comma-separated list for IP roles, with no space after the commas.

--force_add_ip

Force the addition of the new IP address even if it could not be verified.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds_ip --sds_ip 192.168.1.4 --new_sds_ip 192.168.1.24
```

disable_sds_rmcache

Disable read RAM cache on the specified SDS.

Syntax

```
scli --disable_sds_rmcache ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --disable_sds_rmcache --sds_name sds23 --protection_domain_name pd01
```

NOTE: To disable the read RAM cache for all SDSs in a protection domain using a single command, add the relevant protection domain name or ID to the `--disable_sds_rmcache` command. For example:

```
scli --disable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain pd25
```

enter_maintenance_mode

Place one or more SDSs in maintenance mode.

Syntax

```
scli --enter_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
  [--force_insufficient_spare_or_free_space]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--force_insufficient_spare_or_free_space
Allow entry into maintenance mode, even without enough available capacity.

Example

```
scli --enter_maintenance_mode --sds_name sds10 --force_insufficient_spare_or_free_space
```

exit_maintenance_mode

Cancel maintenance mode for one or more SDSs.

NOTE: The system displays an error message if you attempt to remove an SDS in Maintenance Mode, ("The task failed since there is an SDS in the Protection Domain currently in Maintenance Mode"). However, you can force an SDS out of Maintenance Mode by using one of the flags, and then removing the SDS.

Syntax

```
scli --exit_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
  [--force_failed_device_id <IDs>]
  [--force_failed_sds]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--force_failed_device_id *<IDs>*
Allow exit from maintenance mode, even when the specified devices are in an error state.

--force_failed_sds
Allow exit from maintenance mode, even when an SDS is down.

NOTE: If you attempt to remove an SDS in maintenance mode, the system displays an error message: `The task failed since there is an SDS in the Protection Domain currently in Maintenance Mode`. Using the `force_failed_device_id` or `force_failed_sds` flag forces the SDS out of maintenance mode.

Example

```
scli --exit_maintenance_mode --sds_ip 10.76.1.10 --sds_port 7072
```

enable_sds_rmcache

Enable read RAM cache on the specified SDS.

For a read to be stored in the RAM of a specific SDS, the read RAM cache feature on that SDS must be enabled, and the relevant storage pool and the relevant volume must both be configured to use read RAM cache. Caching only begins after one or more devices are added to the SDS.

Syntax

```
scli --enable_sds_rmcache ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --enable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain_name pd01
```

NOTE: To enable read RAM cache for all SDSs in a protection domain using a single command, add the relevant protection domain name or ID to the `--enable_sds_rmcache` command. For example:

```
scli --enable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain pd25
```

enter_protected_maintenance_mode

When entering protected maintenance mode (PMM), a third copy of the SDS is created, which ensures that if there is a node failure where the second copy of SDS is required, there is still a full backup of the SDS. Move one or more SDSs to protected maintenance mode.

Syntax

```
scli --enter_protected_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
```

Parameters

None.

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_id *<ID>*
Fault set ID

--fault_set_name *<NAME>*
Fault set name

Example

```
scli --enter_protected_maintenance_mode --mdm_ip 192.168.1.200 --sds_name sds120
```

exit_protected_maintenance_mode

Move one or more SDSs out of protected maintenance mode.

Syntax

```
scli --exit_protected_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> |
--protection_domain_name <NAME>) --fault_set_name <NAME>))) [--force_failed_device_id
<IDs>] [--force_failed_sds] [--abort_entering_maintenance_mode]
```

Parameters

None.

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

--force_failed_device_id *<IDs>*

Allow exiting protected maintenance mode when the specified devices are in an error state.

--force_failed_sds

Allow exiting protected maintenance mode when the SDS is down.

--abort_entering_maintenance_mode

Acknowledge in case the process of entering protected maintenance mode is not yet complete.

NOTE: In fine granularity storage pools, the exit-protected maintenance mode process is delayed until the counter, `CAPACITY_IN_RECLAMATION_IN_KB`, reaches zero for all storage pools involved with the storage data server that is exiting protected maintenance mode.

Example

```
scli --exit_protected_maintenance_mode --mdm_ip 192.168.1.200 --sds_name sds120 --
force_failed_sds
```

modify_sds_ip_role

Modify the role of an SDS's IP address. The IP address can be used for SDC only, SDS only, or both.

Syntax

```
scli --modify_sds_ip_role (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) [--sds_ip_to_modify <IP>] --new_sds_ip_role {sdc_only | sds_only | all} [--
force_role_modification]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--sds_ip_to_modify *<IP>*

IP address of the SDS being modified

--new_sds_ip_role {sdc_only | sds_only | all}

New role to assign the SDS IP: SDC only, SDS only, or all

--force_role_modification

Force the modification of the SDS's role even if the IP address could not be verified.

Example

```
scli --modify_sds_ip_role --sds_name sds01 --sds_ip_to_modify 192.168.1.4 --
new_sds_ip_role all
```

modify_sds_port

Modify the port used by the SDS for communication purposes.

Syntax

```
scli --modify_sds_port (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_sds_port <PORT>
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--new_sds_port *<PORT>*

New port to be associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --modify_sds_port --sds_ip 192.168.1.4 --new_sds_port 6801
```

query_all_sds

Retrieve information about all SDSs in the system.

Syntax

```
scli --query_all_sds
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sds
```

Sample output

```
Query-all-SDS returned 5 SDS nodes.

Protection Domain f11d784d00000000 Name: pd1
SDS ID: 7a3cf71e00000004 Name: sds_idd_1 State: Connected, Joined IP: 10.55.137.114
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71c00000003 Name: sds_idd_2 State: Connected, Joined IP: 10.55.137.115
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71d00000002 Name: sds_idd_4 State: Connected, Joined IP: 10.55.137.116
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71b00000001 Name: sds_idd_5 State: Connected, Joined IP: 10.55.137.117
Port: 7072 Version: 3.6.0
Protection Domain f11d784e00000001 Name: pd2
SDS ID: 7a3cf71a00000000 Name: sds_idd_3 State: Connected, Joined IP: 10.55.137.119
Port: 7072 Version: 3.6.0
Protection Domain f11d9f5d00000002 Name: dfg
```

query_sds_connectivity_status

Retrieve information regarding the connectivity status of the SDSs in the specified protection domain.

Syntax

```
scli --query_sds_connectivity_status (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

Example

```
scli --query_sds_connectivity_status --protection_domain_name pd_18
```

Sample output

```
Successfully queried protection domain pd1
Got responses from 4 SDSs:
        ID: 06b108a100000001 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a100000001 <===> ID: 06b108a100000001  Connected Count 8
                10.55.137.116  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a300000003  Connected Count 8
                10.55.137.115  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119  : Connected
        ID: 06b108a000000000 <===> ID: 06b108a000000000  Connected Count 8
                10.55.137.117  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a300000003  Connected Count 4
                10.55.137.115  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a400000004  Connected Count 8
                10.55.137.114  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119  : Connected
```

query_sds

Retrieve detailed information about the specified SDS.

Syntax

```
scli --query_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>])
[--show_oscillating_failures]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--show_oscillating_failures
Show extended oscillating failure information, if such failures have occurred

Example

```
scli --mdm_ip 192.168.1.200 --query_sds --sds_id 06b108a400000004
```

Sample output

```
SDS 06b108a400000004 Name: sds_idd_1 Version: 3.6.0
Protection Domain: fcc9af5500000000, Name: pd1
Fault Set: 36c9d76800000000, Name: fs1_mg
Medium granularity DRL mode: Volatile
Authentication error: NONE
IP information (total 1 IP Addresses):
         1: 10.55.137.114    Role: All (SDS and SDC)
        Port: 7072
I/O flow control is enabled
Fine granularity number of concurrent writes: 1000
RAM Read Cache information:
        128.0 MB (131072 KB) total size
        Cache is enabled
        RAM Read Cache memory allocation state is pending
FGL metadata cache information:
        FGL metadata cache is disabled
        FGL metadata cache size: 0 MB

Storage Pool information (total of 1 Storage Pools):
         1: Storage Pool: sp1  inflight requests factor: 115, inflight bandwidth factor
115

Device information (total 2 devices):
         1: Name: N/A  Path: /dev/sdb  Original-path: /dev/sdb  ID: c8bea95800040000
                Storage Pool: sp1, Capacity: 148 GB, State: Normal
                Scanned 936040 MB, Unresolved Read errors: 0/0, Unresolved Compare
errors: 0/0
         2: Name: N/A  Path: /dev/sdc  Original-path: /dev/sdc  ID: c8bea95900040001
                Storage Pool: sp1, Capacity: 148 GB, State: Normal
                Scanned 936134 MB, Unresolved Read errors: 0/0, Unresolved Compare
errors: 0/0

Acceleration device information (total 0 devices):

Membership-state: Joined, Connection-state: Connected, SDS-state: Normal, 0 devices with
error,

Physical data:
        297.9 GB (305000 MB) total capacity
        280.5 GB (287184 MB) unused capacity
        17.4 GB (17816 MB) in-use capacity
        0 Bytes unreachable-unused capacity
        0 active moving-in forward-rebuild jobs
        0 active moving-in backward-rebuild jobs
        0 active moving-in rebalance jobs
        0 active moving-in enter protected maintenance mode jobs
        0 active moving-in exit protected maintenance mode jobs
        0 active moving-out forward-rebuild jobs
        0 active moving-out backward-rebuild jobs
        0 active moving-out enter protected maintenance mode jobs
        0 active moving-out exit protected maintenance mode jobs
        0 active moving-out rebalance jobs
        0 pending moving-in forward-rebuild jobs
        0 pending moving-in backward-rebuild jobs
        0 pending moving-in rebalance jobs
        0 pending moving-in enter protected maintenance mode jobs
        0 pending moving-in exit protected maintenance mode jobs
        0 pending moving-out forward-rebuild jobs
        0 pending moving-out backward-rebuild jobs
        0 pending moving-out rebalance jobs
        0 pending moving-out enter protected maintenance mode jobs
        0 pending moving-out exit protected maintenance mode jobs
```

```
        0 Bytes decreased capacity

Bandwidth:
        Primary-reads                                 114 IOPS 81.2 MB (83148 KB) per-second
        Primary-writes                                150 IOPS 75.6 MB (77450 KB) per-second
        Secondary-reads                               0 IOPS 0 Bytes per second
        Secondary-writes                              142 IOPS 69.8 MB (71516 KB) per-second
        Backward-rebuild-reads                        0 IOPS 0 Bytes per second
        Backward-rebuild-writes                       0 IOPS 0 Bytes per second
        Forward-rebuild-reads                         0 IOPS 0 Bytes per second
        Forward-rebuild-writes                        0 IOPS 0 Bytes per second
        Rebalance-reads                               0 IOPS 0 Bytes per second
        Rebalance-writes                              0 IOPS 0 Bytes per second
        Volume-migration-reads                        0 IOPS 0 Bytes per second
        Volume-migration-writes                       0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads        0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes       0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads         0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes        0 IOPS 0 Bytes per second
```

query_sds_connectivity_status

Retrieve information regarding the connectivity status of the SDSs in the specified protection domain.

Syntax

```
scli --query_sds_connectivity_status (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --query_sds_connectivity_status --protection_domain_name pd_18
```

Sample output

```
Successfully queried protection domain pd1
Got responses from 4 SDSs:
        ID: 06b108a100000001 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117   : Connected
        ID: 06b108a100000001 <===> ID: 06b108a100000001  Connected Count 8
                10.55.137.116   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a300000003  Connected Count 8
                10.55.137.115   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119   : Connected
        ID: 06b108a000000000 <===> ID: 06b108a000000000  Connected Count 8
```

```
            10.55.137.117   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a000000000  Connected Count 4
            10.55.137.117   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a100000001  Connected Count 4
            10.55.137.116   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a300000003  Connected Count 4
            10.55.137.115   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a400000004  Connected Count 8
            10.55.137.114   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a200000002  Connected Count 4
            10.55.137.119   : Connected
```

query_sds_network_test_results

Retrieve the results of the last network test performed between an SDS and its peers.

Syntax

```
scli --query_sds_network_test_results (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>])
```

Parameters

--sds_id *<ID>*
: SDS ID

--sds_name *<NAME>*
: SDS name

--sds_ip *<IP>*
: SDS IP address

--sds_port *<PORT>*
: Port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --query_sds_network_test_results --sds_ip 192.168.1.4
```

query_sdc_to_sds_disconnections

Retrieve details on any disconnections between SDCs and SDSs currently in the system.

Syntax

```
scli --query_sdc_to_sds_disconnections
```

Parameters

None.

Example

```
scli --query_sdc_to_sds_disconnections
```

remove_sds

Initiate the removal of an SDS. The command removes an SDS object and disconnects the SDS from the MDM.

You can remove an SDS at any time, with no downtime required. During execution of this command, the associated data is replicated to different nodes. Therefore, the removal process is asynchronous and might take a long time.

NOTE: If the capacity of this SDS is still used by volumes, and the capacity cannot be replaced due to lack of available free space, the command will fail.

Syntax

```
scli --remove_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>])
```

Parameters

--sds_id *<ID>*
ID of the SDS to remove

--sds_name *<NAME>*
Name of the SDS to remove

--sds_ip *<IP>*
IP address of the SDS to remove

--sds_port *<PORT>*
Port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds --sds_ip 192.168.1.4
```

Related operations

The removal operation runs in the background. To track its progress, run the `query_all_sds` command periodically.

To abort the removal, run the `abort_remove_sds` command.

remove_sds_ip

Remove an IP address from an SDS.

Possible scenarios for using this command are:

- In PowerFlex setups that include an SDS installed on a server with more than one NIC (network interface card), you might want to divert the usage of one of the NICs to a different purpose.
- An IP range may have changed. For example, an SDS may be using an IP address in a certain range, but the IP range has been changed to a different one in order to free up the previously used addresses.

Syntax

```
scli --remove_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --sds_ip_to_remove <IP>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--sds_ip_to_remove *<IP>*
IP address to remove from the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds_ip
 --sds_ip 192.168.1.4 --sds_ip_to_remove 192.168.1.24
```

rename_sds

Assign a new name to the specified SDS.

Syntax

```
scli --rename_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_name <NAME>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--new_name *<NAME>*
New, unique name to assign to the specified SDS

Example

```
scli --mdm_ip 192.168.1.200 --rename_sds --sds_ip 192.168.1.4 --new_name sds_new_name
```

SDS names

Assign the SDS a meaningful name in order to facilitate future object identification. The name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

set_drl_properties

Set the properties of the dirty-region-logging used by one or all SDSs.

NOTE: The settings apply to medium granularity layout devices only.

Syntax

```
scli --set_drl_properties ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | --set_all) (--volatile | --hardened)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--set_all
Apply the properties to all SDSs. Overrides previous settings

--volatile
Dirty-region-logging information will be stored in memory only. This is the default.

--hardened
Dirty-region-logging information will be stored both in memory and on disk.

Example

```
scli --set_drl_properties --sds_ip 192.168.1.4 --hardened
```

set_fgl_metadata_cache_size

Configure the fine granularity metadata cache size for the specified SDS.

Syntax

```
scli --set_fgl_metadata_cache_size (--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--metadata_cache_size_mb <SIZE> | --
use_protection_domain_default_size)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--metadata_cache_size_mb *<SIZE>*
Define the default size of the metadata cache for the SDS in MB

--use_protection_domain_default_size
Use the default size of the protection domain

Example

```
scli --mdm_ip 192.168.1.200 --set_fgl_metadata_cache_size --sds_name SDS1 --
metadata_cache_size_mb 10248
```

set_sds_network_limits

Limit the network bandwidth used by all the SDSs in the specified protection domain for various traffic types, including I/O.

SDS nodes transfer data among themselves. This data consists of user-data being replicated as part of the RAID protection, and data copied for internal rebalancing and recovery from failures.

You can modify the balance between these types of data loads by limiting the data copy bandwidth. This change affects all SDSs in the specified protection domain.

CAUTION: Contact Dell Technologies Support before you modify this configuration.

Syntax

```
scli --set_sds_network_limits (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
[--rebuild_limit <LIMIT>] [--rebalance_limit <LIMIT>] [--volume_migration_limit <LIMIT>]
[--protected_maintenance_mode_limit <LIMIT>] [--overall_limit <LIMIT>]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--rebuild_limit *<LIMIT>*

Limit the network bandwidth for rebuild in MB/s. 0 is unlimited

--rebalance_limit *<LIMIT>*

Limit the network bandwidth for rebalance in MB/s. 0 is unlimited

--volume_migration_limit *<LIMIT>*

Limit the network bandwidth for V-Tree migration in MB/s. 0 is unlimited

--protected_maintenance_mode_limit *<LIMIT>*

Limit the network bandwidth for protected maintenance mode in MB/s. 0 means unlimited bandwidth.

--overall_limit *<LIMIT>*

Limit the overall network bandwidth in MB/s. This includes the rebuild, rebalance, and application I/O bandwidth. 0 is unlimited

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.68.1.200 --set_sds_network_limits --protection_domain_name pd1 --
rebuild_limit 100
```

set_sds_rmcache_size

Set the RMcache size of the specified SDS.

By default, the RMcache size is set to 128 MB in all SDSs. The amount of RAM you can allocate for cache is limited by the amount of RAM on the SDS server:

- If the RAM is less than 32 GB, 50 percent of memory can be used for cache.
- If the RAM is more than 32 GB, 75 percent of memory can be used for cache.

The maximum amount of RAM cache is 128 GB.

NOTE: In cases of Non-Uniform Memory Access (NUMA), the total RAM refers to the first node only.

Syntax

```
scli --set_sds_rmcache_size ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>)) --
rmcache_size_mb <SIZE>
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--rmcache_size_mb *<SIZE>*

New size of RMcache in MB. Range: 128 MB-300 GB.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_sds_rmcache_size --protection_domain_name pd01 --sds_name sds10 --
rmcache_size_mb 256
```

show_certificate

Display the certificate properties for the specified SDS or SDR.

Syntax

```
scli --show_certificate ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) | --all_sds | (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) | --all_sdr | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--days_until_expiration <DAYS>]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--all_sds

Show certificates for all SDSs.

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

Port associated with the SDR

--all_sdr

Show certificates for all SDRs.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--days_until_expiration *<DAYS>*

Only show certificates expiring within a given number of days.

Example

```
scli --show_certificate --all_sds --days_until_expiration 7
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Protection Domain fcc9af5600000001 Name: pd2
Protection Domain fcc9d66500000002 Name: rrr
```

start_sds_network_test

Initiate a network test between an SDS and all of its peers.

Syntax

```
scli --start_sds_network_test (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>])
  [--parallel_messages <NUMBER>]
  [--network_test_size_gb <SIZE>]
  [--network_test_length_secs <SECONDS>]
  [--test_associated_protection_domains]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--parallel_messages *<NUMBER>*

Number of parallel messages sent during the test. Range is 1 - 16. Default: 4.

--network_test_size_gb *<SIZE>*

Amount of data sent between the tested SDS and every other SDS. Default size: 1 GB.

--network_test_length_secs *<SECONDS>*

Maximum amount of time spent testing the SDS, regardless of the number of SDSs in the system. Default value: 0 (unlimited).

--test_associated_protection_domains

Test associated protection domains. Protection domains can be connected during V-Tree migration between different protection domains.

Example

```
scli --mdm_ip 192.168.1.200 --start_sds_network_test
--sds_ip 192.168.1.4 --parallel_messages 8 --network_test_size_gb 2 --
network_test_length_secs 120
```

update_sds_original_paths

Change the original path configuration of all devices to the path currently assigned to the devices in the SDS.

Syntax

```
scli --update_sds_original_paths (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) [--force_failed_devices]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

SDS port

--force_failed_devices

Force path update even for devices in error

Example

```
scli --update_sds_original_paths --sds_name sds1 --force_failed_devices
```

set_sds_device_path

Configure the path of the given storage device.

Syntax

```
scli --set_sds_device_path (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
new_device_path <PATH>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path. The file path is used only when the device is available at this path

--new_device_path *<PATH>*
The new path to configure

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SDT commands

This section contains commands for managing storage data targets (SDT).

add_sdt

Add an SDT IP address.

Syntax

```
scli --add_sdt --sdt_ip <IP> [--sdt_ip_role <ROLE>] [--storage_port <PORT>] [--nvme_port
<PORT>] [--discovery_port <PORT>] [--sdt_name <NAME>] (--protection_domain_id <ID> | --
protection_domain_name <NAME>) [--profile <PROFILE>] [--force_clean [--i_am_sure]]
```

Parameters

--sdt_ip *<IP>*

A comma separated list of IP addresses assigned to the SDT

--sdt_ip_role *<ROLE>*

A comma separated list of roles assigned to each SDT IP address

Role options: storage_only, host_only, or storage_and_host

--storage_port *<PORT>*

Port assigned to the SDT (default: 12200)

--nvme_port *<PORT>*

Port used by the NVMe hosts (default: 4420)

--discovery_port *<PORT>*

Port used by the NVMe hosts for discovery (default: 8009). Set to 1 in order to indicate no use of discovery port.

--sdt_name *<NAME>*

A name assigned to the SDT

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

--profile *<PROFILE>*

Set the performance profile from the following options: compact | high_performance

The default is high_performance

--force_clean

Clean a previous SDT configuration

--i_am_sure

Preemptive approval

Example

```
scli --add_sdt --sdt_name sdt1 --sdt_ip 10.0.0.1,10.0.1.1 --sdt_ip_role
storage_and_host, storage_and_host --protection_domain_name pd1
```

remove_sdt

Start the removal of an SDT.

Syntax

```
scli --remove_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--force
Force removing IP regardless the number of SDTs left on protection domains.

Example

```
scli --remove_sdt --sdt_name sdt1
```

rename_sdt

Name or rename an SDT.

Syntax

```
scli --rename_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --new_name <NAME>
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--new_name *<NAME>*

A new name to be assigned to the SDT

Example

```
scli --rename_sdt sdt1 --new_name someSdt
```

add_sdt_ip

Add an IP address to an SDT.

Syntax

```
scli --add_sdt_ip (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --new_sdt_ip <IP> [--sdt_ip_role <ROLE>]
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--new_sdt_ip *<IP>*

New IP address to be assigned to the SDT

--sdt_ip_role *<ROLE>*

The role assigned to the new SDT IP address

Role options: storage_only, host_only, or storage_and_host

Example

```
scli --add_sdt_ip  --sdt_name sdt1 --new_sdt_ip 10.0.3.1 --sdt_ip_role storage_and_host
```

remove_sdt_ip

Remove an IP address from an SDT.

Syntax

```
scli --remove_sdt_ip (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --sdt_ip_to_remove <IP> --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--sdt_ip_to_remove *<IP>*
IP address to remove from the SDT

--force
Force removing IP regardless the number of hosts connections

Example

```
scli --remove_sdt_ip --sdt_name sdt1 --sdt_ip_to_remove 10.0.3.1
```

modify_sdt_ip_role

Modify the role of an SDT IP address.

Syntax

```
scli --modify_sdt_ip_role (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--sdt_ip_to_modify <IP>] --sdt_ip_role <ROLE> --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--sdt_ip_to_modify *<IP>*
IP address for which to modify the role. Required if --sdt_ip is not specified

--sdt_ip_role *<ROLE>*

The new role assigned to the modified SDT IP address

Role options: storage_only, host_only, or storage_and_host

--force

Force modifying IP role regardless the number of hosts connections

Example

```
scli --modify_sdt_ip_role --sdt_name sdt1 --sdt_ip_to_modify 10.0.3.1 -sdt_ip_role
host_only
```

modify_sdt_port

Modify the port of an SDT.

Syntax

```
scli --modify_sdt_port (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--storage_port <PORT>] [--nvme_port <PORT>] [--discovery_port <PORT>]
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--storage_port *<PORT>*

Port assigned to the SDT

--nvme_port *<PORT>*

Port used by the NVMe hosts

--discovery_port *<PORT>*

Port used by the NVMe hosts for discovery.

Example

```
scli --modify_sdt_port (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--storage_port <PORT>] [--nvme_port <PORT>] [--discovery_port <PORT>]
```

query_sdt

Retrieve information about an SDT.

Syntax

```
scli --query_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>])
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

Example

```
scli --quert_sdt --sdt_name sdt1
```

query_all_sdt

Retrieve information about all SDTs.

Syntax

```
scli --query_all_sdt
```

enter_sdt_maintenance_mode

Move the SDT into maintenance mode.

Syntax

```
scli --enter_sdt_maintenance_mode (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--
sdt_port <PORT>]) --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--force

Force entering maintenance mode regardless the number of SDTs in a protection domain.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

Example

```
scli --enter_sdt_maintenance_mode --sdt_name sdt1
```

exit_sdt_maintenance_mode

Move the SDT out of maintenance mode.

Syntax

```
scli --exit_sdt_maintenance_mode (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--
sdt_port <PORT>])
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

Example

```
scli --exit_sdt_maintenance_mode --sdt_name sdt1
```

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Snapshot commands

This section contains commands for managing snapshots.

clear_snapshot_security

Clear the snapshot security for the specified volume to enable editing or removing the volume. This command can only be run with a signature file obtained from customer support. If you need to clear the snapshot security, contact Dell Technologies Support to obtain the signature file.

Syntax

```
scli --clear_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
signature_file <FILE> [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--signature_file *<FILE>*

Signature file provided by customer support.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --clear_snapshot_security --volume_name vol1 --
signature_file /opt/vol-dd6e34001.sig --i_am_sure
```

remove_snapshots_consistency_group

Remove all snapshots that belong to a specific consistency group. Depending on the size of the consistency group, this command may take a long time.

NOTE: You cannot remove the consistency group if it contains auto snapshots.

Syntax

```
scli --remove_snapshots_consistency_group --scg_id <ID>
  [--i_am_sure]
```

Parameters

--scg_id *<ID>*

The consistency group's ID number.

NOTE: The consistency group ID is displayed when you create the consistency group. To find this ID at a later time, use the `query_all_volumes` command.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli  --remove_snapshots_consistency_group
--scg_id 783c7f00000000 --i_am_sure
```

set_snapshot_security

Set the specified snapshot as secure. The snapshot is not editable or removable for the specified retention period. If the specified snapshot is already secure, the security expiration date may be postponed, but not configured to an earlier point in time.

Syntax

```
scli --set_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
retention_period <PERIOD> [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--retention_period *<PERIOD>*

The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m` | `h` | `D` | `W` | `M` | `Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days).

--i_am_sure

Preemptive approval

Example

```
scli --mdm_ip 192.168.1.200 --set_snapshot_security --volume_id 022beb2500000006 --
retention_period 2Y --i_am_sure
```

snapshot_volume

Creates a snapshot of one or more volumes.

When you specify more than one volume (a list), a consistency group is generated and the ID associated with it is displayed. The snapshots under the consistency group are taken simultaneously for all listed volumes, thus ensuring their consistency.

Secure snapshots, created using the `--retention_period` parameter, prohibit anyone from deleting the snapshot until the expiration time that is defined.

NOTE: The consistency group is for convenience purposes only. There are no protective measures to conserve the consistency group, and you can delete members from it.

Syntax

```
scli --snapshot_volume (--volume_id <ID>| --volume_name <NAME> | --volume_id_from_file
<FILE> | --volume_name_from_file <FILE> | --volume_class <CLASS>)
  [--snapshot_name <NAME> | --snapshot_name_from_file <FILE>] [--read_only] [--
retention_period <PERIOD>] [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Comma-separated list of volume IDs. Omit the space after each comma.

--volume_name *<NAME>*

Comma-separated list of volume names. Omit the space after each comma.

--volume_id_from_file *<FILE>*

File containing new, line-separated list of volume IDs

--volume_name_from_file *<FILE>*

File containing a new, line-separated list of volume names

--snapshot_name *<NAME>*

Comma-separated list of names to be associated with the created snapshots. The names in this list correspond to the names in the volume name or volume ID list. Omit the space after each comma.

--snapshot_volume_class *<Class>*

Allows creating snapshot in special cases

--snapshot_name_from_file *<FILE>*

File containing a new, line-separated list of names to be associated with the created snapshots

--read_only

All mappings of the created volume will be limited to read-only access

--retention_period *<PERIOD>*

Use this parameter to create the snapshots as secure. The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m|h|D|W|M|Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days)

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --snapshot_volume
--volume_name vol_1 --snapshot_name snap_1 --retention_period 2Y
```

Snapshot names

Assign the snapshot a meaningful name. When a name has not been defined, the system may display default system-defined names that use the ID of the volume.

Assign snapshot names according to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

add_snapshot_copy_task

Adds an asynchronous task to copy a snapshot image to a target volume.

Syntax

```
scli --add_snapshot_copy_task (--remote_system_id <ID> | --remote_system_name
<NAME>) (--source_volume_id <ID> | --source_volume_name <NAME> | --
source_volume_id_from_file <FILE> | --source_volume_name_from_file <FILE>) [--
base_volume_id <ID> | --base_volume_name <NAME> | --base_volume_id_from_file
<FILE> | --base_volume_name_from_file <FILE>]  (--destination_volume_nsid <NSID>
| --destination_volume_nsid_from_file <FILE>) [--snapshot_copy_task_name <NAME> | --
snapshot_copy_task_name_from_file <FILE>] [--copy_allocated_blocks_only]
```

Parameters

--snapshot_copy_task_name *<NAME>*
The snapshot copy task name

--remote_system_id *<ID>*
The ID of the remote system

--remote_system_name *<NAME>*
The name of the remote target system

--source_volume_id *<ID>*
A comma-separated list of the source volume IDs

--source_volume_name *<NAME>*
A comma-separated list of the source volume names

--source_volume_id_from_file *<FILE>*
A file containing new line separated list of the source volume IDs

--source_volume_name_from_file *<FILE>*
A file containing new line separated list of the source volume names

--base_volume_id *<ID>*
A comma-separated list of the base volume IDs

--base_volume_name *<NAME>*
A comma-separated list of the base volume names

--base_volume_id_from_file *<FILE>*
A file containing new line separated list of the base volume IDs

--base_volume_name_from_file *<FILE>*
A file containing new line separated list of the base volume names

--destination_volume_nsid *<NSID>*

A comma-separated list of the destination volume NSIDs

--destination_volume_nsid_from_file *<FILE>*

A file containing new line separated list of the destination volume NSIDs

--snapshot_copy_task_name *<NAME>*

A comma-separated list of the snapshot copy task names

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the snapshot copy task names

--copy_allocated_blocks_only

Skip the copy of unallocated blocks (in full snapshot copy tasks only)

query_snapshot_copy_task

Retrieve information about the snapshot copy task.

Syntax

```
scli --query_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME>)
```

Parameters

--snapshot_copy_task_id *<ID>*

The snapshot copy task ID

--snapshot_copy_task_name *<NAME>*

The snapshot copy task name

query_all_snapshot_copy_tasks

Retrieve information about all snapshot copy tasks.

Syntax

```
cli --query_all_snapshot_copy_tasks
```

pause_snapshot_copy_task

Pause the snapshot copy task.

Syntax

```
scli --pause_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME> | --snapshot_copy_task_id_from_file <FILE> | --
snapshot_copy_task_name_from_file <FILE>)
```

Parameters

--snapshot_copy_task_id *<ID>*

A comma-separated list of the task IDs

--snapshot_copy_task_name *<NAME>*

A comma-separated list of the task names

--snapshot_copy_task_id_from_file *<FILE>*

A file containing new line separated list of the task IDs

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the task names

resume_snapshot_copy_task

Resume the snapshot copy task.

Syntax

```
scli --resume_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME> | --snapshot_copy_task_id_from_file <FILE> | --
snapshot_copy_task_name_from_file <FILE>)
```

Parameters

--snapshot_copy_task_id *<ID>*

A comma-separated list of the task IDs

--snapshot_copy_task_name *<NAME>*

A comma-separated list of the task names

--snapshot_copy_task_id_from_file *<FILE>*

A file containing new line separated list of the task IDs

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the task names

remove_snapshot_copy_task

Remove the snapshot copy task.

Syntax

```
scli --remove_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME> | --snapshot_copy_task_id_from_file <FILE> | --
snapshot_copy_task_name_from_file <FILE>) [--force]
```

Parameters

--snapshot_copy_task_id *<ID>*

A comma-separated list of the task IDs

--snapshot_copy_task_name *<NAME>*

Protection domain name

--snapshot_copy_task_id_from_file *<FILE>*

A file containing new line separated list of the task IDs

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the task names

--force

Allow removal of non-completed tasks

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Storage pool commands

This section contains commands for managing storage pools.

add_storage_pool

Add a storage pool to a protection domain.

A storage pool is a group of storage devices within a protection domain. Each time that you add devices to the system, you must map them to either storage pools or acceleration pools. Create storage pools before you start adding SDSs and storage devices to the system. You can modify storage pools post-installation using all management clients, except for OpenStack.

Syntax

```
scli --add_storage_pool (--protection_domain_id <ID> | --protection_domain_name <NAME>)
  [--storage_pool_name <NAME>]
  [--use_rmcache | --dont_use_rmcache]
  [--rmcache_write_handling_mode  {cached | passthrough}]
  [--media_type {HDD | SSD}]
  [--external_acceleration <TYPE>] {none | read | write | read_and_write}]
  [--data_layout <LAYOUT> {medium_granularity |fine_granularity}
  [--compression_method <METHOD> {none | normal}]
  (--fine_granularity_acceleration_pool_name <NAME> | --
fine_granularity_acceleration_pool_id <ID>)]]
  [--enable_inflight_checksum | --disable_inflight_checksum]
  [--persistent_checksum_enabled | --persistent_checksum_disabled]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Name to be associated with the new storage pool

--use_rmcache

Use RMcache on all SDSs in the storage pool. You can change this setting later on specific SDSs (they can be individually configured not to use RMcache).

--dont_use_rmcache

Do not use read RAM cache on any SDSs in the storage pool (default)

--rmcache_write_handling_mode *<MODE>* **{cached | passthrough}**

Determines whether the system stores the data of this storage pool writes in the SDS read RAM cache, or not. The default is to store the write data in cache (`cached`). You can change this setting later, using the command `set_rmcache_write_handling_mode`.

NOTE: Only I/Os that are multiples of 4k bytes can be cached.

--media_type *<TYPE>* **{HDD | SSD}**

Media type of the devices in the storage pool, where the devices will all be HDD or SSD.

--external_acceleration *<TYPE>*

External acceleration for the device where *<TYPE>* can be one of the following:

none

All devices are not accelerated by a non-PowerFlex read or write cache (default).

read

All devices are accelerated by a non-PowerFlex read cache.

write

All devices are accelerated by a non-PowerFlex write cache.

read_and_write

All devices are accelerated by both non-PowerFlex read and write cache.

--data_layout *<LAYOUT>* **{medium_granularity | fine_granularity}**

Indicates the layout of the data for devices in the storage pool, where *<LAYOUT>* can be medium granularity or fine granularity. The default is medium granularity.

--compression_method *<METHOD>*

When using fine granularity data layout, this Indicates the compression method to be used. *<METHOD>* can be one of the following:

none

No compression applied.

normal

Normal compression applied.

--fine_granularity_acceleration_pool_name *<NAME>*

Name of the NVDIMM acceleration pool to be used for fine granularity.

--fine_granularity_acceleration_pool_id *<ID>*

ID of the NVDIMM acceleration pool to be used for fine granularity.

--enable_inflight_checksum

Enable inflight checksum protection.

--disable_inflight_checksum

Disable inflight checksum protection.

--persistent_checksum_enabled

Enable persistent checksum protection.

--persistent_checksum_disabled

Disable persistent checksum protection.

Example

```
scli --add_storage_pool --protection_domain_name pd1 --storage_pool_name sp2
--dont_use_rmcache --media_type ssd --external_acceleration read_and_write --
persistent_checksum_enabled
```

Storage pool names

Assign each storage pool a meaningful name. When a name has not been defined, the system may display default system-defined names that use the ID of the storage pool.

Each volume name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

disable_background_device_scanner

Disable the background device scanner on the specified storage pool.

Syntax

```
scli --disable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --disable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp180
```

disable_persistent_checksum

Disable persistent checksum for the specified storage pool.

Syntax

```
scli --disable_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --disable_persistent_checksum --storage_pool_id
d089bdc100000000
```

enable_background_device_scanner

Enable the background device scanner on the devices in the specified storage pool to check for errors (not supported in fine granularity storage pools).

Syntax

```
scli --enable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --
read_error_action <ACTION>  --compare_error_action <ACTION>
  [--scanner_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--read_error_action

Read error handling. Can be one of the following:

--report_and_fix

Perform read operations and automatically fix from peer on errors.

--report_only

Perform read operations and report errors without fixing them.

--compare_error_action

Compare error handling between primary and secondary copies of data.

NOTE: Zero padding must be enabled in order to set the background device scanner to data compare mode.

Can be one of the following:

--no_compare

Perform read operations without comparing data content with peer.

--report_and_fix

Perform read operations, compare data content with peer, and automatically fix.

--report_only

Perform read operations, compare data content with peer, and report mismatches without fixing them.

--scanner_bandwidth_limit *<LIMIT>*

Bandwidth limit in KB per second per device. The value should be in the range of 10 KB to 10 MB. Default: 3 MB.

NOTE: High bandwidth should be used very carefully for extreme cases only (such as an urgent need to check some devices), as it may create negative impact on the system performance. Setting the background device scanner bandwidth should take into account maximum bandwidth of the devices.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --enable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp18 --read_error_action --report_only --compare_error_action --
report_only --scanner_bandwidth_limit 2000
```

Errors

To check for errors detected by the background device scanner, query SDSs using the `--query_sds` command. Error information is provided in event reports.

Bandwidth limits

High bandwidth may create a negative impact on system performance and should be used carefully and in extreme cases only—for example, when there is an urgent need to check certain devices. When setting the background device scanner bandwidth, you should consider the maximum bandwidth of the devices.

enable_persistent_checksum

Enable a persistent checksum for the specified storage pool. Persistent checksum can be used to support the medium granularity layout in protecting the storage device from data corruption.

Syntax

```
scli --enable_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) [--
validate_on_read <MODE>] [--builder_bandwidth_limit <LIMIT>]
```

Parameters

`--protection_domain_id <ID>`
Protection domain ID

`--protection_domain_name <NAME>`
Protection domain name

`--storage_pool_name <NAME>`
Storage pool name

`--storage_pool_id <ID>`
Storage pool ID

`--validate_on_read <MODE>`
Validate checksum upon reading data: `enabled` or `disabled`

`--builder_bandwidth_limit <LIMIT>`

Bandwidth limit in KB/s for the checksum building process. Valid range: 1024-10240. Default: 3072

Example

```
scli --mdm_ip 192.168.1.200 --enable_persistent_checksum --storage_pool_id
d089bdc100000000 --validate_on_read enabled --builder_bandwidth_limit 6144
```

modify_compression_method

Modify the compression method of a storage pool or a V-Tree. Compression can only be used for fine granularity layouts.

NOTE: When fine granularity layouts are not used, use the `none` compression option.

Syntax

```
scli --modify_compression_method ( (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) | --
vtree_id <ID> | --volume_id <ID> | --volume_name <NAME> ) --compression_method <METHOD>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

--storage_pool_id *<ID>*

Storage pool ID

--vtree_id *<ID>*

V-Tree ID

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--compression_method *<NAME>*

Indicates the compression method to be used on the storage pool or V-Tree. Compression is only relevant for fine granularity data layouts. Set to `none` if fine granularity is not used. Options include:

`none`- No compression is applied

`normal`- Normal compression is applied

Example

```
scli --modify_compression_method --protection_domain_name pd2 --storage_pool_name sp21 --
compression_method normal
```

modify_persistent_checksum

Modify the persistent checksum configuration for the specified storage pool.

Syntax

```
scli --modify_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) [--
validate_on_read <MODE>] [--builder_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--validate_on_read *<MODE>*
Validate checksum upon reading data: `enabled` or `disabled`

--builder_bandwidth_limit *<LIMIT>*
Bandwidth limit in KB/s for the checksum building process. Valid range: 1024-10240. Default: 3072

Example

```
scli --mdm_ip 192.168.1.200 --modify_persistent_checksum --storage_pool_id
d089bdc100000000 --validate_on_read disabled --builder_bandwidth_limit 1024
```

modify_zero_padding_policy

Modify the zero padding policy for the storage pool. Use the zero-padded policy when the storage pool data layout is fine granularity (FG). The zero-padded policy ensures that every read from an area that is previously not written to, returns zeros.

Syntax

```
scli --modify_zero_padding_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) (--
enable_zero_padding | --disable_zero_padding)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--enable_zero_padding

Enable zero padding in this storage pool

--disable_zero_padding

Disable zero padding in this storage pool

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --modify_zero_padding_policy --protection_domain_name pd10 --storage_pool_name sp1
--zero_padding_disabled
```

query_inaccessible_user_address_space

List all inaccessible address spaces for the specified storage pool.

Syntax

```
scli --query_inaccessible_user_address_space (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id <ID>

Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_inaccessible_user_address_space --storage_pool_id
d089bdc100000000
```

query_storage_pool

Retrieve aggregated information about all objects in the specified storage pool.

Syntax

```
scli --query_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --query_storage_pool --protection_domain_name pd --storage_pool_name sp_822jIk
```

Sample output

```
Storage Pool sp_822jIk (Id: 2a2a65ce00000000) has 24 volumes and 20.5 TB (20969 GB) free
net capacity, 365.9 TB (374688 GB) Remaining volume allocation limit, 413.9 TB (423840
GB) Total volume allocation limit
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using No-Limit policy
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: none, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 21% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Doesn't use Read Flash Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Fine granularity
```

```
        Compression method: Normal
        Fine granularity acceleration pool: 68a806e600000000
        Fine granularity advanced configuration:
                Over provisioning factor: 20
                Write atomicity size: 64KB
                NVDIMM write cache size: 16MB
                NVDIMM metadata amortization 1.50
                Max compression ratio: 10
                Performance profile: high_performance
        Replication Capacity Max Allowed Ratio: 0%
        Replication Capacity Max Allowed: 0 Bytes
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0


Physical data:
        52.4 TB (53651 GB) total capacity
                493.2 GB (505026 MB) in-use capacity
                        362.7 GB (371424 MB) user data
                        39.8 GB (40745 MB) metadata
                        90.7 GB (92856 MB) fine granularity spares
                        47.1 GB (48205 MB) capacity in reclamation
                41.0 TB (41938 GB) unused capacity
                11.0 TB (11219 GB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                493.2 GB (505026 MB) protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity
                0 Bytes inaccessible capacity

User data:
        152.4 TB (156032 GB) maximum user data
        354.5 GB (363034 MB) total user data
                354.5 GB (363034 MB) compressed data (1.9:1 compression ratio)
                0 Bytes uncompressed data
        0 Bytes trimmed capacity
        354.5 GB (363034 MB) user data without trimmed capacity
        2.0:1 total compression ratio
        138.6:1 thin and snapshot saving
        271.0:1 overall saving
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                354.5 GB (363034 MB) at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                           7431 IOPS 116.1 MB (118896 KB)
per-second
        Primary-writes                          7475 IOPS 116.8 MB (119593 KB)
per-second
```

```
        Secondary-reads                             0 IOPS 0 Bytes per second
        Secondary-writes                            7404 IOPS 115.7 MB (118464 KB)
per-second
        Backward-rebuild-reads                      0 IOPS 0 Bytes per second
        Backward-rebuild-writes                     0 IOPS 0 Bytes per second
        Forward-rebuild-reads                       0 IOPS 0 Bytes per second
        Forward-rebuild-writes                      0 IOPS 0 Bytes per second
        Rebalance-reads                             0 IOPS 0 Bytes per second
        Rebalance-writes                            0 IOPS 0 Bytes per second
        Volume-migration-reads                      0 IOPS 0 Bytes per second
        Volume-migration-writes                     0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads      0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes     0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads       0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes      0 IOPS 0 Bytes per second

Volumes summary:
        24 thin-provisioned volumes. Total user data size: 354.5 GB (363034 MB)
        48.0 TB (49152 GB) all volumes size
        48.0 TB (49152 GB) base volumes size
        24 user volumes
```

remove_storage_pool

Remove a storage pool from a protection domain.

NOTE: Each protection domain must contain at least one storage pool, which can be either the default or a user-defined storage pool. You can only remove all storage pools from a protection domain if they are empty and do not contain any devices.

Syntax

```
scli --remove_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --remove_storage_pool
--protection_domain_name rack_1.1
--storage_pool_name sp_1.1_rack_1.1
```

rename_storage_pool

Name, or rename, a storage pool.

Syntax

```
scli --rename_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--new_name *<NAME>*
New name to associate with the storage pool

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --rename_storage_pool --protection_domain_name rack_1.1 --
storage_pool_name sp_1.1_rack_1.1 --new_name sp_2.2_rack_1.1
```

Storage pool names

Assign the storage pool a meaningful name in order to facilitate future object identification. The name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

reset_scanner_error_counters

Reset background device scanner error counters for the specified storage pool or device. You can use this command to reset counters for comparison errors, read errors, or both counter types.

Syntax

```
scli --reset_scanner_error_counters ((((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) |
(--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) (--device_name <NAME> | --device_path <PATH>))) )
```

```
  [--reset_compare_errors]
  [--reset_read_errors]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<SDS IP ADDRESS>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path

--storage_pool_id *<ID>*
Storage pool ID

--reset_read_errors
Reset the read error counter.

--reset_compare_errors
Reset the compare error counter.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

NOTE: If you use the `device_id` parameter in the command, you do not need to also supply the SDS ID or SDS name.

Example

```
scli --reset_scanner_error_counters --protection_domain_name pd120 --storage_pool_name
sp120  --sds_name sds13 --reset_compare_errors
```

set_capacity_alerts_threshold

Set the threshold for triggering capacity usage alerts.

Alerts thresholds are calculated from each storage pool capacity after deducting the defined amount of spare capacity. You can set thresholds for a single storage pool or for all storage pools.

Syntax

```
scli --set_capacity_alerts_threshold --capacity_high_threshold <PERCENT> --
capacity_critical_threshold <PERCENT> (--all_storage_pools | --system_default | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
|--storage_pool_id <ID>))
```

Parameters

--capacity_high_threshold *<PERCENT>*

Threshold of the non-spare capacity of the Storage Pool that will trigger a high-priority alert, expressed as a percentage. Setting a zero threshold for a storage pool resets it to use the global threshold.

--capacity_critical_threshold *<PERCENT>*

Threshold of the non-spare capacity of the storage pool that triggers a critical-priority alert, expressed as a percentage. Setting a zero threshold for a storage pool resets it to use the global threshold.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--all_storage_pools

Thresholds apply to all storage pools

--system_default

Thresholds apply to newly created storage pools

Example

```
scli --mdm_ip 192.168.1.200 --set_capacity_alerts_threshold --system_default --
capacity_high_threshold 80 --capacity_critical_threshold 90
```

set_rebalance_mode

Enable or disable rebalancing of the specified storage pool.

CAUTION: Rebalancing is an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. Disabling rebalance may cause the system to become unbalanced even if no capacity is added or removed—for example, during recovery from an SDS or device failure.

Syntax

```
scli --set_rebalance_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebalance | --
disable_rebalance)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--enable_rebalance

Enable rebalancing of the specified storage pool.

--disable_rebalance

Disable rebalancing of the specified storage pool.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebalance_mode --protection_domain_name pd10 --storage_pool_name sp20 --
enable_rebalance
```

set_rebalance_policy

Set the rebalance I/O priority policy for a storage pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_rebalance_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for rebalance I/O priority. Can be one of the following:

no_limit

Rebalance I/Os are not limited.

limit_concurrent_io

Limit the number of allowed concurrent rebalance I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

(Default) Limit the number of allowed concurrent rebalance I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent rebalance I/Os per device. Default: 1.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of rebalance I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebalance_policy --protection_domain_name pd1 --storage_pool_name sp1 --
policy favor_application_io --bandwidth_limit 5120
```

set_rebuild_mode

Enable or disable rebuilds in the specified storage pool.

CAUTION: Rebuilds are an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. If rebuilds are disabled, redundancy will not be restored after failures.

Syntax

```
scli --set_rebuild_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebuild | --
disable_rebuild)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--enable_rebuild

Enable rebuilds in the specified storage pool.

--disable_rebuild

Disable rebuilds in the specified storage pool.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebuild_mode --protection_domain_name pd10 --storage_pool_name sp20 --
enable_rebuild
```

set_rebuild_policy

Set the rebuild I/O priority policy for a storage pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_rebuild_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for rebuild I/O priority. Can be one of the following:

no_limit

Rebuild I/Os are not limited.

limit_concurrent_io

(Default) Limit the number of allowed concurrent rebuild I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

Limit the number of allowed concurrent rebuild I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent rebuild I/Os per device. Default: 1.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of rebuild I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebuild_policy --protection_domain_name pd1 --storage_pool_name sp1 --policy
favor_application_io --bandwidth_limit 5120
```

set_rebuild_rebalance_parallelism

Set the maximum number of concurrent rebuild and rebalance activities on all SDSs within the specified storage pool.

Syntax

```
scli --set_rebuild_rebalance_parallelism (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--
limit <LIMIT>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--limit *<LIMIT>*

The maximum number of concurrent rebuild and rebalance activities (1..10) on SDSs in the storage pool.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebuild_rebalance_parallelism --protection_domain_name pd_18 --
storage_pool_name flash_18 --limit 5
```

set_rmcache_usage

Control the use of read RAM cache in the specified Storage Pool.

When using read RAM cache, you must also enable caching in each SDS in the storage pool using the `enable_sds_rmcache` command. Caching only begins once devices have been added to the SDSs.

NOTE: You can configure read RAM cache for a storage pool and then disable caching on one or more SDSs individually.

Syntax

```
scli --set_rmcache_usage (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--use_rmcache | --
dont_use_rmcache)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--use_rmcache

Use read RAM cache in the storage pool.

--dont_use_rmcache

Do not use read RAM cache in the storage pool (the default).

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rmcache_usage --protection_domain_name pd10 --storage_pool_name spflash --
use_rmcache
```

set_rmcache_write_handling_mode

Set the RMcache write handling mode of the specified storage pool. Writes typically skip the cache, but in certain circumstances you may want to cache writes, and reads.

Syntax

```
scli --set_rmcache_write_handling_mode (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
rmcache_write_handling_mode {passthrough | cached}
[--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--rmcache_write_handling_mode {passthrough | cached}

The write-handling mode used by the RMcache. One of:

passthrough

Writes skip the cache and are stored in storage only.

cached

Writes are stored in both cache and storage (the default).

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rmcache_write_handling_mode --protection_domain_name pd1 --storage_pool_name
spflash --rmcache_write_handling_mode cached
```

set_storage_pool_external_acceleration_type

Set the external acceleration type for devices in the storage pool.

Syntax

```
scli --set_storage_pool_external_acceleration_type ((((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
external_acceleration <TYPE>
```

```
  [--override_sds_device_specific_external_acceleration | --
keep_sds_device_specific_external_acceleration]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--external_acceleration *<TYPE>*

External acceleration for the device, where TYPE is one of the following:

none

All devices are not accelerated by a non-PowerFlex read or write cache.

read

All devices are accelerated by a non-PowerFlex read cache.

write

All devices are accelerated by a non-PowerFlex write cache.

read_and_write

All devices are accelerated by both non-PowerFlex read and write cache.

--override_sds_device_specific_external_acceleration

Override SDS device-specific external acceleration.

NOTE: The external acceleration type of the device is overridden according to the external acceleration type set for the `--external_acceleration` parameter. If there is a mismatch between the external acceleration type of the storage pool and the external acceleration type of the device, the command fails with informative error message.

--keep_sds_device_specific_external_acceleration

Keep SDS device-specific external acceleration.

Example

```
scli --set_storage_pool_external_acceleration_type --protection_domain_name
pd1 --storage_pool_name sp2 --external_acceleration read_and_write --
override_sds_device_specific_external_acceleration
```

set_storage_pool_media_type

Set the media type of the devices in the storage pool.

Syntax

```
scli --set_storage_pool_media_type (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
media_type <TYPE>
  [--override_sds_device_media_type]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--media_type *<TYPE>*
Media type of the device where TYPE is one of the following:

HDD
All devices in this storage pool are HDDs.

SSD
All devices in this storage pool are SSDs.

transitional
Media type is defined per device

--override_sds_device_media_type
Override SDS device media type

Example

```
scli --set_storage_pool_media_type --protection_domain_name pd2 --storage_pool_name sp2
--media_type ssd --override_sds_device_media_type
```

set_checksum_mode

Inflight checksum protection mode can be used to validate data reads and writes in storage pools, in order to protect data from data corruption. Modify the checksum protection policy of a storage pool.

Syntax

```
scli --set_checksum_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) (--
enable_inflight_checksum | --disable_inflight_checksum)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--enable_inflight_checksum

Enable inflight checksum for the storage pool.

--disable_inflight_checksum

Disable inflight checksum for the storage pool.

Example

```
scli --set_checksum_mode --protection_domain_name PD_1 --storage_pool_name SP_2 --
enable_inflight_checksum
```

set_protected_maintenance_mode_policy

Set the protected maintenance mode I/O priority policy for a storage pool. This feature affects system performance, and only advanced users should modify it.

Syntax

```
scli --set_protected_maintenance_mode_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for protected maintenance mode I/O priority. Can be one of the following:

no_limit

Protected maintenance mode I/Os are not limited.

limit_concurrent_io

(Default) Limit the number of allowed concurrent protected maintenance mode I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

Limit the number of allowed concurrent protected maintenance mode I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent protected maintenance mode I/Os per device. Default: 1.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of protected maintenance mode I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_protected_maintenance_mode_policy --protection_domain_name pd1 --
storage_pool_name sp1 --policy favor_application_io --bandwidth_limit 5120
```

test_inaccessible_data

After identifying where there is inaccessible data using the `--query_inaccessible_user_address_space` command, manually clear the inaccessible data from the storage pool. You can then test the inaccessible data, which triggers a rebuild operation, and the data is accessible again.

Syntax

```
scli --test_inaccessible_data (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
```

Parameters

`--sds_id <ID>`
SDS ID

`--sds_name <NAME>`
SDS name

`--sds_ip <IP>`
SDS IP address

`--sds_port <PORT>`
Port assigned to the SDS

`--device_id <ID>`
Device ID

`--device_name <NAME>`
Device name

`--device_path <PATH>`
SDS storage device path or filepath

Example

```
scli --mdm_ip 192.168.1.200 --test_inaccessible_data --device_id fceff00700040001
```

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System commands

This section contains commands for managing the system.

abort_upgrade

Stop an upgrade that is in process.

Syntax

```
scli --abort_upgrade
```

Parameters

None.

Example

```
scli --abort_upgrade
```

add_certificate

Add a certificate authority (CA) signed certificate to the trusted certificates list of the CLI. CLI automatically trusts all applied certificates that the CA has signed.

Syntax

```
scli ---add_certificate --certificate_file <FILE>
```

Parameters

--certificate_file *<FILE>*

File name of the root or trusted certificate

Example

```
scli --add_certificate --certificate_file CAcert-cacert.pem
```

add_standby_mdm

Add a standby MDM to the system. This MDM can later be added to the cluster.

Syntax

```
scli --add_standby_mdm --mdm_role {manager    | tb} --new_mdm_ip <IP>
  [--new_mdm_port <PORT>]
  [--new_mdm_management_ip <IP>]
  [--new_mdm_virtual_ip_interface <INTF>]
  [--new_mdm_name <NAME>]
  [--allow_asymmetric_ips]
  [--force_clean]
  [--i_am_sure]
```

Designate the standby MDM's role, either manager or tiebreaker (default). You can give the standby a unique name.

Parameters

--mdm_role {manager | tb}

Designates whether the MDM role is a manager (primary MDM of the cluster) or will only serve as a tiebreaker in the cluster primary MDM election process

--new_mdm_ip *<IP>*

Comma-separated list of up to four IP addresses assigned to the new MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_port *<PORT>*

Port of the new MDM. Default: 9011.

--new_mdm_management_ip *<IP>*

Comma-separated list of up to four IP addresses used to manage the MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of virtual IP addresses assigned to the cluster. Omit the space after each comma.

--new_mdm_name *<NAME>*

Name of the new MDM

--allow_asymmetric_ips

Allow the added node to have a different number of IP addresses from the primary node.

--force_clean

Clean the previous configuration of the MDM.

--i_am_sure

Skip the safety questions for command execution, and give preemptive approval.

Example

```
scli --add_standby_mdm --mdm_role manager --new_mdm_ip 192.168.1.153 --
new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth4
```

allow_commands_during_upgrade

Allows cluster commands to be sent during an upgrade for this session.

Syntax

```
scli --allow_commands_during_upgrade
```

Parameters

None.

Example

```
scli --allow_commands_during_upgrade
```

create_mdm_cluster

Create an MDM cluster from the selected primary MDM.

This command does not require the user to be logged in before running it. When the command is executed, the MDM cluster operates in single mode.

Syntax

```
scli --create_mdm_cluster --primary_mdm_ip <IP>
  [--primary_mdm_management_ip <IP>]
  [--cluster_virtual_ip <IP>]
  [--primary_mdm_virtual_ip_interface <INTF>]
  [--primary_mdm_name <NAME>]
  [--accept_license]
  [--disable_client_secure_communication]
  [--approve_certificate_once]
  [--approve_certificate]
```

Parameters

--primary_mdm_ip*<IP>*

Comma- separated list of IP addresses assigned to the primary MDM, to be used for MDM cluster internal control communications. Omit the space after each comma.

--primary_mdm_management_ip *<IP>*

Comma-separated list of IP addresses, used to manage the MDM. Omit the space after each comma.

--cluster_virtual_ip *<IP>*

Comma-separated list of virtual IP addresses to be used for the cluster. Omit the space after each comma.

--primary_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of the virtual IP addresses assigned to the cluster. Omit the space after each comma.

--primary_mdm_name*<NAME>*

MDM name. Each MDM name must be unique. Ensure that each server on which the MDM is installed has a unique hostname.

--accept_license

Accept the license agreement.

--disable_client_secure_communication

Create the cluster without management client secure communications.

--approve_certificate_once

One-time approval of the MDM's certificate (without adding the certificate to the truststore).

--approve_certificate

Automatic approval of the MDM's certificate for the next commands (adds the certificate to the truststore).

Example

```
scli --create_mdm_cluster --primary_mdm_ip 192.168.1.152 --cluster_virtual_ip
192.168.100.152 --primary_mdm_virtual_ip_interface eth4
```

finalize_upgrade

Finalize the upgrade process.

Syntax

```
scli --finalize_upgrade
```

Parameters

None.

Example

```
scli --finalize_upgrade
```

generate_certificate

Generate a new certificate and private key for the SDS or SDR or SDT. The certificate and key are signed by the system.

Syntax

```
scli --generate_certificate ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) | (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>]))
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--sdr_id *<ID>*

SDR ID

--sdr_port <PORT>

Port assigned to the SDR

--sdr_name *<NAME>*

SDR name

--sdt_id *<ID>*

SDT ID

--sdr_ip *<IP>*

SDR IP address

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

generate_mdm_certificate

Create a new self-signed certificate and private key for the specfied MDM.

NOTE: Running this command on the primary MDM causes the MDM to restart. This may cause the MDM to switch ownership.

Syntax

```
scli --generate_mdm_certificate (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*

ID of the MDM for which the certificate and private key are required

--target_mdm_ip *<IP>*

IP address of the MDM for which the certificate and private key are required

--target_mdm_port *<PORT>*

Port of the MDM for which the certificate and private key are required

--target_mdm_name *<NAME>*

Name of the MDM for which the certificate and private key are required

Example

```
scli --generate_mdm_certificate --target_mdm_ip 198.168.1.2 --target_mdm_port 9011
```

generate_mdm_csr_file

Create a certificate signing request (CSR) file for the specified MDM.

The file is used to obtain security certificates from a certificate authority and is saved in the specified MDM directory:

- Linux: `/opt/emc/scaleio/mdm/cfg`
- Windows: `C:\Program Files\emc\scaleio\mdm\cfg`

Syntax

```
scli --generate_mdm_csr_file (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
  [OPTIONS]
```

Parameters

--target_mdm_id *<ID>*

ID of the MDM for which the certificate and private key are required

--target_mdm_ip *<IP>*

IP address of the MDM for which the certificate and private key are required

--target_mdm_port *<PORT>*

Port of the MDM for which the certificate and private key are required

--target_mdm_name *<NAME>*

Name of the MDM for which the certificate and private key are required

--common_name *<NAME>*

Common name to appear in the CSR. The default is the host name of the machine.

Options

NOTE: If you use any of these options, the default values will be disabled for all of them.

--country *<NAME>*

Country to appear in the CSR

--state *<NAME>*

State to appear in the CSR

--location *<NAME>*

Location to appear in the CSR

--organization *<NAME>*

Organization to appear in the CSR

--organizational_unit *<NAME>*

Organizational unit to appear in the CSR

--email_address *<EMAIL>*

Email address to appear in the CSR

Example

```
scli --generate_mdm_csr_file --target_mdm_ip 192.168.1.2 --target_mdm_port 9011 --
country australia --state victoria --location melbourne --organization example_pty_ltd
--organizational_unit sales --email_address sales@example.com.au
```

modify_cluster_virtual_ips

Set or modify the virtual IP address of an MDM cluster.

NOTE: After changing or adding a virtual IP address, you are required to update the SDC with the new virtual IP address.

Notes

For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM (when the MDM is installed on a Linux server), disable the IPv6 DAD setting on the server's relevant interfaces, using the command: `sysctl net.ipv6.conf.<interface_name>.dad_transmits=0`

Syntax

```
scli --modify_cluster_virtual_ips (--cluster_virtual_ip <IP> | --clear_all)
```

Parameters

--cluster_virtual_ip *<IP>*

Comma-separated list of IP addresses to be used for the cluster. Omit the space after each comma.

--clear_all

Clear all virtual IP addresses.

Example

```
scli --modify_cluster_virtual_ips --cluster_virtual_ip 192.168.100.152,192.168.100.153
```

modify_management_ip

Modify an MDM node's management IP addresses.

If you are using this command to add additional management IP addresses, you must enter all of the existing IP addresses in the command, and add the new ones at the end. Up to four IP addresses are supported.

Syntax

```
scli --modify_management_ip (--new_mdm_management_ip <IP> | --clear_all) (--
target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --target_mdm_name
<NAME>)
  [--allow_duplicate_management_ips]
  [--i_am_sure]
```

Parameters

--new_mdm_management_ip *<IP>*
Comma-separated list of IP address to be used to manage the MDM. Omit the space after each comma.

--clear_all
Clear all management IP addresses.

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

--allow_duplicate_management_ips
Allow duplicate management IP addresses.

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --modify_management_ip --target_mdm_id 0x34e8f1df6c84a410 --new_mdm_management_ip
fd00::192:168:1:17,10.103.110.17
```

Implementing the changes

After modifying MDM IP addresses, ensure that you implement the change in all components and interfaces that are mapped to the MDM:

1. PowerFlex GUI: In the **Login** window, log in using the new IP address, then clear the old IP address from the history if it is now obsolete.
2. REST Gateway and OpenStack: In the REST Gateway, modify the mapping to the new management IP address.
3. On each SDC, perform the following locally, in command line with root access permissions:
   - If for any reason, the management IP address is not updated in the SDC, run:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
     ```

   - If the SDC is mapped to more than one PowerFlex system, run:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
     ```

4. vSphere: Using the vSphere web plug-in interface, unregister PowerFlex and reregister it with the new IP address.

modify_spare_policy

Modify the current spare capacity reservation policy.

To ensure data protection during server failures, PowerFlex reserves 10 percent of its capacity by default, instead of allowing this capacity to be used for volume allocation. To ensure full system protection in the event of a node failure, the spare capacity must be at least equal to the amount of capacity in the node containing the maximum capacity or the maximum Fault Set capacity.

If all nodes contain equal capacity, you should set the capacity value to at least 1/N of the total capacity (where N is the number of SDS nodes).

Syntax

```
scli --modify_spare_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --spare_percentage
<PERCENT>
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--spare_percentage *<PERCENT>*

Percentage of the total capacity to be set aside as spare

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --modify_spare_policy
--protection_domain_name rack_1.1 --storage_pool_name --spare_percentage 10
```

modify_virtual_ip_interfaces

Set the virtual IP interfaces of a node in the MDM cluster.

Syntax

```
scli --modify_virtual_ip_interfaces (--new_mdm_virtual_ip_interface <INTF> | --
clear_all) (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --
target_mdm_name <NAME>)
```

Parameters

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP interfaces. Omit the space after each comma. The order of interfaces must match the order of virtual IPs assigned to the cluster.

--clear_all

Clear all virtual IP interfaces.

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

Example

```
scli --modify_virtual_ip_interfaces --target_mdm_ip 192.168.1.152 --
new_mdm_virtual_ip_interface eth4
```

query_all

Retrieve general information about the system.

Syntax

```
scli --query_all
```

Parameters

None.

Example

```
scli --query_all
```

Sample output

```
System Info:
        Product: DellTechnologies PowerFlex Version: R3_6.7000.123
        ID: 7d4d7fa03275000f
        Manager ID: 0000000000000000
        Name:     system_e0aUWp

License info:
        Installation ID: 6c01ff8058a72813
        SWID:
        Maximum capacity: Unlimited
        Usage time left: 90 days *** Non-Production License ***
        Enterprise features: Enabled
        The system was activated 0 days ago

System settings:
        Capacity alert thresholds: High: 80, Critical: 90
        MDM restricted SDC mode: NONE
        Management Clients secure communication: Enabled
        TLS version: TLSv1.2
        CLI preemptive login banner acceptance via shell: Enabled
        CLI login banner acceptance by password: Enabled
```

```
        User authentication method: Native
        MDM connection authentication: Disabled
        Defragmentation: Enabled
        SDC refresh with MDM IP addresses: disabled

SDC-SDS connectivity status: All connected

SDC-SDR connectivity status: All connected

SDR-SDS connectivity status: All connected

Query all returned 1 Protection Domain:
Protection Domain pd (Id: bbf165fb00000000) has 1 storage pools, 1 acceleration pools, 0
Fault Sets, 5 SDS nodes and 24 volumes
        Free net capacity: 20.5 TB (20963 GB), Remaining Volume allocation limit: 365.2
TB (373992 GB), Total Volume allocation limit: 413.9 TB (423840 GB)
        Operational state is Active
        RFcache enabled, Mode: Write miss, Page Size 64 KB, Max IO size 128 KB,
Acceleration Pool ID: N/A
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Max Replication remote capacity ratio: 50%
        Max Replication remote capacity: 0 Bytes

Storage Pool sp_822jIk (Id: 2a2a65ce00000000) has 24 volumes and 20.5 TB (20963 GB) free
net capacity, 365.2 TB (373992 GB) Remaining volume allocation limit, 413.9 TB (423840
GB) Total volume allocation limit
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using No-Limit policy
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: none, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 21% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Doesn't use Read Flash Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Fine granularity
        Compression method: Normal
        Fine granularity acceleration pool: 68a806e600000000
        Fine granularity advanced configuration:
                Over provisioning factor: 20
                Write atomicity size: 64KB
                NVDIMM write cache size: 16MB
                NVDIMM metadata amortization 1.50
                Max compression ratio: 10
                Performance profile: high_performance
        Replication Capacity Max Allowed Ratio: 0%
        Replication Capacity Max Allowed: 0 Bytes
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0

Acceleration Pool ap_ORFKOk (Id: 68a806e600000000)
        Media Type: NVDIMM
        Used for RFcache: no

SDR-SDS connectivity status: All connected
```

```
SDS Summary:
        Total 5 SDS Nodes
        5 SDS nodes have membership state 'Joined'
        5 SDS nodes have connection state 'Connected'
        1 SDS nodes have maintenance state 'Exiting-Maintenance-Mode'

Physical data:
        52.4 TB (53651 GB) total capacity
                526.5 GB (539114 MB) in-use capacity
                        387.2 GB (396494 MB) user data
                        42.5 GB (43496 MB) metadata
                        96.8 GB (99123 MB) fine granularity spares
                        86.8 GB (88840 MB) capacity in reclamation
                40.9 TB (41927 GB) unused capacity
                555.2 GB (568502 MB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                519.1 GB (531587 MB) protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                10.4 TB (10642 GB) unreachable-unused capacity
                0 Bytes semi-protected capacity
                7.4 GB (7527 MB) in-maintenance capacity
                0 Bytes temporary capacity

User data:
        378.5 GB (387588 MB) total user data
                0 Bytes medium granularity user data
                378.5 GB (387588 MB) fine granularity user data
                        378.5 GB (387588 MB) compressed data (2.0:1 compression ratio)
                        0 Bytes uncompressed data
        0 Bytes trimmed capacity
        378.5 GB (387588 MB) user data without trimmed capacity
        2.0:1 total compression ratio
        129.9:1 thin and snapshot saving
        253.9:1 overall saving
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                10.4 GB (10698 MB) exit protected maintenance mode capacity
                368.1 GB (376889 MB) at-rest capacity
                10.4 GB (10698 MB) total moving capacity
                        622.0 MB (636961 KB) active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                622.0 MB (636961 KB) active exit protected maintenance
mode capacity
                        9.8 GB (10076 MB) pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                9.8 GB (10076 MB) pending exit protected maintenance
mode capacity

Bandwidth:
        Primary-reads                           7029 IOPS 109.8 MB (112457 KB)
per-second
        Primary-writes                          7140 IOPS 111.6 MB (114236 KB)
per-second
        Secondary-reads                         0 IOPS 0 Bytes per second
        Secondary-writes                        7071 IOPS 110.5 MB (113123 KB)
per-second
        Backward-rebuild-reads                  0 IOPS 0 Bytes per second
        Backward-rebuild-writes                 0 IOPS 0 Bytes per second
        Forward-rebuild-reads                   0 IOPS 0 Bytes per second
        Forward-rebuild-writes                  0 IOPS 0 Bytes per second
```

```
        Rebalance-reads                              0 IOPS 0 Bytes per second
        Rebalance-writes                             0 IOPS 0 Bytes per second
        Volume-migration-reads                       0 IOPS 0 Bytes per second
        Volume-migration-writes                      0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads       0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes      0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads        0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes       0 IOPS 0 Bytes per second

Volumes summary:
        24 thin-provisioned volumes. Total user data size: 378.5 GB (387588 MB)
        48.0 TB (49152 GB) all volumes size
        48.0 TB (49152 GB) base volumes size
        0 snapshot policies
        24 user volumes

SDR Summary:
        Total 0 SDR Nodes

Replication Peer System Summary:
        Total 0 Replication Peer System Nodes
```

Output description

The following tables explain the output for `query_all`.

Table 174. System information

System information	Description
Product	The PowerFlex version of the system.
ID	The system ID number.
Manager ID	The ID of the management application that controls the MDM.

Table 175. License information

License information	Description
Installation ID	The Installation ID number, which is required for licensing purposes.
SWID	The SWID number, which is required for licensing purposes.
Maximum capacity	The maximum capacity permitted by the system's license.
Usage time left	The amount of time left until the license expires.
Enterprise features	Indicates whether enterprise features are enabled.
The system was activated *n* days ago	The number of days since the system was activated .

Table 176. System settings

System settings	Description
Volumes are/are not obfuscated	The obfuscation setting for all volumes in the system. Default: not obfuscated.
Capacity alert thresholds	The alert thresholds, as percentages, currently configured for capacity usage.
Thick volume reservation	The amount of capacity, as a percentage, reserved for thick volumes.
MDM restricted SDC mode	Indicates whether users on remote clients are restricted to read-only access: • enabled: read-only access • disabled: full read-write access

Table 176. System settings (continued)

System settings	Description
Management clients secure communication	The status of the enabled/disabled
TLS version	
User authentication method	
SDS connection authentication	The status of the ..... : enabled/disabled

Table 177. Protection domain information

Protection domain	-
Protection domain summary	The following items are displayed separately for each protection domain: ● Name ● ID ● Number of storage pools in the protection domain ● Number of fault sets in the protection domain ● Number of SDS nodes in the protection domain ● Number of volumes associated with the protection domain ● Amount of capacity in the protection domain that is available for volume allocation
Operational state	Indicates the protection domain's current state, such as active, inactive, and so on.

Table 178. Storage pool information

Storage pool	-
Storage pool summary	Lists the following information for each storage pool: ● Name ● ID ● Number of volumes associated with the storage pool ● Amount of capacity in the storage pool that is available for volume allocation
The number of parallel rebuild/rebalance jobs	The number of parallel rebuild and rebalance jobs currently existing in the system
Rebuild is enabled/disabled	Indicates whether the rebuild feature is enabled or disabled. (In normal production conditions, rebuild should be enabled.) When enabled, the rebuild policy configured in the system is displayed, along with the relevant parameters.
Rebalance is enabled/disabled	Indicates whether the rebalance feature is enabled or disabled. (In normal production conditions, rebalance should be enabled.) When enabled, the rebalance policy configured in the system is displayed, along with the relevant parameters.
Background device scanner	Displays the mode and the bandwidth limit per device.
Zero padding is enabled/disabled	Indicates whether zero padding is enabled or disabled.
Spare policy	The percentage of total capacity reserved as spare.
Read RAM cache	Indicates whether the read RAM cache feature is used in the storage pool. If the read RAM cache is used, it displays the write handling mode currently in use (passthrough or cached).
Capacity alert thresholds	The alert thresholds, as percentages, configured for capacity usage. Thresholds may be configured for the entire system, or per storage pool.

Table 179. SDS summary

SDS summary	-
Total	The total number of SDS nodes.
n SDS nodes have membership state 'Joined'	The number of SDS nodes that are connected to the cluster, can receive I/O, and hold primary and secondary copies of data. It takes some time from actual disconnection until the SDS is disconnected from the cluster, or the reverse. There are also up-pending and down-pending states.
n SDS nodes have connection state 'Connected'	The number of SDS nodes currently connected to the MDM.
x TB (*y* GB) total capacity	The total amount of available raw storage. This does not represent the total capacity available for volume allocation.
x TB (*y* GB) unused capacity	The quantity of raw capacity in the system that can be earmarked for specific purposes, such as Spare, or used for new volume creation.
x Bytes snapshots capacity	The quantity of capacity used for storing snapshots.
x TB (*y* GB) in-use capacity	The total quantity of healthy, degraded, and failed capacity.
x Bytes thin capacity	The quantity of capacity currently needed for storage purposes.
x TB (*y* GB) protected capacity	The quantity of capacity that is fully protected (primary and secondary copies of the data exist).
x Bytes failed capacity	The quantity of capacity that is not available at all (neither primary, nor secondary copies).
x Bytes degraded-failed capacity	The quantity of degraded-failed capacity. When an SDS fails, all of its capacity is defined as degraded-failed, and the secondary copies (which are spread across the protection domain) are defined as degraded-healthy.
x Bytes degraded-healthy capacity	The quantity of degraded-healthy capacity. When an SDS fails, all of its capacity is defined as degraded-failed, and the secondary copies (which are spread across the protection domain) are defined as degraded-healthy.
x Bytes unreachable-unused capacity	The quantity of capacity not configured for any use type in the system that is currently unavailable.
x Bytes active rebalance capacity	The quantity of capacity that is currently being migrated to a different location for load balancing purposes.
x Bytes pending rebalance capacity	The quantity of capacity that is waiting in the job queue for migration to a different location for load balancing purposes.
x Bytes active forward-rebuild capacity	The quantity of capacity for which one copy of data exists and a second copy is currently being created.
x Bytes pending forward-rebuild capacity	The quantity of capacity for which one copy of data exists, and a job for the creation of a second copy is waiting in the job queue.
x Bytes active backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are currently being synchronized in that copy.
x Bytes pending backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are waiting in the job queue to be synchronized in that copy.

Table 179. SDS summary (continued)

SDS summary	-
x Bytes rebalance capacity	The total quantity of capacity that is either currently rebalancing or is pending rebalance.
x Bytes forward-rebuild capacity	The total quantity of degraded capacity that is either currently in forward rebuild state or is pending forward rebuild.
x Bytes backward-rebuild capacity	The total quantity of degraded capacity that is either currently in backward rebuild state or is pending backward rebuild.
x Bytes active moving capacity	The quantity of capacity that is currently being migrated from one location to another.
x Bytes pending moving capacity	The quantity of capacity that is waiting in the job queue for migration from one location to another.
x Bytes total moving capacity	The total quantity of active and pending moving (migrating) capacity.
x TB (*y* GB) spare capacity	The quantity of capacity that is reserved for system use when recovery from failure is required. This capacity cannot be used for storage purposes.
x TB (*y* GB) at-rest capacity	The quantity of capacity that is fully protected and not in a rebuild or rebalance state.
x Bytes semi-protected capacity	
x Bytes in-maintenance capacity	
x Bytes decreased capacity	The quantity of decreased capacity that was deducted from devices (using the **Set Device Capacity Limit** GUI command or `modify_sds_device_capacity` CLI command) and cannot be used for any purpose
Primary-reads *x* IOPS *y* bytes per second	Number of primary data copy read IOPS, and read bandwidth.
Primary-writes *x* IOPS *y* bytes per second	Number of primary data copy write IOPS, and write bandwidth.
Secondary-reads *x* IOPS *y* bytes per second	Number of secondary data copy (protection) read IOPS, and read bandwidth.
Secondary-writes *x* IOPS *y* bytes per second	Number of secondary copy (protection) write IOPS, and write bandwidth.
Backward-rebuild-reads *x* IOPS *y* bytes per second	Number of backward rebuild read IOPS, and read bandwidth.
Backward-rebuild-writes *x* IOPS *y* bytes per second	Number of backward rebuild write IOPS, and write bandwidth.
Forward-rebuild-reads *x* IOPS *y* bytes per second	Number of forward rebuild read IOPS, and read bandwidth.
Forward-rebuild-writes *x* IOPS *y* bytes per second	Number of forward rebuild write IOPS, and write bandwidth.
Rebalance-reads *x* IOPS *y* bytes per second	Number of rebalance read IOPS, and read bandwidth.
Rebalance-writes *x* IOPS *y* bytes per second	Number of rebalance write IOPS, and write bandwidth.

Table 180. Volumes summary

Volumes summary	-
n thick-provisioned volumes	The number of thick-provisioned volumes, and the total size in TB and GB.
n volumes mapped to all SDC nodes	The number of volumes mapped to SDC nodes.

query_cluster

Retrieve MDM cluster information.

Syntax

```
scli --query_cluster [--show_certificate_info]
```

Parameters

--show_certificate_info
Display certificates for all the MDMs

Example

```
scli --mdm_ip 192.168.1.200 --query_cluster --show_certificate_info
```

Sample output

The following is sample output for a five-node cluster:

```
scli --query_cluster
Cluster:
    Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
    Virtual IP Addresses: N/A
Primary MDM:
    Name: MDM_116, ID: 0x155c8db3025d5f01
        IP Addresses: 10.55.137.116, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Issuer:     /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: 2E:68:A4:48:A3:BC:DF:77:D9:8B:1C:77:01:99:04:5B:1D:F9:77:B6
Secondary MDMs:
    Name: MDM_114, ID: 0x75c68cb264723f00
        IP Addresses: 10.55.137.114, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Issuer:     /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: F7:B6:37:3C:45:EF:98:5B:64:57:5E:D3:7D:A1:B7:AD:E4:7B:5A:EA
    Name: MDM_115, ID: 0x7386c66b551fdf02
        IP Addresses: 10.55.137.115, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Issuer:     /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Valid From: Jan 30 09:34:18 2021 GMT
            Valid To:   Jan 29 10:34:18 2031 GMT
            Thumbprint: A8:FB:1E:A2:6C:92:32:6C:42:FB:A0:A3:50:1D:6F:90:06:E2:1B:A0
```

Possible values for items in the sample output

Table 181. query_cluster output values

Item	Possible values
Mode	Cluster mode: • 1_node • 3_node • 5_node
State	State of the cluster: • Normal • Degraded (one or more cluster members are down or inactive)
Active	The number of cluster members that are active. If all members are in normal state, this number equals the number of cluster members.
Replicas	The number of MDM repository holders (the primary MDM, plus the number of secondary MDMs in normal state)
Secondary MDM status	• Normal • Error • Disconnected • Not synchronized
Tiebreaker status	• Normal • Error • Disconnected
Standby MDM	Role (manager or tiebreaker)

query_performance_parameters

Retrieve performance parameters. The output displays the configurations of performance-related parameters for the performance profile that is currently active. When no parameters are added to the command, the name of the active profile is shown.

Syntax

```
scli --query_performance_parameters
  [--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_ip <IP>] | --all_sds]
[--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_name <NAME> | --
all_sdc] [--peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--
peer_system_port <PORT>] | --peer_system_object_id <ID> | --all_peer_system] [--sdr_id
<ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] | --all_sdr] [--print_all]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--all_sds

Query SDS performance parameters from all SDSs.

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC Global Unique Identifier

--sdc_ip *<IP>*

SDC IP address

--all_sdc

Query SDC performance parameters from all SDCs.

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system Name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--all_peer_system

Query replication peer system performance parameters from all replication peer systems.

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

Port assigned to the SDR

--all_sdr

Query SDR performance parameters from all SDRs.

--print_all

Print all performance parameters, including those with active profile default values.

Example

```
scli --query_performance_parameters --print_all
```

Sample output

```
MDM Performance Parameters:
        Active profile: high_performance
                                                                    Current Value
Profile Value
        MDM_NUMBER_SDC_RECEIVE_UMT                                          5
5
```

```
        MDM_NUMBER_SDS_RECEIVE_UMT                          20
20
        MDM_NUMBER_SDS_SEND_UMT                             10
10
        MDM_NUMBER_SDS_KEEPALIVE_RECEIVE_UMT                10
10
        MDM_SDS_CAPACITY_COUNTERS_UPDATE_INTERVAL            1
1
        MDM_SDS_CAPACITY_COUNTERS_POLLING_INTERVAL           5
5
        MDM_SDS_VOLUME_SIZE_POLLING_INTERVAL                15
15
        MDM_SDS_VOLUME_SIZE_POLLING_RETRY_INTERVAL           5
5
        MDM_NUMBER_SDS_TASKS_UMT                         1536
1536
        MDM_INITIAL_SDS_SNAPSHOT_CAPACITY                1024
1024
        MDM_SDS_SNAPSHOT_CAPACITY_CHUNK_SIZE             5120
5120
        MDM_SDS_SNAPSHOT_USED_CAPACITY_THRESHOLD           50
50
        MDM_SDS_SNAPSHOT_FREE_CAPACITY_THRESHOLD          200
200
        MDM_NET_ALLOC_RCV_BUFFER_WAIT_MS                  500
500
        MDM_NET_BREAK_DO_IO_LOOP                            5
5
        MDM_FGL_INITIAL_SNAPSHOT_CAPACITY                5120
5120
        MDM_FGL_SNAPSHOT_CAPACITY_CHUNK_SIZE             5120
5120
        MDM_FGL_SNAPSHOT_USED_CAPACITY_THRESHOLD           50
50
        MDM_FGL_SNAPSHOT_FREE_CAPACITY_THRESHOLD          200
200
        MDM_RENAME_DISABLE_MODE_ON_SNAP_DELETION            0
0
        MDM_PORT_FLAP_FEATURE_ENABLED                       1
1
        MDM_DISQUALIFICATION_EPOCH_SEC                    120
120
        MDM_DISCONNECTIONS_ERT                              3
3
        MDM_DISCONNECTIONS_ERTF_MS                      60000
60000
        MDM_HIGH_LATENCY_ERT                               50
50
        MDM_HIGH_LATENCY_ERTF_MS                        60000
60000
        MDM_PACKET_LOSS_ERT                                50
50
        MDM_PACKET_LOSS_ERTF_MS                         60000
60000
        RPL_MDM_PORT_FLAP_FEATURE_ENABLED                   0
0
        RPL_MDM_DISQUALIFICATION_EPOCH_SEC                120
120
        RPL_MDM_DISCONNECTIONS_ERT                          3
3
        RPL_MDM_DISCONNECTIONS_ERTF_MS                  60000
60000
        RPL_MDM_HIGH_LATENCY_ERT                           50
50
        RPL_MDM_HIGH_LATENCY_ERTF_MS                    60000
60000
        RPL_MDM_PACKET_LOSS_ERT                            50
50
        RPL_MDM_PACKET_LOSS_ERTF_MS                     60000
60000
        RPL_MDM_CAP_MGR_JOURNAL_VOLUME_ALLOC_SIZE_GB        8
8
        RPL_MDM_SLEEP_TIME_FOR_CE_ANALYZER_MS             500
```

```
500
        RPL_MDM_SYNCED_OPERATION_TIMEOUT_SECONDS                        30
30
        RPL_MDM_CAP_MGR_MONITOR_TIMEOUT_MS                            1000
1000
        RPL_MDM_NUM_CG_TASKS_UMT                                         1
1
        RPL_MDM_NUM_PAIR_TASKS_UMT                                       1
1
        RPL_MDM_NUM_SDR_RECEIVE_UMT                                     10
10
        RPL_MDM_NUM_SDR_SEND_UMT                                        10
10
        RPL_MDM_NUM_SDR_KEEPALIVE_RECEIVE_UMT                           10
10
        RPL_MDM_SDR_COUNTERS_POLLING_INTERVAL_SECONDS                    1
1
        RPL_MDM_CAP_MGR_INCREASE_THRESHOLD_SIZE_GB                       4
4
        RPL_MDM_CAP_MGR_DECREASE_CAP_TIMEOUT_MS                       2000
2000
        RPL_MDM_RPO_DRIFT_UNTIL_BARRIER                                 64
64
        RPL_MDM_NUM_SDR_TASKS_UMT                                      512
512
        RPL_MDM_NUM_PEER_MDM_TASKS_UMT                                 126
126
        RPL_MDM_NUM_PEER_MDM_RECEIVE_UMT                                10
10
        RPL_MDM_NUM_PEER_MDM_SEND_UMT                                   10
10
        RPL_MDM_CURRENT_BARRIER_MAX_NUM_IOS_THRESHOLD           1999000000
1999000000
        RPL_MDM_CURRENT_BARRIER_MAX_SIZE_MB_THRESHOLD              65536
65536
        RPL_MDM_SDR_SESSION_CAPACITY_LOW_WATERMARK_MB              20480
20480
        RPL_MDM_SDR_SESSION_CAPACITY_HIGH_WATERMARK_MB              4096
4096
        RPL_MDM_CAP_MGR_CRITICAL_CAPACITY_THRESHOLD_MB                32
32
        RPL_MDM_CAP_MGR_SWITCH_SESSIONS_INTERVAL_MS                 120000
120000
        RPL_MDM_CAP_MGR_MINOR_UTILIZATION_THRESHOLD                     75
75
        RPL_MDM_CAP_MGR_MAJOR_UTILIZATION_THRESHOLD                     80
80
        RPL_MDM_CAP_MGR_VERY_HIGH_UTILIZATION_THRESHOLD                 90
90
        RPL_MDM_CAP_MGR_CRITICAL_UTILIZATION_THRESHOLD                  95
95
```

query_properties

Retrieve any set of properties on any set of objects of the specified type.

The properties that can be queried depend on the type of the object that is being queried. For example, SDS has different attributes than volume.

Several preset combinations are available for use with this command. A preset is a predefined set of properties, usually with some logical relationship, which specifies a group of commonly queried properties. The available presets are also dependent on the object type. All object types have at least one preset called `all`, which queries all the properties for that type.

Syntax

```
scli --query_properties --object_type <TYPE> (--object_id <ID> | --all_objects) (--
properties <PROPS> | --preset <PRESET>)
```

```
   [--group_by_property]
   [--display_property_options]
   [--flat_list]
```

Parameters

--object_type *<TYPE>*

Object type to query. One of:

SYSTEM

PROTECTION_DOMAIN

STORAGE_POOL

FAULT_SET

SDS

HOST

VOLUME

DEVICE

VTREE

MDM

SP_SDS

ACCELERATION_POOL

SNAPSHOT_POLICY

SDR

PEER_MDM

REPLICATION_PAIR

REPLICATION_CONSISTENCY_GROUP

SDT

NOTE: To view detailed presets and properties options for a type, enter `scli --query_properties --object_type <TYPE>`, where *<TYPE>* is one of the above options.

--object_id *<ID>*

Comma-separated list of object ID used for querying one or more specific objects. Omit the space after each comma.

--all_objects

Query all objects of the specified type.

--properties *<PROPS>*

Query a comma-separated list of properties. Omit the space after each comma. See the scli help output for a list of properties that can be queried per object type.

--preset *<PRESET>*

Query a predefined set of properties. Omit the space after each comma. See the scli help output for the list of presets that can be used per object type.

--group_by_property

Group results by property type, rather than by object (the default).

--display_property_options

Display all available properties grouped by object type.

--flat_list

Print results as a flat list rather than by group (better for searching).

NOTE: Use `scli --query_properties --object_type <TYPE>` with one of the valid types above, to get detailed presets and properties options.

Example

```
scli --query_properties --object_type SDS --all_objects --properties
IPS,PORT,PROTECTION_DOMAIN_ID,RMCACHE_ENABLED
```

Sample output

```
# scli --query_properties --object_type SDS --all_objects --properties
IPS,PORT,PROTECTION_DOMAIN_ID,RMCACHE_ENABLED
SDS 5fcafa4300000001:
        IPS                                             10.55.105.40,10.76.30.11
        PORT                                            7072
        PROTECTION_DOMAIN_ID                            a080539500000000
        RMCACHE_ENABLED                                 Yes

SDS 5fcafa4600000004:
        IPS                                             10.55.105.43,10.76.30.15
        PORT                                            7072
        PROTECTION_DOMAIN_ID                            a080539500000000
        RMCACHE_ENABLED                                 Yes
```

Object presets

The following table provides a list of presets that can be used per object type.

Table 182. Object presets for query_properties

Object type	ALL	CAPACITY	RPL_JOURNAL_CAPACITY	RPL_BW_COUNTERS	IO	RAM_CACHE	PERFORMANCE_PARAMETERS	RF_CACHE
SYSTEM	x	x	N/A	N/A	x	x	x	x
PROTECTION_DOMAIN	x	x	N/A	N/A	x	x	N/A	x
STORAGE_POOL	x	x	x	N/A	x	x	x	x
FAULT_SET	x	x	N/A	N/A	x	x	N/A	x
SDS	x	x	N/A	N/A	x	x	x	x
SDC	x	N/A	N/A	N/A	x	N/A	x	N/A
VOLUME	x	N/A	N/A	N/A	x	x	N/A	N/A
DEVICE	x	x	N/A	N/A	x	N/A	N/A	x
VTREE	x	x	N/A	N/A	N/A	N/A	N/A	N/A
MDM	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
SP_SDS	x	x	N/A	N/A	x	N/A	N/A	x
ACCELERATION_POOL	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
SNAPSHOT_POLICY	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
SDR	x	N/A	x	x	N/A	N/A	x	N/A

Table 182. Object presets for query_properties (continued)

Object type	ALL	CAPACITY	RPL_JOURNAL_CAPACITY	RPL_BW_COUNTERS	IO	RAM_CACHE	PERFORMANCE_PARAMETERS	RF_CACHE
PEER_MDM	x	N/A	N/A	N/A	N/A	N/A	x	N/A
REPLICATION_PAIR	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
REPLICATION_CONSISTENCY_GROUP	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A

query_remote_read_only_limit_state

Query the MDM's remote access restriction state. If the state is enabled, remote users may only issue read-only commands to the MDM. If the state is disabled, all command types may be issued to the MDM by remote clients.

Syntax

```
scli --query_remote_read_only_limit_state
```

Parameters

None.

Example

```
scli --query_remote_read_only_limit_state
```

query_system_limits

Retrieve information about the system limits.

Syntax

```
scli --query_system_limits
```

Parameters

None.

Example

```
scli --query_system_limits
```

Sample output

```
System limits (45 entries)
  Number of protections domains:  Max: 256
  Number of storage pools:  Max: 1024
  Number of fault sets:  Max: 16384
  Number of SDSs:  Max: 512
  Number of SDCs:  Max: 1024
  Number of volume trees:  Max: 131072
  Number of volumes:  Max: 131072
  Number of snapshots:  Max: 133120
  Number of system users:  Max: 256
  Number of LDAP services:  Max: 8
  Number of sessions:  Max: 128
  Number of rsyslog clients:  Max: 16
  Number of storage pools per protection domain:  Max: 64
  Number of SDSs per protection domain:  Max: 128
  Number of fault sets per protection domain:  Max: 64
  Number of SDSs per fault set:  Max: 64
  Number of devices per SDS:  Max: 64
  Number of devices per storage pool:  Max: 300
  Number of volumes per medium-granularity volume tree:  Max: 127
  Number of volumes per fine-granularity volume tree:  Max: 127
  Number of snapshots per medium-granularity volume tree:  Max: 254
  Number of snapshots per fine-granularity volume tree:  Max: 254
  Number of volumes mapped to a single SDC:  Max: 8192
  Number of volumes per protection domain:  Max: 32768
  Number of LDAP groups per LDAP service:  Max: 8
  Number of IPs per MDM:  Min: 1  Max: 4
  Number of IPs per SDS/SDC:  Min: 1  Max: 8
  Single volume size in GB:  Min: 8  Max: 1048576
  Total size of all volumes in Storage Pool in GB:  Max: 4194303
  Total size of all volumes in System (GB):  Max: 16777216
  Number of VTrees per protection domain:  Max: 32768
  Number of acceleration devices in acceleration pool:  Max: 512
  Number of acceleration pools per protections domain:  Max: 32
  Number of acceleration devices per acceleration pool that belong to the same SDS:
Max: 8
  Capacity per medium granularity device in GB:  Min: 90
  Capacity per fine granularity device in GB:  Min: 128
  Capacity per SDS in GB:  Max: 131072
  Number of Replication Consistency Groups:  Max: 1024
  Number of Replication Destination Systems:  Max: 1
  Total Number of SDRs per system:  Max: 128
  Total Number of Replication Pairs in system:  Max: 32768
  Total Number of concurrent Move Pairs operations:  Max: 512
  Single replication volume size in GB:  Min: 8  Max: 65536
  Number of mappings between volumes and SDCs:  Max: 262143
```

query_upgrade

Retrieve information about the upgrade process.

Syntax

```
scli --query_upgrade
   [--show_sdc_info]
```

Parameters

--show_sdc_info

Include SDC information

Example

```
scli --query_upgrade --show_sdc_info
```

refresh_mdm_cluster_capabilities

Refresh the exposed cluster node capabilities.

This command is usually used to instruct the cluster nodes to check whether LDAP and SSL libraries are installed.

Syntax

```
scli --refresh_mdm_cluster_capabilities
```

Parameters

Example

```
scli --refresh_mdm_cluster_capabilities --mdm_port 6611
```

remove_standby_mdm

Remove a standby MDM or standby tiebreaker, that is not part of the cluster, from the system.

Syntax

```
scli --remove_standby_mdm (--remove_mdm_id <ID> | --remove_mdm_ip <IP> [--
remove_mdm_port <PORT>] | --remove_mdm_name <NAME>)
```

Parameters

--remove_mdm_id *<ID>*
: ID of the MDM to remove

--remove_mdm_ip *<IP>*
: IP address of the MDM to remove

--remove_mdm_port *<PORT>*
: Port associated with the MDM to remove

--remove_mdm_name *<NAME>*
: Name of the MDM to remove

Example

```
scli --remove_standby_mdm --remove_mdm_name mdm153
```

rename_mdm

Rename an MDM cluster node.

Syntax

```
scli --rename_mdm (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port
<PORT>] | --target_mdm_name <NAME>) --new_mdm_name <NAME>
```

Parameters

--target_mdm_id *<ID>*

ID of the MDM to be renamed

--target_mdm_ip *<IP>*

IP address of the MDM to be renamed

--target_mdm_port *<PORT>*

Port of the MDM to be renamed

--target_mdm_name *<NAME>*

Current name of the MDM to be renamed

--new_mdm_name *<NAME>*

New MDM name. The MDM name must be unique. If a new name is not provided, the old name is deleted.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --rename_mdm --new_mdm_name mdm3 --target_mdm_name mdm2
```

rename_system

Name or rename the system.

To determine the current system name, use the `query_cluster` or `query_all` command.

Syntax

```
scli --rename_system --new_name <NAME>
```

Parameters

--new_name *<NAME>*

New name to associate with the system.

Example

```
scli --rename_system --new_name sio-prod-sys2
```

query_all_system_networks

Retrieve information about all system networks.

query_system_network

Retrieve information about a system network.

Syntax

```
scli --query_system_network (--network_id <ID> | --network_name <NAME>)
```

Parameters

--network_id *<ID>*
System network ID

--network_name *<NAME>*
System network name

add_system_network

Add a system network to the system.

Syntax

```
scli --add_system_network --network_ip <IP> (--network_mask <MASK> | --network_prefix
<PREFIX>) [--network_name <NAME>] [(--host_group_id <ID> | --host_group_name <NAME>)]
```

Parameters

--network_ip *<ID>*
The new network IP

--network_mask *<MASK>*
The new network mask

--network_prefix *<PREFIX>*
The new network prefix

--network_name *<NAME>*
A name to be assigned to the system network

--network_prefix *<PREFIX>*
The new network prefix

--host_group_id *<ID>*

A comma separated list of host group IDs that have access to the system network

--host_group_name *<NAME>*

A comma separated list of host group names that have access to the system network

rename_system_network

Name or rename a system network.

Syntax

```
scli --rename_system_network (--network_id <ID> | --network_name <NAME>) --new_name
<NAME>
```

Parameters

--network_id *<ID>*

System network ID

--network_name *<NAME>*

System network name

--new_name *<NAME>*

A new name to be assigned to the system network

remove_system_network

Remove a system network from the system.

Syntax

```
scli --remove_system_network (--network_id <ID> | --network_name <NAME>)
```

Parameters

--network_id *<ID>*

System network ID

--network_name *<NAME>*

System network name

replace_cluster_mdm

Replace members of the MDM cluster. Up to half (rounded down) of the cluster members can be replaced at one time.

Syntax

```
scli --replace_cluster_mdm ((--add_secondary_mdm_id <ID> |
        --add_secondary_mdm_ip <IP> [--add_secondary_mdm_port <PORT>] |
        --add_secondary_mdm_name <NAME>)| (--remove_secondary_mdm_id <ID> |
        --remove_secondary_mdm_ip <IP> [--remove_secondary_mdm_port <PORT>] |
```

```
        --remove_secondary_mdm_name <NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP>
        [--add_tb_port <PORT>] | --add_tb_name <NAME>) | (--remove_tb_id <ID> | --
remove_tb_ip
        <IP> [--remove_tb_port <PORT>] | --remove_tb_name <NAME>)) [--allow_leave_failed]
        [--i_am_sure]
```

Parameters

--add_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --replace_cluster_mdm --add_secondary_mdm_name mdm33 --remove_secondary_mdm_name
mdm31 --add_tb_name tb44 --remove_tb_name tb41
```

replace_mdm_security_files

Replace the security certificate and key of the MDM. The replacement certificate can be either a self-generated and self-signed certificate, or a certificate generated and signed by an external certificate authority. Security permissions are required in order to run this command.

NOTE: Running the command on the primary MDM causes the MDM to restart. It may also cause a brief, single point of failure period due to an MDM switch over.

Syntax

```
scli --replace_mdm_security_files (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

Example

```
scli --replace_mdm_security_files --target_mdm_name mdm153
```

set_component_authentication_properties

Enable or disable component authentication properties.

Syntax

```
scli --set_component_authentication_properties (--use_authentication | --
dont_use_authentication)
```

Parameters

None.

--use_authentication

Use authentication between system components

--dont_use_authentication

Do not use authentication between system components

Example

```
scli --set_component_authentication_properties --use_authentication
```

set_management_client_communication

Enable or disable secure communications between the MDM and the management client.

Syntax

```
scli --set_management_client_communication (--enable_client_secure_communication | --
disable_client_secure_communication)
```

Parameters

--enable_client_secure_communication

Enable the use of security when communicating with management clients.

--disable_client_secure_communication

Disable the use of security when communicating with management clients.

Example

```
scli --set_management_client_communication --enable_client_secure_communication
```

set_performance_parameters

Apply performance profiles to system components. You can apply separate profiles for SDSs, SDCs, the MDM cluster, replication peer systems, and SDRs.

When a container is provided in the command (protection domain, fault set, all SDSs, all SDCs), the command configures all objects in the container. Objects that are added later will not be configured.

NOTE: If you change the performance profile of an SDS located on an SVM, you must afterwards perform manual memory allocation on the SVM.

Syntax

```
scli --set_performance_parameters ([--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>] | --protection_domain_id <ID> | --protection_domain_name
<NAME> | --fault_set_id <ID> | ((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --fault_set_name <NAME>) | --all_sds] [--sdc_id <ID> | --sdc_name <NAME>
| --sdc_guid <GUID> | --sdc_ip <IP> | --all_sdc] [(((--protection_domain_id <ID>
| --protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>)] [--peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP>
[--peer_system_port <PORT>] | --peer_system_object_id <ID> | --all_peer_system] [--
sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] | --all_sdr] [--
apply_to_mdm]) (--profile <PROFILE>)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_id *<ID>*
Fault set ID

--fault_set_name *<NAME>*
Name of the new fault set

--all_sds
Set SDS-related parameters on all SDSs.

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC global unique identifier

--sdc_ip *<IP>*
SDC IP address

--all_sdc
Set SDS-related parameters on all SDCs.

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--all_peer_system

Set Replication peer system related parameters to all replication peer systems.

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

SDR port

--all_sdr

Set SDR-related parameters on all SDRs.

--apply_to_mdm

Apply the performance profile to the MDM. This flag is only applicable when setting the profile.

--profile {compact | high_performance}

Set the performance profile as compact or high_performance. The high performance profile configures a predefined set of parameters for very high performance use cases.

NOTE: For a complete list of parameters controlled by the profiles, see the *PowerFlex Performance Fine-Tuning Technical Notes*.

Example

```
scli --set_performance_parameters --all_sds --all_sdc --profile high_performance
```

start_upgrade

Start the upgrade process.

Syntax

```
scli --start_upgrade
```

Parameters

None.

Example

```
scli --start_upgrade
```

switch_cluster_mode

Switch the MDM cluster from one operating mode or configuration to another.

Syntax

```
scli --switch_cluster_mode --cluster_mode <MODE> ((--add_secondary_mdm_id <ID> |
        --add_secondary_mdm_ip <IP> [--add_secondary_mdm_port <PORT>] |
        --add_secondary_mdm_name <NAME>)| (--remove_secondary_mdm_id <ID> |
        --remove_secondary_mdm_ip <IP> [--remove_secondary_mdm_port <PORT>] |
        --remove_secondary_mdm_name <NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP>
        [--add_tb_port <PORT>] | --add_tb_name <NAME>) | (--remove_tb_id <ID> | --
remove_tb_ip
        <IP> [--remove_tb_port <PORT>] | --remove_tb_name <NAME>)) [--allow_leave_failed]
        [--i_am_sure]
```

Parameters

--cluster_mode *<MODE>*

Cluster operation mode. Can be one of the following:

1_node

The MDM cluster will operate in single-node mode.

NOTE: Single-node mode should only be used temporarily during maintenance activities. Single-node mode does not provide failover protection for the MDM.

3_node

The MDM cluster will operate in 3-node mode.

5_node

The MDM cluster will operate in 5-node mode.

--add_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --switch_cluster_mode --cluster_mode 5_node --add_secondary_mdm_ip 192.168.1.164 --
add_tb_ip 192.168.1.134
```

switch_mdm_ownership

Switch MDM cluster ownership from the current primary MDM to a different MDM.

Syntax

```
scli --switch_mdm_ownership (--new_primary_mdm_id <ID> | --new_primary_mdm_ip
       <IP> [--new_primary_mdm_port <PORT>]| --new_primary_mdm_name <NAME>)
```

Parameters

--new_primary_mdm_id *<ID>*

MDM ID. Must be an MDM with manager role.

--new_primary_mdm_ip *<IP>*

MDM IP. Must be an MDM with manager role.

--new_primary_mdm_port *<PORT>*

MDM port. Must be an MDM with manager role.

--new_primary_mdm_name *<NAME>*

MDM name. Must be an MDM with manager role.

Example

```
scli --switch_mdm_ownership --new_primary_mdm_ip 192.168.1.164
```

generate_mdm_mno_csr

Generate a certificate signing request (CSR) file for the given address. The file is saved at <generated CSR path> and also locally in the configuration directory of the MDM.

Syntax

```
scli --generate_mdm_mno_csr --csr_file_path <generated CSR path>
```

Parameters

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

replace_mdm_mno_certificate

Replace the old MDM certificate with a new certificate at a given address.

Syntax

```
scli --replace_mdm_mno_certificate (--target_mdm_id <ID> | --target_mdm_ip
<IP> [--target_mdm_port <PORT>] | --target_mdm_name <NAME>) --certificate_file
<MDM_CERT_FILENAME>
```

Parameters

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

query_all_network_sets

Retrieve information about all network sets.

query_network_set

Retrieve information about a network set.

Syntax

```
scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

modify_network_set

Modify the list of system networks of an existing network set.

Syntax

```
scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [--
network_id <ID> | --network_name <NAME>]
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

--network_id *<ID>*
A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*
A comma separated list of system network names to be replaced with the old one

rename_network_set

Name or rename a network set.

Syntax

```
scli --rename_network_set (--network_set_id <ID> | --network_set_name <NAME>) --new_name
<NAME>
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

--new_name *<NAME>*
A new name to be assigned to the network set

remove_network_set

Remove a network set from the system.

Syntax

```
scli --remove_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
ID of the protection domain to which the fault set is added

--network_set_name *<NAME>*
Network set name

55

Volume commands

This section contains commands for managing volumes.

add_volume

Add a volume to the PowerFlex system.

You can create a volume when the requested capacity is available and system capacity is balanced. To start allocating volumes, at least three SDS nodes must be configured in the system.

The created volume cannot be used until it is mapped to at least one SDC.

Syntax

```
scli --add_volume (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
storage_pool_name <NAME>) | --storage_pool_id <ID>) --size_gb <SIZE>
  [--volume_name <NAME>]
  [--compression_method none | normal]
  [--thin_provisioned | --thick_provisioned | <BLANK>]
  [--use_rmcache | --dont_use_rmcache | <BLANK>]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--size_gb *<SIZE>*

Volume size, in GB. Basic allocation granularity is 8 GB.

--volume_name *<NAME>*

Name to be associated with the added volume

--compression_method {none | normal}

Indicates the compression method to be used, relevant only for fine granularity data layouts.

--thin_provisioned | --thick_provisioned | <BLANK>

Specifies whether the specified volume will be thin provisioned or thick provisioned (the default). If left blank, the default of the storage pool is used (thick provisioned for medium granularity, thin provisioned for fine granularity).

--use_rmcache | --dont_use_rmcache | <BLANK>

Specifies whether the specified volume will use read RAM cache for caching. Default: `--use_rmcache`. If left blank, use the value of the storage pool.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --add_volume --size_gb 8 --volume_name vol_1 --
protection_domain_name pd1 --storage_pool_name sp1
```

Volume names

Assign each volume a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the volume's ID.

Each volume name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

clear_scsi2_reservation

Clear SCSI2 reservation from an SDC.

Syntax

```
scli --clear_scsi2_reservation (--volume_ID <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id *<ID>*

ID of the volume where SCSI2 reservation from the SDC will be reset

--volume_name *<NAME>*

Name of the volume where SCSI2 reservation from the SDC will be reset

--wait

Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --clear_scsi2_reservation --volume_name public --sdc_ip 10.76.60.233 --wait
```

clear_scsi3_reservation

Clear SCSI3 reservations from all SDCs.

Syntax

```
scli --clear_scsi3_reservation (--volume_id <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id ***<ID>***

ID of the volume for which SCSI3 reservation will be reset

--volume_name ***<NAME>***

Name of the volume for which SCSI3 reservation will be reset

--wait

Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --clear_scsi3_reservation --volume_name temp1 --i_am_sure
```

map_volume_to_host

Map a volume to SDC or to NVMe host.

Syntax

```
scli --map_volume_to_host (--volume_id <ID> | --volume_name <NAME>) (--host_id <ID>
| --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip <IP>) [--
access_mode <ACCESS_MODE>] [--allow_multi_map]
```

Add the attribute, `--volume class` to map volumes to host:

```
scli --map_volume_to_host --volume_id <ID> --sdc_guid <GUID> --volume_class nascluster
```

For example:

```
scli --map_volume_to_host --volume_id 2f362c3800000100 --sdc_guid 01D16FC2-342B-4CC2-
A050-DCD97B4E0GH7 --volume_class nascluster
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--access_mode *<ACCESS_MODE>*

Mapping access mode. Can be one of the following: read_write (default), read_only, no_access.

--allow_multi_map

Allow this volume to be mapped to more than one host.

unmap_volume_from_host

Unmap a volume from one host or all hosts.

Syntax

```
scli --unmap_volume_from_host (--volume_id <ID> | --volume_name <NAME>) (--host_id <ID>
| --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip <IP> | --
all_host) [--i_am_sure]
```

Add the attributes `--volume class` and `--sdc_guid` to unmap volumes from the host:

```
scli --unmap_volume_from_host --volume_id <ID> --sdc_guid <GUID> --volume_class
nascluster
```

For example:

```
scli --unmap_volume_from_host --volume_id 2f362c3800000100 --sdc_guid 01D16FC2-342B-4CC2-
A050-DCD97B4E0GH7 --volume_class nascluster
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe Host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--all_host

Unmap the volume from all hosts.

--i_am_sure

Preemptive approval

modify_volume_capacity

Increase the capacity of a volume. You can increase (but not decrease) a volume capacity at any time, as long as there is enough capacity for the volume size to grow.

Syntax

```
scli --modify_volume_capacity (--volume_id <ID> | --volume_name <NAME>) --size_gb <SIZE>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--size_gb *<SIZE>*

New volume size, in GB. Basic allocation granularity is 8 GB.

Example

```
scli --mdm_ip 192.168.1.200 --modify_volume_capacity
--volume_name vol_1 --size_gb 150000
```

overwrite_volume_content

Overwrite the destination volume content with the content of the source volume. Both volumes must belong to the same V-Tree.

Syntax

```
scli --overwrite_volume_content (--source_vol_id <ID> | --source_vol_name <NAME>) (--
destination_vol_id <ID> | --destination_vol_name <NAME>)
[--overwrite_locked_auto_snapshot]
```

Parameters

--source_vol_id *<ID>*

Source volume ID

--source_vol_name *<NAME>*

Source volume name

--destination_vol_id *<ID>*

Destination volume ID

--destination_vol_name *<NAME>*

Destination volume name

--overwrite_locked_auto_snapshot

Use this flag to overwrite volume content when the destination volume is a locked auto snapshot

Example

```
scli --overwrite_volume_content --source_vol_name vol1 --destination_vol_name vol2
```

query_all_volumes

Retrieve information about all volumes in the system. You can optionally filter the query results by protection domain, storage pool, or snapshot consistency group ID.

Syntax

```
scli --query_all_volumes
   [(((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
storage_pool_name <NAME>) | --storage_pool_id <ID>) | --scg_id <ID>]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--scg_id *<ID>*

Snapshot consistency group ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_volumes --protection_domain_id dc65bd9900000000
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Storage Pool 3df1cd9600000000 Name: sp1
 Volume ID: f6351e3700000003 Name: vol1_114 Size: 16.0 GB (16384 MB) Mapped to 1 SDC
Thin-provisioned  intercepted for replication
 Volume ID: f6351e3800000004 Name: vol2_114 Size: 8.0 GB (8192 MB) Not mapped Thin-
provisioned
 Volume ID: f6351e3900000005 Name: vol3_pol1 Size: 8.0 GB (8192 MB) Not mapped Thin-
provisioned
 Volume ID: f6683c2700000002 Name: N/A Size: 8.0 GB (8192 MB) Not mapped Auto Snapshot
of vol3_pol1

Protection Domain fcc9af5600000001 Name: pd2
Storage Pool 3df1cd9700000001 Name: sp2
 <No volumes defined>
```

```
Protection Domain fcc9d66500000002 Name: rrr
 <No volumes defined>
Query-all-volumes returned 4 volumes
```

query_volume

Retrieve information about the specified volume.

Syntax

```
scli --query_volume (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_volume --volume_id f6351e3900000005
```

Sample output

```
>> Volume ID: f6351e3900000005 Name: vol3_pol1
   Provisioning: Thin
   Size: 8.0 GB (8192 MB)
   Total base user data: 0 Bytes
   Storage Pool 3df1cd9600000000 Name: sp1
   Protection Domain fcc9af5500000000 Name: pd1
   Creation time: 2021-01-31 13:00:40
   Number of Direct Snapshots: 2
   Reads:  0 IOPS 0 Bytes per second
   Writes: 0 IOPS 0 Bytes per second
   Data layout: Medium granularity
   Mapping:
      SDC ID: c27ab52600000001 IP: 10.55.137.119 Name: N/A Access mode: READ_WRITE
   No SCSI reservations
   The volume is a source volume of Snapshot Policy: aaa4278700000000
   VTree migration info:
      Migration status: Paused
      Pause Reason: No space in destination Storage Pool
      Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain: pd1 ID:
fcc9af5500000000
      Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain: pd2 ID:
fcc9af5600000001
      Conversion type: No conversion
      Queue position: 1
      Progress percentage : 0%
   Doesn't use RAM Read Cache
   Replication info:
      Replication state: Not being Replicated
```

query_volume_sdc_meters

Query the IOPS, bandwidth, and latency counters for volume mappings. You can either specify a volume, an SDC, or both. For volume only, display metrics for all mappings of the specified volume. For SDC only, display metrics for all mappings of volumes to the specified SDC. For both volume and SDC, display metrics for a specific mapping.

Syntax

```
scli --query_volume_sdc_meters [--volume_id <ID> | --volume_name <NAME>] [--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>]
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC global unique identifier

--sdc_ip *<IP>*
SDC IP address

Example

```
scli --query_volume_sdc_meters --volume_id f6351e3700000003
```

Sample output

```
Information on 1 mapping:

Volume: f6351e3700000003 SDC: c27ab52600000001
Reads:  426 IOPS 212.8 MB (217907 KB) per-second
        213 IOPS 22.033 ms average latency
Writes: 450 IOPS 224.6 MB (230000 KB) per-second
        225 IOPS 261.137 ms average latency
```

query_volume_sdt_meters

Query the IOPS, bandwidth, and latency counters for volume mappings. You can either specify a volume, an SDT, or both. For volume only, display metrics for all mappings of the specified volume. For SDT only, display metrics for all mappings of volumes to the specifed SDT. For both volume and SDT, display metrics for a specific mapping.

Syntax

```
scli --query_volume_sdt_meters [(--volume_id <ID> | --volume_name <NAME>)] [(--sdt_id
<ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>])]
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--sdt_id *<ID>*
SDC ID

--sdt_name *<NAME>*
SDC name

--sdt_ip *<IP>*
SDC IP address

--sdt_port *<PORT>*
Port assigned to the SDT

remove_volume

Remove a PowerFlex volume.

NOTE: Removing a volume erases all the data on the corresponding volume.

Before removing a volume, you must ensure that it is not mapped to any SDCs. If the volume is mapped to SDCs, unmap the volume using the `unmap_volume_from_sdc` command or the PowerFlex GUI, or the vSphere plug-in (where applicable) before removing it.

Syntax

```
scli --remove_volume (--volume_id <ID> | --volume_name <NAME>)
{--remove_entire_snapshot_tree | --remove_with_descendant_snapshots | --
remove_descendant_snapshots_only}
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*
ID of the volume to remove

--volume_name *<NAME>*
Name of the volume to remove

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

[Options]

Additional optional parameters:

--remove_entire_snapshot_tree

Remove the entire volume tree (V-Tree) attached to the specified volume, including parents and siblings.

--remove_with_descendant_snapshots

Remove the specified volume, and all volumes that were created as snapshots of the specified volume or one of its descendants.

--remove_descendant_snapshots_only

Remove snapshots created from the specified volume, but do not remove the specified volume.

NOTE: Before removing a V-Tree, you must unmap every volume in the V-Tree.

Example

```
scli --mdm_ip 192.168.1.200 --remove_volume --volume_name vol_1
```

rename_volume

Changes the name of a volume. You can use this command at any time.

Syntax

```
scli --rename_volume --volume_id <ID>| --volume_name <NAME> --new_name <NAME>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--new_name *<NAME>*

New name to assign to the volume

Example

```
scli --mdm_ip 192.168.1.200 --rename_volume
--volume_name vol1 --new_name vol_new_1
```

set_volume_rmcache_usage

Control the use of RMcache in the specified volume.

You can only configure one volume at a time. If you want to ensure that all I/Os for this volume are cached, the relevant storage pool should be configured to use RMcache, and the relevant SDSs should all have RMcache enabled.

NOTE: You can also enable RMcache at the volume level during initial creation of the volume using the `--add_volume` command.

Syntax

```
scli --set_volume_rmcache_usage (--volume_id <ID>| --volume_name <NAME>)(--use_rmcache |
--dont_use_rmcache)
  [--i_am_sure]
```

Parameters

`--volume_id <ID>`

Volume ID

`--volume_name <NAME>`

Volume name

`--use_rmcache`

Use read RAM cache on the volume.

`--dont_use_rmcache`

Do not use RMcache on the volume. This is the default setting.

`--i_am_sure`

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_volume_rmcache_usage --volume_name vol_10 --use_rmcache
```

set_volume_mapping_access_mode

Set the mapping access mode for the specified volume.

Syntax

```
scli --set_volume_mapping_access_mode (--volume_id <ID> | --volume_name <NAME>) (--
host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip
<IP> | --all_host) --access_mode <ACCESS_MODE>
```

Parameters

`--volume_id <ID>`

Volume ID

`--volume_name <NAME>`

Volume name

`--host_id <ID>`

Host ID

`--host_name <NAME>`

host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--sdc_guid *<GUID>*
SDC GUID

--sdc_ip *<IP>*
SDC IP address

--all_host
Use this flag to apply the volume access mode to all hosts

--access_mode {read_write | read_only | no_access}
The access mode

set_volume_access_mode_limit

Set the highest mapping access mode allowed for a volume.

Syntax

```
scli --set_volume_access_mode_limit (--volume_id <ID> | --volume_name <NAME>) --
access_mode_limit <ACCESS_MODE>
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--access_mode_limit {read_write | read_only}
Access mode limit.

Example

```
scli --mdm_ip 192.168.1.200 set_volume_access_mode_limit --volume_id a4b5862000000000 --
access_mode_limit read_only
```

query_auto_provisioned_destination_volume_suffix

Query the suffix added to auto-provisioned destination volumes.

Syntax

```
scli --query_auto_provisioned_destination_volume_suffix
```

modify_auto_provisioned_destination_volume_suffix

Modify the suffix added to auto-provisioned destination volumes.

Syntax

```
scli --modify_auto_provisioned_destination_volume_suffix --
destination_volume_name_suffix <suffix>
```

Parameters

--destination_volume_name_suffix <suffix>

The new suffix. The suffix value is mandatory but an empty string ("") is permitted.

add_snapshot_policy

Create a new snapshot policy to enable you to define the number of snapshots taken at a given time for one or more volumes.

Volumes snapshots are taken according to the rule defined in the snapshot policy. In a multi-level structure, define the time interval in between two rounds of snapshots as well as the number of snapshots to retain. For example, take snapshots every x minutes/hours/days/weeks. There are one to six levels, with the first level having the most frequent snapshots.

Syntax strings

```
scli --add_snapshot_policy [--snapshot_policy_name <NAME>] --snapshot_creation_cadence
<CADENCE> --number_of_snapshots_per_retention_level  <LEVELS> [--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>] [--
paused] [--read_only_snapshots] [--secure_snapshots [--i_am_sure]]
```

Parameters

--snapshot_policy_name *<NAME>*

Snapshot policy name

--snapshot_creation_cadence *<CADENCE>*

The interval in minutes between two snapshots in the policy (minimum is 5).

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are one to six levels, and the first level has the most frequent snapshots. There can be up to 60 snapshots in all levels (two consecutive levels overlap by one).

--paused

Indicates that the snapshot policy should pause on creation.

--read_only_snapshots

Specify all mappings of the created auto snapshots to be limited to read-only access.

--secure_snapshots

Define all auto snapshots created as secure. It will not be possible to edit or remove them before the retention period is complete.

--i_am_sure

Preemptive approval for the creating the snapshots as secure.

--replication_consistency_group_id <ID>

Replication consistency group ID

--replication_consistency_group_name <NAME>

Replication consistency group name

When one of `--replication_consistency_group_id` or `--replication_consistency_group_name` is specified, the snapshot policy applies to that

replication consistency group rather than volumes. Thus, volumes cannot be added to this snapshot policy.

add_source_volume_to_snapshot_policy

Add a source volume to the snapshot policy specified by ID or name.

You can add multiple source volumes to a snapshot policy, but a specific volume can only be the source volume of a single policy. Only one volume per V-Tree may be used as a source volume of a policy.

Syntax

```
scli --add_source_volume_to_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

--source_vol_id *<ID>*

Source volume ID

--source_vol_name *<NAME>*

Source volume name

Example

```
scli --add_source_volume_to_snapshot_policy --snapshot_policy_name snap_pol_1 --
source_vol_id a7e798d100000000
```

align_snapshot_policy_to_current_time

Align the specified snapshot policy's cadence cycles to current time.

Syntax

```
scli --align_snapshot_policy_to_current_time (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Snapshot policy name

Example

```
scli --mdm_ip 192.168.1.200 --align_snapshot_policy_to_current_time --
snapshot_policy_name SNP1
```

lock_auto_snapshot

Lock an auto snapshot to prevent removal by the snapshot policy.

Syntax

```
scli --lock_auto_snapshot (--auto_snapshot_id <ID> | --auto_snapshot_name <NAME>
```

Parameters

--auto_snapshot_id *<ID>*
Auto snapshot ID

--auto_snapshot_name *<NAME>*
Auto snapshot name

Example

```
scli --lock_auto_snapshot --auto_snapshot_name auto_snapshot_1
```

modify_snapshot_policy

Modify the number of snapshots taken per minute and/or the number of snapshots per retention level for a specific policy

Syntax

```
scli --modify_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) ((--snapshot_creation_cadence <CADENCE> --
number_of_snapshots_per_retention_level <LEVELS>) | ([--set_read_only_snapshots] [--
set_secure_snapshots] [--apply_to_existing_snapshots])) [--i_am_sure]
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

--snapshot_creation_cadence *<CADENCE>*
The interval in minutes between two snapshots in the policy

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are six levels, and the first level has the most frequent snapshots. There can be up to 60 snapshots total in all levels (note that two consecutive levels overlap by one).

--set_read_only_snapshots

All mappings of the created auto snapshots will be limited to read-only access

--set_secure_snapshots

The auto snapshots will be created as secure. They cannot be edited or removed prior to their policy expiration time

--apply_to_existing_snapshots

Applies the `--set_read_only_snapshots` and `--set_secure_snapshots` flags to the policy's existing auto snapshots (excluding locked snapshots)

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --modify_snapshot_policy --snapshot_policy_name SP_pol1 --snapshot_creation_cadence
2 --numer of snapshots_per_retention_level 5 --set_read_only_snapshots --i_am_sure
```

pause_snapshot_policy

Pause a snapshot policy so that no auto snapshots will be created or removed.

Syntax

```
scli --pause_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

Example

```
scli --pause_snapshot_policy --snapshot_policy_name snap_pol_1
```

query_all_snapshot_policies

Retrieve information about all snapshot policies.

Syntax

```
scli --query_all_snapshot_policies
```

Parameters

None.

Example

```
scli --query_all_snapshot_policies
```

Sample output

```
Query all snapshot policies returned 3 policies
Snapshot Policy policy1 (Id: 36a39c6200000000)
        Cadence: 30 minutes
        Number of snapshots per retention level: 1 (read-only)
        State: Active
        Time of last auto snapshot: N/A
        Next run time: 2020-11-12 09:56:17
        Number of source volumes: 0
        Number of auto snapshots: 0

Snapshot Policy policy2_secure (Id: 36a3c37200000001)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1 (secure, read-only)
        State: Active
        Time of last auto snapshot: 2020-11-12 09:47:18
        Next run time: 2020-11-12 09:52:17
        Number of source volumes: 1
        Number of auto snapshots: 1

Snapshot Policy policy_read_write (Id: 36a3ea8200000002)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1
        State: Active
        Time of last auto snapshot: 2020-11-12 09:44:18
        Next run time: 2020-11-12 09:49:17
        Number of source volumes: 1
        Number of auto snapshots: 1
```

query_snapshot_policy

Retrieve information about a snapshot policy.

Syntax

```
scli --query_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
[--show_auto_snapshots]
```

Parameters

--snapshot_policy_id *<ID>*
: Snapshot policy ID

--snapshot_policy_name *<NAME>*
: Name of the snapshot policy

--show_auto_snapshots
: Indicate whether to display the full list of auto snapshots

Example

```
scli --query_snapshot_policy --snapshot_policy_name policy1 --show_auto_snapshots
```

Sample output

```
Snapshot Policy policy1 (Id: aaa4278700000000)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1
        State: Active
        Time of last auto snapshot: 2021-02-10 13:39:13
        Next run time: 2021-02-10 13:44:13
        Number of auto snapshots: 2

Source volume information (total 2 volumes):
         1: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB)
         2: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB)
```

remove_snapshot_policy

Remove the snapshot policy specified by name or ID. If this is a replication snapshot policy, specify what should be done with the auto snapshots.

Syntax

```
scli --remove_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--detach_auto_snapshots | (--remove_auto_snapshots [--
detach_locked_auto_snapshots]))
```

Parameters

--snapshot_policy_id *<ID>*
: Snapshot policy ID

--snapshot_policy_name *<NAME>*
: Name of the snapshot policy

--detach_auto_snapshots
: Detach all associated auto snapshots

--remove_auto_snapshots
: Remove all associated auto snapshots

--detach_locked_auto_snapshots
: When the remove option is specified, lock is detached and not removed.

Example

```
scli --remove_snapshot_policy --snapshot_policy_name SP_pol_1
```

remove_source_volume_from_snapshot_policy

Remove a source volume from the specified snapshot policy.

When removing the source volume from the policy, you must choose how to handle the snapshots created by the policy (auto snapshots). Your selection depends on if there are locked auto snapshots.

- If the source volume has no auto snapshots, you don't need to use the `--detach_auto_snapshots` or `--remove_auto_snapshots` options.
- If the source volume has auto snapshots but none of them are locked, you can choose to detach all snapshots with the `--detach_auto_snapshots` flag. They become regular snapshots as if they were created manually. If you use the `--remove_auto_snapshots`, the snapshots are deleted.
- If the source volume has locked auto snapshots, you can use `--detach_auto_snapshots` to turn the auto snapshots into regular snapshots, as if they were created manually. If you use `--remove_auto_snapshots`, the auto snapshots that are not locked are removed, while the auto snapshots that are locked are detached.

Syntax

```
scli --remove_source_volume_from_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>) (--
detach_auto_snapshots | (--remove_auto_snapshots [--detach_locked_auto_snapshots]))
```

Parameters

`--snapshot_policy_id <ID>`
Snapshot policy ID

`--snapshot_policy_name <NAME>`
Name of the snapshot policy

`--source_vol_id <ID>`
Source volume ID

`--source_vol_name <NAME>`
Source volume name

`--detach_auto_snapshots`
Detach all associated auto snapshots

`--remove_auto_snapshots`
Remove all associated auto snapshots

`-- detach_locked_auto_snapshots`
When the `remove_policy_snapshots` option is specified, any locked snapshots will not be removed, but rather detached.

Example

```
scli --remove_source_volume_from_snapshot_policy --snapshot_policy_name snap_pol_1 --
source_vol_id a7e798d100000000 --remove_auto_snapshots --detach_locked_auto_snapshots
```

rename_snapshot_policy

Name or rename the snapshot policy specified by name or ID.

Syntax

```
scli --rename_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>) --new_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

--new_name *<NAME>*
New name of the snapshot policy

Example

```
scli --rename_snapshot_policy --snapshot_policy_name SP_pol_1 --new_name snap_pol_1
```

resume_snapshot_policy

Resume a paused snapshot policy so that auto snapshots will once again be created and removed.

Syntax

```
scli --resume_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

Example

```
scli --resume_snapshot_policy --snapshot_policy_name snap_pol_1
```

unlock_auto_snapshot

Unlock an auto snapshot to allow the snapshot policy to remove it, if and when required.

Syntax

```
scli --unlock_auto_snapshot (--auto_snapshot_id <ID> | --auto_snapshot_name <NAME>) [--
auto_snapshot_will_be_removed]
```

Parameters

--auto_snapshot_id *<ID>*

Auto snapshot ID

--auto_snapshot_name *<NAME>*

Auto snapshot name

--auto_snapshot_will_be_removed

Acknowledge that if the auto snapshot has expired while it was locked, unlocking it will cause it to be removed.

Example

```
scli --unlock_auto_snapshot --auto_snapshot_name auto_snap_1 --
auto_snapshot_will_be_removed
```

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V-Tree commands

This section contains commands for managing V-Trees.

migrate_vtree

Migrate an entire V-Tree (a volume and all its snapshots) from one storage pool to another storage pool, within the same protection domain or between two different protection domains.

During V-Tree migration, you can run other tasks such as creating snapshots, deleting snapshots, and entering maintenance mode.

When a V-Tree migration is requested, the MDM first estimates whether the destination storage pool has enough capacity for the migration to complete successfully. The MDM bases the estimation on its information of the current capacity of the V-Tree. If there is insufficient capacity at the destination based on that estimate, migration does not start. (You can use the `--ignore_destination_capacity` option to force the migration even if there is insufficient capacity at the destination, with the intention to increase the capacity as required during the migration.) The MDM does not reserve the estimated capacity at the destination (since the capacity of the source volume can grow during migration and the reserved capacity does not guarantee success). The MDM does not hold onto source capacity once it has been migrated, but releases it immediately.

V-Tree migration can take a long time, depending on the size of the V-Tree and the system workload. During migration, the V-Tree is fully available for user I/O. V-Tree migration is done volume block by volume block. When a single block has completed its migration, the capacity of the block at the source becomes available, and it becomes active in the destination storage pool. During migration, the V-Tree has some of its blocks active in the source storage pool and the remaining blocks active in the destination storage pool.

When migrating from a medium granularity storage pool to a fine granularity storage pool, volumes must be zero padded.

NOTE: You cannot create snapshots when migrating from a medium granularity storage pool to a fine granularity storage pool.

Syntax

```
scli --migrate_vtree (--volume_id <ID> | --volume_name <NAME>) (((--
destination_protection_domain_id <ID> | --destination_protection_domain_name <NAME>) --
destination_storage_pool_name <NAME>) | --destination_storage_pool_id <ID>)
  [--migration_queue_position {head_of_queue | tail_of_queue}]
  [ --convert_to_thin_provisioned]
  [ --convert_to_thick_provisioned]
  [compression_method {none | normal}
  [--ignore_destination_capacity]
```

Parameters

`--volume_id <ID>`
Volume ID of the V-Tree to be migrated

`--volume_name <NAME>`
Volume name of the V-Tree to be migrated

`--destination_protection_domain_id <ID>`
ID of the destination protection domain

`--destination_protection_domain_name <NAME>`
Name of the destination protection domain

--destination_storage_pool_name *<NAME>*

Name of the destination storage pool

--destination_storage_pool_id *<ID>*

ID of the destination storage pool

NOTE: The migrateVTree command is also used for rollback volume migration while using the `destination_storage_pool_id`.

--migration_queue_position {head_of_queue | tail_of_queue}

Define whether this migration should be placed at the head or the tail of the migration queue. By default, it is placed at the tail or it retains its position if the V-Tree is already in migration.

--convert_to_thin_provisioned

Convert a thick-provisioned V-Tree to be thin-provisioned.

NOTE: SDCs with a version earlier than v3.0 do not fully support converting a thick-provisioned V-Tree to a thin-provisioned V-Tree during migration; after migration, the V-Tree is thin-provisioned, but the SDC is unable to trim it. These volumes can be trimmed by unmapping and then remapping them, or by rebooting the SDC. The SDC version does not affect capacity allocation and a V-Tree converted from thick to thin provisioning is reduced in size accordingly in the system.

--convert_to_thick_provisioned

Convert a thin-provisioned V-Tree to be thick-provisioned.

compression_method {none | normal}

Indicate the compression method to be used, relevant only for fine granularity data layouts. By default, the compression is according to the settings of the destination storage pool.

--ignore_destination_capacity

Force migration, even if currently there is not enough capacity at the destination.

Example

```
scli --set_vtree_migration_queue_position --volume_name vol1_test --
migration_queue_position head_of_queue
```

pause_vtree_migration

Pause migration of a V-Tree (a volume and all its snapshots), either gracefully or forcefully. Once paused, you can choose to resume the V-Tree migration, or to roll back the migration and have all volume blocks returned to the original storage pool.

Syntax

```
scli --pause_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
  [--pause_graceful]
  [--pause_forceful]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--pause_graceful

The pause is handled gracefully, allowing all data currently being moved to complete the migration.

--pause_forceful

The pause is handled forcefully, aborting unfinished data and leaving it at the source, if possible.

Example

```
scli --pause_vtree_migration --volume_name vol1_test --pause_ graceful
```

query_all_vtree_migrations

Retrieve the current status of all V-Tree migrations.

Syntax

```
scli --query_all_vtree_migrations
```

Parameters

None.

Example

```
scli --query_all_vtree_migrations
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Storage Pool 3df1cd9600000000 Name: sp1
VTree ID: 40e7f23600000005 (Name: vol3_pol1 Volume ID: f6351e3900000005)
VTree migration info:
        Migration status: Pending retry
        Pause Reason: No space in destination Storage Pool
        Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain: pd1 ID:
fcc9af5500000000
        Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain: pd2 ID:
fcc9af5600000001
        Conversion type: No conversion
        Queue position: 1
        Progress percentage : 0%

Protection Domain fcc9af5600000001 Name: pd2
Storage Pool 3df1cd9700000001 Name: sp2
 <No vtrees defined>

Protection Domain fcc9d66500000002 Name: rrr
 <No vtrees defined>
```

query_vtree

Retrieve information about the entire V-Tree (a volume and all of its snapshots) of the specified volume.

Syntax

```
scli --query_vtree (--vtree_id <ID> | --volume_id <ID> | --volume_name
```

Parameters

--vtree_id *<ID>*
V-Tree ID

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_vtree --volume_id f6351e3800000004
```

Sample output

```
VTree ID: 40e7f23500000004 Name: vol2_114
        Storage Pool 3df1cd9600000000 Name: sp1
        Protection Domain fcc9af5500000000 Name: pd1
        Data layout: Medium granularity
        Provisioning: Thin
        Total capacity in use: 0 Bytes
        Total user data: 0 Bytes
                Total base user data: 0 Bytes
                Total snapshots user data: 0 Bytes
        Number of Volumes: 2
        Migration status: Not in migration

>> Volume ID: f6351e3800000004 Name: vol2_114
   Provisioning: Thin
   Size: 8.0 GB (8192 MB)
   Total base user data: 0 Bytes
   Creation time: 2021-01-31 12:59:42
   Number of Direct Snapshots: 1
   Volume is not mapped
   No SCSI reservations
   The volume is a source volume of Snapshot Policy: aaa4278700000000
   Doesn't use RAM Read Cache
   Replication info:
      Replication state: Not being Replicated

   >> Volume ID: f66afb540000000a Name: N/A
      Auto Snapshot of vol2_114
      Size: 8.0 GB (8192 MB)
      Creation time: 2021-02-10 17:56:45
      SCG ID: 49645e1c00000003
      Volume is not mapped
      No SCSI reservations
      The volume is an auto snapshot of Snapshot Policy: aaa4278700000000
         Auto snapshot is unlocked
         Retention levels: 1
```

```
         Expected deletion time: 2021-02-10 18:02:46
      Doesn't use RAM Read Cache
      Replication info:
         Replication state: Not being Replicated
```

query_vtree_migration

Retrieve the current status of a V-Tree migration.

Syntax

```
scli --query_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

None.

--volume_id *<ID>*

Volume ID on VTree to query migration of

--volume_name *<NAME>*

Volume name on VTree to query migration of

Example

```
scli --query_vtree_migration --volume_name vol3_pol1
```

Sample output

```
VTree ID: 40e7f23600000005 Name: vol3_pol1
        Storage Pool 3df1cd9600000000 Name: sp1
        Protection Domain fcc9af5500000000 Name: pd1
        Data layout: Medium granularity
        Provisioning: Thin
        Total capacity in use: 0 Bytes
        Total user data: 0 Bytes
                Total base user data: 0 Bytes
                Total snapshots user data: 0 Bytes
        Number of Volumes: 3
        VTree migration info:
                Migration status: Paused
                Pause Reason: No space in destination Storage Pool
                Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain:
pd1 ID: fcc9af5500000000
                Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain:
pd2 ID: fcc9af5600000001
                Conversion type: No conversion
                Queue position: 1
                Progress percentage : 0%
```

resume_vtree_migration

Resume a paused V-Tree migration.

Syntax

```
scli --resume_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

None.

--volume_id *<ID>*

Volume ID of the V-Tree

--volume_name *<NAME>*

Volume name

Example

```
scli --resume_vtree_migration --volume_name vol1_test
```

rollback_vtree_migration

PowerFlex does not provide a dedicated SCLI command to roll back a V-Tree migration.

To roll back a paused V-Tree migration and return the volume and all associated snapshots to the original storage pool, reverse the migration direction and resume the migration. This workflow provides functionality equivalent to the rollback operation available in the PowerFlex GUI.

Prerequisites

Before you roll back a V-Tree migration:

- Ensure that a V-Tree migration is currently in progress.
- Pause the V-Tree migration before initiating the rollback procedure.
- Identify the original source Storage Pool to which the V-Tree will be migrated back.

Procedure

1. Query the current migration.

   Display the migration details and identify the source and destination storage pools.

   Syntax

   ```
   scli --query_vtree_migration --volume_name <VOL_NAME>
   ```

   Example

   ```
   scli --query_vtree_migration --volume_name vol1
   ```

2. Pause the migration.

   Pause the migration before initiating rollback. To force an immediate pause:

   ```
   scli --pause_vtree_migration --volume_name <VOL_NAME> --pause_forceful
   ```

To allow in-flight operations to complete before pausing:

```
scli --pause_vtree_migration --volume_name <VOL_NAME> --pause_graceful
```

3. Reverse the migration direction.

   Reverse the migration direction by migrating the V-Tree back to the original source storage pool.

   Syntax

   ```
   scli --migrate_vtree --volume_name <VOL_NAME> \
   --destination_storage_pool_name <ORIGINAL_SOURCE_SP>
   ```

   Example

   ```
   scli --migrate_vtree --volume_name vol1 \
   --destination_storage_pool_name sp_source
   ```

4. Optionaly, Move the rollback migration to the head of the migration queue.

   To prioritize the rollback migration:

   ```
   scli --set_vtree_migration_queue_position \
   --volume_name <VOL_NAME> \
   --migration_queue_position head_of_queue
   ```

5. Resume the migration.

   Resume the paused migration to begin moving the V-Tree back to the original storage pool.

   ```
   scli --resume_vtree_migration --volume_name <VOL_NAME>
   ```

6. Monitor rollback progress.

   Verify that the rollback migration is progressing and continue monitoring until the V-Tree is no longer reported as migrating.

   ```
   scli --query_vtree_migration --volume_name <VOL_NAME>
   ```

   NOTE: The rollback operation does not remove the migration object from the migration queue model. It reverses the migration direction and completes the migration back to the source storage pool.

set_vtree_migration_policy

Set the I/O priority policy for V-Tree migration for a specific storage pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_vtree_migration_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
--policy {optimize_migration | limit_concurrent_io | favor_application_io |
dynamic_bandwidth_throttling} (Options)
  [--i_am_sure]
```

Note about V-Tree migration throttling

During V-Tree migration, the source V-Tree is copied to two destination copies. This means that for each device in the source storage pool there are *X* MB/sec of reads, while in the destination storage pool there are 2*X* MB/sec of writes. If you set the `bandwidth_limit` to *X* MB/sec, it remains that at the source, while it increases to 2*X* MB/sec at the destination.

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for I/O priority of V-Tree migration. Can be one of the following:

limit_concurrent_io

Limit the number of allowed concurrent V-Tree migration I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

(Default) Limit the number of allowed concurrent V-Tree migration I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent V-Tree migration I/Os per device. Valid range: 1-20.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of V-Tree migration I/Os, in KB/s, per device. The valid range is 1024-25600 (25 MB). This property is effective only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_vtree_migration_policy --protection_domain_name pd10 --storage_pool_name sp20
--policy favor_application_io --bandwidth_limit 5120
```

set_vtree_migration_queue_position

Set the position of a V-Tree migration to be at the head or the tail of the migration queue.

Syntax

```
scli --set_vtree_migration_queue_position (--volume_id <ID>| --volume_name <NAME>)  --
migration_queue_position {head_of_queue | tail_of_queue}
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--migration_queue_position

Queue position of the V-Tree migration:

- `head_of_queue`
- `tail_of_queue`

Example

```
scli --set_vtree_migration_queue_position --volume_name vol1_test --
migration_queue_position head_of_queue
```